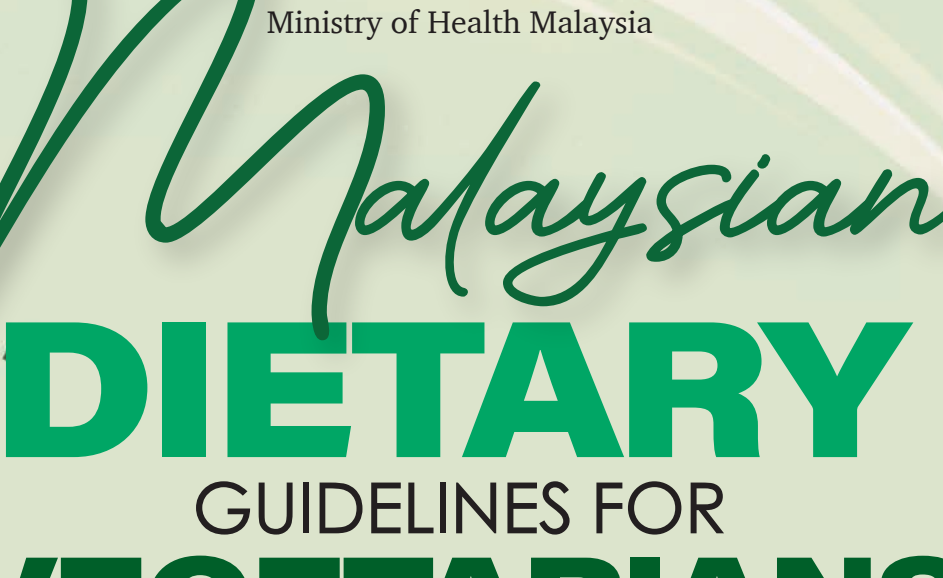




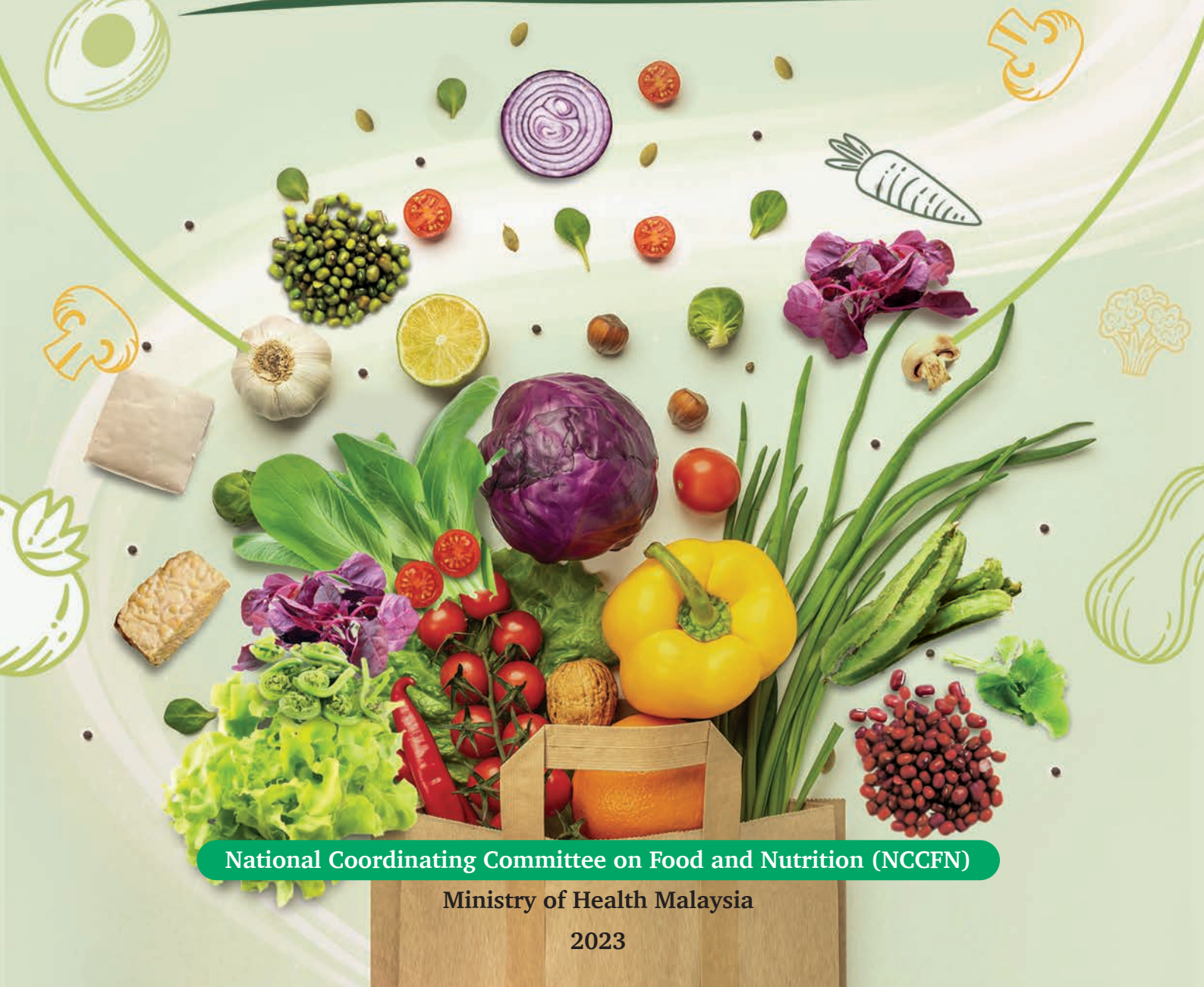
Ministry of Health Malaysia

Malaysian

DIETARY

GUIDELINES FOR

VEGETARIANS



2023



Ministry of Health Malaysia

MALAYSIAN DIETARY GUIDELINES FOR VEGETARIANS

National Coordinating Committee on Food and Nutrition (NCCFN)
Ministry of Health Malaysia
2023

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Ministry of Health Malaysia

MALAYSIAN DIETARY GUIDELINES FOR VEGETARIANS



National Coordinating Committee on Food and Nutrition (NCCFN)
Ministry of Health Malaysia
2023

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Key Message 2 : Eat plenty of vegetables and fruits everyday

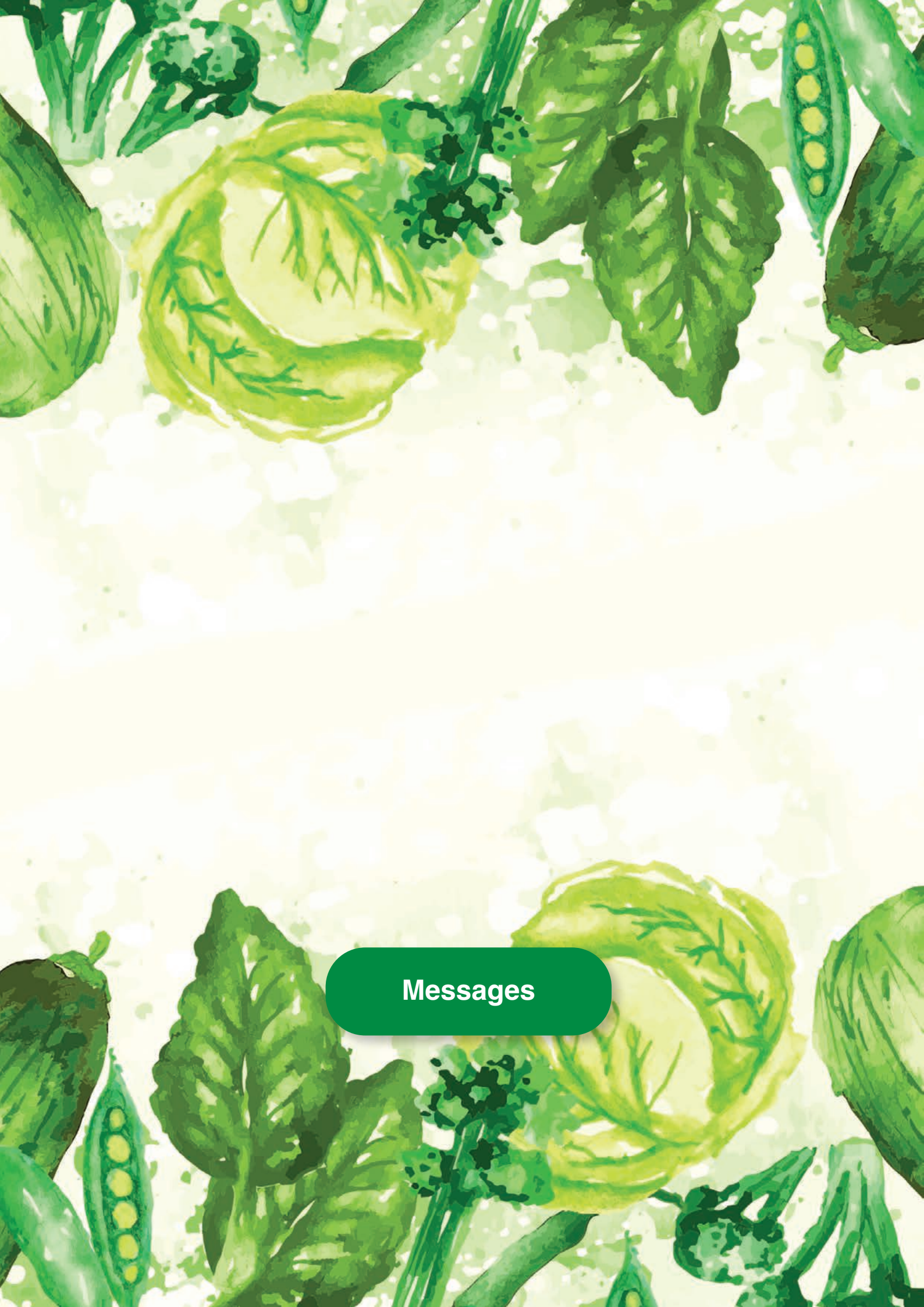
Key Message 3 : Eat adequate amounts of rice, other cereals, whole grain cereal-based products and tubers

Key Message 4 : Consume adequate amounts of legumes and nuts and moderate amounts of eggs

Key Message 5 : Consume adequate amounts of milk and milk products

Key Message 6 : Include appropriate amounts and types of fat in the diet

Key Message 7 : Make effective use of information on food labels



Messages

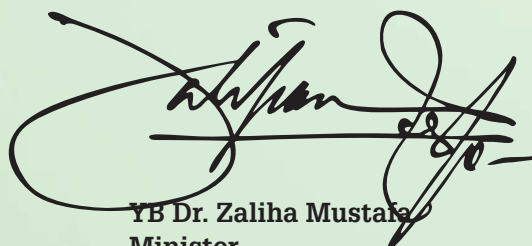
Message by
Minister
Ministry of Health Malaysia



In Malaysia, the prevalence of morbidity and mortality associated with chronic diseases caused by poor dietary patterns and excessive calorie intake is on the rise and this is cause for concern. This situation also poses threats to the environment and future food security due to the unsustainable food system. Studies show that an increasing number of individuals are switching to plant-based diets for ethical and health reasons, such as worries about climate change and animal welfare, which contribute to the sustainability of the food system.

The popularity of plant-based (vegetarian) diets is increasing among Malaysians too, with various forms of vegetarianism gaining recognition for their health benefits and potential in reducing the risk of chronic illnesses. The Malaysian Ministry of Health is committed to improving the health of its population by providing appropriate guidance on nutrition and food preferences. The establishment of the Malaysian Dietary Guidelines for Vegetarians 2023 (MDGV) is timely, with the aim to support vegetarians and assist them in achieving the acceptable recommended nutrient intakes for Malaysian vegetarians. The dietary guideline prioritises health promotion and disease prevention, promoting the adoption of plant-based diets for all age groups. However, it is crucial to approach a vegetarian diet with careful planning.

Therefore, it is anticipated that all the relevant ministries, agencies, academic institutions, hospitals, the healthcare industries, professional organisations, and the food industries will apply the MDGV 2023 to meet the demands of the population for such dietary guidance. The initiative of the Technical Working Group on Nutritional Guidelines, operating under the aegis of the National Coordinating Committee on Food and Nutrition (NCCFN), in developing this vegetarian guideline for practicing vegetarians in the country, is commendable.

A stylized black ink signature of YB Dr. Zaliha Mustafa.

YB Dr. Zaliha Mustafa
Minister
Ministry of Health Malaysia

Foreword by **Director-General of Health Ministry of Health Malaysia**

The release of the Malaysian Dietary Guidelines for Vegetarians 2023 (MDGV) is timely given that Malaysians' eating habits have changed to also include a vegetarian or plant-based diet. Despite the fact that eating and nutrition practises having altered over the past few decades, our community continues to struggle with the double burden of malnutrition and the increased prevalence of non-communicable diseases. The findings of National and Health Morbidity Survey (2019), had presented that 4.9% of Malaysia's adult population were vegetarian for a variety of reasons.

It is important for people and vegetarians in particular to acquire more practical knowledge that is applicable for their daily life. Therefore, comprehensive nutritional guidelines that are focusing on a specific population has been established and disseminated to the public through various platforms. The MDGV is a nutrition education tool that provides scientific evidence to promote healthy eating practices to reduce the risk for chronic diseases and ensure adequate essential micronutrient intake among vegetarians.

It is my fervent hope that all the healthcare professionals will make use of the MDGV as their practical guidance in ensuring that vital nutrition information is effectively communicated and not overlooked. I would also like extend my congratulations to the Technical Working Group on Nutritional Guidelines for their tremendous efforts for the completion of document.



Datuk Dr. Muhammad Radzi Bin Abu Hassan
Director-General of Health
Ministry of Health Malaysia

Preface by
Deputy Director-General of Health (Public Health)
Ministry of Health Malaysia



Ensuring adherence to nutrient recommendations plays a pivotal role in preserving the health and well-being of individuals and specific groups within the country. The government has been actively promoting healthy eating practices as part of its ongoing healthy lifestyle campaign to combat non-communicable diseases linked to nutrition. Across the globe, the vegetarian diet or plant-based diet has gained popularity as a health-oriented choice.

While the prevalence of vegetarianism among Malaysians remains uncertain, there is a noticeable increase in its adoption due to concerns surrounding health, religion, the environment, animal welfare, ethics, and food sustainability. In light of this trend, the Malaysian Dietary Guidelines for Vegetarians (MDGV) has been established to provide appropriate dietary guidance, ensuring a well-balanced diet for everyone. Remarkably, a well-planned vegetarian diet offers numerous health benefits and can be suitable for all life stages.

Special appreciation goes to the Technical Working Group on Nutritional Guidelines and the sub-committee on MDG for Vegetarians for their tremendous efforts in successfully publishing the vegetarian guidelines. My heartfelt gratitude also goes to all the individuals who have contributed in various ways towards the completion of the Malaysian Dietary Guidelines for Vegetarians.

Datuk Dr. Norhayati Binti Rusli
Deputy Director-General (Public Health)
Ministry of Health Malaysia

Preface by Chairman Malaysian Dietary Guidelines for Vegetarians

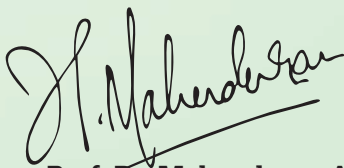
Strengthening a continuous healthy eating habit is one of the fundamental efforts to ensure optimal nutritional requirements that result in overall good quality of life. Healthy eating habits should be practised in any type of diet. In Malaysia, there are about 4.9% who are practising a vegetarian diet due to different reasons such as to support animal rights, religion reasons, to help save the environment and to improve health.

A vegetarian diet will typically closely resemble the healthy eating guidelines for the general population since it is low in saturated fat, rich in fibre (with lots of wholegrains, fruits, and vegetables), provides moderate levels of protein, and includes little additional fats and carbohydrates. A varied and well balanced vegetarian diet can provide all of the nutrients needed for good health. However, a vegetarian diet requires careful planning, particularly for new vegetarians, to ensure nutritional needs are met. Key nutrients should be addressed while designing a vegetarian diet, according to Malaysian Recommended Nutrient Intakes (RNI), 2017, Malaysian Dietary Guidelines 2020 and Malaysian Food Pyramid 2020.

Hence, the Malaysian Dietary Guidelines for Vegetarians were created as a useful tool for delivering nutrition advice and guidance based on a realistic and comprehensible scientific viewpoint. The MDGV comprises 7 Key Messages (KM), prepared by the appointed experts according to each KM, under the coordination of Technical Working Group for Nutritional Guidelines (TWG for Nutritional Guidelines). It is hoped this guidelines will be used as a reference by health professionals and other related stakeholders in order to assist the vegetarian population to maintain their healthy eating habits.

I want to express my gratitude to the Technical Working Group participants, the authors, the participants in the focus groups and consensus workshops, the editorial team, the TWG Secretariat, and all others who contributed to the development of this important nutrition reference book for vegetarians.

Additionally, I would like to express my gratitude to Cik Rusidah Selamat, Director of Nutrition Division, Ministry of Health Malaysia, and Emeritus Professor Dr. Mohd Ismail Noor, Chairman of TWG for Nutritional Guidelines, for their invaluable cooperation and guidance in developing this nutrition guide for Malaysian practicing vegetarianism.



Assoc. Prof. Dr. Mahenderan Appukutty
Chairman
Malaysian Dietary Guidelines for Vegetarians



Acknowledgement

Individuals

from the various Departments and Institutes, the Ministry of Health Malaysia. Academicians from local universities, nutritionists, dietitians, representatives from related professional bodies, representatives from the food manufacturing and trading industry, and consumer bodies are all acknowledged by the Technical Working Group on Nutritional Guidelines. Their invaluable contributions and dedication to complete this document successfully are sincerely appreciated.

A word of gratitude is also conveyed to the:

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- Dean, Faculty of Medicine and Health Sciences, Universiti Putra Malaysia
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for their generous support and cooperation.

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of the Malaysian Dietary Guidelines (MDG) for Vegetarians 2023 was coordinated by the Technical Working Group (TWG) on Nutritional Guidelines, which is under the purview of the National Coordinating Committee on Food and Nutrition (NCCFN),

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Executive Summary

In recent years, there has been a global surge of interest to adopt a vegetarian style of eating. There is no exact definition for the term vegetarian but it is commonly used to describe a plant based food diet which devoid of animal-based foods such as meat, fish and poultry. This practice has been associated with potential health benefits, cultural and religious beliefs, and concern over environment and animal welfare.

Numerous studies and observations have found that vegetarians generally have a higher consumption of plant foods especially fruits and vegetables, which are rich in phytochemicals, known to alter the risk of diet related diseases. Researchers have highlighted that vegetarians have a lower intake of cholesterol, saturated fat and total fat, and a higher intake of fibre as

compared to non-vegetarian counterparts. The incidence of chronic diseases and conditions such as obesity, coronary artery disease, hypertension, diabetes, cancer and mortality rates are thus lower among vegetarians. As there is now a wealth of evidence to support the health benefits of a vegetarian diet, more people are beginning to practise various forms of vegetarian diet.

Although the prevalence of vegetarianism among Malaysians is unknown, increasing concerns over religious, environmental, animal and ethical issues, as well as concerns about food sustainability and world hunger are turning many towards practising a vegetarian diet. Planning and consuming a vegetarian diet is not as simple as just avoiding meat. An individual need to take extra care to ensure that he or she will be able to meet the daily nutritional needs. In planning nutritionally adequate vegetarian meals, special attention is needed to replace meat and other animal products with nutritionally equivalent vegetarian foods.

It is important for people in general and vegetarians in particular to acquire more knowledge, both theoretical and practical, on how to prepare a healthy diet. Lack of appropriate guidelines and knowledge about vegetarian meal planning may lead to potential imbalances in intake of nutrients. NGOs and individuals have been requesting for a vegetarian dietary guideline that would help them plan their daily menu without compromising nutritional values. Therefore, it is timely to produce a well-planned dietary guideline that is especially suited for vegetarians to maintain health, prevent chronic diseases and prevent nutritional deficiencies.

The Malaysian Dietary Guidelines for Vegetarians (MDGV) 2023 was developed by a committee comprising nutritionists and dietitians from various sectors including government agencies, food and nutrition industries, academia and relevant NGOs

such as the Malaysian Vegetarian Society. This document is based on established scientific evidence and complements the Malaysian Dietary Guidelines 2020. The MDGV 2023 is designed in such a way that it is suitable for people with different cultural backgrounds, religions and food practices. The paramount aim of this guideline is to provide practical guidance for planning a well-balanced, plant-based diet without compromising an individual's health.

This vegetarian dietary guideline comprises 7 key messages, with key recommendations and 'how to achieve' that addresses risk of nutrient deficiencies due to potential inadequate intakes of specific nutrients such as vitamin B12, omega-3, iron, calcium and protein. Recommendations on number of servings related to specific vegetarian food items consumed by various ethnic groups are also given in this guideline.

The MDGV 2023 is also useful for planning healthy diets for children (from 6 months and above), adolescents, adult, older persons, pregnant and lactating mothers who are practising a vegetarian diet. Policy makers, nutritionists, dietitians, other related healthcare providers and the public can also benefit from this guideline. The messages in this guideline could be translated into education materials, as well as for planning and implementing nutrition related programmes and policies for vegetarians.

The following is the list of Key Messages for the Malaysian Dietary Guidelines for Vegetarians:

Key Message 1	:	Eat a variety of foods within the recommended servings.
Key Message 2	:	Eat plenty of vegetables and fruits everyday
Key Message 3	:	Eat adequate amounts of rice, other cereals, whole grain cereal-based products and tubers
Key Message 4	:	Consume adequate amounts of legumes and nuts and moderate amounts of eggs.
Key Message 5	:	Consume adequate amounts of milk and milk products.
Key Message 6	:	Include appropriate amounts and types of fats in the diets.
Key Message 7	:	Make effective use of nutrition information on food labels.



Introduction

History of vegetarianism

Vegetarianism has a long and rich history, predating the 20th century by centuries. Its origins can be traced back to as early as the sixth century, with anthropologists suggesting that ancient humans primarily consumed a plant-based diet due to the resemblance of their digestive system to that of herbivores (Andrew et al., 2020; Philips, 2005). The practice of vegetarianism has sustained civilisations for thousands of years, driven by moral, religious, and economic factors (Hargreaves et al., 2021; Philips, 2005).

Evidence of vegetarianism can be found in various ancient cultures, including Egypt, China, India, the Middle East, Mexico, and Peru. These societies relied on staple foods such as wheat, barley, soya, sorghum, corn, rice, lentils, and other grains and vegetables (Leitzmann, 2014). The adoption and advocacy of a meatless diet can be attributed to the ancient Greek philosopher Pythagoras, who played a significant role in popularising vegetarianism in Greece (Hargreaves et al., 2021; Holler et al., 2021). The meatless diet way of eating became known as the "Pythagorean diet" and persisted for many years.

However, it was during the 19th and 20th centuries that the modern vegetarian movement gained momentum. This movement, which began in the mid-1800s, further promoted and popularised the

idea of abstaining from meat (Hargreaves et al., 2021; Andrew et al., 2020). It marked a significant turning point in the widespread acceptance and recognition of vegetarianism as a viable dietary choice.

In conclusion, vegetarianism has deep historical roots that stretch back to ancient times. Its practice has been sustained by various civilisations for thousands of years, driven by moral, religious, and economic motivations. The influence of figures like Pythagoras and the development of the modern vegetarian movement in the 19th and 20th centuries have further solidified the prominence of vegetarianism as a dietary choice.

Developments in vegetarianism

In the past few decades of the 20th century, extensive research has shed light on the numerous health advantages associated with a plant-based diet. Ecological studies conducted in African, Asian, and Mediterranean countries have consistently shown that populations following predominantly plant-based nutrition in these regions tend to have lower rates of non-communicable diseases related to diet and enjoy longer lifespans (Leitzmann, 2014).

Furthermore, studies conducted on the Californian Seventh Day Adventists (SDAs) in the late 1970s

revealed interesting findings. Approximately half of the SDAs were vegetarians, and their mortality rates from major types of cancer were lower compared to the general Californian population (Kwok et al., 2014). Moreover, research has indicated that vegetarians have a reduced risk of developing ischemic heart disease and cardiovascular disease (excluding stroke) compared to non-vegetarians (Dybvik et al., 2023; Koutentakis et al., 2023). Longevity has also been observed among long-term vegetarians (Marsh, Zeuschner & Saunders, 2012).

Research has also revealed that vegetarians have lower rates of a number of other health problems including type 2 diabetes, cancer, gallstones, kidney diseases, constipation and diverticular disease (Huang et al., 2012; Dinu et al., 2017; Leitzmann, 2005). Other studies have shown that vegetarians have lower rates of several other health conditions, including type 2 diabetes, cancer, gallstones, kidney diseases, constipation, and diverticular disease (Huang et al., 2012; Dinu et al., 2017; Leitzmann, 2005).

In recent decades, advancements in nutrition research have transformed scientists' understanding of the impact of vegetarian diets on human health and diseases. Building upon these findings, Sabaté (2003) proposed three paradigm models (Figure A1, A2 and A3) to

demonstrate the positive effects of a vegetarian diet on nutritional well-being compared to a meat-based diet. The initial model depicts the belief prevalent in the 1960s and 1970s that individuals following a vegetarian diet faced a higher risk of nutrient deficiencies than those adhering to a meat-based diet. During this period, the potential preventive benefits of a vegetarian diet against chronic and degenerative diseases had not yet been recognized (Figure A1). Additionally, Figure A2 portrays the model that prevailed in the 1980s and 1990s, highlighting the health risks and benefits associated with vegetarian and meat-based diets, including the dangers of excessive meat consumption. Finally, a third paradigm model has emerged, supported by a growing body of epidemiological, clinical, and scientific evidence, illustrating the numerous health advantages of vegetarian diets (Figure A3) compared to meat-based diets (Boeing et al., 2012; Marsh et al., 2012; Hu, 2003).

Sabaté (2003) presented a series of models that highlight the progress made in scientific research. These models emphasize that diets primarily composed of plant-based foods, including balanced vegetarian diets, are now perceived as promoting health instead of being linked to diseases, in contrast to meat-based diets.

Figure A1
Early model on the adequacy of vegetarian diets that prevailed through the 1960s.

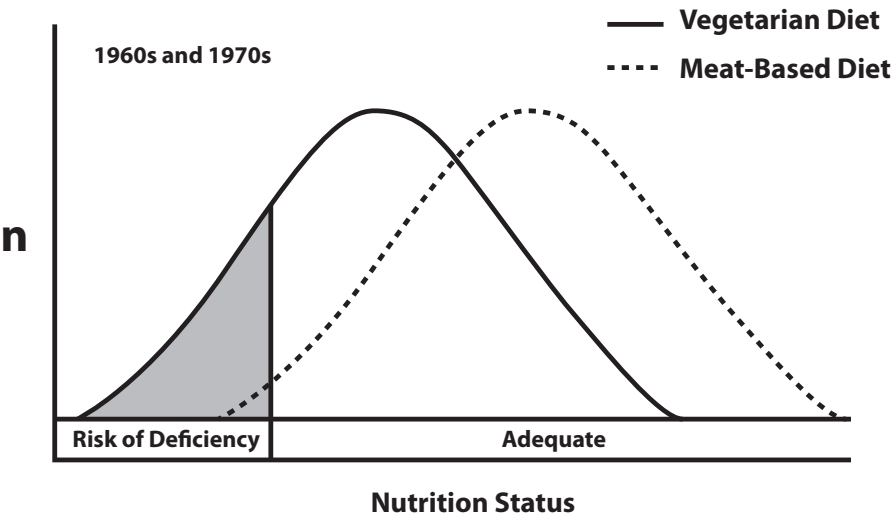


Figure A2
The model on population health risks and benefits of vegetarian and meat-based diets that prevailed during the 1980s and 1990s.

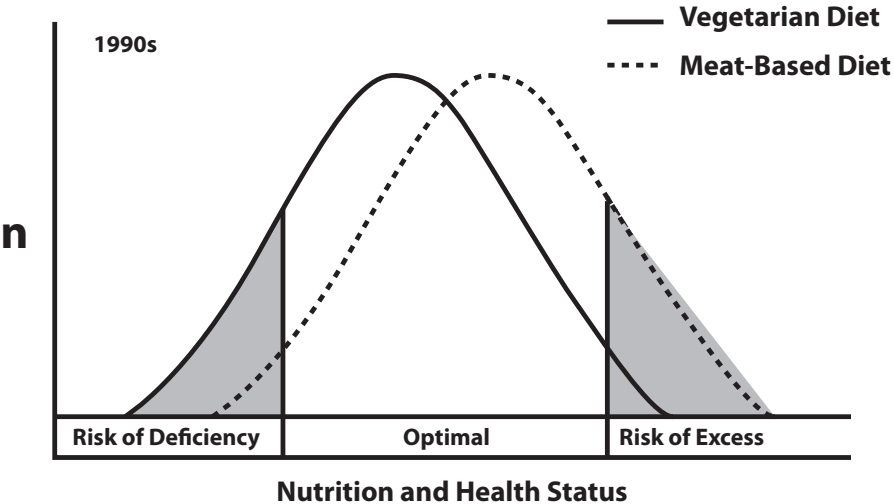
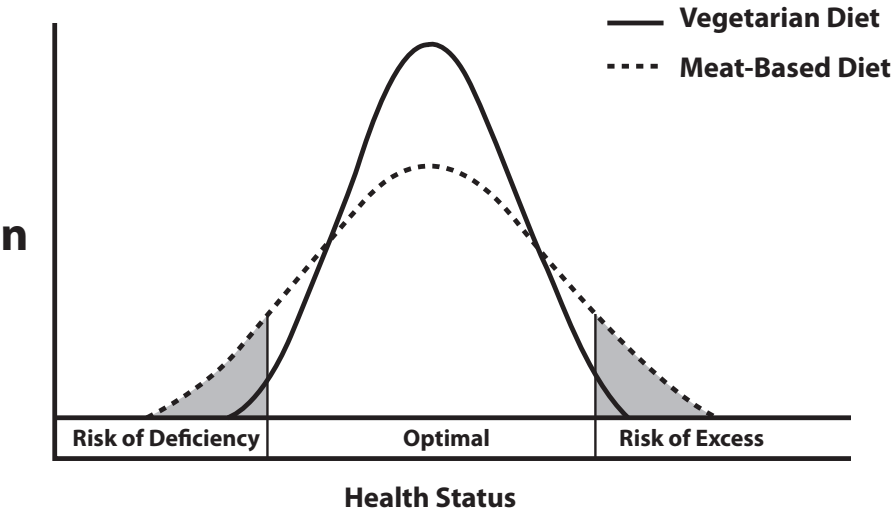


Figure A3
The model that prevails today. The area under the curve represents the proportion of individuals in a population for which a given diet pattern may be a health risk or benefit optimal). At both extremes of the health continuum, there is risk of disease for deficiency or excess of nutrients or other food compounds.



Vegetarian diets are rich in bio-active phytochemicals (phyto means plant in Greek), which are inadequate in a non-vegetarian diet. Phytochemicals are naturally occurring non-nutritive biochemicals found in vegetables, fruits, herbs, flowers, legumes and roots. These compounds not only provide colour, flavour, aroma, and texture but, more importantly, possess various protective and disease-preventive properties that work in conjunction with nutrients (Hollman & Katan, 1997; Khor et al., 2000; Appleby et al., 2002; Kahleova et al., 2017). The primary groups of phytochemicals contributing to the overall antioxidant capacity of plants are polyphenols and vitamins such as vitamin C and E. These components have the potential to prevent the initiation and/or progression of numerous human diseases (Ullah et al., 2020). These findings demonstrate that diets primarily based on plant foods not only prevent nutrient deficiencies but also mitigate the risk of diet-related chronic diseases. Vegetarian diets generally have lower levels of total fat, cholesterol, and protein, while being higher in complex carbohydrates, which offers a protective effect against weight gain.

These findings collectively highlight the potential benefits of adopting a plant-based diet for overall health and well-being.

Global practice of vegetarianism

A plant-based or vegetarian diet revolves around consuming predominantly plant-derived foods, including whole grains, legumes, beans, nuts, seeds, oils, fruits, and vegetables. Notably, numerous influential historical figures, such as Pythagoras, Plato, George Bernard Shaw, Mahatma Gandhi, Albert Einstein, Leonardo da Vinci, and Leo Tolstoy, have embraced vegetarianism (Leitzmann, 2014). The practice of vegetarianism takes on various forms worldwide (Fehér et al., 2020), exhibiting regional and cultural variations. India has one of the highest rates of vegetarianism, while Israel, Italy, Germany, Taiwan, and certain European nations have notably high rates of vegetarianism. In North America, the percentage is 6 percent, while in Europe, it is 5 percent (Memon et al., 2021). Approximately 16 percent of individuals in Africa and the Middle East, and 8 percent in Central and South America, are vegetarians.

Moreover, there has been an increasing number of leaders in the arts, sciences, music and sports who are now supporting the vegetarian movement, which will indirectly influence the general public to follow in their footsteps (Leitzmann., 2014). Table 1 lists the distribution of vegetarians in selected countries.

Table 1: Distribution of vegetarians in selected countries

Country	Population	No. of vegetarians	Proportion of vegetarians
	Million	Million	%
India	1,260	450	35
Italy	61	5.9	9
Great Britain	63	5.4	9
Germany	82	7.4	9
Netherlands	17	0.7	4
United States	320	12.1	4
Canada	35	1.3	4
Austria	8	0.25	3
Switzerland	8	0.23	3
France	64	1.2	2

(Source: Leitzmann, 2014)

There is a growing trend towards vegetarianism in Asia, as consumers increasingly demand for vegetarian and vegan food products. Over the years, the markets for vegan and organic foods in China and India have undergone significant changes (Memon et al., 2021). Furthermore, China's latest dietary guidelines promote the consumption of ample cereals, tubers, whole grains, legumes, vegetables, and fruits, while recommending to limit meat intake (Wang et al., 2016). A similar approach is evident in the current Malaysian dietary guidelines and the Malaysian Healthy Plate concept (NCCFN, 2021).

In Malaysia, the vegetarian community has been growing steadily as evidenced by the increasing number of vegetarian and vegan food outlets. It has been reported that there are 1,185 vegetarian-friendly restaurants throughout Malaysia (The Sun, 2017). Malaysia was ranked as the third most vegetarian-friendly destination in the world by the Global Vegetarian Index, with a score of 311 (Koh, 2016). This index was created as a guide to help meat-free travellers choose their preferred diet. The Global Vegetarian Index for countries surveyed was based on the number of vegetarian restaurants, number of vegetarian restaurants in relation to population size and annual meat consumption per capita (kg).

Types of vegetarian diets

The term "vegetarian" is commonly used to encompass various dietary practices observed by different populations, each with its own level of restrictions. Presently, vegetarians can be categorised into several subgroups based on their specific dietary choices. According to the Quick Survey on Vegetarian Practice in Malaysia, out of the 470 individuals who identified as vegetarians, the majority (56%) follow lacto-vegetarian diets, followed by ovo-vegetarian (16%) and vegan diets (16%) (MOH, 2014). The study also found that the majority of vegetarians (86%) adhere to their vegetarian diets on a full-time basis, while a smaller percentage fall under the category of semi-vegetarians who practice vegetarianism occasionally during selected periods.

In Malaysia, there are several common types of vegetarian diets practiced as below:

- **Lacto-ovo-vegetarian:** Lacto-ovo-vegetarians exclude meat, fish, and poultry from their diet but include dairy products (lacto) and eggs (ovo). Lacto-ovo-vegetarianism is a popular choice among individuals who prefer to include dairy

products and eggs in their vegetarian diet while abstaining from consuming meat, fish, and poultry.

- **Lacto-vegetarian:** This type of vegetarian diet includes dairy products along with plant-based foods. Individuals following a lacto-vegetarian diet abstain from consuming meat, fish, poultry, and eggs, but they do consume dairy products such as milk, cheese, and yogurt.
- **Ovo-vegetarian:** Ovo-vegetarians exclude meat, fish, poultry, and dairy products from their diet but include eggs.
- **Vegan:** Vegans follow a strict plant-based diet and avoid all animal products, including meat, fish, poultry, dairy, eggs, and honey. They rely solely on plant-based foods for their nutrition.

In addition to the common types of vegetarians, there are a few other types of vegetarian diets, such as:

- **Pescatarian:** Pescatarians follow a primarily vegetarian diet but include fish and other seafood in their meals. They abstain from consuming meat and poultry but include plant-based foods, dairy products, eggs, and seafood.
- **Flexitarian or Semi-Vegetarian:** Flexitarians, also known as semi-vegetarians, primarily follow a vegetarian diet but occasionally include small amounts of meat, fish, or poultry in their meals. Their diet mostly consists of plant-based foods, with occasional flexibilities.
- **Raw Vegan:** Raw vegans consume only raw and unprocessed plant-based foods. Their diet primarily consists of fruits, vegetables, nuts, seeds, and sprouted grains. They avoid cooking or heating their food above a certain temperature to preserve its natural enzymes and nutrients.
- **Macrobiotic:** Macrobiotic diets emphasise whole grains, unpolished rice, pulses, legumes, vegetables, fermented foods, nuts, seeds and seaweed, while limiting the consumption of refined sugars, and processed foods. The diet is often followed for its perceived health benefits and adherence to balancing yin and yang elements in food and this diet started in Japan.

It's important to note that the specific details and variations of these diets may differ among individuals, and some individuals may adopt a combination of different vegetarian dietary practices based on their personal dietary preferences and beliefs.

Nutritional adequacy of a vegetarian diet

The key to a healthy vegetarian diet is to enjoy a variety of foods. No single food can provide all the necessary nutrients required by the body. The more restrictive a person's diet is, the more challenging it can be to obtain all the essential nutrients. For example, in a vegan diet, there is absence of natural sources of vitamin B12 and the exclusion of milk and dairy products leads to lower calcium intake.

However, with careful planning, a well-balanced vegetarian or vegan diet can ensure that the diet is complete with all nutrients needed by the body. By ensuring adequate intake of these key nutrients, individuals following a vegetarian diet can maintain a well-rounded and nutritionally complete eating plan. Individuals following a vegetarian diet should pay special attention to the following nutrients:

Protein helps to maintain healthy skin, bones, muscles and organs. Daily consumption of a variety of plant foods can provide an adequate supply of essential amino acids, which are the building blocks of protein. Plant sources of protein include soya products and meat substitutes, legumes, beans, lentils, nuts, seeds and whole grains. The regular intake of a combination of legumes, soya products and other grains will ensure an adequate protein supply with all the essential amino acids and other nutrients for vegetarians.

Vitamin B12 plays a crucial role in red blood cell production and the prevention of anaemia. As plant-based foods do not naturally contain vitamin B12, it is important for individuals following a vegetarian diet to incorporate alternative sources or fortified foods to meet the recommended intake. Vegans, in particular, are recommended to include vitamin supplements, vitamin-enriched cereals, and fortified products in their diet. For example, fortified plant-based milk, breakfast cereals, and nutritional yeast are common sources of vitamin B12 for vegans.

Vitamin D plays an important role in bone health. Sufficient vitamin D can be obtained from food especially fortified foods and sun exposure.

Calcium helps build and maintain strong teeth and bones. While dairy products, which are rich in calcium, are often limited in vegetarian diets, there are alternative sources to consider. In the case of a vegan diet that excludes milk, it is important to incorporate other calcium sources such as dark green vegetables, calcium-enriched and fortified products like juices, cereals, soya milk, soya yogurt, and tofu. Alongside a balanced diet, regular exercise further promotes optimal bone health.

Iodine is a component in thyroid hormones that help to regulate the metabolism, growth and function of key organs. Vegans may not get enough iodine and be at risk of deficiency and possibly, goitre. Soya beans, cruciferous vegetables and sweet potatoes may contain goitrogens, substances that can promote goiter formation. However, incorporating these foods in moderation and focusing on non-goitrogenic options can help vegans meet their iodine requirements. Non-goitrogenic foods rich in iodine include seaweed, kelp, pumpkin, tomato, bell peppers, green beans, peas, cucumber, asparagus, eggplant, carrots, and celery. Some non-goitrogenic fruits vegans should eat include mangoes, citrus fruits, blueberries, dark-skinned grapes, pomegranates, guavas, cherries, apricots, apples, pineapples and kiwis. The use of iodised salt as much as 1/4 teaspoon a day can provide a significant amount of iodine.

Iron is an essential component of red blood cells, and in an omnivorous diet, heme-iron derived from meat, particularly red meat, is easily absorbed by the body. However, in a vegetarian diet, heme-iron is replaced by non-heme iron obtained from sources like dried beans and peas, lentils, enriched cereals, whole-grain products, dark leafy green vegetables, and dried fruits. The bioavailability of non-heme iron depends on the balance between inhibitors (such as phytates and polyphenolics found in certain vegetables and legumes) and enhancers (such as vitamin C, citric acid, and other organic acids). To optimize iron absorption in plant-based diets, it is important to consume iron-rich sources like legumes, whole grains, tofu, nuts, and seeds alongside foods rich in vitamin C such as guava, citrus fruits, tomatoes, cabbage, and broccoli to enhance iron absorption.

Zinc is an essential component of many enzymes and it plays a role in cell division and in the formation of proteins. Plant sources of zinc include whole grains, soy products, legumes, nuts and wheat germ.

Omega-3 fatty acids are important for heart health. While fish is a common source of omega-3 fatty acids, there are vegetarian-friendly options available. Flaxseeds and flaxseed oil, chia seeds, hemp seeds and walnut are excellent plant-based sources of omega-3 fatty acids. They contain alpha-linolenic acid (ALA), a type of omega-3 fatty acid. Algal oil is derived from algae and is another vegetarian source of omega-3 fatty acids, including docosahexaenoic acid (DHA) and eicosapentaenoic acid (EPA). Algal oil supplements are available and can be a convenient way to ensure adequate omega-3 intake. Including these plant-based sources of omega-3 fatty acids in a vegetarian diet can help meet nutritional needs. It's important to note that while ALA can be converted to EPA and DHA in the body, the conversion rate is relatively low. However, the conversion of plant-based omega-3 to the types used by humans is inefficient, it is, therefore, necessary for vegetarians to take fortified products or supplements, or both.

A carefully planned vegetarian diet proven to provide all the necessary nutrients to support overall health and well-being. These diets not only maximise the intake of beneficial phytochemicals found in plant foods but also help maintain body weight and prevent chronic diseases. However, many vegetarians may be unaware of the health benefits associated with plant-based diets, emphasising the need for ongoing nutrition education (Rizzo et al., 2013; Martin et al., 2013).

Vegetarian diets with proper planning can provide all the nutrients needed to support health at every stage of life, including pregnancy, lactation, infancy, childhood, and adolescence, as well as for athletes or other physically active people. For children who practise vegetarianism, attention should be given to plant foods that are rich in protein, calcium, iron and zinc that will provide all required nutrients to support growth and development in children. According to the American Dietetic Association's (ADA) position paper on vegetarian diets, appropriately planned vegan and lacto-ovo-

vegetarian diets satisfy the nutrient needs of infants, children and adolescents, and promote normal growth (Melina et al., 2016; Yen et al., 2008). Nevertheless, parents should consult the help of a nutritionist or dietitian for personalised guidance and support for making informed decisions on the child's dietary choices.

Athletes, in particular, should collaborate with sports nutritionists or sports dietitians, along with trainers or coaches, to develop tailored menu plans and training schedules. Vegetarian diets offer advantages such as higher quantities of carbohydrates, fiber, magnesium, potassium, folate, vitamins C and E, and phytochemicals, along with lower levels of saturated fat, cholesterol, and animal protein (Shaw et al., 2022; American Dietetic Association, & Dietitians of Canada, 2003). Individual assessment of dietary intakes and continuous education on plant-based diets are essential for athletes to optimise their performance (Baroni et al., 2023; Craddock et al., 2016; Fuhrman & Ferreri, 2010).

The Malaysian Dietary Guideline for Vegetarians offers valuable insights into important nutrients from various plant-based foods, as well as dairy products and eggs. By gaining a better understanding of vegetarian food choices and nutrient requirements, individuals can confidently incorporate a vegetarian diet into a healthy lifestyle. In addition to incorporating a variety of foods from different food groups, it is crucial to apply key messages from the Malaysian Dietary Guidelines 2020, such as limiting salt and sugar intake used as flavour enhancers in food preparation.

Vegetarian diets have the potential to provide various health benefits, including weight management and the prevention of chronic diseases. The Malaysian Dietary Guideline for Vegetarians serves as a valuable resource for individuals currently following or considering a vegetarian diet.

References:

- American Dietetic Association, & Dietitians of Canada (2003). Position of the American Dietetic Association and Dietitians of Canada: Vegetarian diets. *Journal of the American Dietetic Association*, 103(6), 748–765. <https://doi.org/10.1053/jada.2003.50142>
- Andrews, P., & Johnson, R. J. (2020). Evolutionary basis for the human diet: Consequences for human health. *Journal of Internal Medicine*, 287(3), 226–237. <https://doi.org/10.1111/joim.13011>
- Appleby, P. N., Davey, G. K., & Key, T. J. (2002). Hypertension and blood pressure among meat eaters, fish eaters, vegetarians and vegans in EPIC-Oxford. *Public Health Nutrition*, 5(5), 645–654. <https://doi.org/10.1079/PHN2002332>
- Baroni, L., Pelosi, E., Giampieri, F., & Battino, M. (2023). The VegPlate for sports: A plant-based food guide for athletes. *Nutrients*, 15(7), 1746. <https://doi.org/10.3390/nu15071746>
- Boeing, H., Bechthold, A., Bub, A., Ellinger, S., Haller, D., Kroke, A., Leschik-Bonnet, E., Müller, M. J., Oberitter, H., Schulze, M., Stehle, P., & Watzl, B. (2012). Critical review: vegetables and fruit in the prevention of chronic diseases. *European Journal of Nutrition*, 51(6), 637–663. <https://doi.org/10.1007/s00394-012-0380-y>
- Craddock, J. C., Probst, Y. C., & Peoples, G. E. (2016). Vegetarian and Omnivorous Nutrition - Comparing Physical Performance. *International Journal of Sport Nutrition and Exercise Metabolism*, 26(3), 212–220. <https://doi.org/10.1123/ijsnem.2015-0231>
- Dinu, M., Abbate, R., Gensini, G.F., Casini, A., Sofi, F. (2017). Vegetarian, vegan diets and multiple health outcomes: A systematic review with meta-analysis of observational studies. *Critical Review in Food Science and Nutrition*. 57(17), 3640–3649. doi: 10.1080/10408398.2016.1138447.
- Dybvik, J.S., Svendsen, M., Aune, D. (2023). Vegetarian and vegan diets and the risk of cardiovascular disease, ischemic heart disease and stroke: a systematic review and meta-analysis of prospective cohort studies. *European Journal of Nutrition*, 62(1), 51–69. doi: 10.1007/s00394-022-02942-8
- Fehér, A., Gazdecki, M., Véha, M., Szakály, M., & Szakály, Z. (2020). A Comprehensive Review of the Benefits of and the Barriers to the Switch to a Plant-Based Diet. *Sustainability*, 12(10), 4136. <https://doi.org/10.3390/su12104136>
- Fresán, U., & Sabaté, J. (2019). Vegetarian Diets: Planetary health and its alignment with human health. *Advances in Nutrition*, 10(Suppl_4), S380–S388. <https://doi.org/10.1093/advances/nmz019>
- Fuhrman, J., & Ferreri, D. M. (2010). Fueling the vegetarian (vegan) athlete. *Current Sports Medicine Reports*, 9(4), 233–241. <https://doi.org/10.1249/JSR.0b013e3181e93a6f>
- Hargreaves, S. M., Raposo, A., Saraiva, A., & Zandonadi, R. P. (2021). Vegetarian Diet: An Overview through the Perspective of Quality of Life Domains. *International Journal of Environmental Research and Public Health*, 18(8). doi: 10.3390/ijerph18084067
- Holler, S., Cramer, H., Liebscher, D., Jeitler, M., Schumann, D., Murthy, V., Michalsen, A., & Kessler, C. S. (2021). Differences Between Omnivores and Vegetarians in Personality Profiles, Values, and Empathy: A Systematic Review. *Frontiers in Psychology*, 12. <https://doi.org/10.3389/fpsyg.2021.579700>
- Hollman, P. C., & Katan, M. B. (1997). Absorption, metabolism and health effects of dietary flavonoids in man. *Biomedicine & Pharmacotherapy = Biomedecine & pharmacotherapie*, 51(8), 305–310. [https://doi.org/10.1016/s0753-3322\(97\)88045-6](https://doi.org/10.1016/s0753-3322(97)88045-6)
- Hu, F. B. (2003). Plant-based foods and prevention of cardiovascular disease: an overview. *The American Journal of Clinical Nutrition*, 78(3 Suppl), 544S–551S. <https://doi.org/10.1093/ajcn/78.3.544S>

- Huang, T., Yang, B., Zheng, J., Li, G., Wahlqvist, M.L., Li, D. (2012). Cardiovascular disease mortality and cancer incidence in vegetarians: a meta-analysis and systematic review. *Annals of Nutrition and Metabolism*. 60(4), :233-40. doi: 10.1159/000337301
- Kahleova, H., Klementova, M., Herynek, V., Skoch, A., Herynek, S., Hill, M., Mari, A., & Pelikanova, T. (2017). The Effect of a Vegetarian vs Conventional Hypocaloric Diabetic Diet on Thigh Adipose Tissue Distribution in Subjects with Type 2 Diabetes: A Randomized Study. *Journal of the American College of Nutrition*, 36(5), 364–369. <https://doi.org/10.1080/07315724.2017.1302367>.
- Khor, G.L., Voon, P.C., Tee, E.S., Hsu-Hage, B.H. & Wahlqvist, M.L. (2000). Cardiovascular risk factors among Malaysian urban vegetarians. *Journal of Community Nutrition* 2: 110-118.
- Koh, F. (2016). From: <https://www.straitstimes.com/singapore/singapore-named-second-most-vegan-friendly-city-in-asia-by-peta>. [Retrieved date: 14 Sept 2018].
- Koutentakis, M., Surma, S., Rogula, S., Filipiak, K.J., Gąsecka, A. (2023). The effect of a vegan diet on the cardiovascular system. *Journal of Cardiovascular Development and Disease*. 10(3), 94. doi: 10.3390/jcdd10030094
- Kwok, C. S., Umar, S., Myint, P. K., Mamas, M. A., & Loke, Y. K. (2014). Vegetarian diet, Seventh Day Adventists and risk of cardiovascular mortality: a systematic review and meta-analysis. *International Journal of Cardiology*, 176(3), 680–686. <https://doi.org/10.1016/j.ijcard.2014.07.080>.
- Leitzmann C. (2005). Vegetarian diets: what are the advantages?. *Forum of nutrition*, (57), 147–156. <https://doi.org/10.1159/000083787>.
- Leitzmann C. (2014). Vegetarian nutrition: past, present, future. *The American Journal of Clinical Nutrition*, 100 Suppl 1, 496S–502S. <https://doi.org/10.3945/ajcn.113.071365>
- Marsh K, Zeuschner C & Saunders A (2012). Health implications of a vegetarian diet: A review. *American Journal of Lifestyle Medicine* 6: 250-267. <https://doi.org/10.1177/155982761142576>
- Martin, C., Zhang, Y., Tonelli, C., & Petroni, K. (2013). Plants, diet, and health. *Annual Review of Plant Biology*, 64, 19–46. <https://doi.org/10.1146/annurev-arplant-050312-120142>.
- Melina, V., Craig, W., & Levin, S. (2016). Position of the Academy of Nutrition and Dietetics: Vegetarian Diets. *Journal of the Academy of Nutrition and Dietetics*, 116(12), 1970–1980. <https://doi.org/10.1016/j.jand.2016.09.025>.
- Memon, S.R.; Pawase, V.R.; Pavase, T.R., Soomro, M.A (2021). Investigation of COVID-19 Impact on the Food and Beverages Industry: China and India Perspective. *Foods*, 10, 1069. <https://doi.org/10.3390/foods10051069>
- MOH (2014). Quick Survey on Vegetarian Practice in Malaysia Selected Areas. Nutrition Division, Ministry of Health Malaysia (Unpublished report).
- NCCFN. 2021. Malaysian Dietary Guidelines 2020. National Coordinating Committee on Food and Nutrition, Ministry of Health Malaysia.
- Phillips, F. (2005). Vegetarian nutrition. *Nutrition Bulletin*, 30(2), 132-167. <https://doi.org/10.1111/j.1467-3010.2005.00467.x>
- Rizzo, N. S., Jaceldo-Siegl, K., Sabate, J., & Fraser, G. E. (2013). Nutrient profiles of vegetarian and nonvegetarian dietary patterns. *Journal of the Academy of Nutrition and Dietetics*, 113(12), 1610–1619. <https://doi.org/10.1016/j.jand.2013.06.349>.

- Sabaté J. (2003). The contribution of vegetarian diets to health and disease: a paradigm shift?. *The American Journal of Clinical Nutrition*, 78(3 Suppl), 502S–507S. <https://doi.org/10.1093/ajcn/78.3.502S>.
- Shaw, K. A., Zello, G. A., Rodgers, C. D., Warkentin, T. D., Baerwald, A. R., & Chilibeck, P. D. (2022). Benefits of a plant-based diet and considerations for the athlete. *European Journal of Applied Physiology*, 122(5), 1163–1178. <https://doi.org/10.1007/s00421-022-04902-w>
- The Sun Daily (2017). New index names most vegetarian-friendly countries in the world. From: <http://www.thesundaily.my/news/2017/10/25> [Retrieved date: 14 Sept 2018].
- Ullah, A., Munir, S., Badshah, S. L., Khan, N., Ghani, L., Poulson, B. G., Emwas, A.-H., et al. (2020). Important Flavonoids and Their Role as a Therapeutic Agent. *Molecules*, 25(22), 5243. MDPI AG. Retrieved from <http://dx.doi.org/10.3390/molecules25225243>.
- Wang, S. S., Lay, S., Yu, H. N., & Shen, S. R. (2016). Dietary Guidelines for Chinese Residents (2016): comments and comparisons. *Journal of Zhejiang University. Science. B*, 17(9), 649–656. <https://doi.org/10.1631/jzus.B1600341>
- Yen, C. E., Yen, C. H., Huang, M. C., Cheng, C. H., & Huang, Y. C. (2008). Dietary intake and nutritional status of vegetarian and omnivorous preschool children and their parents in Taiwan. *Nutrition research* (New York, N.Y.), 28(7), 430–436. <https://doi.org/10.1016/j.nutres.2008.03.012>.





Key Message 1

Eat a variety of
foods within the
recommended
servings

Key Message 1

**Eat a variety of foods
within the recommended
servings**

Assoc. Prof. Dr. Amutha Ramadas and Ms. Teh Wai Siew

1.1 Terminology

Adequate diet

An adequate diet provides enough energy, nutrients and fibre to maintain an individual's health. A diet that is adequate for an individual may not be adequate for another.

Balanced diet

A balanced diet is a diet that contains a combination of foods that provide a proper balance of nutrients. The body needs many types of foods in varying amounts to maintain health. The right balance of nutrients needed to maintain health can be achieved by eating proper balance of all healthy foods including fruit, vegetable, cereal, fish, meat, legume and milk.

Fermentation

Food fermentation is a process by which a microorganism transforms food into other products, usually through the production of lactic acid, ethanol, and other metabolic end products. Notwithstanding, the technology has evolved beyond food preservation into a tool for creating desirable organoleptic, nutritional, and functional attributes in food products.

Food groups

A food groups can be generally grouped foods based on similar nutrient content and function. There are five (5) food groups namely, i. vegetables, ii. fruits, iii. rice, other cereals, wholegrain cereal-based products and tubers, iv. eggs and legumes, and v. milk and milk products. These food groups contain foods similar in calories, carbohydrate, protein and fat contents.

Malaysian Food Pyramid for Vegetarians

A food pyramid is a visual tool used as a guide to your daily food intake in order to achieve a recommended servings of food groups in the diet. It provides a useful guide based on types and amounts of food to be eaten in combination to achieve a balanced and healthy diet. A food pyramid consists of four different levels, represented by main five food groups. This indicates that an individual should eat more of the foods at the base of the pyramid and less of the foods at the top of the pyramid. The Malaysian Food Pyramid for Vegetarians is modified from the Malaysian Food Pyramid 2020, that animal sources were removed. The recommended food servings in this food pyramid is used for children aged 7 years and older, adolescents and adults, and it can be practiced by vegetarians such lacto-, lacto-ovo-, ovo- and vegan).

Healthy diet

A healthy diet is a diet that provides a proper combination of energy and nutrients. The four characteristics of a healthy diet are varied, adequate, balanced and moderate.

Legumes

Legumes are plants in the *Leguminosae* family. Legumes comprise pulses and lentils. The terms pulses and lentils are often used interchangeably. Most legumes have nitrogen-fixing bacteria in their root nodules. Peanuts, also known as groundnuts and monkey nuts are legumes despite being called nuts. Example: dhal, tempeh and *tauhu*.

Malaysian Healthy Plate

Malaysian Healthy Plate (Ministry of Health, 2016) is a visual guide to show the total food in each food group that needs to be consumed in a meal to achieve a healthy and balanced diet based on the principle of quarter, quarter, half. The Malaysian Healthy Plate for Vegetarians translates recommendations from the Malaysian Dietary Guidelines for Vegetarians and the Malaysian Food Pyramid for Vegetarians to help Malaysian vegetarians practice healthy eating habits by planning their daily meals.

Moderation

Moderation is key to a healthy diet. Moderation refers to eating the right amount of foods to maintain a healthy weight and to optimise the body's metabolic process.

Plant-based protein products

The newest plant-based protein products are made to have a similar texture, colour, taste or form as meat including poultry and fish. The main function of these products is to replace meat in the diet and used it as a protein alternative. Textured vegetable protein (TVP) and Mycoprotein are the most common plant-based protein products. These products can be classified as ultra-processed foods, though some types of plant-based protein products can be classified as processed foods as they use less refined ingredients such as bean protein flour, chickpea flour and wheat flour (Asgar et al, 2010; He et al, 2020).

Processed foods

Edible parts of plants after separation from nature or modified/preserved by minimal processes or modified with the addition of salt, sugar, oils or fats to preserve and enhance their sensory qualities. These include canned or bottled vegetables or legumes (pulses) preserved in brine; whole fruit preserved in syrup; most freshly baked breads; and simple cheeses to which salt is added (Monteiro et al, 2019a).

Recommended Nutrient Intake (RNI)

Recommended nutrient intakes are nutrient standards used to plan and assess dietary nutrient intakes of healthy individuals or populations. Nutrient recommendations in RNI differ with age, sex, and physical activity level. The range of intakes encompassing the RNI should be sufficient to prevent deficiency and maintain optimal health while avoiding toxicity (NCCFN, 2017).

Serving size

In the Malaysian Food Pyramid for Vegetarians, serving size is the recommended amount of foods consumed daily in household measures used for foods and drinks for example; cup, plate, bowl, tablespoon, teaspoon and glass. However, the serving size defined in the Malaysian Food Pyramid for Vegetarians may not equal a serving size defined on a food label.

Unprocessed and minimal processed foods

Unprocessed (or natural) foods are the edible parts of plants (such as fruit, leaves, stems, seeds, and roots) or from animals (such as eggs and milk), and also fungi, algae and water, after separation from

KM1

Eat a variety of foods within the recommended servings

nature. Minimally processed foods are natural foods altered by methods that include the removal of inedible or unwanted parts, and also processes that include drying, crushing, grinding, powdering, fractioning, filtering, roasting, boiling, non-alcoholic fermentation, pasteurisation, chilling, freezing, placing in containers, and vacuum packaging. Unprocessed and minimally processed foods vary in energy density, their content and balance of fats, carbohydrates, proteins, and their fractions, and vitamins, minerals, and other bioactive compounds (Lane et al., 2021; Monteiro et al., 2019b).

Ultra-Processed Foods (UPFs)

The NOVA classification system characterises ultra-processed foods as industrial formulations generated through compounds extracted, derived or synthesised from food or food substrates. Ultra-processed foods also commonly contain artificial substances such as colours, sweeteners, flavours, preservatives, thickeners, emulsifiers and other additives to promote aesthetics, enhance palatability and increase shelf life. Ready-to-eat food and beverage, spreads, packaged snacks and pastries, cakes, instant noodles, pre-prepared ready-to-heat products, and some types of plant-based protein alternatives are examples of ultra-processed foods which are commonly high in sugar, salt, fat and/or artificial substances (Lane et al., 2021; Monteiro et al., 2019b).

1.2 Introduction

A healthy diet is vital as the source of energy, nutrients and other food components to maintain health, reduce the risk of diseases and in the management of certain diseases. Eating a balanced and healthy diet can help to prevent nutrient deficiencies and excesses such as preventing the excessive weight gain and the risk of chronic diseases such as dyslipidaemia and hypertension (Melina et al., 2016). The Malaysian Dietary Guidelines for Children and Adolescents (NCCFN, 2013) and the Malaysian Dietary Guidelines 2020 (NCCFN, 2021) were the primary references for preparing this guideline. Important considerations were adopted in healthy meal planning, including eating a balanced diet and consuming a wide variety of foods in moderation.

Dietary variety can be characterised by including foods from all major food groups and might be best defined as simply including foods from each of the major groups. Greater diet variation is considered a “net plus” for diet quality (Kennedy, 2004). Evidence has suggested that a wide variety of plant foods can provide adequate nutrients, which is reflected in the

Variety

Variety refers to eating many different types of foods each day to ensure a better selection of healthier foods. By selecting a variety of foods, the chances of consuming the multitude of nutrients the body needs are optimised.

Vegetarian diets

The term vegetarian is often used to describe a range of diets practised by a particular group of populations with varying degrees of food restrictions by omitting animal products. The four common categories of vegetarian diets are as below:

- **Vegan diets** exclude meat, poultry, fish, seafood, eggs and dairy products and foods that contain these products.
- **Lacto-vegetarian diets** exclude meat, poultry, fish, seafood and eggs, as well as foods that contain them. Dairy products such as milk, cheese, yogurt and butter are included in the diet.
- **Ovo-vegetarian diets** exclude meat, poultry, fish, seafood and dairy products but allow eggs.
- **Lacto-ovo-vegetarian diets** exclude meat, poultry, fish and seafood but allow dairy products and eggs.

position statements of nutrition groups around the world (Melina et al., 2016; Thomas, Erdman & Burke, 2016; Agnoli et al., 2017). The dietary pattern of vegetarians was found to be of higher quality than omnivores. The vegan diet, for example, although seemingly restrictive, received the highest Healthy Eating Index (HEI-2010) and Mediterranean Diet Score (MDS) values as opposed to omnivorous diets (Clarys et al., 2014; Parker & Vadiveloo, 2019). In adults, the variety of foods consumed by the vegetarians was found to be closer to nutritional recommendations than omnivorous diets (Deriemaeker et al., 2010; Clarys et al., 2013).

Limited local evidence suggests a positive association between higher diet quality and anthropometry and clinical outcomes (Ramadas et al., 2021). Higher diet quality has also been consistently associated with a lower risk of all-cause mortality, especially cardiovascular disease (CVD), type 2 diabetes and cancer (Miller et al., 2020; Morze et al., 2020). Specifically, an inverse association between mortality was found with greater adherence to plant-based dietary patterns that

emphasise consumption of whole fruits and vegetables, whole grains, low-fat dairy, legumes and nuts (Kim et al., 2019; Keaver et al., 2021). Diverse vegetarian diet may offer higher diet quality (Parker & Vadiveloo, 2019) and nutritional advantages such as: lower total energy intake, lower cholesterol and animal protein, and higher quantities of carbohydrate, phytochemicals, antioxidants, fibre, vitamins such as C and E, as well as micronutrients such as magnesium and folic acids (Appleby et al., 1999; Craig, 2010; Rizzo et al., 2013; Clarys et al., 2014; García Maldonado, Gallego-Narbón & Vaquero, 2019; Bowman, 2020).

Although there was some degree of variations, most of the recent observational and interventional studies have shown protective effects of plant-based or vegetarian diets against lifestyle-related diseases and disorders such as cancer, CVD, metabolic syndrome, diabetes, obesity, dyslipidaemia, and elevated blood pressure compared to omnivorous diets (Dinu et al., 2017; Benatar & Stewart, 2018; Lee et al., 2020; Oussalah

et al., 2020; Pinheiro et al., 2020; Tran et al., 2020; Marrone et al., 2021; Pollakova et al., 2021; Remde et al., 2022). A wide ranges of vegetarian eating pattern has been associated with lowered risk of chronic diseases, suggesting the importance of eating a variety of plant-based foods or vegetarian diets. The dietary habits of vegetarians demonstrate a notable inclination towards reduced consumption of processed foods and a preference for the consumption of natural, whole foods (Hargreaves et al., 2020). This dietary pattern holds potential benefits for their overall health, as evidenced by studies indicating that the intake of highly palatable foods, such as ultra-processed foods, processed meats, and artificial flavourings, heightens the risk of all-cause mortality (English et al., 2021). Consequently, the dietary choices of vegetarians align with a healthier approach that may contribute to improved well-being and longevity. In addition, a growing body of evidence also suggests the potential effect of vegetarian diets on gut microbiota, resulting in positive health outcomes (Losno et al., 2021).

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1.3 Scientific Basis

1.3.1 Varied vegetarian diets

A variety of healthy foods is crucial for human health because no single food can provide all the required nutrients (World Health Organization, 2000). Promotion strategy of nutritious diets that is primarily based on plant-based food products have been introduced and recommended as part of the Countrywide Integrated Non-Communicable Disease Intervention (CINDI) program (WHO, 2000). The CINDI program aimed to reduce the risk of chronic diseases by reducing the common risk factors including unhealthy diet.

Vegetarian diets that are varied and balanced are healthy and nutritionally adequate and may provide positive health effects for the prevention of certain chronic diseases (Glenn et al., 2019; Rocha et al., 2019; Jabri et al., 2021). One of the major nutritional concerns of vegetarian eating pattern, especially a restrictive and monotonous vegetarian eating pattern tends to increase the risk of nutrient deficiencies (Rocha et al., 2019). It is worthwhile to elaborate this to highlight the point of restrictive and monotonous. Hence, proper vegetarian eating strategy should be encouraged to ensure that a diverse diet that is nutritionally adequate is available for vegetarians, especially for vulnerable groups (McEvoy, Temple & Woodside, 2012; Bakaloudi et al., 2021).

A varied and well-balanced vegetarian diet rich in wholegrain, legumes, nuts and seeds, green vegetables, fresh and dried fruits and fortified food products provides an adequate supply of various micronutrients (Saunders et al., 2013a; Saunders, Craig & Baines, 2013b; Saunders, Davis & Garg, 2013c; Foster et al., 2015). However, Rammohan, Awofeso and Robitaille (2012) argued that most vegetarians from developing countries such as India are unable to afford varied vegetarian foods and most traditional vegetarian meals do not consider diversity and lack certain essential nutrients such as iron.

Some vegetarian subgroups may have to further diversify their diet due to their stringent dietary requirements. Nutritional requirements are the same for vegetarians and non-vegetarians, but some nutrients require special attention (Eveleigh et al., 2020; Bakaloudi et al., 2021; Rashid, Meier & Patrick, 2021). Eating a variety of foods must be a priority for vegetarians, especially vegans, who have been shown to have the lowest energy, calcium and protein intake of all diet groups (Bakaloudi et al., 2021). For example, the prevalence of vitamin B12 deficiency is higher among those who had adhered to a vegetarian diet since birth compared with those who had adopted vegetarianism later (Pawlak et al., 2013), and this is a matter of concern among vegans

Eat a variety of foods within the recommended servings

(Woo, Kwok & Celermajer, 2014). Consumptions of vitamin B12-fortified foods, regular B12 supplementations, seaweed, milk, fruit and vegetable intake are useful to improve vitamin B12 status among vegetarians, emphasising the need for diet diversity in this group (Lee & Krawinkel, 2011; Kwok et al, 2012; Zeuschner et al, 2013; Naik et al, 2013; Pawlak, Lester & Babatunde, 2014; Woo et al, 2014; Rizzo et al, 2016; Gallego-Narbón et al, 2019).

The nutrients intake of children on a vegetarian diet is considered adequate and well-balanced when the diet contains dairy products and eggs. Without proper guidelines, intense growth and development of children can be impeded by severe or strict vegetarian diets such as vegan diet (Kersting, 2018; Cofnas, 2019; Sutter & Bender, 2021). After such regimens, severe deficiency states have been described, such as the particular risk of malnutrition, growth failure, and developmental delay (Kiely, 2021). Adequate dietary advice from a dietitian, nutritionist or medical doctor is required for infants and children who are vegan diet practitioners. A well-balanced lacto-ovo-vegetarian diet, including dairy products, can satisfy all the nutritional needs of the growing child. In contrast, a vegan diet, excluding all animal food sources, must at least be supplemented with vitamin B12, with particular attention to adequate intakes of calcium and zinc, and energy-dense foods containing enough high-quality protein for young children (Van Winckel et al, 2011; Müller et al, 2020).

High consumption of plant-based diet could facilitate high carbohydrate food intakes, which is essential to support prolonged exercise. While a vegan diet may pose a challenge to athletes'

performance due to issues surrounding energy and protein sufficiency, adequacy of micronutrients such as vitamin B12, iron, zinc and calcium, and lack of specific fatty acids, a nutritive vegan diet can still be designed for athletes to meet their dietary needs via strategic management of foods and appropriate supplementation (Rogerson, 2017). A well-planned vegetarian diet can provide athletes with adequate amounts of known nutrients, although the potential lack of other minerals such as iron, zinc, iodine, trace elements and protein intake exist if the diet is too restrictive (Rodriguez et al, 2009; Fuhrman & Ferreri, 2010; Vitale & Hueglin, 2021).

Cross-sectional studies of vegetarians and vegans have shown that, on average, they have a relatively low body mass index (BMI), and a low plasma cholesterol concentration and a meta-analysis has also shown higher plasma homocysteine concentrations than in non-vegetarians (Obersby et al, 2013). Overall, studies showed that a vegetarian diet may result in more health benefits, such as reducing the risk of all cancers and lowering the death rates compared to a non-vegetarian diet (Rocha et al, 2019; Jabri et al, 2021).

1.3.2 Plant-based protein products

Some traditional plant-based protein products typically used in vegetarian dishes are tofu, tempeh and seitan. With the increasing popularity of vegetarian diets, plant-based protein products have been expanded to include various meat substitutions while ensuring a supply of protein and other essential nutritional components (Malav et al, 2015). Plant-based protein products are known by different names such as vegan meat, fake meat,

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Eat a variety of foods within the recommended servings



vegan food product, plant-based meat option/alternatives, meat substitutes, mock meat and meat analogue. Common ingredients in these products are textured soy protein (TSP), fermentation of *Fusarium venenatum* (Mycoprotein), konjac, mushroom, beetroot, beans, lentils, seeds, seaweed, cane, various types of flour, modified starch and yam powder. Furthermore, these plant-based protein products should not include animal flavouring, gelatine, chitin or those containing animal products or extractions.

The TSP is also known as texture vegetable protein (TVP) which is a trademark of Archer Daniels Midland (Shurtleff & Aoyagi, 2016), and three common types of TSP are soy flour, soy concentrate and soy isolates (Figure 1.1). Of these three types, soy isolate contains the highest proportion of protein (90%) (Riaz, 2020). TSP is used as a meat extender because of its important properties, such as good water and oil absorption and meat-like texture. The soy meat or soy chunks are defatted soy flour or soy isolate products in dry form and have to be soaked till softened before cooking and are suitable for quick cooking. On the other hand, Mycoprotein is an alternative nutritious protein source with a meat-like texture made by fermenting *Fusarium venenatum*, a naturally occurring fungus found in soil and converting wheat into protein when added glucose, oxygen, nitrogen and minerals.

Both TSP and Mycoprotein are high in protein and fibre, but low in fat, cholesterol, sodium, and sugar. Each portion of TSP (30g) contains 60 kcal of energy, 5.9g of carbohydrate, 4.7g of protein, 2.4g of fat and 138mg of sodium. In comparison, each portion of Mycoprotein (100g) contains 85 kcal of energy, 3.0g of carbohydrate, 11.0g of protein, 2.9g of fat and 5mg of sodium (Finnigan et al., 2019). Mycoprotein also contains several micronutrients, including vitamin B9 (folate), vitamin B12, calcium, phosphorous, magnesium and zinc.

The Malaysian random market survey revealed that almost 70% of main vegan meats ingredients are TSP, such as vegetarian chicken, fish, goose and meat slices (MOH, 2014a). Besides TSP, an estimated 24% of meat alternatives in Malaysia use konjac, mushrooms or wheat flour as the main ingredients in foods such as vegetarian prawn, mutton, gizzard, fish balls and burger patties. It is important to note that these products have higher carbohydrate content than protein.

The traditional techniques of making plant-based products such as tofu, yuba and tempeh use less refined ingredients such as bean protein flour and chickpea flour with mineral concentrations. However,

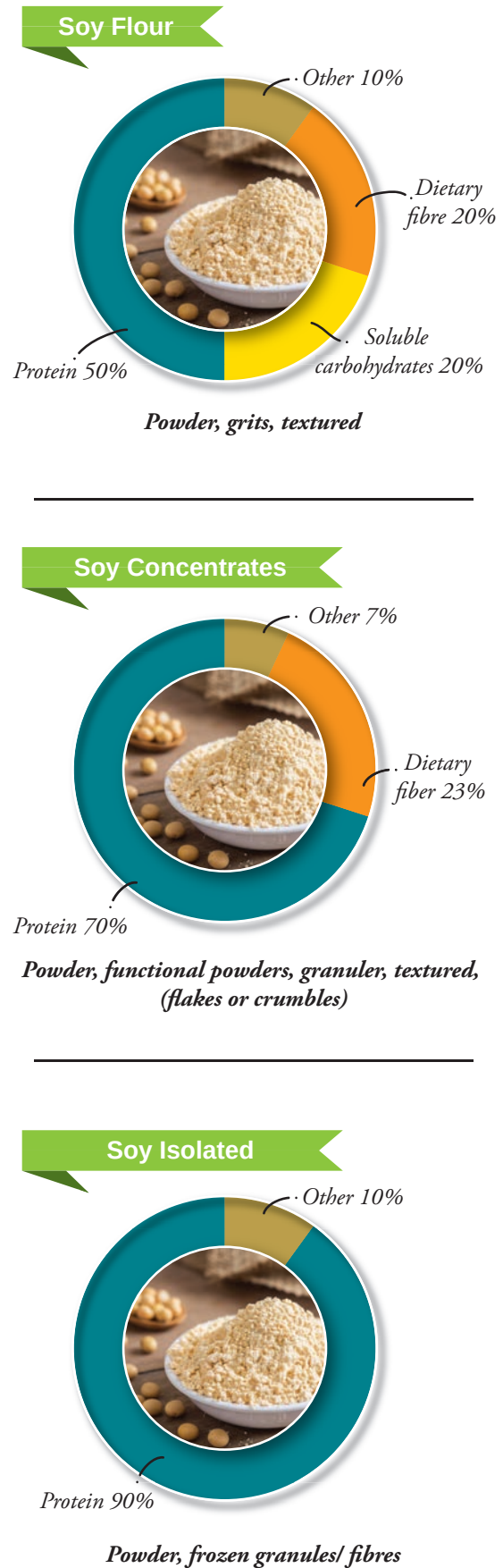


Figure 1.1: Different Types of Textured Soy Protein (TSP)

these products become nutritionally unbalanced when excessive oil, sugar or salt are added or transformed into unhealthy dishes with unhealthy preparation methods, such as deep-frying or cooking in thick gravy. While some types of plant-based protein products can be regarded as processed foods with the addition of culinary ingredients (sugar, salt, fat) similar to texture and form of meat, some types are classified as ultra-processed foods due to the use of colouring agents, flavourings, additives or in the form of ready-to-consume and pre-prepared ready-to-heat products (Sadler, 2004; Monteiro et al., 2019b; Michel et al., 2021).

The US Food and Drug Administration (US FDA) concluded that 25g of soy protein a day, as part of a diet low in saturated fat and cholesterol, may reduce the risk of heart disease (US FDA, 1999). Several reviews also concluded that Mycoprotein can help maintain healthy blood cholesterol levels, promote muscle synthesis, have a lesser incidence of allergic reactions and increase satiety, though the evidence was less conclusive for blood glucose levels (Derbyshire & Ayoub, 2019; Finnigan et al., 2019; Hashempour-Baltork et al., 2020; Derbyshire & Delange, 2021).

Overall, the dry forms of TSP and Mycoprotein are good sources of protein that can be used to prepare healthy dishes if we adhere to healthier cooking methods that use low sugar, salt and fat with added fibre.

1.3.3 Food groups

A food pyramid acts as a visual guide to provide a framework for the types and amounts of foods that can be eaten in combination to provide a nutritious and healthy daily diet. The levels in the food pyramid from the bottom to the top indicate that a person should eat more of the foods at the base and fewer foods intake from the top of the pyramid. Similarly, the Malaysian Food Pyramid for Vegetarians (Figure 1.2) is a practical visual guide for vegetarians to choose and eat a variety of foods of their choice. This pyramid was modelled after the Malaysian Food Pyramid with the meat sources removed (NCCFN, 2021).

The principles of the vegetarian pyramid are similar to the general population food pyramid, with some modifications to meet the requirements of vegetarian individuals. The recommended number of servings indicates an average amount that individuals should choose to eat each day, and it

differs according to life stages and sex. The Malaysian Recommended Nutrient Intake (RNI) (NCCFN, 2017) has recommended that the total carbohydrate contributes 50 to 65%, total fat 25 to 30% and protein 10 to 20% of the total caloric intake per day for adults. The total calories calculated were based on this percentage distribution that will be converted to the exchange list for macronutrients, and then converted to servings based on calorie intake. The number of servings for the macronutrient intake for adults are calculated for 1,500 kcal, 1,800 kcal and 2,000 kcal per day according to lacto-ovo-vegetarians, lacto-vegetarians, ovo-vegetarians and vegans, respectively. Additionally, similar conversions were employed for infants and toddlers (Table 1.1), children, adolescents, pregnant and lactating women (Table 1.2 to Table 1.5).

It is crucial to include all food groups in the daily diet, considering the nutritional values of each of food groups. Insufficient or excessive intake of these foods should be avoided as both situations have significant implications on an individual's health. Balanced vegetarian diets meet the nutritional needs of all life cycle stages, including pregnancy, lactation, infancy, childhood, adolescence and the elderly (Melina et al., 2016). Well-planned vegetarians tend to consume fewer overall calories; a lower proportion of calories from fat (particularly saturated fat) and cholesterol; and higher intakes of fruits, vegetables, whole grains, nuts, soy products, fibre and phytochemicals than non-vegetarians. Because of the variability in vegetarian diets, individuals need to become familiar with their individual nutritional needs and potential dietary deficiencies.

1.3.4 Number of servings of each food group

Fruits and vegetables are placed at the base of the food pyramid. Fruits are recommended to be consumed between 2-4 servings, while ≥ 3 servings of vegetables are recommended per day. Fruits and vegetables, generally low in energy density and when consumed in variety, are sources of fibre, vitamins, minerals and other bioactive compounds such as phytochemicals. Increased intake of fruits and vegetables, in addition to a healthy lifestyle, has been shown to decrease the risk of chronic diseases such as cardiovascular disease (Hartley et al., 2013; Lamb et al., 2017; Toh et al., 2020), diabetes (Halvorsen et al., 2021), hypertension (Li et al., 2016) and some types of cancers (Norat et al., 2014). More information on fruits and vegetables can be found in Key Message 2.

As carbohydrate-rich foods such as rice, other cereals, wholegrain cereal-based products and tubers form the bulk of caloric needs, this food group is placed at the second level of the food pyramid and should be eaten the most, which is ranges between 3 to 7 servings per day. It is recommended to increase the consumption of whole grains such as whole wheat, whole oats, oatmeal, brown and wild rice. The basis for most meals should include unrefined or minimally processed cereals and grains (wholegrain or wholemeal) where possible and choose fortified cereals and grains when available. This is in tandem with evidence pointing towards the protective effects of wholegrains and fortified cereals and grains for vegetarians (Agnoli et al., 2017; Aune, 2019). More information on grains and cereal products can be found in Key Message 3.

Protein sources are placed at the third level of the Malaysian Food Pyramid for Vegetarians. These foods provide nutrients, which are vital for the health and maintenance of the body, and include milk, dairy products, eggs, grains, legumes, pulses and various soy foods such as tauhu, tempeh and seitan (wheat gluten). Legumes are recommended to be consumed between 1 to 3 servings. The recommended serving sizes make it possible to consume enough protein by consuming a vegan or vegetarian diet (Mariotti & Gardner, 2019). Lacto-ovo and ovo-vegetarians are recommended to consume 1-2 servings of eggs. Unprocessed complex carbohydrates and vegetables can supplement vegetarians' diets with some essential amino acids. Hence, the complementary amino acid composition may not be necessary if a combination

of different protein sub-groups is consumed in a vegetarian diet (McDougall, 2002).

Milk and milk products are also essential as rich sources of riboflavin, vitamin B12 and calcium, while eggs and pulses contain iron. Lacto-ovo and lacto-vegetarians are recommended to consume 2-3 servings of milk and milk products. Although the iron found in eggs and pulses are more difficult to be absorbed, consumption of vitamin C-rich foods together would improve the rate of absorption. The addition of green and red capsicum to scrambled eggs is an example. Moderate consumption of these foods is advised, as excessive intake can result in undesirable outcomes. For example, excessive intake of nuts and seeds should be avoided due to their high fat content. (See more information on eggs, legumes and nuts in Key Messages 4 and 5.

Fats, oils, sugar and salt are placed at the tip of the food pyramid. This includes ultra-processed foods which contain artificial substances such as colours, sweeteners, flavours, preservatives and other additives. Fats and oils, sugars and salt are not considered food groups, but fats/ oils and especially sugar and salt, are often used during cooking or in food preparations. Vegetarians are recommended to consume these foods and ingredients the least or sparingly compared to the other food groups in the food pyramid. The recommendation for fat intake is 5 servings including nuts. However, it is well documented in the literature that excessive intake of fats/ oils, sugar and salt contributes to chronic diseases (Bowen et al., 2018). More information on fats can be found in Key Message 6.

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Eat a variety of foods within the recommended servings



1.4 Current Status

There is a lack of data on the dietary patterns of vegetarians in Malaysia. In addition to Cheah et al. (2018)'s study among vegetarians in Sarawak, several unpublished local studies have previously examined the dietary intake of vegetarians, primarily among Buddhists, including monks (Wong, 1999), nuns (Chen, 2001), Chinese adults (Lee, 2000), adolescents (Chong, 2013) and Indians adults (MOH, 2014b). Collectively, these studies' findings suggest no significant differences in nutrient adequacy between vegetarian practices, such as lacto-vegetarians, lacto-ovo-vegetarians and vegans.

Based on three-day food records, male and female vegetarians' mean daily energy intake ranged between 1,840 – 2,020 kcal and 1,450 – 1,670 kcal, respectively (Wong, 1999; Lee, 2000; Chen, 2001). These studies showed similar energy intakes compared to the data obtained from MANS 2014, where the median total energy intake of 1,489 kcal and 1,445 kcal were reported for males and females, respectively (IPH, 2014). However, the findings were markedly different among Sarawakian vegetarians' energy intake as assessed with a food frequency questionnaire; a total energy intake of 2,349 kcal was reported (Cheah et al, 2018). For children and adolescents aged 10-18 years, the reported energy intakes averaged 1,700 kcal for females and 2,150 kcal for males (Chong, 2013), based on a 24-hour diet recall for three days.

A recent study among Chinese and Indian vegetarians in Kuala Lumpur and Selangor revealed that about two in five vegetarians (43.5%) exceeded the recommended percentage of energy intake derived from fat. A possible reason for excessive fat intake could be the high-frequency intakes of plant-

based protein products among Malaysian vegetarians, which were popular foods. More than half of the vegetarians consumed these foods at least once a week. While plant-based protein products can be part of a good source of plant protein, it should be noted that they are categorized as processed foods or ultra-processed foods, and are commonly added with seasoning and additives. Furthermore, "deep frying" is the most common cooking method for plant-based protein products, which may contribute to high fat intake among vegetarians (Khor et al, 2000; Ching et al, 2019).

There is limited evidence locally to indicate the issue of nutrient inadequacy or lack of diet variety among vegetarians. One of the vital micronutrients to be explored among vegetarians will be iron, as haemoglobin concentrations fell below the recommended level of 12 mg/dl (World Health Organization, 1993) among 70%, 48% and 60% of vegan, lacto-vegetarian and lacto-ovo vegetarian participants, respectively. While Cheah et al. (2018) reported that Hindu vegetarians have a greater intake of iron and postulated that this could be due to their increased consumption of non-heme iron-rich foods such as legumes and lentils, another study (Chong, 2013) highlighted the possible poor iron status among lacto-vegetarians in Selangor. This study reported a high prevalence of anaemia (52%) among the 44 vegetarian adolescents. The dietary intake of iron was reported to be significantly higher among the semi-vegetarians, defined as those who practice vegetarianism occasionally, than the full-time vegetarians in this study (Chong, 2013). The findings of this study corroborate those of Chen (2001), which was conducted among 53 female vegetarian adults.

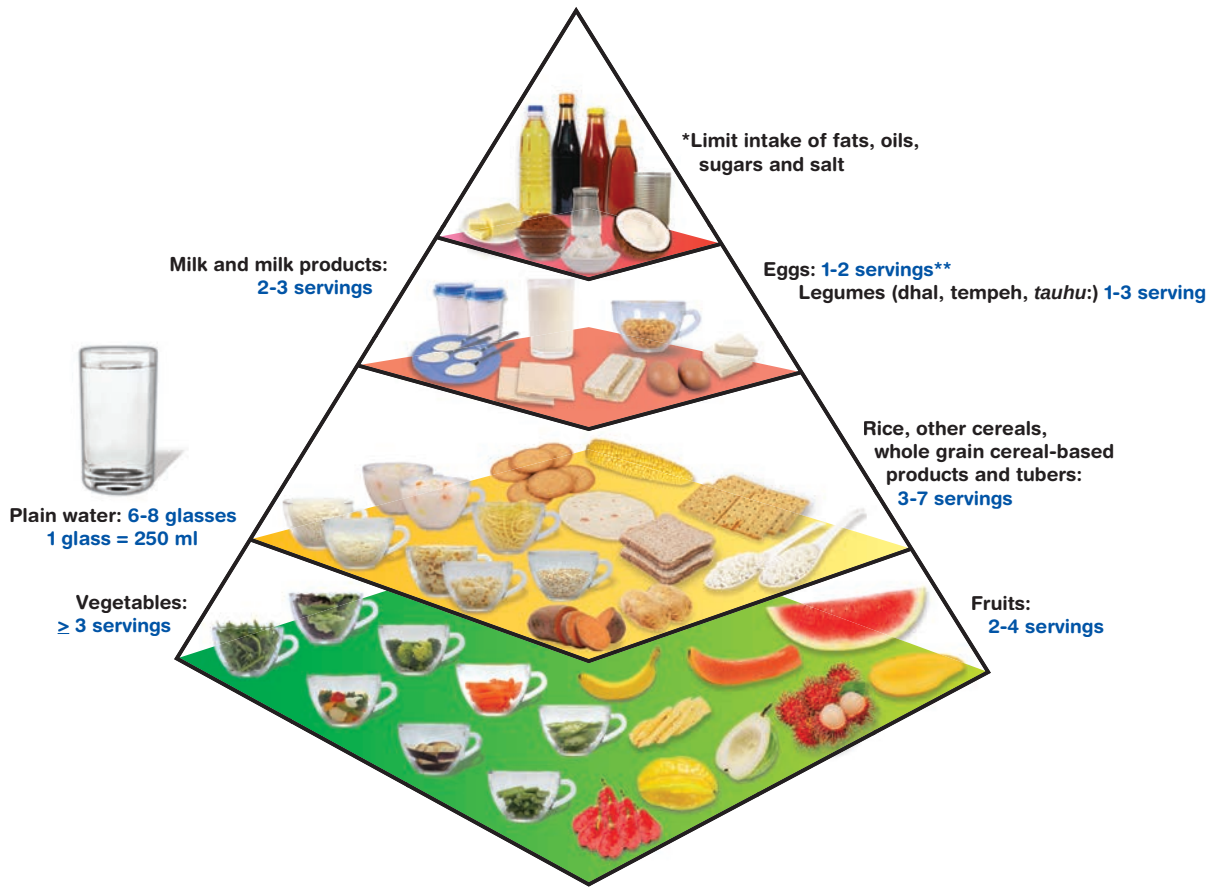
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Eat a variety of foods within the recommended servings



MALAYSIAN FOOD PYRAMID FOR VEGETARIANS

Guide to Your **DAILY** Food Intake



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Figure 1.2: Malaysian Food Pyramid for Vegetarians

Notes:

- The number of servings is calculated based on 1,500 kcal to 2,300 kcal.
- This pyramid is meant for children aged 7 years and older, adolescents and adults. Refer to Recommendations of Food Servings for Infants and Children for serving size recommendations for children younger than 7 years (Table 1.1).
- For children and adolescents aged 7-<18 years old (lacto-ovo- and lacto-vegetarians), 3 servings of milk are recommended.
- For adolescent boys aged 13-<18 years old (lacto-ovo-, lacto-, ovo-vegetarians and vegan), 3-4 servings of fruits, 7 servings of rice, other cereals, whole-grain cereal-based products and tubers and 2-3 servings of legumes are recommended.
- Refer to Recommendations of Food Servings for lacto-ovo-, lacto-, ovo-vegetarians and vegan (Table 1.2 – Table 1.5). A general guide for additional food recommendations when omitting eggs and/ or milk is:
 - Add 1 serving of legume for lacto-vegetarian.
 - Add 1 serving of fruit, 1 serving of rice, other cereals, whole grain cereal-based products and 1-2 servings of legumes for ovo-vegetarian and vegan.

*This includes ultra-processed foods which contain artificial substances such as colours, sweeteners, flavours, preservatives and other additives.

** Omit eggs servings for lacto-vegetarian and vegan; and omit milk and milk products serving for ovo-vegetarian and vegan.

Eat a variety of foods within the recommended servings

1.5 Key Recommendations

Key Recommendation 1

Choose your daily food intake based on the recommended number of servings in the Malaysian Food Pyramid for Vegetarians (Figure 1.2).

How to achieve:

1. Choose all food groups in the Malaysian Food Pyramid for Vegetarians and eat within the recommended amount to ensure the body gets all the nutrients needed.
2. Choose your vegetarian type and eat within the recommended number of food serving based on daily caloric needs (Table 1.1-Table 1.5). For the food serving equivalent list, please refer Table 1.6 - Table 1.11. (see Appendices).
3. Plan your daily menu based on your recommended number of servings for each food group. (Refer Table 1.12, see Appendices).
4. Limit intakes of fats and oils, salts and sugars in your daily diet.
5. Limit intake of processed foods such as vegetarian fish balls or meatballs and sausages, and ultra-processed foods such as soft drinks, sweetened breakfast cereals, salty fatty packaged snacks (junk foods) and instant noodles, which are nutritionally unbalanced.

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Eat a variety of foods within the recommended servings



Table 1.1a: Recommended food portions for vegetarian infants and toddlers (ages between 6 - 8 months)

6 – 8 months (boy & girl)								
Age group	*Energy (kcal)	**Vegetables	Fruits	Rice, other cereals, wholegrain cereal-based products and tubers	Eggs	Legumes	***Milk & milk products	Fats (including nuts)
Lacto-ovo	630	2 dsp (20 g) of cooked vegetables	1½ dsp (30 g) of banana/ papaya/ apple	1½ cup (250 g) of porridge + 1 pc (5 g) of fortified biscuit	¼ egg (grade A)	1½ dsp (45 g) of soft <i>tauhu</i>	Breast milk on demand	1.5 tsps (7.5 g) of added oil
Lacto	630	2 dsp (20 g) of cooked vegetables	1½ dsp (30 g) of banana/ papaya/ apple	1½ cup (250 g) of porridge + 1 pc (5 g) of fortified biscuit	0	3 dsp (90 g) of soft <i>tauhu</i>	Breast milk on demand	1.5 tsps (7.5 g) of added oil
Ovo	630	2 dsp (20 g) of cooked vegetables	1½ dsp (30 g) of banana/ papaya/ apple	1½ cup (250 g) of porridge + 1 pc (5 g) of fortified biscuit	¼ egg (grade A)	1½ dsp (45 g) of soft <i>tauhu</i>	Breast milk on demand	1.5 tsps (7.5 g) of added oil
Vegan	630	2 dsp (20 g) of cooked vegetables	1½ dsp (30 g) of banana/ papaya/ apple	1½ cup (250 g) of porridge + 1 pc (5 g) of fortified biscuit	0	3 dsp (90 g) of soft <i>tauhu</i>	Breast milk on demand	1.5 tsps (7.5 g) of added oil

* Recommended amount of each food group refers to the maximum requirement of each age group.

** The recommended amount or portion size of vegetables to meet the vitamin C requirement.

*** For age 2 years old and below, total energy intake is divided into 2 sources: complementary feeding and milk feeding. For infants fed with infant formula, the amount intake of formula milk must not exceed 413 kcal/day and 379 kcal/day for infants aged 6 to 8 months and 9 to 11 months respectively. (Dewey et al., 2004); or not more than 5 feedings per day and 4 feedings for infants aged 6 to 8 months and 9 to 11 months respectively. (FAO/WHO, 2020). If baby is not breastfed, give in addition : 1-2 cups of milk per day (WHO & UNICEF, 2021).

Note:

- Malaysian Breastfeeding Policy states that all mothers are encouraged to breastfeed their children with breast milk from birth until six months old continuing until two years of age. Complementary foods should be given from 6 months old.
- Added sugar is not allowed for children below 2 years old, WHO 2015 recommends that intake of added sugar for children 3 years old not more than 2 teaspoons per day.
- Recommended amount of fats/ oil intake per day is including oil used in cooking and food preparation.
- Refer to figure 1.5 for standard household measurements used in this dietary guidelines.

Table 1.1b: Recommended food portions for vegetarian infants and toddlers (ages between 9 - 11 months)

9 - 11 months (boy & girl)							
Age group	*Energy (kcal)	**Vegetables	Fruits	Rice, other cereals, wholegrain cereal-based products and tubers	Eggs	Legumes	***Milk & milk products
Lacto-ovo	720	3 dsp (30 g) of cooked vegetables	2 dsp (40 g) of banana/ papaya/ apple	1½ cups (250 g) of porridge + 1 pc (5 g) of fortified biscuit + ½ whole (40 g) medium sized potato	¼ egg (grade A)	1½ dsp (45 g) of soft <i>tauhu</i>	Breast milk on demand
Lacto	720	3 dsp (30 g) of cooked vegetables	2 dsp (40 g) of banana/ papaya/ apple	1½ cups (250 g) of porridge + 1 pc (5 g) of fortified biscuit + ½ whole (40 g) medium sized potato	0	3 dsp (90 g) of soft <i>tauhu</i>	Breast milk on demand
Ovo	720	3 dsp (30 g) of cooked vegetables	2 dsp (40 g) of banana/ papaya/ apple	1½ cups (250 g) of porridge + 1 pc (5g) of fortified biscuit + ½ whole (40 g) medium sized potato	¼ egg (grade A)	1½ dsp (45 g) of soft <i>tauhu</i>	Breast milk on demand
Vegan	720	3 dsp (30 g) of cooked vegetables	2 dsp (40 g) of banana/ papaya/ apple	1½ cups (250 g) of porridge + 1 pc (5 g) of fortified biscuit + ½ whole (40 g) medium sized potato	0	3 dsp (90 g) of soft <i>tauhu</i>	Breast milk on demand

* Recommended amount of each food group refers to the maximum requirement of each age group.

** The recommended amount or portion size of vegetables to meet the vitamin C requirement.

*** For age 2 years old and below, total energy intake is divided into 2 sources: complementary feeding and milk feeding. For infants fed with infant formula, the amount intake of formula milk must not exceed 413 kcal/day and 379 kcal/day for infants aged 6 to 8 months and 9 to 11 months respectively. (Dewey et al., 2004); or not more than 5 feedings per day and 4 feedings for infants aged 6 to 8 months and 9 to 11 months respectively. (FAO/WHO, 2020). If baby is not breastfed, give in addition : 1-2 cups of milk per day (WHO & UNICEF, 2021).

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- Recommended amount of fats/ oil intake per day is including oil used in cooking and food preparation.
- Refer to figure 1.5 for standard household measurements used in this dietary guidelines.

Table 1.1c: Recommended food portions for vegetarian infants and toddlers (ages between 1 - 2 years)

1-2 years (boy & girl)								
Age group	*Energy (kcal)	**Vegetables	Fruits	Rice, other cereals, wholegrain cereal-based products and tubers	Eggs	Legumes	***Milk & milk products	Fats (including nuts)
Lacto-ovo	980	4 dsp (40 g) of cooked vegetables	1 banana, berangan/ apple/ 1 small slice of papaya (60 g)	1 cup (100 g) of soft rice + 1 slice of fortified bread (30 g) + 1 whole medium sized (80 g) potato	½ egg (grade A)	½ pc (50 g) of <i>taukua</i>	3 glasses of full cream milk/ Breast milk on demand	4 tsps (20 g) of cooking fat/ added oil/ finely chopped nuts
Lacto	980	4 dsp (40 g) of cooked vegetables	1 banana, berangan/ apple/ 1 small slice papaya (60 g)	1 cup (100 g) of soft rice + 1 slice of fortified bread (30 g) + 1 whole medium sized (80 g) potato	0	½ pc (50 g) of <i>taukua</i>	3 glasses of full cream milk/ Breast milk on demand	4 tsps (20 g) of cooking fat/ added oil/ finely chopped nuts
Ovo	980	4 dsp (40 g) of cooked vegetables	1 banana, berangan/ apple/ 1 small slice papaya (60 g)	1 cup (100 g) of soft rice + 1 slice of fortified bread (30 g) + 1 whole medium sized (80 g) potato	½ egg (grade A)	½ pc (50 g) <i>taukua</i> + 4 dsp (90 g) finely chopped chickpea + 3 glasses of soya bean milk	Breast milk on demand	4 tsps (20 g) of cooking fat/ added oil/ finely chopped nuts
Vegan	980	4 dsp (40 g) of cooked vegetables	1 banana, berangan/ apple/ 1 small slice papaya (60 g)	1 cup (100 g) of soft rice + 1 slice of fortified bread (30 g) + 1 whole medium sized (80 g) potato	0	½ pc (50 g) <i>taukua</i> + 4 dsp (90 g) finely chopped chickpea + 3 glasses of soya bean milk	Breast milk on demand	4 tsps (20 g) of cooking fat/ added oil/ finely chopped nuts

* Recommended amount of each food group refers to the maximum requirement of each age group.

** The recommended amount or portion size of vegetables to meet the vitamin C requirement.

*** For age 2 years old and below, total energy intake is divided into 2 sources: complementary feeding and milk feeding. For infants fed with infant formula, the amount intake of formula milk must not exceed 413 kcal/day and 379 kcal/day for infants aged 6 to 8 months and 9 to 11 months respectively. (Dewey et al., 2004); or not more than 5 feedings per day and 4 feedings for infants aged 6 to 8 months and 9 to 11 months respectively. (FAO/WHO, 2020). If baby is not breastfed, give in addition : 1-2 cups of milk per day (WHO & UNICEF, 2021).

Note:

- Malaysian Breastfeeding Policy states that all mothers are encouraged to breastfeed their children with breast milk from birth until six months old continuing until two years of age. Complementary foods should be given from 6 months old.
- Added sugar is not allowed for children below 2 years old. WHO 2015 recommends that intake of added sugar for children 3 years old not more than 2 teaspoons per day.
- Recommended amount of fats/ oil intake per day is including oil used in cooking and food preparation.
- Refer to figure 1.5 for standard household measurements used in this dietary guidelines.

Eat a variety of foods within the recommended servings

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Table 1.1d: Recommended food portions for vegetarian infants and toddlers (age 3 years)

3 years (boy & girl)								
Age group	*Energy (kcal)	**Vegetables	Fruits	Rice, other cereals, wholegrain cereal-based products and tubers	Eggs	Legumes	***Milk & milk products	Fats (including nuts)
Lacto-ovo	980	4 dsp (40 g) of cooked vegetables	1 banana, berangan/ apple/ 1 small slice of papaya (60 g)	1 cup (100 g) of rice + 1 slice of fortified bread (30 g) + 1 whole medium sized (80 g) potato	½ egg (grade A)	½ pc (50 g) of <i>taukua</i>	3 glasses of full cream milk/ Breast milk on demand	4 tsps (20 g) of cooking fat/ chopped nuts
Lacto	980	4 dsp (40 g) of cooked vegetables	1 banana, berangan/ apple/ 1 small slice of papaya (60 g)	1 cup (100 g) of rice + 1 slice of fortified bread (30 g) + 1 whole medium sized (80 g) potato	0	½ pc (50 g) of <i>taukua</i>	3 glasses of full cream milk/ Breast milk on demand	4 tsps (20 g) of cooking fat/ chopped nuts
Ovo	980	4 dsp (40 g) of cooked vegetables	1 banana, berangan/ apple/ 1 small slice of papaya (60 g)	1 cup (100 g) of rice + 1 slice of fortified bread (30 g) + 1 whole medium sized (80g) potato	½ egg (grade A)	½ pc (50 g) of <i>taukua</i> + 4 dsp (90 g) chopped chickpea + 3 glasses of soya bean milk	Breast milk on demand	4 tsps (20 g) of cooking fat/ chopped nuts
Vegan	980	4 dsp (40 g) of cooked vegetables	1 banana, berangan/ apple/ 1 small slice of papaya (60 g)	1 cup (100 g) of rice + 1 slice of fortified bread (30 g) + 1 whole medium sized (80 g) potato	0	½ pc (50 g) of <i>taukua</i> + 4 dsp (90 g) chopped chickpea + 3 glasses of soya bean milk	Breast milk on demand	4 tsps (20 g) of cooking fat/ chopped nuts

* Recommended amount of each food group refers to the maximum requirement of each age group.

** The recommended amount or portion size of vegetables to meet the vitamin C requirement.

*** For age 2 years old and below, total energy intake is divided into 2 sources: complementary feeding and milk feeding. For infants fed with infant formula, the amount intake of formula milk must not exceed 413 kcal/day and 379 kcal/day for infants aged 6 to 8 months and 9 to 11 months respectively. (Dewey et. al., 2004); or not more than 5 feedings per day and 4 feedings for infants aged 6 to 8 months and 9 to 11 months respectively. (FAO/WHO, 2020). If baby is not breastfed, give in addition : 1-2 cups of milk per day (WHO & UNICEF, 2021).

Note:

- Malaysian Breastfeeding Policy states that all mothers are encouraged to breastfeed their children with breast milk from birth until six months old continuing until two years of age. Complementary foods should be given from 6 months old.
- Added sugar is not allowed for children below 2 years old, WHO 2015 recommends that intake of added sugar for children 3 years old not more than 2 teaspoons per day.
- Recommended amount of fats/ oil intake per day is including oil used in cooking and food preparation.
- Refer to figure 1.5 for standard household measurements used in this dietary guidelines.

Table 1.2: Recommendation of food servings for lacto-ovo-vegetarians by age group

Age group	Energy (kcal)	Vegetable ¹	Fruit ²	Rice, other cereals, wholegrain cereal-based products and tubers ³	Eggs ⁴	Legumes ⁵	Milk & Milk products ⁶	Fat (including nuts) ⁷	Sugar ⁸
4-6 yrs (Girl)	1,210	3	2	3	½	1	2	3	1
4-6 yrs (Boy)	1,300	3	2	3	½	1	2	4	1
7-9 yrs (Girl)	1,610	3	2	3	1	1	3	5	2
7-9 yrs (Boy)	1,750	3	2	4	1	1	3	5	2
10-12 yrs (Girl)	1,710	3	2	4	1	1	3	5	2
10-12 yrs (Boy)	1,930	3	3	5	1	1	3	5	2
13-15 yrs (Girl)	1,810	3	2	5	1	1	3	5	2
13-15 yrs (Boy)	2,210	3	3	7	1	1	3	5	2
16-<18 yrs (Girl)	1,890	3	2	5	1	1	3	5	2
16-<18 yrs (Boy)	2,340	4	4	7	1	1	3	5	2
18-59 yrs (Adult)	1,500	3	2	3	1	2	2	5	1
18- 59 yrs (Adult)	1,800	3	2	5	1	2	2	5	2
18-59 yrs (Adult)	2,000	3	3	5	1	2	2	5	2
18-59 yrs (Pregnant 1 st trimester)	1,740	3	3	4	2	1	2	5	1
18-59 yrs (Pregnant 2 nd trimester)	1,940	3	3	5	2	2	2	5	1
18-59 yrs (Pregnant 3 rd trimester);	2,130	3	3	6	2	2	2	5	2
18-29 yrs (1 st 6mths lactation)									
30-59 yrs (1 st 6mths lactation)	2,160	3	3	6	2	2	2	5	2

Note:
¹Calorie free

²Based on 15 g carbohydrate and 60 kcal per serving;

³Based on 30 g carbohydrate, 4 g protein, 1 g fat and 150 kcal per serving;

⁴Based on 14 g protein, 8 g fat and 130 kcal per serving;

⁵Based on 40 g carbohydrate, 14 g protein, 0.5 g fat and 220 kcal per serving.

⁶Based on 15 g carbohydrate, 8 g protein, 1 g fat and 90 kcal per serving;

⁷Based on 5 g fat and 45 kcal;

⁸Based on 15 g CHO and 60 kcal per serving.

Sources: Suzana et al. (2015); *RNI (2017)

Table 1.3: Recommendation of food servings for lacto-vegetarians by age group

Age group	Energy (kcal)	Vegetable ¹	Fruit ²	Rice, other cereals, wholegrain cereal-based products and tubers ³	Eggs ⁴	Legumes ⁵	Milk & Milk products ⁶	Fat (including nuts) ⁷	Sugar ⁸
4-6 yrs (Girl)	1,210	3	2	3	0	1	2	3	1
4-6 yrs (Boy)	1,300	3	2	3	0	1	2	4	1
7-9 yrs (Girl)	1,610	3	2	3	0	2	3	5	2
7-9 yrs (Boy)	1,750	3	2	4	0	2	3	5	2
10-12 yrs (Girl)	1,710	3	2	4	0	2	3	5	2
10-12 yrs (Boy)	1,930	3	3	5	0	2	3	5	2
13-15 yrs (Girl)	1,810	3	2	5	0	2	3	5	2
13-15 yrs (Boy)	2,210	3	3	7	0	2	3	5	2
16-<18 yrs (Girl)	1,890	3	2	5	0	2	3	5	2
16-<18 yrs (Boy)	2,340	4	4	7	0	2	3	5	2
18-59 yrs (Adult)	1,500	3	2	3	0	2	2	5	1
18- 59 yrs (Adult)	1,800	3	2	5	0	2	2	5	2
18-59 yrs (Adult)	2,000	3	3	5	0	2	2	5	2
18-59 yrs (Pregnant 1 st trimester)	1,740	3	3	4	0	2	2	5	1
18-59 yrs (Pregnant 2 nd trimester)	1,940	3	3	5	0	3	2	5	1
18-59 yrs (Pregnant 3 rd trimester); 18-29 yrs (1 st 6mths lactation)	2,130	3	3	6	0	3	2	5	2
30-59 yrs (1 st 6mths lactation)	2,160	3	3	6	0	3	2	5	2

Note:

¹Calorie free

²Based on 15 g carbohydrate and 60 kcal per serving;

³Based on 30 g carbohydrate, 4 g protein, 1 g fat and 150 kcal per serving;

⁴Based on 14 g protein, 8 g fat and 130 kcal per serving;

⁵Based on 40 g carbohydrate, 14g protein, 0.5 g fat and 220 kcal per serving.

⁶Based on 15 g carbohydrate, 8g protein, 1 g fat and 90 kcal per serving;

⁷Based on 5 g fat and 45 kcal;

⁸Based on 15 g CHO and 60 kcal per serving.

Sources: Suzana et al. (2015); *RNI (2017)

Table 1.4: Recommendation of food servings for ovo-vegetarians by age group

Age group	Energy (kcal)	Vegetable ¹	Fruit ²	Rice, other cereals, wholegrain cereal-based products and tubers ³	Eggs ⁴	Legumes ⁵	Milk & Milk products ⁶	Fat (including nuts) ⁷	Sugar ⁸
4-6 yrs (Girl)	1,210	3	2	3	1	1	0	6	1
4-6 yrs (Boy)	1,300	3	2	3	1	2	0	6	1
7-9 yrs (Girl)	1,610	3	2	4	1	2	0	6	2
7-9 yrs (Boy)	1,750	3	3	5	1	2	0	6	2
10-12 yrs (Girl)	1,710	3	3	1	1	2	0	6	2
10-12 yrs (Boy)	1,930	3	3	6	2	2	0	6	2
13-15 yrs (Girl)	1,810	3	3	6	1	2	0	6	2
13-15 yrs (Boy)	2,210	3	3	7	2	2	0	7	2
16-<18 yrs (Girl)	1,890	3	3	6	1	2	0	6	2
16-<18 yrs (Boy)	2,340	4	4	7	2	2	0	7	2
18-59 yrs (Adult)	1,500	3	2	4	2	2	0	6	1
18- 59 yrs (Adult)	1,800	3	3	6	1	2	0	6	2
18-59 yrs (Adult)	2,000	3	3	6	2	2	0	7	2
18-59 yrs (Pregnant 1 st trimester)	1,740	3	3	5	2	2	0	6	1
18-59 yrs (Pregnant 2 nd trimester)	1,940	3	3	6	2	2	0	6	1
18-59 yrs (Pregnant 3 rd trimester); 18-29 yrs (1 st 6mths lactation)	2,130	4	4	6	2	2	0	7	2
30-59 yrs (1 st 6mths lactation)	2,160	4	4	6	2	2	0	7	2

Note:
¹Calorie free

²Based on 15 g carbohydrate and 60 kcal per serving;

³Based on 30 g carbohydrate, 4 g protein, 1 g fat and 150 kcal per serving;

⁴Based on 14 g protein, 8 g fat and 130 kcal per serving;

⁵Based on 40 g carbohydrate, 14 g protein, 0.5 g fat and 220 kcal per serving.

⁶Based on 15 g carbohydrate, 8 g protein, 1 g fat and 90 kcal per serving;

⁷Based on 5 g fat and 45 kcal;

⁸Based on 15 g CHO and 60 kcal per serving.

Sources: Suzana et al. (2015); *RNI (2017)

Table 1.5: Recommendation of food servings for vegan by age group

Age group	Energy (kcal)	Vegetable ¹	Fruit ²	Rice, other cereals, wholegrain cereal-based products and tubers ³	Eggs ⁴	Legumes ⁵	Milk & Milk products ⁶	Fat (including nuts) ⁷	Sugar ⁸
4-6 yrs (Girl)	1,210	3	2	3	0	2	0	6	1
4-6 yrs (Boy)	1,300	3	2	3	0	2	0	6	1
7-9 yrs (Girl)	1,610	3	2	4	0	3	0	6	2
7-9 yrs (Boy)	1,750	3	3	5	0	3	0	6	2
10-12 yrs (Girl)	1,710	3	3	5	0	3	0	6	2
10-12 yrs (Boy)	1,930	3	3	6	0	3	0	6	2
13-15 yrs (Girl)	1,810	3	3	6	0	3	0	6	2
13-15 yrs (Boy)	2,210	3	3	7	0	3	0	7	2
16-<18 yrs (Girl)	1,890	3	3	6	0	3	0	6	2
16-<18 yrs (Boy)	2,340	4	4	7	0	3	0	7	2
18-59 yrs (Adult)	1,500	3	2	4	0	3	0	6	1
18- 59 yrs (Adult)	1,800	3	3	6	0	3	0	6	2
18-59 yrs (Adult)	2,000	3	3	6	0	3	0	7	2
18-59 yrs (Pregnant 1 st trimester)	1,740	3	3	5	0	3	0	6	1
18-59 yrs (Pregnant 2 nd trimester)	1,940	3	3	6	0	3	0	6	1
18-59 yrs (Pregnant 3 rd trimester); 18-29 yrs (1 st 6mths lactation)	2,130	4	4	6	0	3	0	7	2
30-59 yrs (1 st 6mths lactation)	2,160	4	4	6	0	3	0	7	2

Note:

¹Calorie free

²Based on 15 g carbohydrate and 60 kcal per serving;

³Based on 30 g carbohydrate, 4 g protein, 1 g fat and 150 kcal per serving;

⁴Based on 14 g protein, 8 g fat and 130 kcal per serving;

⁵Based on 40 g carbohydrate, 14 g protein, 0.5 g fat and 220 kcal per serving.

⁶Based on 15 g carbohydrate, 8 g protein, 1 g fat and 90 kcal per serving;

⁷Based on 5 g fat and 45 kcal;

⁸Based on 15 g CHO and 60 kcal per serving.

Sources: Suzana et al. (2015); *RNI (2017)

Key Recommendation 2

Eat your main meals (breakfast, lunch and dinner) as recommended by the Malaysian Healthy Plate for Vegetarians. (Figure 1.3).



Figure 1.3: Malaysian Healthy Plate for Vegetarians

How to achieve:

- Use the quarter-quarter-half concept to plan your main meals.
 - Fill in the first quarter of your plate with rice or other cereals (e.g. *meehoon*) or wholegrain cereal-based products (e.g. wholegrain bread) or tubers (e.g. sweet potato). Preferably fill your plate with whole grains.
 - Fill in the second quarter of your plate with egg/ legumes/ nuts (e.g. dhal, tempeh, soya bean curd)/ dairy products.
 - Fill half of your plate with vegetables and fruits.
- Drink plain water or unsweetened beverages with the meal.
- Consume milk and/ or milk products as recommended. Ovo-vegetarians and vegans should make a special effort to consume foods with good calcium sources such as calcium-fortified foods (cereals, soya bean milk), legumes (cowpea black gram lentils, dhal, soya bean), vegetables (*bok choy*, okra, amaranth) and condiments and spices (cardamom, coriander, cumin and fenugreek seeds). Refer to Key Message 5 for further information.

4. Eat a variety of foods with protein sources daily such as whole grains (e.g. millet or bajra (pearl millet) or foxtail millet and quinoa), soya and soya products (e.g. *tauhu*, tempeh, beancurd sheet or *fucuk keping*, soya chip or *dou bao*), beans (broad bean), lentils (dhal), seeds and nuts, milk and eggs.
5. Limit adding condiments such as soy sauce, tomato sauce, chili sauce and gravies that is high in salt, sugar and fat to your main meal (breakfast, lunch and dinner).

Key Recommendation 3

Choose micronutrient-rich food in your main meal.

How to achieve:

1. Consume egg or drink milk daily as sources of Vitamin B12. Vegans may choose fortified foods (added Vitamin B12, e.g. cereals, soya milk or biscuit) and shiitake mushrooms.
2. Eat more foods rich in iron (e.g. red/ black dates, raisins, fungus, black beans or kidney beans) daily and together with foods rich in vitamin C (e.g. guava, banana, *kedondong*, papaya or pineapple) in the same meal.
3. Eat other vegetables rich in minerals (e.g. seaweed, white fungus (*xue er*), black fungus (*mu er*), mushroom, konjac (*ju rou*), herbs and spices, as part of the daily total vegetable intake (Figure 1.4).
4. Encourage to eat fermented food products such as tempeh, miso and yoghurt.

Key Recommendation 4

Limit intakes of processed and ultra-processed foods.

How to achieve:

1. Prioritize fresh foods instead of processed foods, including plant-based protein products (such as *tauhu*, tempeh, beancurd sheet or *fucuk keping*, soya chip or *dou bao*) in daily menu planning.
2. Prepare or choose natural ingredients for cooking instead of using ingredients made from processed foods (such as unseasoned textured vegetable protein, isolated soya protein and minimally processed mushroom) or ultra-processed foods (such as vegetarian fish balls, meatballs, salami or sausage).
3. Limit intake of ultra-processed foods such as soft drinks, sweetened biscuits, salty fatty packaged snacks, and instant noodles, which are nutritionally unbalanced.
4. If processed plant-based protein products (such as textured soya protein or mycoprotein in meat alternative form) are consumed, limit them to 1 to 2 times per month and 3-4 pieces of cutlets in each meal. Healthier cooking methods should be emphasized.
5. Read food labels about ingredients and nutrition information panels before purchasing.
6. Choose processed plant-based protein products made from 2 or 3 ingredients without colouring, flavouring and additives. Take note that some of these products may contain eggs and/ or milk.
7. Reduce the frequency of ready-to-consume (e.g. muffin, *pau* and bun) and pre-prepared ready-to-heat (e.g. frozen meal and frozen *kueh*) food intake.
8. Be aware with advertisement of ultra-processed products which often conveys misleading information about diet and health.
9. Shop mindfully. Limit purchasing of processed and ultra-processed products.

KM1

Eat a variety of foods within the recommended servings



Seaweed



Fungus



Mushroom



Konjac



Herb & Spices

Figure 1.4: Examples of vegetables rich in mineral

1.5.1 Additional recommendations

1.5.1.1 Nutritional supplements

Eating a variety of foods daily as guided by the food pyramid should provide all the nutrients needed by the body. Furthermore, good sources of vitamins, minerals or dietary fibre include wholegrain cereals, vegetables, fruits, legumes, beans, lentils, nuts and seeds that should be obtained from natural foods, rather than dietary supplements (in the form of liquids, tablets or powder), because these foods provide other micronutrients and other essential components such as amino acids and phytonutrients, which provide their own nutritional benefits (NCCFN, 2021). Therefore, supplements are not necessary for most lacto or lacto-ovo vegetarians.

Previous studies had reported that supplement use among vegetarians in Malaysia in selected areas ranged widely from 15% to 70% (Lee, 2000; Chong, 2013). A more recent quick survey on Malaysian vegetarianism reported that 41% of vegetarians are on supplementation (MOH, 2014b). The most commonly consumed supplements are vitamin B (23.3%), calcium (20.1%), omega-3 (11.5%), iron (10.2%) and zinc (8.1%). Some vegetarians, especially vegans, may have limited food choices due to practice or preference and may have tried to complement their diet with nutritional supplements. However, appropriately planned vegetarian diets that include fortified foods are nutritionally adequate for adults and children (Craig, 2010). These vegetarian variations represent healthy eating patterns and vegetarians can achieve adequate nutrients and calories by eating various foods. Appropriate dietary planning can prevent certain nutrient deficiencies (Melina et al., 2016; Thomas et al., 2016; Agnoli et al., 2017). The Food Pyramid for Malaysian Vegetarians provides an excellent general guideline for vegetarians to accomplish this.

However, supplements may be needed for certain health conditions in clinical practice to meet specific nutrient requirements, such as during convalescence (recovery and illness), in pregnant and lactating women and in the elderly. Reviews and community-based studies have shown that low vitamin B12 concentrations and hyperhomocysteinaemia to be common among vegetarians (Huang et al., 2003; Yajnik et al., 2006; Obersby et al., 2013; Trilok-Kumar, 2020). Serum zinc concentration was also significantly lower among vegetarians than non-vegetarians (Foster et al., 2013). This may result from dietary restrictions imposed by strict vegetarians, such as vegans who may consume limited vitamin B12 and zinc sources in their daily diet (Pawlak et al., 2013; Shridhar et al., 2014; Naik et al., 2018; Bakaloudi et al., 2021). Specifically, born-vegan infants, children, adolescents, pregnant, lactating women and the elderly may be vulnerable. Hence, it is particularly important for the diet of these groups to be monitored against the recommendations (Table 1.1 – 1.5). The growth of young children also needs to be monitored using standard practice to avoid issues of malnutrition, growth failure and developmental delay as described in the literature (Kiely, 2021).

Additionally, research and nutrition groups have recommended fortified foods or supplementation in such cases (Melina et al., 2016; Agnoli et al., 2017; Larpin et al., 2019). Certain dietary practices such as pre-soaking of legumes and modern food processing methods could improve the bioavailability of zinc in particular (Foster et al., 2013; Gibson et al., 2014). Vegetarians have to read food labels (more in Key Message 7) and choose fortified foods to obtain sufficient vitamin B12 and zinc. Vegetarians may need vitamin B12 and zinc supplementations if there are no suitable fortified foods available, and they are advised to seek the recommendations of nutritionists, dietitians or medical doctors.

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Eat a variety of foods within the recommended servings

1.5.1.2 Limit the intake of ultra-processed foods, salt and sugar in a vegetarian diet

Higher avoidance of animal-based foods has been associated with a higher intake of ultra-processed foods (UPFs). UPFs contribute up to 39.5% of the total energy intake of vegans (Gehring et al, 2021). Analysis of nationally representative samples has shown UPFs are inversely associated with overall diet quality with increased free sugars, total fats, saturated fats, and decreased fibre, protein and micronutrient intakes (Martini et al, 2021). Consumption of UPFs has been consistently associated with increased risk for weight gain and related comorbidities (Crimarco et al, 2022). Hence, vegetarians must be mindful of the inclusion of UPFs in their diet and keep the intake minimum.

Salt and sugar are both flavour enhancers in foods and/ or beverages. Besides being flavour-enhancing agents, salt and sugar provide some health and nutrition benefits. However, excessive salt and sugar consumption in foods and beverages has been linked to chronic non-communicable diseases. For example, high intakes of salt/ sodium have been strongly linked to hypertension (WHO, 2012) and stomach cancer (D'Elia et al, 2012), while high sugar intake has been associated with overweight and obesity, including dental caries in children. Therefore, it is vital to limit salt and the use and intake of salt and sugar in a vegetarian diet. With that, vegetarians are recommended to apply the key messages and related key recommendations on salt and sugar, which are available in the Malaysian Dietary Guidelines.

1.5.1.3 Increase the nutrient intakes and improve absorption level

- Add whole grains (such as wheat, oat, brown rice or barley), lentils (including dhals such as *masoor* and *urad*) or beans into the rice.
- Soak dried beans and discard (drain off) the soaking water before cooking.
- Get sufficient exposure (at least 20 minutes) to sunlight in the morning, especially for vegans.

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Eat a variety of foods within the recommended servings



1.6 References

- Agnoli, C., Baroni, L., Bertini, I., Ciappellano, S., Fabbri, A., Papa, M., Pellegrini, N., Sbarbati, R., Scarino, M. L., Siani, V., & Sieri, S. (2017). Position paper on vegetarian diets from the working group of the Italian Society of Human Nutrition. *Nutrition, Metabolism & Cardiovascular Diseases*, 27(12), 1037–1052. <https://doi.org/10.1016/j.numecd.2017.10.020>.
- Appleby, P. N., Thorogood, M., Mann, J. I., & Key, T. J. (1999). The Oxford Vegetarian Study: an overview. *American Journal of Clinical Nutrition*, 70(3 Suppl), 525S–531S. <https://doi.org/10.1093/ajcn/70.3.525s>.
- Asgar, M.A., Fazilah, A., Huda, N., Bhat, R. & Karim, A.A. (2010). Nonmeat protein alternatives as meat extenders and meat analogs. *Comprehensive Review in Food Science and Food Safety*. 9(5):513-529. <https://doi.org/10.1111/j.1541-4337.2010.00124.x>.
- Aune D. (2019). Plant foods, antioxidant biomarkers, and the risk of cardiovascular disease, cancer, and mortality: A review of the evidence. *Advances in Nutrition*, 10(Suppl_4), S404–S421. <https://doi.org/10.1093/advances/nmz042>.
- Bakaloudi, D. R., Halloran, A., Rippin, H. L., Oikonomidou, A. C., Dardavesi, T. I., Williams, J., Wickramasinghe, K., Breda, J., & Chourdakis, M. (2021). Intake and adequacy of the vegan diet. A systematic review of the evidence. *Clinical Nutrition*, 40(5), 3503–3521. <https://doi.org/10.1016/j.clnu.2020.11.035>.
- Benatar, J. R., & Stewart, R. A. H. (2018). Cardiometabolic risk factors in vegans; A meta-analysis of observational studies. *PloS One*, 13(12), e0209086. <https://doi.org/10.1371/journal.pone.0209086>.
- Bowen, K. J., Sullivan, V. K., Kris-Etherton, P. M., & Petersen, K. S. (2018). Nutrition and cardiovascular disease-an update. *Current Atherosclerosis Reports*, 20(2), 8. <https://doi.org/10.1007/s11883-018-0704-3>.
- Bowman, S.A. (2020). A vegetarian-style dietary pattern is associated with lower energy, saturated fat, and sodium intakes; and higher whole grains, legumes, nuts, and soy intakes by adults: National Health and Nutrition Examination Surveys 2013-2016. *Nutrients*. 12(9):2668. <https://doi.org/10.3390/nu12092668>.
- Cheah, W.L., Chang, F.H., Tang, M.F., Yuvaraj, L., Aldrin, R.D. & Ab Llah, N.I. (2018). Vegetarian practices, body composition and dietary intake among Hindus and Buddhists in Kuching, Sarawak, Malaysia. *South East Asia Journal of Public Health*. 8(1):3-9. <https://doi.org/10.3329/seajph.v8i1.42266>.
- Chen, Y.F. (2001). Nutritional status and dietary habits among Buddhist vegetarian women. [BSc Thesis]. Universiti Kebangsaan Malaysia.
- Ching, Y. K., Chin, Y. S., Appukutty, M., Ramanchadran, V., Yu, C. Y., Ang, G. Y., Gan, W. Y., Chan, Y. M., Teh, L. K., & Salleh, M. Z. (2019). Interaction of dietary linoleic acid and α -linolenic acids with rs174547 in FADS1 Gene on metabolic syndrome components among vegetarians. *Nutrients*, 11(7), 1686. <https://doi.org/10.3390/nu11071686>.
- Chong, G.Y. (2013). *Study on iron status and nutritional status among vegetarian in Selangor*. [BSc Thesis]. Universiti Kebangsaan Malaysia.
- Clarys, P., Deriemaeker, P., Huybrechts, I., Hebbelinck, M., & Mullie, P. (2013). Dietary pattern analysis: a comparison between matched vegetarian and omnivorous subjects. *Nutrition Journal*, 12, 82. <https://doi.org/10.1186/1475-2891-12-82>.
- Clarys, P., Deliens, T., Huybrechts, I., Deriemaeker, P., Vanaelst, B., De Keyzer, W., Hebbelinck, M., & Mullie, P. (2014). Comparison of nutritional quality of the vegan, vegetarian, semi-vegetarian, pesco-vegetarian and omnivorous diet. *Nutrients*, 6(3), 1318–1332. <https://doi.org/10.3390/nu6031318>.

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Eat a variety of foods within the recommended servings

- Craig W. J. (2010). Nutrition concerns and health effects of vegetarian diets. *Nutrition in Clinical Practice*, 25(6), 613–620. <https://doi.org/10.1177/0884533610385707>.
- Crimarco, A., Landry, M. J., & Gardner, C. D. (2022). Ultra-processed foods, weight gain, and co-morbidity risk. *Current Obesity Reports*, 11(3), 80–92. <https://doi.org/10.1007/s13679-021-00460-y>.
- Cofnas N. (2019). Is vegetarianism healthy for children?. *Critical Reviews in Food Science and Nutrition*, 59(13), 2052–2060. <https://doi.org/10.1080/10408398.2018.1437024>.
- D'Elia, L., Rossi, G., Ippolito, R., Cappuccio, F. P., & Strazzullo, P. (2012). Habitual salt intake and risk of gastric cancer: a meta-analysis of prospective studies. *Clinical Nutrition*, 31(4), 489–498. <https://doi.org/10.1016/j.clnu.2012.01.003>.
- Derbyshire, E.J. & Delange, J. (2021). Fungal protein – what is it and what is the health evidence? A systematic review focusing on mycoprotein. *Frontiers in Sustainable Food Systems*. 5:581682. <https://doi.org/10.3389/fsufs.2021.581682>.
- Derbyshire, E. & Ayoob, K.T. (2019). Mycoprotein: Nutritional and health properties. *Nutrition Today*. 54(1):7-15. <https://doi.org/10.1097/NT.0000000000000316>.
- Deriemaeker, P., Alewaeters, K., Hebbelinck, M., Lefevre, J., Philippaerts, R., & Clarys, P. (2010). Nutritional status of Flemish vegetarians compared with non-vegetarians: a matched samples study. *Nutrients*, 2(7), 770–780. <https://doi.org/10.3390/nu2070770>.
- Dinu, M., Abbate, R., Gensini, G. F., Casini, A., & Sofi, F. (2017). Vegetarian, vegan diets and multiple health outcomes: A systematic review with meta-analysis of observational studies. *Critical Reviews in Food Science and Nutrition*, 57(17), 3640–3649. <https://doi.org/10.1080/10408398.2016.1138447>.
- English, L. K., Ard, J. D., Bailey, R. L., Bates, M., Bazzano, L. A., Boushey, C. J., Brown, C., Butera, G., Callahan, E. H., de Jesus, J., Mattes, R. D., Mayer-Davis, E. J., Novotny, R., Obbagy, J. E., Rahavi, E. B., Sabate, J., Snetselaar, L. G., Stoody, E. E., Van Horn, L. V., Venkatramanan, S., ... Heymsfield, S. B. (2021). Evaluation of dietary patterns and all-cause mortality: A systematic review. *JAMA Network Open*, 4(8), e2122277. <https://doi.org/10.1001/jamanetworkopen.2021.22277>.
- Eveleigh, E. R., Coneyworth, L. J., Avery, A., & Welham, S. J. M. (2020). Vegans, vegetarians, and omnivores: How does dietary choice influence iodine intake? A systematic review. *Nutrients*, 12(6), 1606. <https://doi.org/10.3390/nu12061606>.
- Finnigan, T. J. A., Wall, B. T., Wilde, P. J., Stephens, F. B., Taylor, S. L., & Freedman, M. R. (2019). Mycoprotein: The future of nutritious nonmeat protein, A symposium review. *Current Developments in Nutrition*, 3(6), nzz021. <https://doi.org/10.1093/cdn/nzz021>.
- Foster, M., Chu, A., Petocz, P., & Samman, S. (2013). Effect of vegetarian diets on zinc status: a systematic review and meta-analysis of studies in humans. *Journal of the Science of Food and Agriculture*, 93(10), 2362–2371. <https://doi.org/10.1002/jsfa.6179>.
- Fuhrman, J., & Ferreri, D. M. (2010). Fueling the vegetarian (vegan) athlete. *Current Sports Medicine Reports*, 9(4), 233–241. <https://doi.org/10.1249/JSR.0b013e3181e93a6f>.
- Gallego-Narbón, A., Zapatera, B., Barrios, L., & Vaquero, M. P. (2019). Vitamin B12 and folate status in Spanish lacto-ovo vegetarians and vegans. *Journal of Nutritional Science*, 8, e7. <https://doi.org/10.1017/jns.2019.2>.
- García Maldonado, E., Gallego-Narbón, A., & Vaquero, M.^aP. (2019). ¿Son las dietas vegetarianas nutricionalmente adecuadas? Una revisión de la evidencia científica [Are vegetarian diets nutritionally adequate? A revision of the scientific evidence]. *Nutrición Hospitalaria*, 36(4), 950–961. <https://doi.org/10.20960/nh.02550>.

- Gehring, J., Touvier, M., Baudry, J., Julia, C., Buscail, C., Srouf, B., Hercberg, S., Péneau, S., Kesse-Guyot, E. & Allès, B. (2021). Consumption of ultra-processed foods by pesco-vegetarians, vegetarians, and vegans: Associations with duration and age at diet initiation. *Journal of Nutrition*. 151(1):120-131. <https://doi.org/10.1093/jn/nxaa196>.
- Gibson, R. S., Heath, A. L., & Szymlek-Gay, E. A. (2014). Is iron and zinc nutrition a concern for vegetarian infants and young children in industrialized countries?. *American Journal of Clinical Nutrition*, 100 Suppl 1, 459S–68S. <https://doi.org/10.3945/ajcn.113.071241>.
- Glenn, A. J., Viguiliouk, E., Seider, M., Boucher, B. A., Khan, T. A., Blanco Mejia, S., Jenkins, D. J. A., Kahleová, H., Rahelić, D., Salas-Salvadó, J., Kendall, C. W. C., & Sievenpiper, J. L. (2019). Relation of vegetarian dietary patterns with major cardiovascular outcomes: a systematic review and meta-analysis of prospective cohort studies. *Frontiers in Nutrition*, 6, 80. <https://doi.org/10.3389/fnut.2019.00080>.
- Halvorsen, R. E., Elvestad, M., Molin, M., & Aune, D. (2021). Fruit and vegetable consumption and the risk of type 2 diabetes: a systematic review and dose-response meta-analysis of prospective studies. *BMJ Nutrition, Prevention & Health*, 4(2), 519–531. <https://doi.org/10.1136/bmjnp-2020-000218>.
- Hargreaves, S. M., Araújo, W. M. C., Nakano, E. Y., & Zandonadi, R. P. (2020). Brazilian vegetarians diet quality markers and comparison with the general population: A nationwide cross-sectional study. *PloS One*, 15(5), e0232954. <https://doi.org/10.1371/journal.pone.0232954>.
- Hartley, L., Igbinedion, E., Holmes, J., Flowers, N., Thorogood, M., Clarke, A., Stranges, S., Hooper, L., & Rees, K. (2013). Increased consumption of fruit and vegetables for the primary prevention of cardiovascular diseases. *Cochrane Database of Systematic Reviews*, 2013(6), CD009874. <https://doi.org/10.1002/14651858.CD009874.pub2>.
- Hashempour-Baltork, F., Khosravi-Darani, K., Hosseini, H., Farshi, P. & Reihani, F.S. (2020). Mycoproteins as safe meat substitutes. *Journal of Cleaner Production*. 253: 119958. <https://doi.org/10.1016/j.jclepro.2020.119958>.
- He, J., Evans, N. M., Liu, H., & Shao, S. (2020). A review of research on plant-based meat alternatives: Driving forces, history, manufacturing, and consumer attitudes. *Comprehensive Reviews in Food Science and Food Safety*, 19(5), 2639–2656. <https://doi.org/10.1111/1541-4337.12610>.
- Huang, Y. C., Chang, S. J., Chiu, Y. T., Chang, H. H., & Cheng, C. H. (2003). The status of plasma homocysteine and related B-vitamins in healthy young vegetarians and nonvegetarians. *European Journal of Nutrition*, 42(2), 84–90. <https://doi.org/10.1007/s00394-003-0387-5>.
- Institute for Public Health (IPH). 2014. *National Health and Morbidity Survey 2014: Malaysian Adult Nutrition Survey (MANS) Vol. II: Survey Findings*.
- Jabri, A., Kumar, A., Verghese, E., Alameh, A., Kumar, A., Khan, M. S., Khan, S. U., Michos, E. D., Kapadia, S. R., Reed, G. W., & Kalra, A. (2021). Meta-analysis of effect of vegetarian diet on ischemic heart disease and all-cause mortality. *American Journal of Preventive Cardiology*, 7, 100182. <https://doi.org/10.1016/j.ajpc.2021.100182>.
- Keaver, L., Ruan, M., Chen, F., Du, M., Ding, C., Wang, J., Shan, Z., Liu, J., & Zhang, F. F. (2021). Plant- and animal-based diet quality and mortality among US adults: a cohort study. *British Journal of Nutrition*, 125(12), 1405–1415. <https://doi.org/10.1017/S0007114520003670>.
- Kennedy, E. (2004). Dietary diversity, diet quality, and body weight regulation. *Nutrition Reviews*. 62(7 Pt 2):S78-81. <https://doi.org/10.1111/j.1753-4887.2004.tb00093.x>.
- Kersting, M., Kalhoff, H., Melter, M. & Lucke, T. (2018). Vegetarian diets in children? - An assessment from pediatrics and nutrition science. *Deutsche Medizinische Wochenschrift*. 143 (4): 279-286. <https://doi.org/10.1055/s-0043-119864>.

- Khor, G.L., Voon, P.C., Tee, E.S., Hage, B.H.H. & Wahlqvist, M.L. (2000). Cardiovascular risk factors among Malaysian urban vegetarians. *Journal of Community Nutrition*. 2(2):110–8.
- Kiely M. E. (2021). Risks and benefits of vegan and vegetarian diets in children. *The Proceedings of the Nutrition Society*, 80(2), 159–164. <https://doi.org/10.1017/S002966512100001X>.
- Kim, H., Caulfield, L. E., Garcia-Larsen, V., Steffen, L. M., Coresh, J., & Rebholz, C. M. (2019). Plant-based diets are associated with a lower risk of incident cardiovascular disease, cardiovascular disease mortality, and all-cause mortality in a general population of middle-aged adults. *Journal of the American Heart Association*, 8(16), e012865. <https://doi.org/10.1161/JAHA.119.012865>.
- Kwok, T., Chook, P., Qiao, M., Tam, L., Poon, Y. K., Ahuja, A. T., Woo, J., Celermajer, D. S., & Woo, K. S. (2012). Vitamin B-12 supplementation improves arterial function in vegetarians with subnormal vitamin B-12 status. *Journal of Nutrition, Health & Aging*, 16(6), 569–573. <https://doi.org/10.1007/s12603-012-0036-x>.
- Lamb, M. J., Griffin, S. J., Sharp, S. J., & Cooper, A. J. (2017). Fruit and vegetable intake and cardiovascular risk factors in people with newly diagnosed type 2 diabetes. *European Journal of Clinical Nutrition*, 71(1), 115–121. <https://doi.org/10.1038/ejcn.2016.180>.
- Lane, M. M., Davis, J. A., Beattie, S., Gómez-Donoso, C., Loughman, A., O'Neil, A., Jacka, F., Berk, M., Page, R., Marx, W., & Rocks, T. (2021). Ultraprocessed food and chronic noncommunicable diseases: A systematic review and meta-analysis of 43 observational studies. *Obesity Reviews*, 22(3), e13146. <https://doi.org/10.1111/obr.13146>.
- Larpin, C., Wozniak, H., Genton, L., & Serratrice, J. (2019). Alimentations végétariennes et véganes : quelles conséquences sur la santé ? [Vegetarian and vegan diets and their impact on health]. *Revue médicale suisse*, 15(667), 1849–1853.
- Lee, A.C. (2000). *Assessment of nutritional status and dietary habits among Chinese vegetarians*. [BSc Thesis]. Universiti Kebangsaan Malaysia.
- Lee, K. W., Loh, H. C., Ching, S. M., Devaraj, N. K., & Hoo, F. K. (2020). Effects of vegetarian diets on blood pressure lowering: A systematic review with meta-analysis and trial sequential analysis. *Nutrients*, 12(6), 1604. <https://doi.org/10.3390/nu12061604>.
- Lee, Y., & Krawinkel, M. (2011). The nutritional status of iron, folate, and vitamin B-12 of Buddhist vegetarians. *Asia Pacific Journal of Clinical Nutrition*, 20(1), 42–49.
- Li, B., Li, F., Wang, L., & Zhang, D. (2016). Fruit and vegetables consumption and risk of hypertension: A meta-analysis. *Journal of Clinical Hypertension*, 18(5), 468–476. <https://doi.org/10.1111/jch.12777>.
- Losno, E. A., Sieferle, K., Perez-Cueto, F. J. A., & Ritz, C. (2021). Vegan diet and the gut microbiota composition in healthy adults. *Nutrients*, 13(7), 2402. <https://doi.org/10.3390/nu13072402>.
- Malav, O. P., Talukder, S., Gokulakrishnan, P., & Chand, S. (2015). Meat analog: a review. *Critical Reviews in Food Science and Nutrition*, 55(9), 1241–1245. <https://doi.org/10.1080/10408398.2012.689381>.
- Malik, A., & Trilok-Kumar, G. (2020). Status of Vitamin B12 among healthy adult and elderly population in India: A review. *Journal of Nutritional Science and Vitaminology*, 66(Suppl), S361–S368. <https://doi.org/10.3177/jnsv.66.S361>.
- Mariotti, F., & Gardner, C. D. (2019). Dietary protein and amino acids in vegetarian diets-a review. *Nutrients*, 11(11), 2661. <https://doi.org/10.3390/nu11112661>.
- Marrone, G., Guerriero, C., Palazzetti, D., Lido, P., Marolla, A., Di Daniele, F., & Noce, A. (2021). Vegan diet health benefits in metabolic syndrome. *Nutrients*, 13(3), 817. <https://doi.org/10.3390/nu13030817>.

- Martini, D., Godos, J., Bonaccio, M., Vitaglione, P., & Grosso, G. (2021). Ultra-processed foods and nutritional dietary profile: A meta-analysis of nationally representative samples. *Nutrients*, 13(10), 3390. <https://doi.org/10.3390/nu13103390>.
- McDougall J. (2002). Plant foods have a complete amino acid composition. *Circulation*, 105(25), e197. <https://doi.org/10.1161/01.cir.0000018905.97677.1f>.
- McEvoy, C. T., Temple, N., & Woodside, J. V. (2012). Vegetarian diets, low-meat diets and health: a review. *Public Health Nutrition*, 15(12), 2287–2294. <https://doi.org/10.1017/S1368980012000936>.
- Melina, V., Craig, W., & Levin, S. (2016). Position of the Academy of Nutrition and Dietetics: Vegetarian Diets. *Journal of the Academy of Nutrition and Dietetics*, 116(12), 1970–1980. <https://doi.org/10.1016/j.jand.2016.09.025>.
- Michel, F., Hartmann, C. & Siegrist, M. (2021). Consumers' associations, perceptions and acceptance of meat and plant-based meat alternatives. *Food Quality & Preference*. 87: 104063. <https://doi.org/10.1016/j.foodqual.2020.104063>.
- Miller, V., Webb, P., Micha, R., Mozaffarian, D., & Global Dietary Database (2020). Defining diet quality: a synthesis of dietary quality metrics and their validity for the double burden of malnutrition. *The Lancet. Planetary Health*, 4(8), e352–e370. [https://doi.org/10.1016/S2542-5196\(20\)30162-5](https://doi.org/10.1016/S2542-5196(20)30162-5).
- Ministry of Health, MOH (2014a). *Survey on nutrition information panel from food products in market*. [Unpublished report]. Nutrition Division, Ministry of Health Malaysia.
- Ministry of Health, MOH (2014b). *Quick survey on vegetarian practice in Malaysia selected areas*. [Unpublished report]. Nutrition Division, Ministry of Health Malaysia.
- Ministry of Health, MOH (2016). *Panduan Pinggan Sihat Malaysia*. Ministry of Health Malaysia.
- Monteiro, C. A., Cannon, G., Levy, R. B., Moubarac, J. C., Louzada, M. L., Rauber, F., Khandpur, N., Cediel, G., Neri, D., Martinez-Steele, E., Baraldi, L. G., & Jaime, P. C. (2019). Ultra-processed foods: what they are and how to identify them. *Public Health Nutrition*, 22(5), 936–941. <https://doi.org/10.1017/S1368980018003762>.
- Monteiro, C.A., Cannon, G., Lawrence, M., Costa Louzada, M. & Pereira Machado, P. (2019b). *Ultra-processed foods, diet quality, and health using the NOVA classification system*. Rome, FAO.
- Morze, J., Danielewicz, A., Hoffmann, G., & Schwingshackl, L. (2020). Diet quality as assessed by the Healthy Eating Index, Alternate Healthy Eating Index, Dietary Approaches to Stop Hypertension Score, and Health Outcomes: A second update of a systematic review and meta-analysis of cohort studies. *Journal of the Academy of Nutrition and Dietetics*, 120(12), 1998–2031.e15. <https://doi.org/10.1016/j.jand.2020.08.076>.
- Müller, P. (2020). *Vegan Diet in Young Children*. Nestle Nutrition Institute Workshop Series 93:103–110.
- Naik, S., Bhide, V., Babhulkar, A., Mahalle, N., Parab, S., Thakre, R., & Kulkarni, M. (2013). Daily milk intake improves vitamin B-12 status in young vegetarian Indians: an intervention trial. *Nutrition Journal*, 12, 136. <https://doi.org/10.1186/1475-2891-12-136>.
- Naik, S., Mahalle, N., & Bhide, V. (2018). Identification of vitamin B12 deficiency in vegetarian Indians. *British Journal of Nutrition*, 119(6), 629–635. <https://doi.org/10.1017/S0007114518000090>.
- National Coordinating Committee on Food and Nutrition (NCCFN). (2013). *Malaysian Dietary Guidelines for Children and Adolescents*. Ministry of Health, Putrajaya, Malaysia.

- National Coordinating Committee on Food and Nutrition (NCCFN). (2017). *Recommended Nutrient Intake for Malaysians*. Ministry of Health, Putrajaya, Malaysia.
- National Coordinating Committee on Food and Nutrition (NCCFN). (2021). *Malaysian Dietary Guidelines 2020*. Ministry of Health, Putrajaya, Malaysia.
- Norat, T., Aune, D., Chan, D., & Romaguera, D. (2014). Fruits and vegetables: updating the epidemiologic evidence for the WCRF/AICR lifestyle recommendations for cancer prevention. *Cancer Treatment & Research*, 159, 35–50. https://doi.org/10.1007/978-3-642-38007-5_3.
- Obersby, D., Chappell, D. C., Dunnett, A., & Tsiami, A. A. (2013). Plasma total homocysteine status of vegetarians compared with omnivores: a systematic review and meta-analysis. *British Journal of Nutrition*, 109(5), 785–794. <https://doi.org/10.1017/S000711451200520X>.
- Oussalah, A., Levy, J., Berthezène, C., Alpers, D. H., & Guéant, J. L. (2020). Health outcomes associated with vegetarian diets: An umbrella review of systematic reviews and meta-analyses. *Clinical Nutrition*, 39(11), 3283–3307. <https://doi.org/10.1016/j.clnu.2020.02.037>.
- US Food and Drug Administration. (1999). Title 21 of the Code of Federal Regulations (1999). *Chapter I-- Food and Drug Administration. Part 101 Food Labeling. Subpart E - Specific Requirements for Health Claims. Sec. 101.82 Health Claims: Soy Protein and Risk of Coronary Heart Disease (CHD).21*, Volume 2. §21CFR101.82. Available: [Www.Accessdata.Fda.Gov/Scripts/Cdrh/Cfdocs/Cfcr/Cfrsearch.Cfm?Fr=101.82](http://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfcfr/cfrsearch.cfm?Fr=101.82).
- Parker, H. W., & Vadiveloo, M. K. (2019). Diet quality of vegetarian diets compared with nonvegetarian diets: a systematic review. *Nutrition Reviews*, 77(3), 144–160. <https://doi.org/10.1093/nutrit/nuy067>.
- Pawlak, R., Lester, S. E., & Babatunde, T. (2014). The prevalence of cobalamin deficiency among vegetarians assessed by serum vitamin B12: a review of literature. *European Journal of Clinical Nutrition*, 68(5), 541–548. <https://doi.org/10.1038/ejcn.2014.46>.
- Pollakova, D., Andreadi, A., Pacifici, F., Della-Morte, D., Lauro, D., & Tubili, C. (2021). The impact of vegan diet in the prevention and treatment of type 2 diabetes: A systematic review. *Nutrients*, 13(6), 2123. <https://doi.org/10.3390/nu13062123>.
- Pinheiro, C., Leite, J. C., Negrão, R., & Keating, E. (2020). Vegetarian diets as a possible therapeutic approach to patients with metabolic syndrome: A brief review. *Porto Biomedical Journal*, 5(6), e098. <https://doi.org/10.1097/j.pbj.0000000000000098>.
- Ramadas, A., Tham, S. M., Lalani, S. A., & Shyam, S. (2021). Diet quality of Malaysians across lifespan: A scoping review of evidence in a multi-ethnic population. *Nutrients*, 13(4), 1380. <https://doi.org/10.3390/nu13041380>.
- Rammohan A, Awofeso N & Robitaille MC (2012). Addressing female iron-deficiency anaemia in India: Is vegetarianism the major obstacle? *ISRN Public Health* 765476: 1-8. <https://doi.org/10.5402/2012/765476>.
- Rashid, S., Meier, V., & Patrick, H. (2021). Review of Vitamin B12 deficiency in pregnancy: a diagnosis not to miss as veganism and vegetarianism become more prevalent. *European Journal of Haematology*, 106(4), 450–455. <https://doi.org/10.1111/ejh.13571>.
- Remde, A., DeTurk, S. N., Almardini, A., Steiner, L., & Wojda, T. (2022). Plant-predominant eating patterns - how effective are they for treating obesity and related cardiometabolic health outcomes? - a systematic review. *Nutrition Reviews*, 80(5), 1094–1104. <https://doi.org/10.1093/nutrit/nuab060>
- Riaz, M.N. (2020). *Soy ingredients processing techniques to produce quality soy products*. Food Protein R & D Center, Texas A & M University.

- Rizzo, N. S., Jaceldo-Siegl, K., Sabate, J., & Fraser, G. E. (2013). Nutrient profiles of vegetarian and nonvegetarian dietary patterns. *Journal of the Academy of Nutrition and Dietetics*, 113(12), 1610–1619. <https://doi.org/10.1016/j.jand.2013.06.349>.
- Rizzo, G., Laganà, A. S., Rapisarda, A. M., La Ferrera, G. M., Buscema, M., Rossetti, P., Nigro, A., Muscia, V., Valenti, G., Sapia, F., Sarpietro, G., Zigarelli, M., & Vitale, S. G. (2016). Vitamin B12 among vegetarians: status, assessment and supplementation. *Nutrients*, 8(12), 767. <https://doi.org/10.3390/nu8120767>.
- Rocha, J. P., Laster, J., Parag, B., & Shah, N. U. (2019). Multiple health benefits and minimal risks associated with vegetarian diets. *Current Nutrition Reports*, 8(4), 374–381. <https://doi.org/10.1007/s13668-019-00298-w>.
- Rodriguez, N. R., DiMarco, N. M., Langley, S., American Dietetic Association, Dietitians of Canada, & American College of Sports Medicine: Nutrition and Athletic Performance (2009). Position of the American Dietetic Association, Dietitians of Canada, and the American College of Sports Medicine: Nutrition and athletic performance. *Journal of the American Dietetic Association*, 109(3), 509–527. <https://doi.org/10.1016/j.jada.2009.01.005>.
- Rogerson D. (2017). Vegan diets: practical advice for athletes and exercisers. *Journal of the International Society of Sports Nutrition*, 14, 36. <https://doi.org/10.1186/s12970-017-0192-9>.
- Sadler, M.J. (2004). Meat alternatives - market developments and health benefits. *Trends in Food Science & Technology*. 15 (5): 250-60. <https://doi.org/10.1016/j.tifs.2003.09.003>.
- Saunders, A. V., Craig, W. J., Baines, S. K., & Posen, J. S. (2013). Iron and vegetarian diets. *Medical Journal of Australia*, 199(S4), S11–S16. <https://doi.org/10.5694/mja11.11494>.
- Saunders, A. V., Craig, W. J., & Baines, S. K. (2013). Zinc and vegetarian diets. *Medical Journal of Australia*, 199(S4), S17–S21. <https://doi.org/10.5694/mja11.11493>.
- Saunders, A. V., Davis, B. C., & Garg, M. L. (2013). Omega-3 polyunsaturated fatty acids and vegetarian diets. *Medical Journal of Australia*, 199(S4), S22–S26. <https://doi.org/10.5694/mja11.11507>.
- Shridhar, K., Dhillon, P. K., Bowen, L., Kinra, S., Bharathi, A. V., Prabhakaran, D., Reddy, K. S., & Ebrahim, S. (2014). Nutritional profile of Indian vegetarian diets--the Indian Migration Study (IMS). *Nutrition Journal*, 13, 55. <https://doi.org/10.1186/1475-2891-13-55>.
- Shurtleff, W. & Aoyagi, A. (eds) (2016). History of Modern Soy Protein Ingredients - Isolates, Concentrates, and Textured Soy Protein Products (1911-2016): Extensively Annotated Bibliography and Sourcebook. SoyInfo Center, USA. pp. 666.
- Sutter, D. O., & Bender, N. (2021). Nutrient status and growth in vegan children. *Nutrition Research*, 91, 13–25. <https://doi.org/10.1016/j.nutres.2021.04.005>.
- Thomas, D. T., Erdman, K. A., & Burke, L. M. (2016). Position of the Academy of Nutrition and Dietetics, Dietitians of Canada, and the American College of Sports Medicine: Nutrition and Athletic Performance. *Journal of the Academy of Nutrition and Dietetics*, 116(3), 501–528. <https://doi.org/10.1016/j.jand.2015.12.006>.
- Toh, D. W. K., Koh, E. S., & Kim, J. E. (2020). Incorporating healthy dietary changes in addition to an increase in fruit and vegetable intake further improves the status of cardiovascular disease risk factors: A systematic review, meta-regression, and meta-analysis of randomized controlled trials. *Nutrition Reviews*, 78(7), 532–545. <https://doi.org/10.1093/nutrit/nuz104>.
- Tran, E., Dale, H. F., Jensen, C., & Lied, G. A. (2020). Effects of plant-based diets on weight status: a systematic review. *Diabetes, Metabolic Syndrome and Obesity: Targets and Therapy*, 13, 3433–3448. <https://doi.org/10.2147/DMSO.S272802>.

- Van Winckel, M., Vande Velde, S., De Bruyne, R., & Van Biervliet, S. (2011). Clinical practice: vegetarian infant and child nutrition. *European Journal of Pediatrics*, 170(12), 1489–1494. <https://doi.org/10.1007/s00431-011-1547-x>.
- Vitale K & Hueglin S (2021). Update on vegetarian and vegan athletes: a review. *Journal of Physical Fitness and Sports Medicine*.10(1): 1-11. <https://doi.org/10.7600/jpfsm.10.1>.
- Wong, Y.W. (1999). *Evaluation of nutritional status and dietary intake of Buddhist monks*. [BSc Thesis]. Universiti Kebangsaan Malaysia.
- World Health Organization. (1993). *Indicators and Strategies for Iron Deficiency and Anemia Programmes*. Report of the WHO/UNICEF/UNU Consultation. Geneva, Switzerland.
- World Health Organization. (2000). *The Countrywide Integrated Non-Communicable Disease Intervention (CINDI) Dietary Guide*. World Health Organization, Geneva, Switzerland.
- World Health Organization. (2012). *Sodium Intake for Adults and Children*. World Health Organization, Geneva, Switzerland.
- Woo, K. S., Kwok, T. C., & Celermajer, D. S. (2014). Vegan diet, subnormal vitamin B-12 status and cardiovascular health. *Nutrients*, 6(8), 3259–3273. <https://doi.org/10.3390/nu6083259>.
- Yajnik, C. S., Deshpande, S. S., Lubree, H. G., Naik, S. S., Bhat, D. S., Uradey, B. S., Deshpande, J. A., Rege, S. S., Refsum, H., & Yudkin, J. S. (2006). Vitamin B12 deficiency and hyperhomocysteinemia in rural and urban Indians. *Journal of the Association of Physicians of India*, 54, 775–782.
- Zeuschner, C. L., Hokin, B. D., Marsh, K. A., Saunders, A. V., Reid, M. A., & Ramsay, M. R. (2013). Vitamin B₁₂ and vegetarian diets. *Medical Journal of Australia*, 199(S4), S27–S32. <https://doi.org/10.5694/mja11.11509>.

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Food Equivalent with Serving Size

Table 1.6: Examples of vegetables equivalent to one serving

Groups of Vegetables	*Serving size	Weight
Vegetables, cruciferous, cooked ^a	½ cup	50 g
Vegetables, green leafy vegetables with edible stem, cooked ^b	½ cup	40 g
Vegetables, fruit, cooked ^c	½ cup	40 g
Vegetables, leafy ulam, raw ^d	1 cup	50 g
Vegetables, beans, cooked ^e	½ cup	50 g
Vegetables, flowers, cooked ^f	½ cup	50 g
Vegetables, sprouting, cooked ^g	½ cup	50 g
Vegetables, starchy ^h	½ cup	60 g
Vegetable juice, with pulp, no sugar added	½ glass	125 ml
Other vegetables ⁱ :		
Mushroom (white/ brown/ shitaki/ enoki, erygii, button, oyster mushroom), fresh/ soaked, cooked	½ cup/ 6 pieces	40 g
Fungus (black fungus, white fungus), cooked	½ cup	40 g
Mixed vegetables, cooked ^j	½ cup	50 g
Wolfberry (Goji berry), dried	1 cup	50 g

Note:

*Serving size and food weight are measured in edible portions.

^a Cruciferous such as cabbage, broccoli and cauliflower.

^b Green leafy-vegetables with edible stem such as *kangkung*, *sawi*, *pucuk manis*, *bayam*, *sayur meranti*, *kau kee*, *daun keledak*, drumstick leaf, gongura, tropical amaranths, sessile joyweed, *tong ho*, *kalian* and *mak choi*.

^c Fruit vegetables such as brinjal, tomato, chayote (*fo shou gua*), capsicum, angled loofah (*petola*), bitter gourd, sour eggplant (*terung asam Dayak*), squash, snake gourd, pointed gourd (*parwal*), ridged gourd (*turai*), bottle gourd.

^d Leafy ulam such as *pegaga*, *ulam raja*, lettuce, garden salad.

^e Vegetables, beans such as string bean, snow pea, lady fingers, and french bean.

^f Vegetables, flowers such as daylily (golden needles/ *jin zhen cai*) asparagus.

^g Vegetables, sprouting such as soya sprout, green bean sprout (*taugeh*), pea sprout (*dou miao*), alfalfa.

^h Vegetables, starchy such as carrot, radish, pumpkin, beet root, lotus root, yam bean (*sengkuang*), water chestnut (*sengkuang Cina*), Chinese yam (*huai san*), arrowroot (*fen ge*).

ⁱ Other vegetables such as mushroom, fungus and seaweed are rich sources of micronutrients such as iron, iodine and Vitamin B12 but low in fibre. Consume with other vegetable groups such as cruciferous, green leafy-vegetables and fruit vegetables to obtain optimum phytonutrients.

^j Mixed vegetables refers to a combination of several types of vegetables such as cabbage + carrot + baby corn + french bean.

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Table 1.7: Examples of fruits equivalent to one serving

Fruits	*Serving size	Weight
Ambarella (<i>kedondong</i>)	6 whole medium sized	200 g
Guava	1 big slice	110 g
Banana, <i>berangan</i>	1 whole medium sized	60 g
Banana, Cavendish	1/2 whole medium sized	60 g
Banana, <i>mas</i>	2 whole medium sized	60 g
<i>Ciku</i>	1 whole large sized	80 g
Dragon fruit, red	1 slice/ 1/3 whole medium sized	135 g
<i>Langsat</i>	20 whole fruits	180 g
Mango	1/2 whole large sized	100 g
Papaya	1 slice/ 1/4 whole medium sized	110 g
Pineapple	1 slice/ 1/4 whole medium sized	150 g
Water apple (<i>jambu air</i>)	10 whole fruits	500 g
Watermelon, without skin	1 big slice	250 g
Apple	1 whole medium sized	110 g
<i>Durian</i> (7 x 3 cm)	5 <i>ulas</i>	40 g
Grapes	8 small	90 g
Jackfruit (<i>cempedak</i>), without seed	4 pieces	60 g
Jackfruit (<i>nangka</i>), without seed	5 pieces	190 g
Pear, green	1/2 whole medium sized	100 g
Pear, yellow/ orange	1 whole medium sized	160 g
<i>Rambutan</i> (4.5 X 3 cm)	6 whole fruits	110 g
Dates, pitted, dried	2 pieces	20 g
Raisin	1 heap dessert spoon	20 g
Prunes, pitted, dried	4 pieces	20 g
Figs, dried	3 pieces	25 g
Fruit juice, with pulp & without added sugar	1/2 glass	125 ml

Note:

*Serving size and food weight are measured in edible portions.

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Table 1.8: Examples of rice, other cereals, whole grain cereal-based products and tubers equivalent to one serving

Rice, other cereals, whole grain cereal-based products and tubers	*Serving size	Weight
Rice, brown/ multigrain/ unpolished/ parboiled/ white, cooked	1 cup/ 2 rice scoops	100 g
Noodles, <i>mee-hoon</i> made of brown/ white rice flour/ sago noodles (<i>tang-hoon</i>), soaked	1½ cups	150 g
Noodles, <i>mee/ kuih-teow/ laksa</i> , wet	1 cup	100 g
Pasta/ spaghetti/ macaroni, cooked	1½ cups	150 g
Porridge, brown/ white rice, plain, cooked	2 cups	330 g
Corn, without skin and cob, cooked	⅓ medium ear	40 g
Corn kernel, without margarine, cooked	3 dessert spoons / ⅓ commercial cup	40 g
Cornflake, without milk and added sugar	1 cup/ 8 dessert spoons	30 g
Bran, coarse, uncooked	1 cup/ 10 dessert spoons	100 g
Oats (rolled/ processed)/ wheat germ, uncooked	6 dessert spoons	40 g
Muesli, without milk	4 dessert spoons	45 g
Quinoa, cooked	1 cup/ 2 rice scoops	150 g
Wheat (<i>gandum</i>)/ barley, without gravy, cooked	12 dessert spoons/ ¾ cup	150 g
Potatoes, without skin, raw	2 whole medium sized	160 g
Sweet potato/ yam (taro)/ tapioca, without skin, raw (1 cm cube)	½ cup/ ⅓ of a whole medium sized	70 g
Bread, wholemeal/ ryemeal/ white	2 square slices	60 g
Bread, pita, wholemeal	½ piece	40 g
<i>Roti canai</i>	½ piece	50 g
Bun ^a	1 rectangular bun/ 2 small buns	50 g
<i>Capati</i> (D= 20 cm)	½ piece	50 g
<i>Dosai/ rawa dosai</i> (D=20 cm)	1 piece	80 g
<i>Idli</i> (D= 6 cm)	2 small pieces	110 g
<i>Pau</i> , with filling	½ piece	40 g
<i>Putu mayam</i>	2 pieces	100 g
Biscuit, wholemeal crackers/ Marie/ milk ^b	5 pieces	30 g
Biscuits, plain/ cream crackers/ oatmeal ^b	4 pieces	45 g

Note:

*Serving size and food weight are measured in edible portions.

**Preferably choose whole grain foods in your daily diet.

^a Choose healthier bun such as less sweet, no cream or sweet filling and fortified with micronutrients or healthy ingredients.

^b Choose healthier biscuits such as low fat, sugar, no cream or sweet filling and low sodium.

D = diameter

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Table 1.9: Examples of legumes, nuts and seeds equivalent to one serving

Legumes, nuts and seeds	*Serving size	Weight
Soya bean, yellow/ black, cooked	1 cup	180 g
Soya bean cake, fermented (<i>tempeh</i>)	2 rectangular pieces	90 g
Soya bean curd, tau-kua/ hard/ traditional/ fried <i>tauhu</i> , compressed <i>tauhu</i> (<i>dou fu gan</i>), (6 x 6 cm)	1½ pieces	150 g
Soya bean curd, soft <i>tauhu</i> (<i>tauhu lembut</i>), (18 x 5 cm)	1 box	240 g
Soya bean curd, <i>tau-hoo-pok</i> , (D= 3 cm)	4 round pieces	60 g
Egg tau-hoo (<i>tauhu telur</i>)	1½ packet	200 g
Soya bean chip (<i>dou bao</i>), fresh (5 x 5 cm)	5 pieces	50 g
Soya bean sheet (<i>fucok</i> sheets), dried (42.5 x 14 cm)	1 sheet	30 g
Soya bean milk, fresh, unsweetened	1½ glasses	375 ml
Soya bean, powder	6 dessert spoons	40 g
Soya bean dessert, <i>tau-hoo-fah</i> , unsweetened	2 containers	700 g
Gram, chickpea/ black-eyed pea/ kidney bean, without gravy, cooked	1 cup	180 g
Gram, red beans/ mung beans, without gravy, cooked	1½ cups	260 g
Dhal, various split pea/ lentils (chana dhal/ <i>kadalei</i> , red lentils/ <i>masoor</i> , green lentils, <i>urad</i>), cooked	1 cup	180 g
Baked bean/ green pea, beans only, canned	1½ cups/ 1 whole medium canned	400 g
Lotus seed ^a	1 cup	120 g
Nuts, peanuts/ almonds/ cashew nuts, without shell ^b	10 pieces/ 2 dessert spoons	15 g
Nut, walnut, without shell ^b	1 piece	15 g
Seed, chia seed/ flax seed/ pumpkin seed/ sesame seed/ watermelon seed/ sunflower seed ^b	2 dessert spoons	15 g
Seed, <i>kuaci</i> , without shell ^b	10 pieces/ 2 dessert spoons	15 g

Note:

*Serving size and food weight are measured in edible portions.

^a Lotus seed is categorised as nuts and seeds group, but considering this food is a good source of protein and low in fat. Therefore, the serving size of lotus seed was calculated based on protein content instead of fat.

^b Nuts and seeds were calculated based on fat exchange. 1 serving of nuts/ seeds = 5 g of fat.

D = diameter

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Table 1.10: Examples of egg equivalent to one serving

Poultry, meat and egg	*Serving size	Weight
Egg, century/ duck	2 whole	120 g
Egg, hen, grade A, without shell	2 whole	100 g
Egg, quail, without shell	12 whole	120 g

Note:

*Serving size and food weight are measured in edible portions.

Table 1.11: Examples of milk & milk products equivalent to one serving

Milk & milk products	*Serving size	Weight
Milk, full cream/ fresh/ low fat	1 glass	250 ml
Milk, evaporated	1/2 glass	125 ml
Milk, powdered, full cream/ skimmed (heap)	4 heap dessert spoons	30 g
Cheese, cottage/ spread	3 heap dessert spoons	40 g
Cheese, processed, slice (8 x 8 cm)	2 square slices	40 g
Yoghurt, natural/ fat free/ low fat, plain	2 yoghurt pots	270 g
Yoghurt, Greek style	1 yoghurt pot	135 g

Note:

*Serving size and food weight are measured in edible portions.

Standard household measurements used in this dietary guideline are as follows:

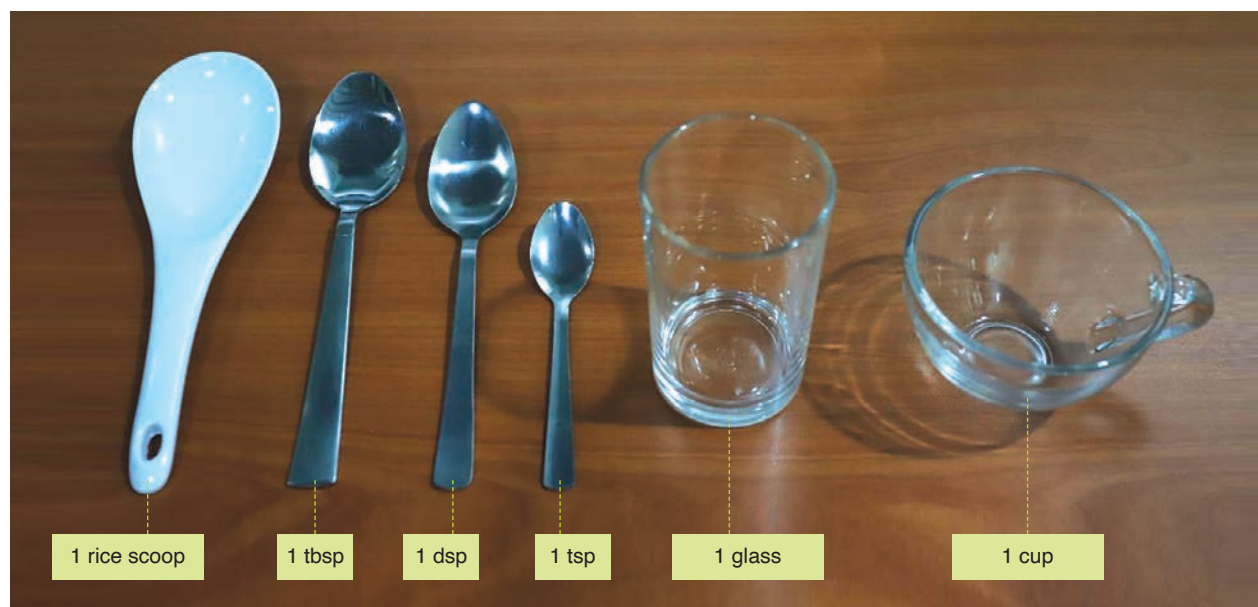


Figure 1.5 : Standard household measurements used in this dietary guideline

1 tablespoon (tbsp)	=	15 ml	1 glass	=	250 ml
1 dessert spoon (dsp)	=	10 ml	1 cup	=	200 ml
1 teaspoon (tsp)	=	5 ml	1 rice scoop	=	50 g

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Appendices

Table 1.12: Example of Vegetarian Menu for 2,000 kcal

Lacto-ovo-vegetarian			Lacto vegetarian		
Menu	Portion size	Calorie (kcal)	Menu	Portion size	Calorie (kcal)
Breakfast			Breakfast		
• Wholemeal sandwich with scramble egg, cheese and lettuce	1 set	(130 g)	• Wholemeal sandwich with cheese and lettuce	1 set	(90 g)
• Chocolate milk	1 glass	(250 ml)	• Chocolate milk	1 glass	(250 ml)
Tea break			Tea break		
• <i>Angku kueh</i> with mung bean filling	1 piece	(40 g)	• <i>Angku kueh</i> with mung bean filling	1 piece	(40 g)
• Sweet potatoes bombs kueh	1 piece	(40 g)	• Granola bar with oat and nuts	1 piece	(50 g)
• Red date, longan and wolfberry drink	1 glass	(250 ml)	• Red date, longan and wolfberry drink	1 glass	(250 ml)
Lunch			Lunch		
• Brown rice	1½ cups	(150 g)	• Brown rice	1½ cups	(150 g)
• Braised mushrooms with broccoli	½ cup	(60 g)	• Braised mushrooms with broccoli and sesame	½ cup	(60 g)
• Stir-fried romaine lettuce with sesame	½ cup	(60 g)	• Stir-fried snow pea, baby corn and cashew nut	½ cup	(60 g)
• Spicy fried <i>tauhu</i>	1 piece	(100 g)	• Spicy fried <i>tauhu</i>	1 piece	(100 g)
• Guava	1 slice	(120 g)	• Guava	1 slice	(120 g)
• Plain water	1 glass	(250 ml)	• Plain water	1 glass	(250 ml)
Tea Break			Tea Break		
• Vegetable <i>pakor</i> as	1 piece	(40 g)	• Vegetable <i>pakor</i> as	1 piece	(40 g)
• Mung bean porridge	1 bowl	(150 g)	• Mung bean porridge	1 bowl	(150 g)
Dinner			Dinner		
• Brown rice	1½ cups	(150 g)	• Brown rice	1½ cups	(150 g)
• Chickpea masala	½ cup	(60 g)	• Chickpea masala	½ cup	(60 g)
• <i>Aloo gho</i> bi (potatoes & cauliflower)	½ cup	(60 g)	• <i>Aloo gho</i> bi (potato & cauliflower)	½ cup	(60 g)
• <i>Palak paneer</i> (spinach & cheese)	½ cup	(60 g)	• <i>Palak</i> (spinach)	½ cup	(60 g)
• Banana	1 whole	(60 g)	• Banana	1 whole	(60 g)
• Plain water	1 glass	(250 ml)	• Plain water	1 glass	(250 ml)
Total calorie		2,000	Total calorie		1,985

Ovo vegetarian			Vegan		
Menu	Portion size	Calorie (kcal)	Menu	Portion size	Calorie (kcal)
Breakfast			Breakfast		
▪ Wholemeal sandwich with avocado, egg and lettuce	1 set	(120 g)	▪ Wholemeal sandwich with avocado, tomato and lettuce	1 set	(80 g)
▪ Soya milk	1 glass	(250 ml)	▪ Soya milk	1 glass	(250 ml)
Tea break			Tea break		
▪ <i>Angku kueh</i> with mung bean filling	1 piece	(40 g)	▪ <i>Angku kueh</i> with mung bean filling	1 piece	(40 g)
▪ Granola bar with oat and nuts	1 piece	(50 g)	▪ Granola bar with oat and nuts	1 piece	(50 g)
▪ Sweet potatoes, red date, longan and white fungus dessert	1 bowl	(150 ml)	▪ Sweet potatoes, red date, longan and white fungus dessert	1 bowl	(150 ml)
Lunch			Lunch		
▪ Brown rice	1½ cups	(150 g)	▪ Brown rice	1½ cups	(150 g)
▪ Braised mushrooms with broccoli, soy chip (<i>dou bao</i>) and sesame	1 cup	(90 g)	▪ Braised mushrooms with broccoli and sesame	1 cup	(60 g)
▪ Corn, Chinese yam (<i>huai san</i>) and red date soup	1 bowl	(150 g)	▪ Corn, Chinese yam (<i>huai san</i>) and red date soup	1 bowl	(150 g)
▪ Guava	1 slice	(120 g)	▪ Spicy fried tempeh and peanut	½ cup	(40 g)
▪ Plain water	1 glass	(250 ml)	▪ Guava	1 slice	(120 g)
			▪ Plain water	1 glass	(250 ml)
Tea Break			Tea Break		
▪ Vegetable <i>pakor</i> as	1 piece	(40 g)	▪ Vegetable <i>pakor</i> as	1 piece	(40 g)
▪ Mung bean porridge	1 bowl	(150 g)	▪ Mung bean porridge	1 bowl	(150 g)
Dinner			Dinner		
▪ Brown rice	1½ cups	(150 g)	▪ Brown rice	1½ cups	(150 g)
▪ Chickpea masala	½ cup	(60 g)	▪ Chickpea masala	½ cup	(60 g)
▪ <i>Aloo ghobi</i> (potato & cauliflower)	½ cup	(60 g)	▪ <i>Aloo ghobi</i> (potato & cauliflower)	½ cup	(60 g)
▪ <i>Palak</i> (spinach)	½ cup	(60 g)	▪ <i>Tauhu varuval</i>	½ cup	(60 g)
▪ Banana	1 whole	(60 g)	▪ Banana	1 whole	(60 g)
▪ Plain water	1 glass	(250 ml)	▪ Plain water	1 glass	(250 ml)
Total calorie		2,035	Total calorie		2,045

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Eat a variety of foods within the recommended servings



Key Message 2

Eat plenty of
vegetables and
fruits everyday

Key Message 2

Eat plenty of vegetables and fruits everyday

Assoc. Prof. Dr. Foo Leng Huat and Prof. Dr. Mirnalini Kandiah (Until Sept 2021)

2.1 Terminology

Fruits

The term fruit is generally used to describe the sweet, fleshy edible portion of a plant that arises from the base of the flower and surrounds the seeds. Examples of fruits are bananas, papayas, apples, oranges and others. Most fruits are eaten fresh and raw when they are ripe which give them a sweet taste. In some cases, they are consumed before the fruits ripen, usually with a spicy or savoury dip. Some ripe and unripe fruits are used in cooking or incorporated into salads which can offer a tasty alternative. Fruits can also be consumed as canned fruits, dried fruits and fruit juice, preferably without added sugar and preservatives. Concentrated fruit juice and fruit jam which are usually added with sugar during processing are excluded as they do not retain any nutritional values of the original food.

Vegetables

Vegetables refer to the edible portions of plants such as leaves, stems, tubers, roots, flowers and bulbs. Vegetables include all fresh, green leafy vegetables (such as spinach and lettuce), coloured vegetables (such as red spinach), fruit vegetables [also known as gourds or melons, such as pumpkin, loofah (petola) and cucumber], bean vegetables (such as long beans), cruciferous vegetables (such as cabbages and broccoli), ulam-ulam [such as pegaga (*Centella asiatica*) and ulam raja (*Cosmos caudatus*)] and edible plant stems (such as celery and asparagus). There are fruits, in a botanical sense, which are treated as vegetables in cooking, for example tomatoes and brinjals. This definition does not include tubers (such as potatoes, tapioca and yam) (WHO, 2003). Vegetables are also sometimes

sold as canned or frozen. Some vegetables are eaten raw; others are cooked to make them more palatable and digestible, sometimes in combination with other

food groups (such as eggs, meat, fish or legumes). Vegetables can also be consumed as juice, preferably without added sugar and/ or salt.

2.2 Introduction

Vegetables and fruits are a significant component of the human diet, including all types of vegetarian diets, because they are important sources of vitamins, minerals, fibre, phytochemicals and other yet unknown bioactive substances (Miller et al., 2000; Benzie & Choi, 2004; Zhu et al., 2018). Table 2.1 lists the several categories of vegetables and fruits that share some similar nutritional characteristics. Although vegetables and fruits are commonly known for their non-calories content of vitamins, minerals and fibre, they also have health benefits beyond supplying these nutrients such as enhancing and improving the bioavailability of other dietary components consumed concurrently such as iron, zinc and other trace minerals. In addition, vegetables and fruits with different colour pigmentations offer a wide variety of phytochemical compounds, including polyphenols, antioxidants, flavonoids and other non-nutrient bioactive substances that could confer many health benefits (Heber, 2004; Benzie et al., 2004; Zhu et al., 2018). The type and content of phytochemicals vary greatly depending upon the colour of vegetables and fruits. For example, orange vegetables and fruits are high

in carotenoids, whereas green vegetables and fruits contain high levels of glucosinolates. Table 2.2 summarises the types of phytochemicals based on colour code of commonly consumed vegetables and fruits. Thus, consumption of a wide range of colourful vegetables and fruits is an important way of increasing the level of various phytochemicals in the body.

Up to now, there are no much and limited information from the large-scale population-based studies to determine the types and quantity of dietary fruit and vegetable intakes between vegetarians and non-vegetarians. There is some scientific literature have compared between vegetarians and non-vegetarians such as omnivorous and found that vegetarians in any types of vegetarians are relatively consume highly amounts of vegetables and fruits than that of the non-vegetarians (Farmer et al., 2011; Clarys et al., 2013; Orlich et al., 2014), even though there was no much differences found in terms of total energy intake and/or overall diet quality between vegetarians and non-vegetarians.

KM2

Eat plenty of vegetables and fruits everyday

Table 2.1: Common types of vegetables and fruits

Vegetables	Examples
Green leafy vegetable	Mustard leaves, spinach, kale, <i>bok choy</i> , lettuce, agathi, drumstick leaves, curry leaves, amaranth leaves
Cruciferous vegetable	Broccoli, cabbage and brussels sprout, cauliflower
Marrow	Pumpkin, bittergourd, <i>loofah (petola)</i> and cucumber
Edible plant stems	Celery, asparagus, leek, rhubarb
Whole plants	<i>Pegaga</i> (Indian Pennywort), <i>ulam raja</i> , mint,
Allium	Onion, garlic, shallot, beetroot
Bean vegetables	Long beans, french beans, okra
Fruits	Examples
Tropical and exotic fruits	Bananas, mangoes, guava, papaya, pineapple, etc.
Citrus fruits	Oranges, grapefruits, mandarins, lemons, limes
Melons	Watermelon, honeydew, cantaloupe
Berry	Strawberries, kiwifruits, passion fruits, blueberries, raspberries, tomatoes, grapes, dates

Table 2.2: Colour code groups of vegetables and fruits

Colour	Phytochemical	Vegetables and fruits
Red	Lycopene	Tomatoes and tomato products (soup, juice, pasta sauce), watermelon, strawberries, cherries, raspberries, red chilies.
Orange	Carotenoids, α and β -carotene	Carrots, mangoes, pumpkin
Orange/ yellow	Beta-cryptoxanthin and flavonoids	Papaya, peaches, cantaloupe, oranges, pineapple, apricots, peaches
Green	Lutein, Zeaxanthin, Glucosinolates	Spinach, broccoli, brussels sprouts, green cabbage, green peppers, mustard greens, avocado, honeydew, okra, celery, asparagus
White	Ally Sulphides,	Leeks, onion, garlic, chives, cauliflower
Purple/ blue	Anthocyanins and polyphenols	Berries (black & purple), grapes, plums, prunes, eggplant, raisins, purple figs

Modified from Heber (2004)

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Eat plenty of vegetables and fruits everyday

It is not surprisingly to found that higher intakes of dietary fibre by almost double and other vitamins and minerals such as vitamin C and magnesium for the vegetarians compared to the non-vegetarians (Clarys et al, 2013). Indeed, vegetables and fruits is not only an important dietary fiber source, but also provide wide ranges of other important nutrients such as vitamins, minerals, dietary phytochemicals and bioactive compounds.

Compelling evidence from epidemiological and experimental studies indicates that diets rich in vegetables and fruits exert significant beneficial effects on general health and wellbeing, and lower the risk of non-communicable diseases such as cardiovascular diseases, type-II diabetes and certain types of cancers (Dauchet et al, 2009; Boeing et al, 2012; Slavin & Lloyd, 2012; Zhu et al, 2018; Wallace et al, 2020). Hence, vegetables and fruits are clearly an important part of a healthy diet meal in vegetarians of all types. It should be noted that consumption of a variety of vegetables and fruits is as important as the quantity, because no single fruits or vegetable provides all essential nutrients. Therefore, regardless of types of vegetarian practices, consuming adequate amounts of a variety of different vegetables and fruits each day is highly recommended to ensure an adequate intake of fibre, vitamins, minerals and a multitude of other important bioactive compounds.

In general, the establishment of healthy dietary practices such as adequate and higher daily intake of vegetables and fruits in vegetarians of any types is an important and effective strategy to prevent and reduce the disease burden associated with insufficient fruits and vegetable intake. Although it is presumably that vegetarians get most of their nutrients from a wide variety of vegetables and fruits than that of non-vegetarians, but proper and adequate consumption of vegetables and fruits every day should be highly needed for all types of vegetarians across all age groups in order to achieve optimal health and wellbeing, and consequently to prevent and reduce the risk of non-communicable diseases (Slavin et al, 2012; Wallace et al, 2020). In a recent published results from 95 different studies that involving a total of approximately 2 million people showed that 5 servings of daily vegetables and fruits (approximately 200 g) intake was significantly associated with reduced risk heart disease (16%), stroke (18%) and cardiovascular disease (13%) (Aune et al, 2017). However, it is also noted that the greatest benefit came from eating 800 g a day (roughly equivalent to ten servings of fruits or vegetables in a day) (Aune et al, 2017).

2.3 Scientific Basis

2.3.1 The beneficial effect of vegetables and fruits on human health

It is generally established that vegetarians in general, regardless of types of vegetarians, consume higher amounts of plant-origin foods such as vegetables and fruits, legumes, whole grains, soya-based foods, plant-based protein products (meat analogues), nuts and seeds when compared with non-vegetarians (Orlich et al., 2014). It is well established that higher high daily intake of vegetables and fruits exerts significant protective effect against some non-communicable diseases (Slavin et al., 2012; Aune et al., 2017; Wallace et al., 2020). A report of the WHO (2009) indicated that about 2.9% of all deaths and 1.1% of all disability-adjusted life years (DALYs) each year globally are attributed to inadequate intakes of vegetables and fruits. In addition, inadequate consumption of vegetables and fruits (<400 grams per day that equivalent to 5 servings of vegetables and fruits) is regarded as one of the top 10 risk factors for global mortality, which is estimated to cause up to 2.6 million deaths a year worldwide (WHO, 2004). This is equivalent to about 1.8% of the total global burden of disease, which is comparable with other known risk factors such as low physical activity (1.3% of the total burden of disease), overweight (2.3%), high cholesterol level (2.8%) and tobacco (4.1%). On the contrary, a high consumption of vegetables and fruits up to 600 grams per day could significantly reduce the burden of coronary heart disease and stroke by 31% and 19%, respectively, and a potential reduction of up to 31% for gastrointestinal cancers such as stomach (19%), oesophageal (20%) and colorectal cancer (2%), respectively (Lock et al., 2005). There is an increasing evidence from well-designed population-based epidemiological studies throughout the world that consumption of high level of vegetables and fruits per day of more than 5 servings to up to 10 servings of vegetables and fruits is significantly associated with better general and functional wellbeing (WHO, 2003; Tomey et al., 2008; Wallace et al., 2020) and reduce wide ranges of non-communicable diseases such as hypertension, cardiovascular disease, type-II diabetes and certain types of cancers (Aune et al., 2017; Wallace et al., 2020). These findings suggest that adequate intake of a variety of vegetables and fruits per day from at least 5 serving up to ten servings are crucial in promoting general wellness and good health throughout life.

Although there is now substantial evidence to show the beneficial health effects of higher intakes for vegetables and fruits among the general populations as well as vegetarian populations, but the precise

contribution of high intakes of vegetables and fruits in protecting against chronic diseases is not fully understood. Several plausible explanations have been postulated. The first is the benefit from indigestible fibre contained in vegetables and fruits. An adequate intake of both soluble and insoluble dietary fibre exerts a wide range of effects on normal physiological functions (Anderson et al., 2009), such as gastrointestinal health by adding bulk and softness to stools that promote bowel regularity, modulate low-density lipoprotein cholesterol (LDL-C) and blood pressure, improve blood glucose homeostasis and promote a feeling of satiety that reduces hunger and assists with weight management (Ello-Martin et al., 2007; Cho et al., 2013). Another mechanism could possibly be attributed to the individual or combined effects of many different bioactive phytochemicals present in vegetables and fruits (Miller et al., 2000; Van Buyn & Piyonka, 2000; Heber, 2014; Zhu et al., 2018; Reis et al., 2021), that can prevent the processes involved in the pathogenesis of various diseases. Free radical damage that induces oxidative stress can cause significant DNA damage with consequently leads to base mutation, DNA cross-linking, and chromosomal breakage and rearrangement. This damage may be limited by dietary antioxidants and phytochemicals in vegetables and fruits, which act by increasing detoxification and antioxidant enzyme activity, scavenging of oxidative agents, stimulation of the immune system, hormone metabolism, regulation of gene expression in cell proliferation and apoptosis (Hubbard et al., 2006; Erlund et al., 2008; McCall et al., 2009; Zhu et al., 2018). Another possible mechanism to explain the beneficial effects of vegetables and fruits on health outcomes may be via the low energy density and high non-digestible fibre content. Vegetables and fruits are all relatively low in both fat and energy density and high levels of water and non-digestible fibre, it may help to promote satiety and consequently reduce the overconsumption of energy (Ello-Martin et al., 2007). It has been demonstrated that adults who consumed higher levels of vegetables and fruits, in comparisons with those with a lower intake, had a lower risk of weight gain (Ello-Martin et al., 2007; Vioque et al., 2008), as a major risk factor for most chronic diseases (WHO, 2003). Lastly, vegetables and fruits consumption might simply be a surrogate for healthy dietary and lifestyle practices. For instance, greater consumption of vegetables and fruits as part of a meal may displace or limit the overconsumption of total energy-dense foods such as meat products, saturated fat and refined

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carbohydrates (Ello-Martin et al, 2007; de Oliveira et al, 2008), lower smoking and alcohol consumption (Joshipura et al, 2001; Boffetta et al, 2010), and increasing physical activity level (Boffetta et al, 2010). Therefore, recommendation of an increased consumption of vegetables and fruits up to 10 servings is an effective public health policy with considerable potential to prevent and reduce the burden of non-communicable diseases among vegetarians throughout lifespans.

2.3.2 Obesity

Epidemiologic evidence has consistently shown an inverse relationship between vegetarian diets and BMI, specifically the diet high in vegetables and fruits could help in maintaining a healthy weight and consequently prevent excessive weight gain, obesity and subsequent risk of chronic diseases. From a total of 40 studies that compared the weight status between adults who adhered to either a vegetarian or non-vegetarian (omnivorous) majority found that adults who adhered to vegetarian diet had significantly lowered levels of BMI and/or body weight than that of their omnivorous counterparts (Berkow & Barnard, 2006). This was further supported by a recent study by Jaacks and her colleagues (2016) among vegetarian adults from Caucasian and South Asian-origins in the US which showed lowered risks of both overweight and/or obesity and abdominal obesity compared to their non-vegetarian counterparts.

2.3.3 Gastrointestinal health

Dietary fibre is an important nutrient which determines the overall gastrointestinal health status (Anderson et al, 2009). Adequate intake of fibre is important to ensure the normal function of the digestive system. Numerous studies conducted in children and adults have shown that a low intake of fibre is associated with an increased risk of constipation (Dukas et al, 2003; Lee et al, 2008). A follow-up study of 3327 adult women aged 36 to 61 years found that a high intake of daily fibre of approximately 20 g per day reduced the risk of constipation by 3-fold compared to those having a daily intake of 7 g of fibre (Dukas et al, 2003). Hence, a simple and easy strategy to increase dietary fibre intake could be to increase the consumption of a variety of vegetables and fruits each day.

High intakes of vegetables and fruits rich in dietary fibre play an important role in gastrointestinal health. Individuals who consumed high-fibre diets high in vegetables and fruits had a significantly lower risk of gastrointestinal disease than those who on a high-meat diet (Nair et al, 1994). A recent study

on the relationship between diet and the intestinal gut microbiota profiles indicates that gut microbial composition could differ between individuals who practised to either a vegetarian or omnivorous diet, despite of them were living in the same area (Wu et al, 2016). Vegetarian adults had a significantly greater richness of intestinal gut microbiota diversity compared to omnivorous adults. For instance, bacteroidetes related operational taxonomic units were greater in individuals who consumed vegan and vegetarian diet than that of omnivorous diet, which in turn has important implications on a variety of health-related outcomes. Despite many and great advances in the scientific studies of diet on gut microbiome, it still remains unknown to which extent the composition of the intestinal gut microbiota is modulated by differences of food consumed from a vegetarian-based diet in comparison to the intake of animal-based products (Wu, 2016).

2.3.4 Cardiometabolic health

A vast body of evidence has suggested that dietary factors may play a vital role in the development of cardiovascular diseases (Dauchet et al, 2009; Cho et al, 2013; Wallace et al, 2020). Coronary heart disease (CHD) is the most important manifestation of arteriosclerosis and is regarded as a major cause of disease mortality worldwide, including in Malaysia (WHO, 2009). In Malaysia, it was estimated that cardiovascular diseases had accounted for approximately 22.2% of total deaths in 2010 from a total of with 22,701 deaths (WHO, 2014).

Epidemiological evidence from numerous longitudinal and meta-analysis studies have shown that the risk of CHD is significantly increased with lower intakes of vegetables and fruits (Joshipura et al, 2001; He et al, 2007; Dauchet et al, 2009). In an observational longitudinal study involving large populations of over 120,000 adult men and women, it was found that a high consumption of vegetables and fruits was associated with reduced risk of developing CHD, and the effect was greatest amongst those who consumed vegetables and fruits rich in vitamin C and leafy green vegetables (Joshipura et al, 2001). Similar findings were observed in a population-based study in The Netherlands, where higher consumption of vegetables and fruits, either in raw or processed foods, exerted a protective effect on the incidence of CHD (Oude Griep et al, 2010). About 34% lower risk of CHD incidence was seen among adults with a high intake of total vegetables and fruits of more than 475 g per day than those with a low total fruits and vegetable consumption (<241 g/d).

Similar effects of vegetables and fruits were also found in relation to the risk of stroke. In a meta-analysis of seven longitudinal studies by Dauchet and co-workers (2005) the risk of stroke was significantly reduced from 3% to 11% based on individual fruits or vegetables or combined. About 11%, 5% and 3% risk reduction of stroke, respectively, was observed by increases in each serving of fruits, combined vegetables and fruits, and vegetables per day. Similarly, significant inverse relationships were found between total fruits and vegetable intakes and the risk of stroke, cerebral infarction and hemorrhagic stroke in a recent longitudinal study of Swedish adults (Larsson et al., 2013). In these adults, those with the highest fruits and vegetable intake had a 13% lower risk of all causes stroke compared with those in the lowest intake group and the risk of stroke was significantly reduced with increasing consumption of vegetables and fruits of at least 5 servings a day.

Several human studies have shown an inverse association between fruits and vegetable intake and blood pressure (Miura et al., 2004; Dauchet et al., 2007). The Dietary Approaches to Stop Hypertension (DASH) intervention among hypertensive adults also showed that a carbohydrate-rich diet high in vegetables and fruits consumed for 8 weeks, reduced systolic blood pressure by 2.8 mmHg and diastolic blood pressures by 1.1 mmHg when compared to a control diet (Appel et al., 1997). This result is supported by another intervention study of healthy adults aged 25 to 64 years who consumed higher intakes of vegetables and fruits for 6 months had lower systolic and diastolic blood pressure by 4 mmHg and 1.5 mmHg, respectively, compared to a control group (John et al., 2002).

Varying evidence on the beneficial effects of vegetables and fruits are also reported on reduced risk of diabetes. A recent meta-analysis found no significant benefit on the incidence of diabetes found with increased intakes of total fruits, or vegetables alone (Carter et al., 2010). However, the risk of diabetes was significantly reduced by 14% among those who consumed relatively large amounts of green leafy vegetables. In another population-based, longitudinal study of the European Prospective Investigation into Cancer and Nutrition in Norfolk, United Kingdom among 3,704 adults, followed-up after 11 years found that adults who consumed a diet characterized by high amounts of vegetables and fruits of 3.5 portions per day and more, with at least 12 different fruits and vegetable items in a week had significantly a lower risk of type-II diabetes (Cooper et al., 2012). This finding suggests that higher intakes of vegetables

and fruits per day may be independently protective against the diabetes risk. In a later study of black Adventists comparing vegetarians and non-vegetarians, it was shown that vegetarians had lower risk of diabetes compared to non-vegetarians (Fraser et al., 2015). This inverse association between a vegetarian diet and diabetes risk could possibly be due to reduced levels of pro-inflammatory markers and oxidative stress, as precursors for insulin resistance and diabetes that is associated with higher intakes of vegetables and fruits.

2.3.5 Cancers

It is widely recognised that the risk of cancer can be prevented by a plant-based diet high in fruits and vegetable intakes (Heber, 2004). However, inconsistent results from many longitudinal studies have not been able to conclusively establish an inverse association between fruits and vegetable intakes and overall cancer risk. In a revised expert report of the World Cancer Research Fund and American Institute of Cancer Research in 2007, results derived from the meta-analysis of published epidemiological studies, indicated that a high consumption of fruits and non-starchy vegetables probably reduced the risk of mouth, pharynx, larynx, oesophageal, stomach and lung cancers (WCRF/AICR, 2007). Similar observations were also reported among European population, where a high intake of vegetables and fruits exerts a protective effect against cancers of the pharynx, esophageal, stomach and lung (Soerjomataram et al., 2010). High consumption of fruits and vegetable per day was found to be associated with a lower incidence of cancer (Shibata et al., 1992; Takachi et al., 2008; Boffetta et al., 2010), while higher amount of green and yellow vegetables and fruits intakes was significantly associated with lower cancer-related mortality (Sauvaget et al., 2003). On the contrary, there was no association reported of the amount of fruits and vegetable intakes and any cancer risk (Hung et al., 2004; George et al., 2009). The effects of vegetables and fruits were further examined separately on specific types of cancer based on the findings from the largest population-based EPIC studies of 10 European countries showed significant inverse association between higher fruits intake and the risk of cancer of mouth, pharynx, larynx and esophageal and lung cancers (Bradbury et al., 2014). Adults with highest intake of ≥ 356 g of fruits per day were significantly associated with a lower risk of these cancer developments, compared with those at lower intakes. In contrast, no association was found between vegetable intakes and the risk of any cancer sites assessed. Research also suggests that

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lifelong vegetarianism may be associated with a reduction in the risk of breast cancer through its association with a higher intake of vegetables, fruits, soy, and pulses and the avoidance of red meat (Butler et al., 2010).

2.3.6 Nutritional concerns on fruits and vegetable preparation

Although fruits and vegetable intakes in the form of juices are often perceived as healthy dietary practice, it is also convenient and practical way to ensure adequate consumption of vegetables and fruits to exert health benefits. However, there is certain caution needed to bear in mind when high fruits and vegetable juices to prevent necessary high intakes of added sugar on these juices. There is emerging evidence suggesting that high consumption of fruits juice exerts negative health effects such as an increased risk of excess body weight and diabetes, whereas consumption of

whole fruits is associated with lower risk in children and adults (Wojcicki & Heyman, 2012; Muraki et al., 2013). Fruits juice contains high amount of sugar, without any fibre which would provide chewing resistance, thus limiting intake. According to Gill and Satter (2014), “fruits juice contains just as much sugar and calories as a sugary carbonated drink and sometimes even more”. For instance, 250 ml of apple juice typically contains 110 kcal and 26 g of sugar, which is similar to the energy and sugar content of sugar-sweetened soda such as 250 ml of cola that typically contains 105 kcal and 26.5 g of sugar (Gill et al., 2014). It is therefore easy to consume a large amount of calories from fruits juices without getting any actual nutrition. In addition, it is also found that 100% fruits juices have low levels of antioxidants compared to whole oranges and other plant-based foods such as vegetables (Crowe & Murray, 2013). It is therefore advisable to consume whole vegetables and fruits either raw or processed whole, rather than in the form of juices.

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2.4 Current Status

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The consumption of vegetables and fruits plays a vital role in providing a diversified and nutritious diet (WHO, 2003b; Mason-D'Croz et al., 2019). It is generally agreed that high income levels and falling food prices are major determinants of food availability in any country. In Malaysia, limited data is available to indicate the current intake of vegetables and fruits among vegetarians. However, based on the Food Balance Sheets in 2010, the quantity of total vegetables and fruits was 112.3 kg/per capita/year for general Malaysian populations (FAOSTAT, 2010). Based on numerous nationwide population-based surveys conducted among the populations across the lifespans in Malaysia, it has reported that the quantity of vegetables and fruits consumed among populations of different age groups are still far below the recommended levels of at least 2 servings of fruits and 3 servings of vegetables per day, as indicated by the prevalence of inadequate intakes of fruits and/or vegetables, as assessed by the daily recommended servings of 5 among adults in Malaysia has increased from 92.5% in 2011 to 94.9% in 2019 (IPH, 2020). On the other hand, about 95% of Malaysian adults did not consume adequate fruits and/or vegetables as recommended by the WHO (IPH, 2020). Almost similar observation was found between males and females (95.1% vs. 94.7%) and geographical location of urban areas and the rural

areas in Malaysia (94.6% vs. 95.9%). When it was assessed based on vegetables and fruits alone, it found that only 9.4% and 10.0% of adults aged 18 years and above in Malaysia had achieved the recommended levels of at least two servings of fruits and three servings of vegetables, respectively (IPH, 2020). Similar observation was also found in adolescents in Malaysia. In a nationwide NMHS survey conducted in 2017, it found that 76.5% of secondary school students had failed to achieve the recommended levels of at least 5 servings of fruits and/ or vegetables per day in the past one month (IPH, 2017). Nonetheless, there is scant information on the fruit and vegetable consumption status and pattern among all subgroups of vegetarian practices in Malaysia, but a finding of the recent study carried-out amongst 574 Seventh-Day Adventists aged between 18 and 80 years, who were practicing on vegetarian diet found that average intakes of fruits and/ or vegetables were 5 servings daily (Tan et al., 2016). It would be interesting to found that 42% of them consumed vegetables and fruits more than 5 servings in a day (Tan et al., 2016). Based on the current nationwide survey and cross-sectional local studies evidences, it is clear that most Malaysians, including in vegetarian groups of all types in all age and sex groups were consumed substantially fewer vegetables and fruits than is recommended.

2.5 Key Recommendations

Key Recommendation 1

Eat more vegetables and fruits everyday.

How to achieve:

1. Consume at least 3 servings of vegetables and 2 servings of fruits everyday.
2. Eat at least one serving of vegetables and fruits in each main meal in a day (breakfast, lunch and dinner).
3. Eat vegetables and fruits as snack meals.
4. Add a variety of vegetables and/ or fruits to yogurt and salad.
5. When the fresh produce is not available, choose frozen vegetables and/ or fruits without added sugar, syrup, cream sauces, salt or other ingredients.
6. When the fresh produce is not available, choose canned vegetables and/ or fruits that low in added sugar, salts or other ingredients.

Key Recommendation 2

Eat a variety of vegetables and fruits everyday.

How to achieve:

1. Eat different colored vegetables and fruits everyday.
2. Eat a variety of vegetables everyday, including leafy vegetables (mustard leaves, spinach, *bok choy* and kale), and non-leafy vegetables such as bean vegetables (long bean, french bean and okra), cruciferous (broccoli and cabbage) and coloured vegetables (carrots, capsicum, eggplant and tomatoes).
3. Eat a variety of fruits everyday, including different colours of fruits such as tropical and exotic fruits (banana, mangoes and guava), citrus fruits (orange, grapefruit and lemon), melons (watermelon and honeydew) and berries (mulberry, grapes and strawberry).

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2.6 References

- Anderson, J.W., Baird, P., Davis, R.H., Ferreri, S., Knudtson, M., Koraym, A., Waters, V., & Williams, C.L. (2009). Health benefits of dietary fibre. *Nutrition Reviews*, 67, 188-205.
- Appel, L.J., Moore, T.J., Obarzanek, E., Vollmer, W.M., Svetkey, L.P., Sacks, F.M., Bray, G.A., Vogt, T.M., Cutler, J.A., Windhauser, M.M., Lin, P.H., & Karanja, N. (1997). A clinical trial of the effects of dietary patterns on blood pressure. DASH Collaborative Research Group. *New England Journal of Medicine*, 336, 1117-1124.
- Aun, D., Giovannucci, E., Boffetta, P., Fadnes, L.T., Keum, N., Norat, T., Greenwood, D.C., Riboli, E., Vatten, L.J., & Tonstad, S. (2017). Fruit and vegetable intake and the risk of cardiovascular disease, total cancer and all-cause mortality-a systematic review and dose-response meta-analysis of prospective studies. *International Journal of Epidemiology*, 46, 1029-1056.
- Berkow, S.E., & Barnard, N. (2006). Vegetarian diets and weight status. *Nutrition Reviews*, 64, 175-188.
- Benzie, I.F., & Choi, S.W. (2014). Antioxidants in food: content, measurement, significance, action, cautions, caveats, and research needs. *Advances in Food & Nutrition Research*, 71, 1-53.
- Boeing, H., Bechthold, A., Bub, A., Ellinger, S., Haller, D., Kroke, A., Leschik-Bonnet, E., Müller, M.J., Oberritter, H., Schulze, M., Stehle, P., & Watzl, B. (2012). Vegetables and fruits in the prevention of chronic diseases. *European Journal of Nutrition*, 51, 637-663.
- Boffetta, P., Couto, E., Wichmann, J., Ferrari, P., Trichopoulos, D., Bueno-de-Mesquita, H.B., van Duijnhoven, F.J., Büchner, F.L., Key, T., Boeing, H., Nöthlings, U., Linseisen, J., Gonzalez, C.A., Overvad, K., Nielsen, M.R., Tjønneland, A., Olsen, A., Clavel-Chapelon, F., Boutron-Ruault, M.C., Morois, S., Lagiou, P., Naska, A., Benetou, V., Kaaks, R., Rohrmann, S., Panico, S., Sieri, S., Vineis, P., Palli, D., van Gils, C.H., Peeters, P.H., Lund, E., Brustad, M., Engeset, D., Huerta, J.M., Rodríguez, L., Sánchez, M.J., Dorronsoro, M., Barricarte, A., Hallmans, G., Johansson, I., Manjer, J., Sonestedt, E., Allen, N.E., Bingham, S., Khaw, K.T., Slimani, N., Jenab, M., Mouw, T., Norat, T., Riboli, E., & Trichopoulou, A. (2010). Fruits and vegetable intake and overall cancer risk in the European Prospective Investigation into Cancer and Nutrition (EPIC). *Journal of the National Cancer Institute*, 102, 529-537.
- Bradbury, K.E., Appleby, P.N., & Key, T.J. (2014). Fruits, vegetable, and fibre intake in relation to cancer risk: findings from the European Prospective Investigation into Cancer and Nutrition (EPIC). *American Journal of Clinical Nutrition*, 100, 394S-398S.
- Bradbury, K.E., Crowe, F.L., Appleby, P.N., Schmidt, J.A., Travis, R.C., & Key, T.J. (2014). Serum concentrations of cholesterol, apolipoprotein A-I and apolipoprotein B in a total of 1694 meat-eaters, fish-eaters, vegetarians and vegans. *European Journal of Clinical Nutrition* 68, 178-183.
- Butler, L.M., Wu, A.H., Wang, R., Koh, W.P., Yuan, J.M., & Yu, M.C. (2010). A vegetable-fruits-soy dietary pattern protects against breast cancer among postmenopausal Singapore Chinese women. *American Journal of Clinical Nutrition*, 91, 1013-1019.
- Carter, P., Gray, L.J., Troughton, J., Khunti, K., & Davies, M.J. (2010). Fruits and vegetable intake and incidence of type 2 diabetes mellitus: systematic review and meta-analysis. *British Medical Journal* 341, c4229.
- Cho, S.S., Qi, L., Fahey, G.C., Jr & Klurfeld, D.M. (2013). Consumption of cereal fibre, mixtures of whole grains and bran, and whole grains and risk reduction in type 2 diabetes, obesity, and cardiovascular disease. *American Journal of Clinical Nutrition*, 98, 594-619.
- Clarys, P., Deriemaeker, P., Huybrechts, I., Hebbelinck, M. & Mullie, P. (2013). Dietary pattern analysis: a comparison between matched vegetarian and omnivorous subjects. *Nutrition Journal*, 12, 82.

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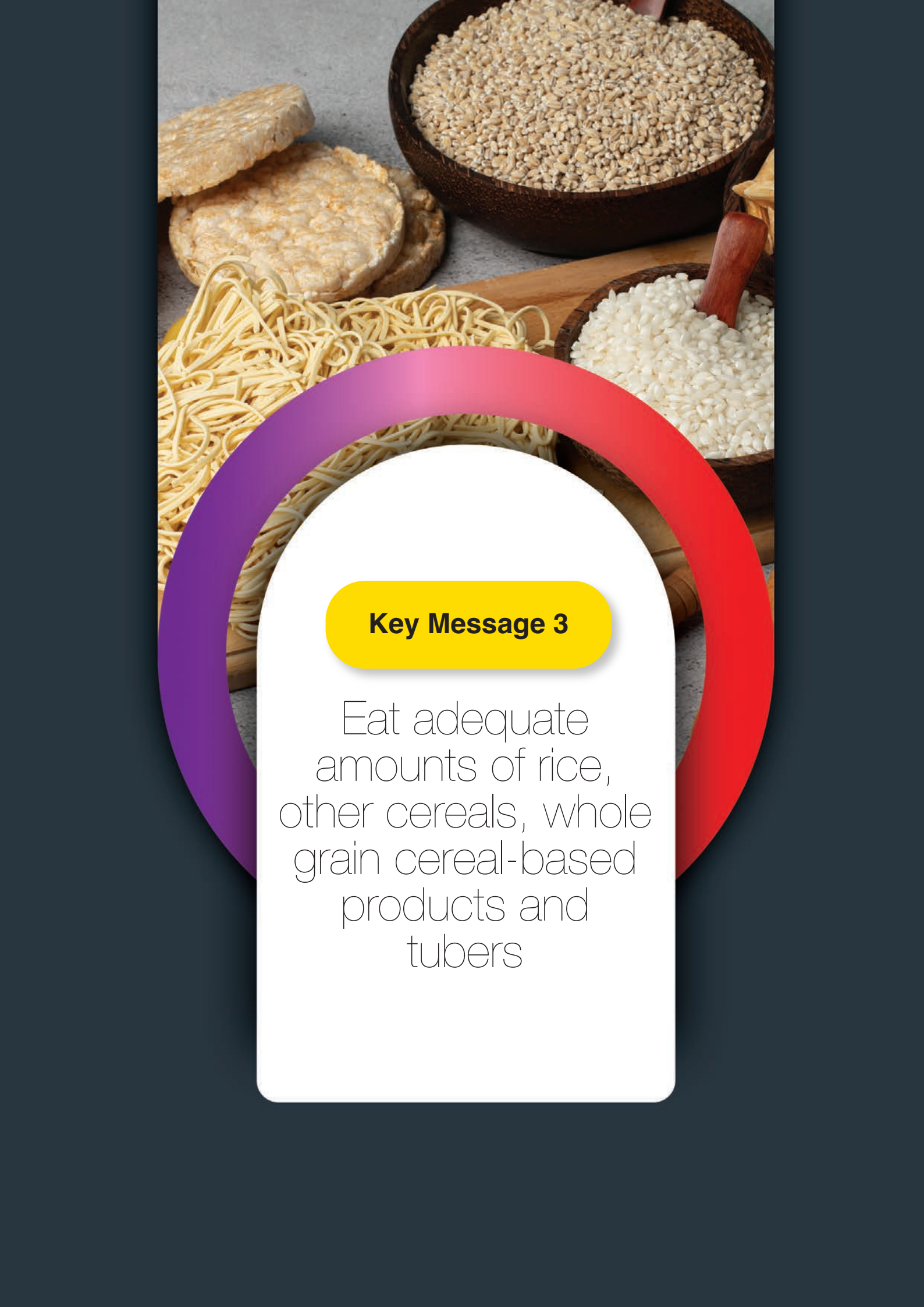
Eat plenty of vegetables and fruits everyday

- Cooper, A.J., Sharp, S.J., Lentjes, M.A., Luben, R.N., Khaw, K.T., Wareham, N.J., & Forouhi, N.G. (2012). A prospective study of the association between quantity and variety of fruits and vegetable intake and incident type 2 diabetes. *Diabetes Care* 35, 1293-1300.
- Crowe, K.M., & Murray, E. (2013). Deconstructing a fruits serving: comparing the antioxidant density of select whole fruits and 100% fruits juices. *Journal of the Academy of Nutrition & Dietetics* 113, 1354-1358.
- Dauchet, L., Amouyel, P., & Dallongeville, J. (2009). Fruits, vegetables and coronary heart disease. *Nature Reviews Cardiology* 6, 599-608.
- Dauchet, L., Amouyel, P., & Dallongeville, J. (2005). Fruits and vegetable consumption and risk of stroke: a meta analysis of cohort studies. *Neurology* 65, 1193-1197.
- Dauchet, L., Kesse-Guyot, E., Czernichow, S., Bertrais, S., Estaquio, C., Péneau, S., Vergnaud, A.C., Chat-Yung, S., Castetbon, K., Deschamps, V., Brindel, P., & Hercberg, S. (2007). Dietary patterns and blood pressure change over 5-y follow-up in the SU.VI.MAX cohort. *American Journal of Clinical Nutrition*, 85, 1650-1656.
- de Oliveira, M.C., Sichieri, R., & Venturim Mozzer, R. (2008). A low-energy-dense diet adding fruits reduces weight and energy intake in women. *Appetite*, 51, 291-295.
- Dukas, L., Willett, W.C., & Giovannucci, E.L. (2003). Association between physical activity, fibre intake, and other lifestyle variables and constipation in a study of women. *American Journal of Gastroenterology*, 14, 1790-1796.
- Ello-Martin, J.A., Roe, L.S., Ledikwe, J.H., Beach, A.M., & Rolls, B.J. (2007). Dietary energy density in the treatment of obesity: a year-long trial comparing 2 weight-loss diets. *American Journal of Clinical Nutrition*, 85: 1465-1477.
- Erlund, I., Koli, R., Alfthan, G., Marniemi, J., Puukka, P., Mustonen, P., Mattila, P., & Jula, A. (2008) Favourable effects of berry consumption on platelet function, blood pressure, and HDL cholesterol. *American Journal of Clinical Nutrition*, 87, 323-331.
- FAOSTAT database, (2010). *Food and Agriculture Organization of the United Nations (Online)*. <<http://faostat.fao.org/site/354/default.aspx>>.
- Farmer, B., Larson, B.T., Fulgoni, V.L., Rainville, A.J., & Liepa, G.U. (2011). A vegetarian dietary pattern as a nutrient-dense approach to weight management: an analysis of the National Health and Nutrition Examination Survey 1999-2004. *Journal of American Dietetic Association*, 111, 819-827.
- Fraser, G., Katuli, S., Anousheh, R., Knutsen, S., Herring, P., & Fan, J. (2015). Vegetarian diets and cardiovascular risk factors in black members of the Adventist Health Study-2. *Public Health Nutrition*, 18, 537-545.
- George, S.M., Park, Y., Leitzmann, M.F., Freedman, N.D., Dowling, E.C., Reedy, J., Schatzkin, A., Hollenbeck, A., & Subar, A.F. (2009). Fruits and vegetable intake and risk of cancer: a prospective cohort study. *American Journal of Clinical Nutrition*, 89, 347-353.
- Gill JM & Sattar N (2014). Fruits juice: just another sugary drink? *Lancet Diabetes Endocrinol* 2: 444-446.
- He FJ, Nowson CA, Lucas M & MacGregor GA (2007). Increased consumption of vegetables and fruits is related to a reduced risk of coronary heart disease: meta-analysis of cohort studies. *J Hum Hypertens* 21: 717-728.
- Heber D (2004). Vegetables, fruits and phytoestrogens in the prevention of diseases. *J Postgrad Med* 50: 145-149.

- Hubbard, G.P., Wolfram, S., de Vos, R., Bovy, A., Gibbins, J.M., & Lovegrove, J.A. (2006). Ingestion of onion soup high in quercetin inhibits platelet aggregation and essential components of the collagen-stimulated platelet activation pathway in man: a pilot study. *British Journal of Nutrition*, 96, 482-488.
- Hung, H.C., Joshipura, K.J., Jiang, R., Hu, F.B., Hunter, D., Smith-Warner, S.A., Colditz, G.A., Rosner, B., Spiegelman, D., & Willett, W.C. (2004). Fruits and vegetable intake and risk of major chronic disease. *Journal of the National Cancer Institute*, 96, 1577-1584.
- IPH (2020). *National Health and Morbidity Survey (NHMS) 2019: Vol. I: NCDs - Non-Communicable Diseases: Risk Factors and other Health Problems*. Institute for Public Health (IPH), National Institutes of Health, Ministry of Health Malaysia, Selangor.
- IPH (2017). *National Health and Morbidity Survey (NHMS) 2017: Adolescent Health Survey 2017*. Institute for Public Health (IPH), Ministry of Health Malaysia, Kuala Lumpur.
- Jaacks, L.M., Kapoor, D., Singh, K., Narayan, K.M., Ali, M.K., Kadir, M.M., Mohan, V., Tandon, N., & Prabhakaran, D. (2016). Vegetarianism and cardiometabolic disease risk factors: differences between South Asian and US adults. *Nutrition*, 32, 975-984.
- John, J.H., Ziebland, S., Yudkin, P., Roe, L.S., & Neil, H.A. (2002). Effects of fruits and vegetable consumption on plasma antioxidant concentrations and blood pressure: a randomised controlled trial. *Lancet*, 359, 1969-1974.
- Joshipura, K.J., Hu, F.B., Manson, J.E., Stampfer, M.J., Rimm, E.B., Speizer, F.E., Colditz, G., Ascherio, A., Rosner, B., Spiegelman, D., & Willett, W.C. (2001). The effect of fruits and vegetable intake on risk for coronary heart disease. *Annals of Internal Medicine*, 134, 1106-1114.
- Larsson, S.C., Virtamo, J., & Wolk, A. (2013). Total and specific fruits and vegetable consumption and risk of stroke: a prospective study. *Atherosclerosis*, 227, 147-152.
- Lee, W.T., Ip, K.S., Chan, J.S., Lui, N.W., & Young, B.W. (2008). Increased prevalence of constipation in pre-school children is attributable to under-consumption of plant foods: A community-based study. *J Paediatrics and Child Health*, 14, 170-175.
- Lock, K., Pomerleau, J., Causer, L., Altmann, D.R., & McKee, M. (2005). The global burden of disease attributable to low consumption of vegetables and fruits: implications for the global strategy on diet. *Bulletin of the World Health Organization*, 83, 100-108.
- McCall, D.O., McGartland, C.P., McKinley, M.C., Patterson, C.C., Sharpe, P., McCance, D.R., Young, I.S., & Woodside, J.V. (2009). Dietary intake of vegetables and fruits improves microvascular function in hypertensive subjects in a dose-dependent manner. *Circulation*, 119, 2153-2160.
- Mason-D'Croz, D., Bogard, J.R., Sulser, T.B., Cenacchi, N., Dunston, S., Herrero, M., & Wiebe, K. (2019). Gaps between fruit and vegetable production, demand, and recommended consumption at global and national levels: an integrated modelling study. *Lancet Planet Health*, 3, e318-e329.
- Miller, H.E., Rigelhof, F., Marquart, L., Prakash, A., & Kanter, M. (2000). Antioxidant content of whole grain breakfast cereal, vegetables and fruits. *Journal of American College of Nutrition*, 19, 312S-319S.
- Miura, K., Greenland, P., Stamler, J., Liu, K., Daviglus, M.L., & Nakagawa, H. (2004). Relation of vegetable, fruits, and meat intake to 7-year blood pressure change in middle-aged men: the Chicago Western Electric Study. *American Journal of Epidemiology*, 159, 572-580.
- Muraki, I., Imamura, F., Manson, J.E., Hu, F.B., Willett, W.C., van Dam, R.M. & Sun, Q. (2013). Fruits consumption and risk of type 2 diabetes: results from three prospective longitudinal cohort studies. *British Medical Journal*, 347, f5001.

- Orlich, M.J., Jaceldo-Siegl, K., Sabaté, J., Fan, J., Singh, P.N., & Fraser, G.E. (2014). Patterns of food consumption among vegetarians and non-vegetarians. *British Journal of Nutrition*, 112, 1644-1653.
- Oude Griep, L.M., Gelejinse, J.M., Kronhout, D., Ocke, M.C., & Verschuren, W.M. (2010). Raw and processed fruits and vegetable consumption and 10-year coronary heart disease incidence in a population-based cohort study in the Netherlands. *PLoS ONE*, 5, e13609.
- Reis, A., Rocha, S., & de Freitas, V. (2021). Going “Green” in the Prevention and Management of Atherothrombotic Diseases: The Role of Dietary Polyphenols. *Journal of Clinical Medicine*, 10, 1490.
- Sauvaget, C., Nagano, J., Hayashi, M., Spencer, E., Shimizu, Y., & Allen, N. (2003). Vegetables and fruits intake and cancer mortality in the Hiroshima/ Nagasaki life span study. *British Journal of Cancer*, 88, 689-694.
- Shibata, A., Paganini-Hill, A., Ross, R.K., & Henderson, B.E. (1992). Intake of vegetables, fruits, beta-carotene, vitamin C and vitamin supplements and cancer incidence among the elderly: a prospective study. *British Journal of Cancer*, 66, 673-679.
- Slavin, J.L., & Lloyd, B. (2012) Health benefits of vegetables and fruits. *Advances in Nutrition*, 3, 506-516.
- Soerjomataram, I., Oomen, D., Lemmens, V., Oenema, A., Benetou, V., Trichopoulou, A., Coebergh, J.W., Barendregt, J., & de Vries, E. (2010). Increased consumption of fruits and vegetable and future cancer incidence in selected European countries. *European Journal of Cancer*, 46, 2563-2580.
- Tan, M.M., Chan, C.K., & Reidpath, D.D. (2016). Religiosity, dietary habit, intake of fruit and vegetable, and vegetarian status among Seventh-Day Adventists in West Malaysia. *Journal of Behavioral Medicine*, 39, 675-686.
- Takachi, R., Inoue, M., Ishihara, J., Kurahashi, N., Iwasaki, M., Sasazuki, S., Iso, H., Tsubono, Y., & Tsugane, S.; JPHC Study Group (2008). Fruits and vegetable intake and risk of total cancer and cardiovascular disease: Japan Public Health Center-Based Prospective Study. *American Journal of Epidemiology*, 167, 59-70.
- Tomey, K.M., Sowers, M.R., Crandall, C., Johnston, J., Jannausch, M., & Yosef, M. (2008). Dietary intake related to prevalent functional limitations in midlife women. *American Journal of Epidemiology*, 167, 935-943.
- Tucker, L.A., & Thomas, K.S. (2009). Increasing total fibre intake reduces risk of weight and fat gains in women. *Journal of Nutrition*, 139, 576-581.
- Van Duyn, M.A., & Pivonka, E. (2000). Overview of the health benefits of fruits and vegetable consumption for the dietetics professional: selected literature. *Journal of American Dietetic Association*, 100, 1511-1521.
- Vioque, J., Weinbrenner, T., Castello, A., Asensio, L., & Garcia de la Hera, M. (2008). Intake of vegetables and fruits in relation to 10-year weight gain among Spanish adults. *Obesity*, 16, 664-670.
- Wallace, T.C., Bailey, R.L., Blumberg, J.B., Burton-Freeman, B., Chen, C.O., Crowe-White, K.M., Drewnowski, A., Hooshmand, S., Johnson, E., Lewis, R., Murray, R., Shapses, S.A., & Wang, D.D. (2020). Fruits, vegetables, and health: A comprehensive narrative, umbrella review of the science and recommendations for enhanced public policy to improve intake. *Critical Reviews in Food Science & Nutrition*, 60, 2174-2211.
- WCRF/ AICR (2007). Food, nutrition, physical activity, and the prevention of cancer: a global perspective. World Cancer Research Fund/ American Institute for Cancer Research, Washington DC: AICR.

- WHO (2003b). *Diet, nutrition and the prevention of chronic diseases. Report of a joint FAO/WHO Expert Consultation*. Technical Report Series, No. 916. Geneva: World Health Organization (WHO).
- WHO (2009). *Global health risks: mortality and burden of disease attributable to selected major risks*. Geneva: World Health Organization (WHO).
- WHO (2003). *The world health report 2003-Shaping the future*. Geneva: World Health Organization (WHO).
- WHO (2014, May 20). Country Profiles: Non-communicable Diseases (NCD). (Online). World Health Organization. <http://www.who.int/nmh/countries/mys_en.pdf>.
- Wojcicki, J.M., & Heyman, M.B. (2012). Reducing childhood obesity by eliminating 100% fruits juice. *American Journal of Public Health*, 102, 1630-1633.
- Wu, G.D. (2016). The Gut Microbiome, Its Metabolome, and Their Relationship to Health and Disease. *Nestle Nutr Inst Workshop Ser*, 84, 103-110.
- Wu, G.D., Compher, C., Chen, E.Z., Smith, S.A., Shah, R.D., Bittinger, K., Chehoud, C., Albenberg, L.G., Nessel, L., Gilroy, E., Star, J., Weljie, A.M., Flint, H.J., Metz, D.C., Bennett, M.J., Li, H., Bushman, F.D., & Lewis, J.D. (2016). Comparative metabolomics in vegans and omnivores reveal constraints on diet-dependent gut microbiota metabolite production. *Gut*, 65, 63-72.
- Zhu, F., Du, B., & Xu, B. (2018). Anti-inflammatory effects of phytochemicals from fruits, vegetables, and food legumes: A review. *Critical Reviews in Food Science & Nutrition*, 58, 1260-1270.



Key Message 3

Eat adequate
amounts of rice,
other cereals, whole
grain cereal-based
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Key Message 3

Eat adequate amounts of rice, other cereals, whole grain cereal-based products and tubers

Assoc. Prof. Dr. Wong Jyh Eiin, Assoc. Prof. Dr. Hanis Mastura Yahya and Prof. Dr. Barakatun Nisak Mohd Yusof

3.1 Terminology

Bioavailability

The rate and extent to which the fraction of ingested dietary components reach their sites of action and exert beneficial effects on the target tissues.

Breads

Leavened or unleavened dough prepared from flour or meal, or a combination of these with water and baked or steamed. Among examples of breads are white or wholemeal bread, bun, pita bread, *roti canai*, *roti Arab*, chapatti, tortilla, bagel and steamed bun or *pau* (MOH, 1985).

Bran

The outer layer, epidermis of the cereal which consists of thin walled, long rectangular cells.

Cereals

Cereals are the edible seeds known as kernel or grains of the grass family, Gramineae or Poaceae (Frølich & Aman, 2010). True cereal grains include wheat, oat, rice, corn, barley, rye, kamut, triticale, sorghum, fonio, millet, teff, and canary seed (AACCI, 2006). All cereal grains have three anatomical components: the bran, endosperm, and germ. Each component contributes to a different nutritional composition. Bran consists primarily of the main outer layers of the grain which is rich in dietary fiber, vitamins, minerals, and phytochemicals (Seal et al., 2016). The endosperm constitutes about 60-85% of

the grain, comprised mainly of carbohydrates in the form of starches with some protein and B vitamins (Frølich & Aman, 2010). The germ is the smallest fraction (2.5-3%) of the grain, containing a high lipid and protein content and some vitamins and minerals (Mathews & Chu, 2020).

Cereal-based products

Any food made from rice, wheat, corn, oats, rye, barley or other cereal grains is a cereal-based product. Examples of cereal-based products include bread, noodles, pasta, oatmeal, breakfast cereals, rice crackers and biscuits.

Dietary fibre

The edible parts of plants or analogous carbohydrates that are resistant to digestion and absorption in the human small intestine with complete or partial fermentation in the large intestine. Dietary fibre includes polysaccharides, oligosaccharides, lignin and associated plant substances.

Dietary phytochemicals

Non-nutritive, bioactive plant metabolites known to possess varying health effects. Examples of dietary phytochemicals include carotenoids, polyphenols, glucosinolates etc.

Fortification

Process of adding one or more essential nutrients to a food by the manufacturers, whether or not it is normally contained in the food, for the purpose of restoration, ensuring nutritional equivalence of substitute foods and enhancing the nutritive value of a food (Hennessy et al, 2013).

Glycemic index (GI)

A way to classify carbohydrate-containing foods based on the ability of different types of carbohydrate-containing foods to raise the blood glucose level. GI is defined as the incremental area under the blood glucose response curve elicited by a 50 g available carbohydrate portion of a food expressed as a percentage of the response to 50 g anhydrous glucose taken by an individual (Wolever, 2013).

Glycemic load (GL)

The GL refers to the cumulative exposure to postprandial glycemia, as a measure of insulin demand, over a specified period of time (Salmeron et al, 1997). It does not take into account the pattern of loading within the specified time e.g., few high – glycemic impact meals versus frequent meals of low glycemic impact. It is calculated indirectly as the

product of the average GI of carbohydrate foods consumed and the total carbohydrate intake over a specified time period (Jenkins et al., 1981).

Noodles and pasta

Products such as mee, bihun, laksa, macaroni and spaghetti that are obtained by extruding or moulding units of dough made of cereal flour (MOH, 1985).

Polyphenols

One of the subgroups of phytochemicals. Each polyphenol molecule is composed of more than one phenol unit. Polyphenols are further divided into several groups including flavonoids, phenolic acids, stilbenes, lignans and other polyphenols based on the carbon skeleton.

Refined grains

Highly processed grains where the outer layers of the grains have been removed during processing (NHMRC, 2013).

Rice

Grains of *Oryza sativa* from which the husk has been removed. White rice are rice grains that have been milled and polished to remove the husk, bran and germ. Among different varieties of rice include brown, wild, basmati, parboiled and fragrant rice (MOH, 1985).

Tubers

A type of starchy plant that stores carbohydrates in underground organs. These include potatoes, sweet potatoes, tapioca (cassava), yam, lotus root, ginger root and sweet turnip (*sengkuang*) (MOH, 1985).

Whole grains

Whole grains consist of the intact, ground, cracked or flaked caryopsis, whose principal anatomical components are the starchy endosperm, germ, and bran, which are present in the same relative proportions as they exist in the intact caryopsis (AACCI, 1999; Jones, 2010). Examples of whole grains include brown and wild rice, barley, whole oats, corn, buckwheat, rye, quinoa and bulgur.

Whole grain products

Food products made with at least one whole grain ingredient (AACC, 2000). The food contains 100% whole grain ingredients for wheat flour, rice flour, rice and grains; a minimum of 60% whole grain ingredients for bread; and a minimum of 25% or 8 g per serving whole grain ingredients for other products (MOH, 2020).

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3.2 Introduction

Cereals, cereal-based products and tubers form the foundation of vegetarian diets (Haddad et al., 1999; Messina et al., 2003). Being rich in carbohydrates, these foods are the major source of food energy. In addition, cereals, especially whole grain cereals, are important sources of dietary fibre, iron, B vitamins and minerals in a vegetarian diet.

Cereal grains, or simply known as cereals, are the edible portion of seeds from the grass family Gramineae. Cereals including rice, corn and wheat are the world's major food crops which supply more than 60 percent of the global food energy intake (FAO, 1995). Other cereal grains are oats, rye, barley, millet, sorghum and triticale. Although not from the Gramineae family, grains such as buckwheat, amaranth and quinoa are largely similar in nutrition content and preparation methods to the 'true' cereals (Grains and Legumes Nutrition Council, 2015). They are therefore considered as cereals in this dietary guideline.

Structurally, each seed of a cereal grain has a protective outer husk, a bran covering, a starchy endosperm and a germ. Because rough husks or chaff of grains are not usually consumed by humans, they are removed to produce whole grains that consist of the bran, germ and endosperm (Figure 3.1). Therefore, whole grains are intact, ground, cracked or flaked kernels after removal of the inedible husks (van der Kamp et al., 2014). Among common types of whole grains are unpolished rice (brown), whole wheat, de-hulled barley, oats, corn, rye, millet and buckwheat.

For improvement of shelf-life, flavour, appearance, palatability and cooking characteristics of cereals and cereal products (Seal et al., 2006), cereals are often milled to remove some or all of the germ and bran, leaving only the endosperm to form refined grain (Figure 3.1). Because of milling, refined grains contain more starch, but less dietary fibre, fats, vitamins, minerals and phytonutrients (Okarter & Liu, 2010).

Among cereals, rice is one of the most common staple foods with nearly 90 percent of the cultivation and consumption by populations in Asia (Steiger et al., 2014). In Malaysia, white rice is consumed daily, almost twice (1.86 times) per day, by the majority of the adult population (Norimah et al., 2008). On average, 297 g (2.48 cups of rice) is consumed daily per person (Mohamad Hasnan et al., 2014). There is a gradual increase in the cereals production over the years to meet the needs of the consumers (FAOSTAT, 2021). Like rice, most cereals such as pearled barley and wheat flour are consumed in the form of refined grains in Malaysia. Some refined grains are added or enriched with vitamins and minerals that were lost during processing, storage or transport.

Cereal-based products are any foods made from wheat, rice, oats, corn, barley, *ragi* (finger millets) or other cereals. They can be made from refined or whole grains and include products such as bread, cakes, noodles, pastas, biscuits and ready-to-eat breakfast cereals. In Malaysia, most cereal-based products, such as bread, noodles and biscuits are made from refined flour. Cereal-based products such

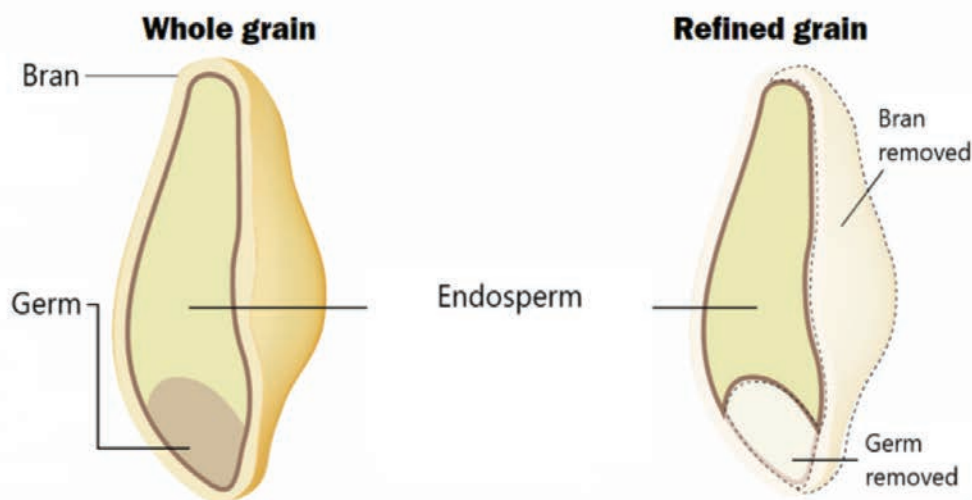


Figure 3.1: Anatomy of a grain

(Source: Illustration adapted with permission from Nutrition Month Malaysia 2018 Infographic kit: Your Only Choice: Eat Healthy, Be Active)

as wheat flour, bread, breakfast cereals and malted milk powder are permitted for voluntary micronutrient fortification under the stipulation of the Food Regulation 1985 (MOH, 1985). Among the commonly added nutrients are vitamin A, thiamin, riboflavin, pyridoxine, biotin, pantothenic acid, niacin, vitamin C, vitamin D, vitamin E, calcium, iodine, phosphorus, folic acid and vitamin B12.

There is no universally accepted definition of 'whole grain products' or criteria on the minimum content of whole grain in a product. In Malaysia, the Food Regulations 1985 has provided legal definitions for the term. A whole grain product should contain 100% whole grain ingredients for wheat flour, rice flour, rice and grains. At least 60% whole grain ingredients must be present for bread; while at least 25% whole grain ingredients or 8 g per serving of whole grain ingredients must be present for other

products. In addition, the regulations have stipulated that the percentage of whole grain in the product must be stated next to the term 'Wholemeal' or 'Whole grain' on the label (MOH, 2020).

Tubers such as potatoes, sweet potatoes, cassava and yam cultivate well in diverse soils and environments, including conditions which are not suitable for cereal production. Traditionally, tubers are widely grown and consumed as subsistence staples in humid and sub-humid tropics, such as Asia, Pacific Islands, Africa and Latin American (FAO, 1998). Tubers have a higher moisture content ranging from 60- 90% compared to 10-15% moisture content in cereal grains. Due to higher moisture content, tubers have lower energy- and nutrient densities relative to cereals (FAO, 1993).

3.3 Scientific Basis

The Malaysian Food Pyramid for Vegetarians suggests that vegetarians consume adequate amounts of cereals, cereal-based products and tubers as part of a healthy diet. This food group contains a myriad of nutrients, including carbohydrates, proteins, dietary fibre, vitamins, minerals and phytochemicals, hence contributing to a balanced vegetarian diet. The recommended serving for this food group is from 3 to 7, with cereal products preferably being whole grain. This recommendation is consistent with many international dietary guidelines for the general population, including the Malaysian Food Pyramid (NCCFN, 2021).

3.3.1 Cereals, cereal-based products, and tubers as the major source of energy, carbohydrate and dietary fibre

Cereal, cereal-based products and tubers provide sources of energy mainly from dietary carbohydrate and protein. They supply more than 50% of the energy consumed by the world population (Stoskopf, 1985; Yu & Tian, 2018). Rice serves as an important source of carbohydrate in the vegetarian diet, especially among the vegans (Farmer et al, 2011; Rizzo et al, 2013). Tubers, being second in importance to cereals as a global source of carbohydrate (Chandrasekara & Josheph Kumar, 2016), are also consumed in a higher proportion among vegetarians compared to the non-vegetarians (Shridhar et al, 2014).

Rice has the highest digestibility among the staples, due to its low dietary fibre and tannin content (FAO,

1993). Polished rice is rapidly digested and absorbed, which results in a marked increase in blood glucose levels after consumption (Barakatun Nisak et al, 2005). The glycemic index (GI) of rice is intermediate to high, depending on its amylose fraction. Rice with a high amylose fraction of about 28% such as Basmati rice have been shown to produce a lower blood glucose response than the other low amylose varieties such as fragrant white rice (Miller et al, 1992). Other cereals and cereal-based products commonly available in Malaysia like white bread, bihun, noodles and biscuits are mostly refined products.

On the contrary, foods that are minimally processed such as whole grains and their products usually have low GI/ GL properties. These foods have slower rates of digestion and absorption, hence reducing the rapid elevation of postprandial hyperglycemia, as well as insulin concentration (Yusof et al, 2009). Whole grains may be eaten whole, cracked, split, flaked or ground. Most often, they are milled into flour and used to make breads, cereals, pasta, crackers and other cereal-based foods.

Cereals and cereal-based products, especially the whole grain varieties, are good sources of dietary fibre. The fibre content of different wholegrain foods can vary considerably, between 0.5 and 4 grams of fibre per serving, depending on the food category and serving size (U.S. Department of Agriculture et al, 2006). Whole grain cereals not only contain high amounts of dietary fibre in the form of resistant starch, but additionally provide a unique blend of bioactive components including pectin, bran,

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arabinoxylan and B-glucan. All of these components have a protective effect on metabolic health (Lattimer & Haub, 2010).

Some cereals that contain a significant or high amount of fibre may not contain a significant amount of whole grain. For example, oat bran or high fibre bran cereals may contain very little or no whole grain, but provide high levels of fibre because they are made with only the bran portion of the grains. It is therefore important to note that high fibre does not equate to whole grain. Similar to cereals, tubers also contain significant amounts of dietary fibre, especially in the skin (Tolessa, 2018). It is recommended that Malaysians consume 20 - 30 g of dietary fibre per day (NCCFN, 2021).

3.3.2 Cereals, cereal-based products and tubers as complementary plant proteins

Protein levels of cereals and tubers range widely; from 1- 2 g per 100 g in tubers to 7-14 g per 100 g in cereals (FAO, 1993). For instance, after cooking, every 150 g of rice and potatoes contain approximately 3.5 g and 2.5 g of protein, respectively (Tee et al., 1997). Similar to legumes, nuts, seeds and vegetables, the concentrations of protein are lower in cereals and tubers compared to animal foods (Young & Pellett, 1994). Therefore, it is not surprising that typical vegetarians have lower protein intake compared to meat-eaters (Cade et al., 2004; Clarys et al., 2014; Gan et al., 2018). Nevertheless, cereals and tubers are important sources of proteins in a vegetarian diet, particularly for vegans who omit milk and eggs in their diets. Cereals and legumes can provide 11% of plant proteins (75% of total protein) in semi-vegetarian diets and up to 13.5% (96% of total protein) in strict vegetarian diets (Rizzo et al., 2013).

Among the staple foods such as wheat, oat and corn, rice has the least protein in terms of its concentration. However, early balance data of whole grains suggest that rice has the highest digestibility, followed by wheat and corn. Compared to reference proteins (egg, milk and meat), the relative digestibility of rice is 93%. This may be due to its low dietary fibre and tannin content (FAO, 1993). Boiling and pressure-cooking may also increase the protein digestibility of rice (Siddhuraju & Becker, 2001).

In terms of protein quality, cereals and tubers are regarded as incomplete proteins as they are deficient in one or more essential amino acids (Young & Pellett, 1994). Based on amino acid analysis, cereals are generally low in lysine, while tubers are limited in sulphur-containing amino acids. For instance, rice as a staple grain is limited

in lysine but rich in methionine. On the contrary, tubers such as yams are higher in lysine, though limiting in cysteine and methionine (FAO, 1990; Young & Pellett, 1994). Therefore, tubers have been used to provide complementary staple foods for rice, wheat and other cereals (FAO, 1990).

The main way to overcome the incomplete amino acid profile in plant proteins is through eating a mixture of plant protein foods (Bressani, 1988). To achieve true protein complementation effect, two plant protein foods, each of which supplies the amino acids missing in the other, are consumed together. For example, cereals are paired with legumes and nuts which lack methionine but are rich in lysine. Consumption of mixed foods with complementary proteins can provide a well-balanced protein source (Friedman, 1996). Therefore, when consumed in appropriate amounts and in combination with nuts and legumes at different meals over the course of the day, cereals and tubers can contribute towards complete proteins in a vegetarian diet (Mariotti & Gardner, 2019; Melina et al., 2016).

3.3.3 Cereals, cereal-based products and tubers provide phytochemicals, essential vitamins and minerals

Cereals, especially whole grains are rich in phytochemicals, which are natural bioactive, non-nutrient compounds that regulate human physiological processes, thereby providing health benefits beyond those conferred by vitamins and minerals (Cherif, 2012). Among examples of phytochemicals are polyphenols, carotenoids, phytosterols, lignans, avenanthramides and alkylresorcinols. In most cereal grains such as rice, wheat, maize, barley, oats and rye, phenolic acids are the major polyphenols contained within the bran and germ. According to Phenol-Explorer, a comprehensive database of polyphenol content, cereals and cereal-based products contain substantial amounts of polyphenols including phenolic acid and alkylphenols (Neveu et al., 2010). The amount of polyphenols in refined cereals and products such as polished rice and white bread can be considered low when compared to germ and bran.

The level of phytochemicals including phenolic compounds, phytosterols, tocols, betaine and folate are higher at the outer layer of the grains, specifically at the pericarp and aleurone (Fardet, 2010). Phenolic acids present in wheat bran are normally bound to the cell wall and this causes difficulty to be digested. However, these compounds can finally reach the colon and be released for digestion with the

assistance of gastrointestinal esterase (Andreasen et al., 2001).

Refined grains lose some phytochemicals, vitamins and minerals during the process of milling, which removes the husk, bran and germ. The percentage of vitamins loss in the refined grains as compared to the whole grain such as wheat, rice and corn can be from 40% up to 95% (Garg et al., 2021). Despite this drawback, processing of grains has shown to increase the bioavailability of phytochemicals (Mateo Anson et al., 2011; Slavin et al., 2000). However, domestic cooking of whole grains can cause a reduction in phytochemical levels (Belobrajdic & Bird, 2013). This emphasises the importance of consuming cereals or cereal-based products with minimal processing such as heating or boiling.

Nonetheless, cereals and tubers are good sources of vitamins and minerals. In general, cereals provide a good source of vitamins A, B, and E but lesser amounts of other vitamins (Garg et al., 2021). Cereals are rich in B vitamins including thiamin, niacin, riboflavin and pantothenic acid (Slavin, 2004), however lacking in vitamin C and vitamin B12 (Cordain, 1999). Minerals such as potassium are found abundant in cereals. Particularly in whole grains, considerable amounts of iron, magnesium and zinc are present (McKevith, 2004).

In the bran and germ of whole grains, a high concentration of carotenoids including lutein, zeaxanthin, β -cryptoxanthin, β -carotene and α -carotene can be found (Adom et al., 2005; Okarter & Liu, 2010). Additionally, vitamin E (i.e. tocopherols and tocotrienols) are found exclusively in the germ fraction of the whole grains. Specifically, corn kernel are rich in vitamins A, B1, B2, B3, B5, B6, B9, E, and K, but are less abundant of vitamins B12, C, and D (Acosta-Estrada et al., 2019). In addition, the highest concentration of mineral and phenolic compounds can be found at pericarp germ, and aleurone layer of corn (Serna-Saldivar, 2016).

Tubers such as potatoes contribute a significant amount of carbohydrate, potassium and ascorbic acid in the diet (Hale et al., 2008). Vitamin C is present in sweet potatoes and yam, while beta-carotene or provitamin A is present in yellow varieties of sweet potatoes and yam (FAO, 1998).

Cereal-based foods such as wholewheat and rye breads, popcorn, cooked cereals and wild rice were reported to be important contributors of daily intake of B vitamins in a large multiethnic cohort study conducted in the United States. These foods provide

vitamins including thiamin, riboflavin, niacin, vitamin B6 and folic acid (Sharma et al., 2013). However, cereal-based products like noodles and pasta should not be cooked in excess water because it may cause nutrient losses especially in vitamin B3 (Ayhan & Köksel, 2019).

3.3.4 Fortified cereal-based products as alternative sources of vitamin B12

Individuals practising vegetarian and vegan diets have a higher tendency of dietary vitamin B12 insufficiency (Niklewicz et al., 2023). A greater risk of vitamin B12 deficiency is associated with pregnant women and older adults. Thus, consumption of vitamin B12 in fortified foods or supplements are important in order to fulfil the dietary recommendation of this vitamin (Allen et al., 2010).

Vitamin B12 is usually added in yeast spread, soy milk and vegetarian meat alternatives (Zeuschner et al., 2012). Fortification is a step to protect the population from becoming deficient in the specific nutrients and not to treat existing nutrient deficiencies (Allen et al., 2010). Specific to vegans, eating fortified cereal-based products such as breakfast cereals, whole grain biscuits, rice beverages and malted drinks becomes one of the alternatives for vitamin B12 source. Fortified foods are recommended because they are well absorbed by the body (Institute of Medicine, 1998). No known adverse effect to health was reported from vitamin B12 fortification and high intake of the vitamin (Institute of Medicine, 2000).

3.3.5 Benefits of whole grain varieties

Consistent with recommendations of the Malaysian Dietary Guidelines (NCCFN, 2021), the Malaysian Food Pyramid for Vegetarians recommends that at least half of the cereal intake should be whole grain. The basis for most vegetarian meals should be to include whole grains where possible, and to choose fortified cereal-based products when available.

Since whole grains contain more dietary fibre, nutrients and phytochemicals (Seal et al., 2021), they are more likely to offer greater beneficial effects on health than refined grains. The association between increased whole grain consumption and lower risks of chronic diseases, such as cardiovascular diseases (CVD), diabetes and cancers, as well as reduced risk of obesity and weight gain, underpins the dietary recommendations encouraging whole grain varieties in the Malaysian Food Pyramid for Vegetarians.

3.3.5.1 Whole grains and reduced risk of CVD

Despite the varying definitions of whole grains and methods used to estimate whole grain intake, epidemiological studies consistently show that consumption of whole grains is associated with reduced risks for CVD. Greater whole grain consumption (2.5 servings/day versus 0.2 servings/day) has been associated with 21% lower CVD events in a meta-analysis of prospective studies (Mellen et al, 2008). A systematic review and dose-response meta-analysis from prospective cohort studies revealed that the whole grain intake (28 g/day) was associated with a 14% lower risk for CVD mortality (Zhang et al, 2018). In the systematic review and meta-analysis of RCTs, consuming whole grain as opposed to refined grain can improve some cardiovascular risk factors in adults with or without CVD risk factors (Marshall et al, 2020).

Protective effects of whole grains on CVD are reported due to various mechanisms (Drewnowski et al, 2021; Seal et al, 2021). Its fermentable carbohydrate including dietary fiber, resistant starch and oligosaccharides lead to the production of short-chain fatty acids (SCFAs), which are known to cause faecal excretion of cholesterol and inhibit cholesterol synthesis (Slavin et al, 1999). Whole grains contain a rich amount of vitamin E, in particular, tocotrienol and β -sitosterol, which act as antioxidants (Slavin et al, 1999). In addition, the unsaturated fatty acids, mainly oleic and linoleic acid in whole grains help to reduce cholesterol levels (Jonnalagadda et al, 2011).

3.3.5.2 Whole grains and reduced risk of type 2 diabetes

Whole grain consumption provides better protection against type 2 diabetes risk than total intake of grains or refined grains. A meta-analysis of prospective studies showed that three servings of whole grain daily is associated with a 32% reduction in the relative risk for type 2 diabetes, compared to 17% relative risk reduction for total grain intake (Aune et al, 2013). A analysis of dietary intake data from the three largest prospective cohorts have also found that whole grains consumptions lower risk of type 2 diabetes (Hu et al, 2020). Interestingly, most of the risk reduction occurred when whole grain intake was up to two servings per day, suggesting that at least two servings of whole grains should be consumed to reduce type 2 diabetes risk. Protective effects of whole grain could be due to its intact

structure and high concentration of antioxidants (vitamin E and selenium), which help to enhance B-cell function, and hence improving insulin regulation as well as reduce inflammatory markers (Knudsen et al, 2016; Malin et al, 2019).

3.3.5.3 Whole grains and reduced risk of cancer

A systematic review of meta-analyses of observational studies reported that whole grain intake was associated with total cancer mortality, where daily intake of 15 g to 90 g of whole grain resulted in 3% to 20% reduction in the risk (Gaesser, 2020). For site-specific cancers, whole grain intake was associated with colorectal, colon, gastric, pancreatic and esophageal cancers (Gaesser, 2020). This finding was supported by another systematic review and meta-analysis (Zhang et al, 2020). Whole grain consumption may offer protection for colorectal cancer by increasing stool bulk, diluting faecal carcinogens, decreasing transit time and production of short chain fatty acids through colonic fermentation.

3.3.5.4 Whole grains and body weight management

In a systematic review of observational studies, high intake of whole grains (approximately three servings/day) has been associated with lower BMI and waist circumference compared to those with low whole grain intake (<0.5 servings/day) (Harland & Garton, 2008). This finding was supported by another two recent systematic reviews and meta-analysis where whole grain intake was associated with lower body weight (Maki et al, 2019; Wang et al, 2020).

Nonetheless, whole grains intake does not affect body weight and obesity measures in a systematic review and meta-analysis of RCTs (Sadeghi et al, 2020). The lack of differences in obesity outcomes could be related to low compliance and short duration of intervention studies (Sadeghi et al, 2020). Although the benefits of whole grains on body weight in RCTs have been inconsistent, a recent systematic review and meta-analysis found that whole grain significantly reduced subjective hunger and desire to eat, and increased fullness and satiety (Sanders et al, 2021). The mechanism might explain the inverse association between whole grain and weight gain.

3.4 Current Status

In Malaysia, there is a lack of local studies on consumption of various food groups among vegetarians. Through the findings of the last two Malaysian Adults Nutrition Surveys, it was known that white rice and biscuits remained as two of the top ten foods consumed by the general population on a daily basis (Norimah et al., 2008; Mohamad Hasnan et al., 2014). Other food items consumed daily were sugar, leafy green vegetables, marine fish, chilies, condensed milk/ creamer, soy sauce and condiments (Mohamad Hasnan et al., 2014). Notably, cereals and cereal-based products are most frequently consumed in their refined forms. Similarly, in small local studies conducted among Chinese vegetarians in Kuala Lumpur (Khor et al., 2000) and Hindus and Buddhist vegetarians in Kuching (Cheah et al., 2018), rice was one of the staple foods consumed daily by almost all of the participants. Other types of cereals and cereal-based products consumed at least once a week included bread, various types of noodles, and root tubers (Khor et al., 2000). However, it is unknown whether the cereal consumption pattern of vegetarians is similar to non-vegetarians.

There is no data on whole grain consumption patterns of Malaysian adults. A national study on whole grain intakes of Malaysian primary and secondary school children showed that only 25% of children and 19% of adolescents are whole grain consumers. The main sources of whole grains were wheat (77.7% of total daily whole grain intake), followed by oat (13.7%), corn (7.4%) and rice (1.2%). Among the whole grain consumers, the daily mean intakes were 9.1 g and 9.2 g for the children and adolescents, respectively. The whole grain intake level was far below the United States whole grain recommendation of 48 g per day (Norimah et al., 2015). It has been suggested that whole grain consumption is influenced by cost, availability and sensory quality of whole grain foods (Kuznesof et al., 2012; Norimah et al., 2015). A cross-sectional study among medical students in Selangor reported that 51% of the participants consumed whole-grain in the past 3 months (Subramanian et al., 2019). However, the liking and acceptance toward whole grain products and other factors influencing whole grain consumption have not been studied in Malaysia.

3.4.1 Areas of concern

In vegetarian diets, non-heme iron and zinc from plant sources are sensitive to inhibitors and are therefore, less bioavailable. One of the major concerns of consuming high amounts of whole grains and tubers is the reduced iron and zinc bioavailability following high intake of iron- and zinc-inhibiting components in these foods (Craig et al., 2009; Melina et al., 2016). Whole grains and tubers store phosphorus in the form of 6-phosphoinositol, better known as phytic acid or phytate, which may bind and hence inhibit the absorption of non-heme iron and zinc that are consumed concurrently (Hunt, 2002, 2003).

Bioavailability of iron and zinc in a vegetarian diet can be increased using some food preparation techniques. Soaking or sprouting grains and leavening bread can hydrolyse phytate, thus reducing the binding of these minerals by phytate (Lonnerdal, 2000). Fermentation can also reduce the levels of phytate in tubers, through the enzyme phytase that is naturally present in tubers or secreted by fermentative microorganism (FAO, 1990). Another effective way to increase iron absorption is by consuming non-heme iron together with fruits and vegetables that are rich in vitamin C and organic acids.

Gluten is a protein that is found in wheat, rye and barley. Gluten-containing cereals are avoided by individuals who are diagnosed with celiac disease or are gluten-sensitive. Vegetarians who are on strict gluten-free diets maybe at risk of deficiency for certain nutrients such as vitamin B12, vitamin D, folate, iron and zinc due to the further elimination of wheat, rye, barley and products derived from these gluten-containing cereals in diets (Vici et al., 2016). Fortified foods or supplements may be used to ensure adequate nutrient intakes in vegetarians who are practising gluten-free diets (Pagano, 2007). For example, gluten-free flours made from whole grains (such as corn, millet, rice, sorghum, quinoa), seeds and beans (such as soya bean, fava bean and garbanzo bean) can be used to provide additional sources of iron, calcium, B vitamins and fibre to a gluten-free vegetarian diet.

KM3
Eat adequate amounts of rice, other cereals, whole grain cereal-based products and tubers

3.5 Key Recommendations

Key Recommendation 1

Eat a variety of cereals, cereal-based products and tubers.

How to achieve:

1. Eat different types of cereals and tubers daily as meals or snacks.
2. Include fortified cereal-based products, such as ready-to-eat breakfast cereal, cereal bars and malted drinks.

Key Recommendation 2

Eat 3-7 servings of rice, other cereals, whole grain cereal-based products, and tubers daily according to your energy needs and physical activity level.

How to achieve:

1. Include at least one serving of cereal, cereal-based products or tubers in every main meal.
2. Choose cereal-based or tuber-based meals or snacks, such as bread, thosai, *idli*, rice crackers, biscuits, and steamed or boiled sweet potatoes or tapioca.
3. Tubers, such as potatoes, yam and tapioca, can be consumed as alternatives or in combination with rice or cereals.

Key Recommendation 3

Choose at least half of your rice, other cereals and cereal-based products from whole grains.

How to achieve:

1. Cook white rice mixed with whole grains such as brown rice, wild rice, hulled barley, whole oats and corn.
2. Choose whole grain alternatives for noodles and pasta, bread, breakfast cereals, cakes, biscuits and other cereal based products.
3. Add whole grains such as corn, hulled barley and oats to soups.
4. Choose whole grain products, such as whole-meal bread, whole-wheat *capati*, *putu mayam ragi* (string hoppers), *ragi* noodles, buckwheat noodles, thosai or oat porridges, over similar refined products.
5. Read the ingredients list of cereal-based product labels. Opt for products with a higher percentage of whole grains.

Key Recommendation 4

Choose cereal-based products that are high in fibre, low in fat, sugar and salt.

How to achieve:

1. Read the Nutrition Information Panel and look for fibre, fat, sugar and salt content.
2. Choose cereal-based products that are higher in fibre and lower in fat, sugar, or salt.

KM3

Eat adequate amounts of rice, other cereals, whole grain cereal-based products and tubers

3.6 References

- AACC. (2000). Whole grain definition. *Cereal Foods World* 45(2), 45-88.
- Acosta-Estrada, B. A., Gutiérrez-Urbe, J. A., & Serna-Saldivar, S. O. (2019). *Minor constituents and phytochemicals of the kernel Corn*. Elsevier.
- Adom, K. K., Sorrells, M. E., & Liu, R. H. (2005). Phytochemicals and antioxidant activity of milled fractions of different wheat varieties. *Journal of Agricultural and Food Chemistry*, 53(6), 2297–2306. <https://doi.org/10.1021/jf048456d>.
- Allen, L. H., Rosenberg, I. H., Oakley, G. P., & Omenn, G. S. (2010). Considering the case for vitamin B12 fortification of flour. *Food and Nutrition Bulletin*, 31(1 Suppl), S36-46. <https://doi.org/10.1177/15648265100311S104>.
- Andreasen, M. F., Kroon, P. A., Williamson, G., & Garcia-Conesa, M. T. (2001). Intestinal release and uptake of phenolic antioxidant diferulic acids. *Free Radical Biology & Medicine*, 31(3), 304–314. [https://doi.org/10.1016/s0891-5849\(01\)00585-8](https://doi.org/10.1016/s0891-5849(01)00585-8).
- Aune, D., Norat, T., Romundstad, P., & Vatten, L. J. (2013). Whole grain and refined grain consumption and the risk of type 2 diabetes: a systematic review and dose-response meta-analysis of cohort studies. *European Journal of Epidemiology*, 28(11), 845–858. <https://doi.org/10.1007/s10654-013-9852-5>.
- Ayhan, D. K., & Köksel, H. (2019). Investigation of the effect of different storage conditions on vitamin content of enriched pasta product. *Quality Assurance and Safety of Crops & Foods*, 11(8), 701–712. <https://doi.org/10.3920/qas2019.1575>.
- Barakatun Nisak, M. Y., Ruzita, A. T., & Norimah, A. K. (2005). Glycaemic index of eight types of commercial rice in Malaysia. *Malaysian Journal of Nutrition*, 11(2), 151-163.
- Belobrajdic, D. P., & Bird, A. R. (2013). The potential role of phytochemicals in wholegrain cereals for the prevention of type-2 diabetes. *Nutrition Journal*, 12(1), 62. <https://doi.org/10.1186/1475-2891-12-62>.
- Bressani, R. (1988). *Protein complementation of foods Nutritional evaluation of food processing*. Springer.
- Cade, J. E., Burley, V. J., Greenwood, D. C., & UK Women's Cohort Study Steering Group. (2004). The UK Women's Cohort Study: comparison of vegetarians, fish-eaters and meat-eaters. *Public Health Nutrition*, 7(7), 871–878. <https://doi.org/10.1079/phn2004620>.
- Chandrasekara, A., & Josheph Kumar, T. (2016). Roots and tuber crops as functional foods: A review on phytochemical constituents and their potential health benefits. *International Journal of Food Science*, 2016, 3631647. <https://doi.org/10.1155/2016/3631647>.
- Cheah, W. L., Chang, K. H., Tang, M. F., Lekha, Y., Aldrin, R.D., & Noorhaswati Izura, A. L. (2018). Vegetarian practices, body composition and dietary intake among Hindus and Buddhists in Kuching, Sarawak, Malaysia. *South East Asia Journal of Public Health*, 8(1), 3-9. <https://doi.org/10.3329/seajph.v8i1.42266>.
- Cherif, A. O. (2012). Phytochemicals components as bioactive foods. In I. Rasooli (Ed.), *Bioactive Compounds in Phytomedicine* (pp. 113-124). InTech.
- Clarys, P., Deliens, T., Huybrechts, I., Deriemaeker, P., Vanaelst, B., De Keyzer, W., Hebbelinck, M., & Mullie, P. (2014). Comparison of nutritional quality of the vegan, vegetarian, semi-vegetarian, pesco-vegetarian and omnivorous diet. *Nutrients*, 6(3), 1318–1332. <https://doi.org/10.3390/nu6031318>.
- Cordain L. (1999). Cereal grains: humanity's double-edged sword. *World review of nutrition and dietetics*, 84, 19–73. <https://doi.org/10.1159/000059677>.

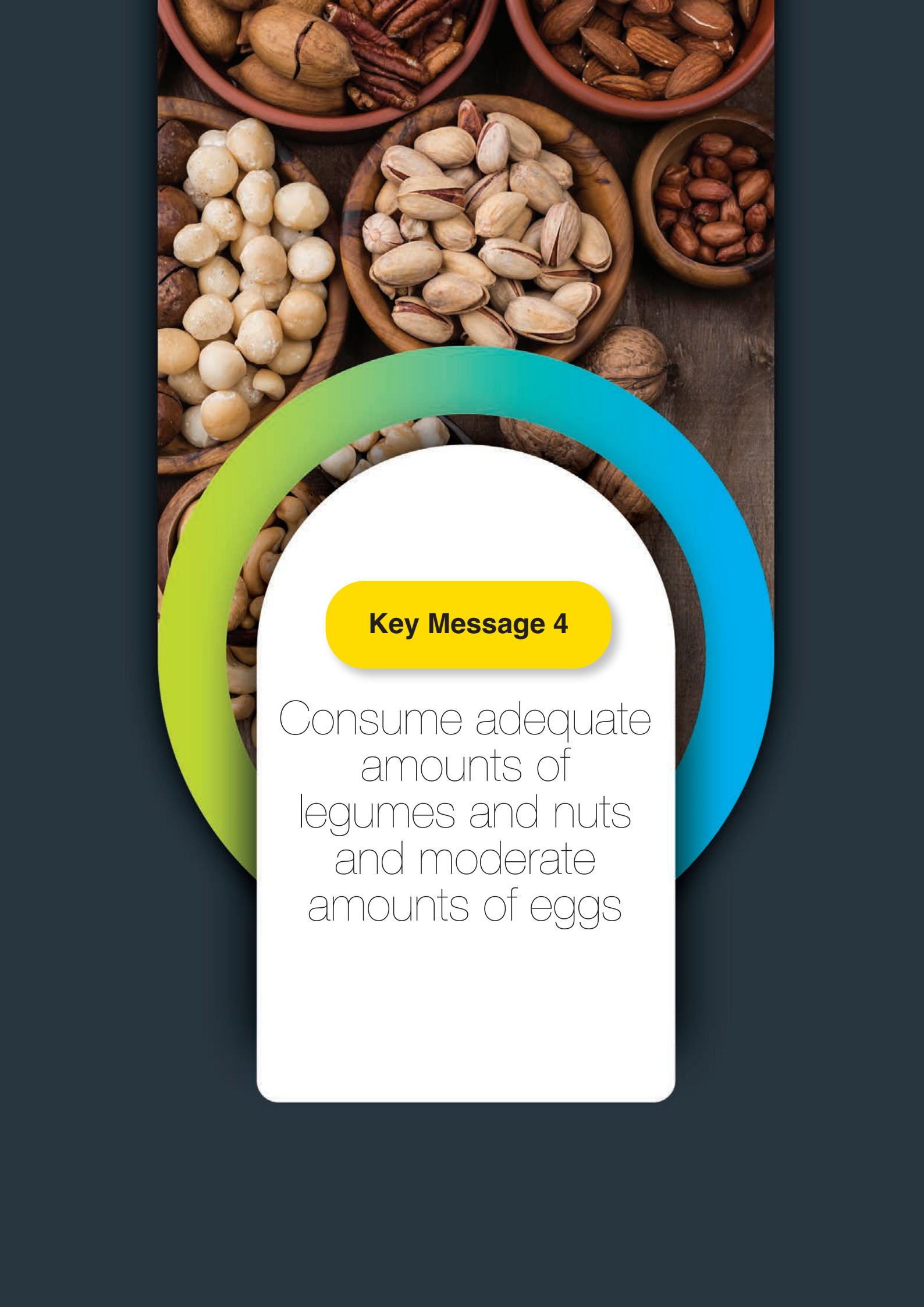
- Craig, W. J., Mangels, A. R., & American Dietetic Association. (2009). Position of the American Dietetic Association: vegetarian diets. *Journal of the American Dietetic Association*, 109(7), 1266–1282. <https://doi.org/10.1016/j.jada.2009.05.027>.
- Drewnowski, A., McKeown, N., Kissonick, K., Beck, E., Mejbom, H., Vieux, F., Smith, J., Masset, G., & Seal, C. J. (2021). Perspective: Why whole grains should be incorporated into nutrient-profile models to better capture nutrient density. *Advances in Nutrition (Bethesda, Md.)*, 12(3), 600–608. <https://doi.org/10.1093/advances/nmaa172>.
- FAO. (1990). Roots, tubers, plantains and bananas in human nutrition. <http://www.fao.org/docrep/t0207e/t0207e00.htm>.
- FAO. (1993). Rice in human nutrition. <http://www.fao.org/docrep/T0567E/T0567E00.htm#Contents>.
- FAO. (1995). Staple foods: what do people eat? <https://www.fao.org/3/u8480e/U8480E07.htm#Staple%20foods%20What%20do%20people%20eat>.
- FAO. (1998). Storage and Processing of Roots and Tubers in the Tropics. <http://www.fao.org/docrep/x5415E/x5415e00.htm#Contents>.
- FAOSTAT. (2014). Cereals total production 1961-2021 in Malaysia. <https://www.fao.org/faostat/en/#country/131>.
- Fardet, A. (2010). New hypotheses for the health-protective mechanisms of whole-grain cereals: what is beyond fibre? *Nutrition Research Reviews*, 23(1), 65–134. <https://doi.org/10.1017/S0954422410000041>.
- Farmer, B., Larson, B. T., Fulgoni, V. L., 3rd, Rainville, A. J., & Liepa, G. U. (2011). A vegetarian dietary pattern as a nutrient-dense approach to weight management: an analysis of the national health and nutrition examination survey 1999-2004. *Journal of the American Dietetic Association*, 111(6), 819–827. <https://doi.org/10.1016/j.jada.2011.03.012>.
- Friedman, M. (1996). Nutritional value of proteins from different food sources. A review. *Journal of Agricultural and Food Chemistry*, 44(1), 6–29. <https://doi.org/10.1021/jf9400167>.
- Gaesser, G. A. (2020). Whole grains, refined grains, and cancer risk: A systematic review of meta-analyses of observational studies. *Nutrients*, 12(12), 3756. <https://doi.org/10.3390/nu12123756>.
- Gan, W. Y., Boo, S., Seik, M. Y., & Khoo, H. E. (2018). Comparing the nutritional status of vegetarians and non-vegetarians from a Buddhist Organisation in Kuala Lumpur, Malaysia. *Malaysian Journal of Nutrition*, 24(1), 89-101.
- Garg, M., Sharma, A., Vats, S., Tiwari, V., Kumari, A., Mishra, V., & Krishania, M. (2021). Vitamins in cereals: A critical review of content, health effects, processing losses, bioaccessibility, fortification, and biofortification strategies for their improvement. *Frontiers in Nutrition*, 8, 586815. <https://doi.org/10.3389/fnut.2021.586815>.
- Grains and Legumes Nutrition Council (2015). Types of grains. <http://www.glnc.org.au/grains/types-of-grains/>.
- Haddad, E. H., Sabaté, J., & Whitten, C. G. (1999). Vegetarian food guide pyramid: a conceptual framework. *American Journal of Clinical Nutrition*, 70(3), S615–S619. <https://doi.org/10.1093/ajcn/70.3.615s>.
- Hale, A. L., Reddivari, L., Nzaramba, M. N., Bamberg, J. B., & Miller, J. C., Jr. (2008). Interspecific variability for antioxidant activity and phenolic content among Solanum species. *American Journal of Potato Research*, 85(5), 332–341. <https://doi.org/10.1007/s12230-008-9035-1>.

- Harland, J. I., & Garton, L. E. (2008). Whole-grain intake as a marker of healthy body weight and adiposity. *Public Health Nutrition*, 11(6), 554–563. <https://doi.org/10.1017/S1368980007001279>.
- Hennessy, Á., Walton, J., & Flynn, A. (2013). The impact of voluntary food fortification on micronutrient intakes and status in European countries: a review. *Proceedings of the Nutrition Society*, 72(4), 433–440. <https://doi.org/10.1017/S002966511300339X>.
- Hu, Y., Ding, M., Sampson, L., Willett, W. C., Manson, J. E., Wang, M., Rosner, B., Hu, F. B., & Sun, Q. (2020). Intake of whole grain foods and risk of type 2 diabetes: results from three prospective cohort studies. *BMJ*, 370, m2206. <https://doi.org/10.1136/bmj.m2206>.
- Hunt, J. R. (2002). Moving toward a plant-based diet: are iron and zinc at risk? *Nutrition Reviews*, 60(5 Pt 1), 127–134. <https://doi.org/10.1301/00296640260093788>.
- Hunt, J. R. (2003). Bioavailability of iron, zinc, and other trace minerals from vegetarian diets. *American Journal of Clinical Nutrition*, 78(3), S633–S639. <https://doi.org/10.1093/ajcn/78.3.633s>.
- Institute of Medicine. (2000). Dietary Reference Intakes: thiamin, riboflavin, niacin, vitamin B6, folate, vitamin B12, pantothenic acid, biotin, and choline. National Academies Press.
- Institute of Medicine. (1998). Dietary Reference Intakes for Thiamin, Riboflavin, Niacin, Vitamin B6, Folate, Vitamin B12, Pantothenic Acid, Biotin, and Choline. National Academies Press.
- Jenkins, D. J., Wolever, T. M., Taylor, R. H., Barker, H., Fielden, H., Baldwin, J. M., Bowling, A. C., Newman, H. C., Jenkins, A. L., & Goff, D. V. (1981). Glycemic index of foods: a physiological basis for carbohydrate exchange. *American Journal of Clinical Nutrition*, 34(3), 362–366. <https://doi.org/10.1093/ajcn/34.3.362>.
- Jonnalagadda, S. S., Harnack, L., Liu, R. H., McKeown, N., Seal, C., Liu, S., & Fahey, G. C. (2011). Putting the whole grain puzzle together: health benefits associated with whole grains--summary of American Society for Nutrition 2010 Satellite Symposium. *Journal of Nutrition*, 141(5), 1011S–22S. <https://doi.org/10.3945/jn.110.132944>.
- Khor, G. L., Voon, P. C., Tee, E. S., Hsu-Hage, B. H., & Wahlqvist, M. L. (2000). Cardiovascular risk factors among Malaysian urban vegetarians. *Journal of Community Nutrition*, 2(2), 110–118.
- Knudsen, K. E. B., Hartvigsen, M. L., Hedemann, M. S., & Hermansen, K. (2016). Mechanisms whereby whole grain cereals modulate the prevention of type 2 diabetes. In *Molecular Nutrition and Diabetes* (pp. 87–103). Elsevier.
- Kuznesof, S., Brownlee, I. A., Moore, C., Richardson, D. P., Jebb, S. A., & Seal, C. J. (2012). WHOLEheart study participant acceptance of wholegrain foods. *Appetite*, 59(1), 187–193. <https://doi.org/10.1016/j.appet.2012.04.014>.
- Lattimer, J. M., & Haub, M. D. (2010). Effects of dietary fiber and its components on metabolic health. *Nutrients*, 2(12), 1266–1289. <https://doi.org/10.3390/nu2121266>.
- Lönnerdal, B. (2000). Dietary factors influencing zinc absorption. *Journal of Nutrition*, 130(5S Suppl), 1378S–83S. <https://doi.org/10.1093/jn/130.5.1378S>.
- Maki, K. C., Palacios, O. M., Koecher, K., Sawicki, C. M., Livingston, K. A., Bell, M., Nelson Cortes, H., & McKeown, N. M. (2019). The relationship between whole grain intake and body weight: Results of meta-analyses of observational studies and randomized controlled trials. *Nutrients*, 11(6), 1245. <https://doi.org/10.3390/nu11061245>.

- Malin, S. K., Kullman, E. L., Scelsi, A. R., Godin, J.-P., Ross, A. B., & Kirwan, J. P. (2019). A whole-grain diet increases glucose-stimulated insulin secretion independent of gut hormones in adults at risk for type 2 diabetes. *Molecular Nutrition & Food Research*, 63(7), e1800967. <https://doi.org/10.1002/mnfr.201800967>.
- Mariotti, F., & Gardner, C. D. (2019). Dietary protein and amino acids in vegetarian diets-A review. *Nutrients*, 11(11), 2661. <https://doi.org/10.3390/nu1112661>.
- Marshall, S., Petocz, P., Duve, E., Abbott, K., Cassettari, T., Blumfield, M., & Fayet-Moore, F. (2020). The effect of replacing refined grains with whole grains on cardiovascular risk factors: A systematic review and meta-analysis of randomized controlled trials with GRADE clinical recommendation. *Journal of the Academy of Nutrition and Dietetics*, 120(11), 1859-1883.e31. <https://doi.org/10.1016/j.jand.2020.06.021>.
- Mateo Anson, N., Aura, A.-M., Selinheimo, E., Mattila, I., Poutanen, K., van den Berg, R., Havenaar, R., Bast, A., & Haenen, G. R. M. M. (2011). Bioprocessing of wheat bran in whole wheat bread increases the bioavailability of phenolic acids in men and exerts antiinflammatory effects ex vivo. *The Journal of Nutrition*, 141(1), 137-143. <https://doi.org/10.3945/jn.110.127720>.
- McKevith, B. (2004). Nutritional aspects of cereals. *Nutrition Bulletin*, 29(2), 111-142. <https://doi.org/10.1111/j.1467-3010.2004.00418.x>.
- Melina, V., Craig, W., & Levin, S. (2016). Position of the Academy of Nutrition and Dietetics: Vegetarian diets. *Journal of the Academy of Nutrition and Dietetics*, 116(12), 1970-1980. <https://doi.org/10.1016/j.jand.2016.09.025>.
- Mellen, P. B., Walsh, T. F., & Herrington, D. M. (2008). Whole grain intake and cardiovascular disease: a meta-analysis. *Nutrition, Metabolism, and Cardiovascular Diseases: NMCD*, 18(4), 283-290. <https://doi.org/10.1016/j.numecd.2006.12.008>.
- Messina, V., Melina, V., & Mangels, A. R. (2003). A new food guide for North American vegetarians. *Journal of the American Dietetic Association*, 103(6), 771-775. <https://doi.org/10.1053/jada.2003.50141>.
- Miller, J. B., Pang, E., & Bramall, L. (1992). Rice: a high or low glycemic index food? *American Journal of Clinical Nutrition*, 56(6), 1034-1036. <https://doi.org/10.1093/ajcn/56.6.1034>.
- MOH (1985). Food Regulations 1985. Ministry of Health Malaysia.
- MOH (2020). Food Regulation 1985. Food (Amendment) (No. 4) Regulations 2020, Regulation 18. Ministry of Health Malaysia.
- Mohamad Hasnan, A., Ahmad Ali, Z., Foo, L. H., Khoo, Y. Y., Anida, Y. S. A. H. Z., & Nur Hidayah, J. (2014). Food Consumption Patterns. In Institute for Public Health (Ed.), *National Health and Morbidity Survey 2014: Malaysian Adult Nutrition Survey (MANS) (Vol. II)*. Ministry of Health Malaysia.
- NHMRC. (2013). Australian dietary guidelines. National Health and Medical Research Council.
- NCCFN. (2021). Malaysian Dietary Guidelines 2020. Technical Working Group on Nutritional Guidelines for National Coordinating Committee on Food and Nutrition.
- Neveu, V., Perez-Jiménez, J., Vos, F., Crespy, V., du Chaffaut, L., Mennen, L., Knox, C., Eisner, R., Cruz, J., Wishart, D., & Scalbert, A. (2010). Phenol-Explorer: an online comprehensive database on polyphenol contents in foods. Database: *Journal of Biological Databases and Curation*, 2010(0), bap024. <https://doi.org/10.1093/database/bap024>.
- Niklewicz, Ali, Smith, A. David, Smith, Alison, Holzer, Andre, Klein, Andrew, McCaddon, Andrew, . . . on behalf of Clu, B. (2023). The importance of vitamin B12 for individuals choosing plant-based diets. *European Journal of Nutrition*, 62(3), 1551-1559. <https://doi.org/10.1007/s00394-022-03025-4>.

- Norimah, A. K., Safiah, M., Jamal, K., Haslinda, S., Zuhaida, H., Rohida, S., Fatimah, S., Norazlin, S., Poh, B. K., Kandiah, M., Zalilah, M. S., Wan Manan, W. M., Fatimah, S., & Azmi, M. Y. (2008). Food consumption patterns: Findings from the Malaysian adult nutrition survey (MANS). *Malaysian Journal of Nutrition*, 14(1), 25–39.
- Norimah, A. K., Koo, H. C., Hamid Jan, J. M., Mohd Nasir, M. T., Tan S. Y., Appukutty, M., Nurliyana, A. R., Thielecke, F., Hopkins, S., Ong, M. K., Ning, C., & Tee, E. S. (2015). Whole grain intakes in the diets of Malaysian children and adolescents - Findings from the MyBreakfast Study. *PLoS One*, 10(10), e0138247. <https://doi.org/10.1371/journal.pone.0142763>.
- Okarter, N., & Liu, R. H. (2010). Health benefits of whole grain phytochemicals. *Critical Reviews in Food Science and Nutrition*, 50(3), 193–208. <https://doi.org/10.1080/10408390802248734>.
- Pagano, A. E. (2007). The gluten-free vegetarian. *Practical Gastroenterology*, 31(5), 94–106.
- Rizzo, N. S., Jaceldo-Siegl, K., Sabate, J., & Fraser, G. E. (2013). Nutrient profiles of vegetarian and nonvegetarian dietary patterns. *Journal of the Academy of Nutrition and Dietetics*, 113(12), 1610–1619. <https://doi.org/10.1016/j.jand.2013.06.349>.
- Ross, A. B., van der Kamp, J.-W., King, R., Lê, K.-A., Mejbourn, H., Seal, C. J., Thielecke, F., & Healthgrain Forum. (2017). Perspective: A definition for whole-grain food products-recommendations from the Healthgrain Forum. *Advances in Nutrition*, 8(4), 525–531. <https://doi.org/10.3945/an.116.014001>.
- Sadeghi, O., Sadeghian, M., Rahmani, S., Maleki, V., Larijani, B., & Esmailzadeh, A. (2020). Whole-grain consumption does not affect obesity measures: An updated systematic review and meta-analysis of randomized clinical trials. *Advances in Nutrition*, 11(2), 280–292. <https://doi.org/10.1093/advances/nmz076>.
- Sanders, L. M., Zhu, Y., Wilcox, M. L., Koecher, K., & Maki, K. C. (2021). Effects of whole grain intake, compared with refined grain, on appetite and energy intake: A systematic review and meta-analysis. *Advances in Nutrition*, 12(4), 1177–1195. <https://doi.org/10.1093/advances/nmaa178>.
- Seal, C. J., Jones, A. R., & Whitney, A. D. (2006). Whole grains uncovered. *Nutrition Bulletin*, 31(2), 129–137. <https://doi.org/10.1111/j.1467-3010.2006.00559.x>.
- Seal, Chris J., Courtin, C. M., Venema, K., & de Vries, J. (2021). Health benefits of whole grain: effects on dietary carbohydrate quality, the gut microbiome, and consequences of processing. *Comprehensive Reviews in Food Science and Food Safety*, 20(3), 2742–2768. <https://doi.org/10.1111/1541-4337.12728>.
- Serna-Saldivar, S. (2016). Role of cereals in human nutrition and health. In *Cereals Grains Properties, Processing, Nutritional Attributes* (pp 565–628). Routledge.
- Sharma, S., Sheehy, T., & Kolonel, L. N. (2013). Ethnic differences in grains consumption and their contribution to intake of B-vitamins: results of the Multiethnic Cohort Study. *Nutrition Journal*, 12(1), 65. <https://doi.org/10.1186/1475-2891-12-65>.
- Shridhar, K., Dhillon, P. K., Bowen, L., Kinra, S., Bharathi, A. V., Prabhakaran, D., Reddy, K. S., & Ebrahim, S. (2014). Nutritional profile of Indian vegetarian diets--the Indian Migration Study (IMS). *Nutrition Journal*, 13, 55. <https://doi.org/10.1186/1475-2891-13-55>.
- Siddhuraju, P., & Becker, K. (2001). Effect of various domestic processing methods on antinutrients and in vitro protein and starch digestibility of two indigenous varieties of Indian tribal pulse, *Mucuna pruriens* Var. utilis. *Journal of Agricultural and Food Chemistry*, 49(6), 3058–3067. <https://doi.org/10.1021/jf001453q>.
- Slavin, J. (2004). Whole grains and human health. *Nutrition Research Reviews*, 17(1), 99–110. <https://doi.org/10.1079/NRR200374>.

- Slavin, J. L., Jacobs, D., & Marquart, L. (2000). Grain processing and nutrition. *Critical Reviews in Food Science and Nutrition*, 40(4), 309-326.
- Slavin, J. L., Martini, M. C., Jacobs, D. R., Jr, & Marquart, L. (1999). Plausible mechanisms for the protectiveness of whole grains. *The American Journal of Clinical Nutrition*, 70(3), S459-S463. <https://doi.org/10.1093/ajcn/70.3.459s>.
- Steiger, G., Müller-Fischer, N., Cori, H., & Conde-Petit, B. (2014). Fortification of rice: technologies and nutrients: Rice fortification in public health. *Annals of the New York Academy of Sciences*, 1324(1), 29-39. <https://doi.org/10.1111/nyas.12418>.
- Stoskopf, N. C. (1985). Cereal grain crops. Reston Publishing.
- Subramanian, S., Patil, S. S., Ponnusamy, S., Hasamnis, A. A., Loh, K. Y., & Santosh, N. (2019). Whole-grain consumption and its determinants in Malaysian medical students: A cross-sectional study. *Indian Journal of Public Health*, 63(3), 220-226. https://doi.org/10.4103/ijph.IJPH_278_18.
- Tee, E. S., Ismail, M. N., Mohd Nasir, A., & Khatijah, I. (1997). *Nutrient Composition of Malaysian Foods* (4th ed.). Institute for Medical Research.
- Tolessa, E. S. (2018). Importance, nutrient content and factors affecting nutrient content of potato. *American Journal of Food, Nutrition and Health*, 3(3), 37-41.
- U.S. Department of Agriculture, Agricultural Research Service, & USDA. (2006). *USDA National Nutrient Database for Standard Reference (Vol. 19)*. United States Department of Agriculture.
- van der Kamp, J. W., Poutanen, K., Seal, C. J., & Richardson, D. P. (2014). The HEALTHGRAIN definition of 'whole grain.' *Food & Nutrition Research*, 58(1), 22100. <https://doi.org/10.3402/fnr.v58.22100>.
- Vici, G., Belli, L., Biondi, M., & Polzonetti, V. (2016). Gluten free diet and nutrient deficiencies: A review. *Clinical Nutrition*, 35(6), 1236-1241. <https://doi.org/10.1016/j.clnu.2016.05.002>.
- Wang, W., Li, J., Chen, X., Yu, M., Pan, Q., & Guo, L. (2020). Whole grain food diet slightly reduces cardiovascular risks in obese/overweight adults: a systematic review and meta-analysis. *BMC Cardiovascular Disorders*, 20(1), 82. <https://doi.org/10.1186/s12872-020-01337-z>.
- Wolever, T. M. S. (2013). Is glycaemic index (GI) a valid measure of carbohydrate quality? *European Journal of Clinical Nutrition*, 67(5), 522-531. <https://doi.org/10.1038/ejcn.2013.27>.
- Young, V. R., & Pellett, P. L. (1994). Plant proteins in relation to human protein and amino acid nutrition. *American Journal of Clinical Nutrition*, 59(5 Suppl), 1203S-1212S. <https://doi.org/10.1093/ajcn/59.5.1203S>.
- Yu, S., & Tian, L. (2017). Breeding major cereal grains through the lens of nutrition sensitivity. *Molecular Plant*, 11(1), 23-30. <https://doi.org/10.1016/j.molp.2017.08.006>.
- Yusof, B. N. M., Abd Talib, R., Karim, N. A., Kamarudin, N. A., & Arshad, F. (2009). Glycaemic index of four commercially available breads in Malaysia. *International Journal of Food Sciences and Nutrition*, 60(6), 487-496. <https://doi.org/10.1080/09637480701804268>.
- Zeuschner, C. L., Hokin, B. D., Marsh, K. A., Saunders, A. V., Reid, M. A., & Ramsay, M. R. (2013). Vitamin B12 and vegetarian diets. *The Medical Journal of Australia*, 199(S4). <https://doi.org/10.5694/mja11.11509>.
- Zhang, B., Zhao, Q., Guo, W., Bao, W., & Wang, X. (2018). Association of whole grain intake with all-cause, cardiovascular, and cancer mortality: a systematic review and dose-response meta-analysis from prospective cohort studies. *European Journal of Clinical Nutrition*, 72(1), 57-65. <https://doi.org/10.1038/ejcn.2017.149>.
- Zhang, X.-F., Wang, X.-K., Tang, Y.-J., Guan, X.-X., Guo, Y., Fan, J.-M., & Cui, L.-L. (2020). Association of whole grains intake and the risk of digestive tract cancer: a systematic review and meta-analysis. *Nutrition Journal*, 19(1), 52. <https://doi.org/10.1186/s12937-020-00556-6>.



Key Message 4

Consume adequate
amounts of
legumes and nuts
and moderate
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Key Message 4

Consume adequate amounts of legumes and nuts and moderate amounts of eggs

*Prof. Dr. Cheah Whye Lian,
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4.1 Terminology

Eggs

Egg usually refers to chicken egg but can also include eggs from other birds, for instance duck and quail.

Legumes

Legumes are plants in the Leguminosae family. Legumes comprise pulses and lentils. The terms pulses and lentils are often used interchangeably. Most legumes have nitrogen fixing bacteria in their root nodules. Peanuts, also known as groundnuts and monkey nuts are legumes despite being called nuts.

Nuts and seeds

In the botanical definition, nuts are fruits that have a hard shell containing a seed within it, such as chestnuts and hazelnuts. Many dried seeds are also called nuts, such as almonds, cashew nuts, walnuts, pistachios, pecans and Brazil nuts. Examples of seeds which are commonly consumed are pine nuts, pumpkin seeds, sunflower seeds, flax seeds, chia seeds and sesame seeds. According to Regulation 252 of the Malaysian Food Regulations 1985, nuts are edible seeds, kernels and other similar parts of plants that are not classified as cereals, vegetables, fruits or spices. Nuts may have their outer integument. They may contain added salt, edible fat or oil and may be roasted.

Textured soy protein (TSP)

TSP is used as one of the meat alternatives. It is de-fatted soya flour product, a by-product of extracting soybean oil. There are three common types with

different protein content which are soya flour (50%), soya concentrates (70%) and soya isolates (90%). Soya isolate has the highest protein content and has meat-like texture (Asgar et al., 2010).

4.2 Introduction

Protein is needed for growth and maintenance of body tissues. It is also necessary for enzymes, hormones, antibodies and for milk production in women who are breastfeeding. Amino acids are building blocks of proteins. Proteins consist of combination of 20 unique amino acids. There are nine essential amino acids that have to be obtained through dietary sources, as these nutrients cannot be synthesized in our body. The remaining eleven amino acids that can be synthesized in our body are nutritionally non-essential (Table 4.1). In a non-vegetarian diet, proteins are obtained from animal sources and plant sources. Animal sources such as eggs, meat and fish provide all essential amino acids required by the body in the correct ratio. Lysine, methionine, cystine, threonine, tryptophan and leucine are essential amino acids because the human body requires them but is unable to synthesize them. In healthy adult humans, the estimated requirement of amino acids is lysine 50 mg/g protein, methionine and cysteine 25 mg/g protein, threonine 25 mg/g protein, tryptophan 10 mg/g protein and leucine 65 mg/g protein (Young, 1994). Methionine and lysine are needed for the endogenous synthesis of carnitine that plays a crucial role in energy production (Krajcovicova-Kudlackova et al., 2000).

Concerns regarding the ability of vegetarian diets to meet protein needs are primarily based on the quality of protein consumed, which is determined by amino acid content and digestibility (Mangels, Messina, & Messina, 2011; Mariotti & Gardner,

2019). Protein of animal origins are considered as high quality because they provide all the essential amino acids adequately, while plant or vegetable proteins are not of the same quality because of low content of some essential amino acids. Nevertheless, a combination of cereals, millets and legumes provides most of the amino acids that complement each other to provide adequate essential amino acids. A healthy vegetarian diet can meet all the body nutrient needs at any stage of life including pregnancy, lactation and for older adults.

For vegans who do not consume eggs and dairy products, they can obtain sources of protein from pulses (red, brown and green lentils), grains (brown/parboiled rice, oatmeal), nuts (hazelnut and chestnuts), seeds (sesame and sunflower seeds), beans (kidney and black beans), soya and soya products (taufu, tempe and fortified soy beverages) and meat alternatives (Bakaloudi et al., 2021). Legumes and nuts are calorie-rich foods that are useful for increasing the energy density and quality of food for vegetarians. However, there are concerns that some meat alternatives or substitutes are ultra-processed (Alcorta et al., 2021). There are also efforts to introduce other plant-based protein sources such as chickpea in combination with TSP to produce imitation meat (Sharima-Abdullah et al., 2018). There might also be ethical issues as demonstrated in a survey of 200 vegetarian samples in Vietnam which were subjected to DNA analysis using multiplex polymerase chain reaction (PCR) assay; findings showed that 68 (34%) of samples contained animal

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Consume adequate amounts of legumes and nuts and moderate amounts of eggs

Table 4.1: Essential and non-essential amino acids

Essential		Non-essential	
Histidine	Threonine	Alanine	Glutamine
Isoleucine	Tryptophan	Arginine	Glycine
Leucine	Valine	Asparagine	Proline
Lysine		Aspartic acid	Serine
Methionine		Cysteine	Tyrosine
Phenylalanine		Glutamic acid	

(Source: IOM, 2005)

derived ingredients (Lao & Le, 2020). The technology to rapidly detect DNA markers is available and could be used in food authentication (Mi et al., 2015).

Protein quality can be measured using the score system, based on a new and improved scoring system for dietary protein quality – Digestible Indispensable Amino Acid Score (DIAAS). Soya has the highest DIAAS of 0.91 among the plant proteins (FAO, 2013), while DIAAS score for milk is 1.22, which is a suitable good quality protein food for lacto-vegetarians. The DIAAS method provides a more accurate measure of amino acids absorbed by the body which contributes to the protein requirement of human beings.

Eggs are excellent source of protein for ovo-vegetarians. They are also suitable for many types of cooking methods. One chicken egg contains about 6 g of protein. Approximately 12.6% of the weight of the chicken egg is protein and this can be found in both the yolk and albumen (Laymen & Rodriguez, 2009). It is also a rich source of leucine, which is lacking in wheat. Eggs are also good source of vitamin B12, riboflavin (vitamin B2), retinol (vitamin A), vitamin D, vitamin E, folate, iodine, selenium, phosphorus, zinc and haem iron (Miranda et al., 2015; Réhault-Godbert & Guyot, 2019). Approximately 9% of the weight of the chicken egg is fat, which is found almost entirely in the yolk. The fatty acid profile is 38% mono-unsaturated fatty acids, 16% polyunsaturated fatty acids and 28% saturated fatty acids. Eggs produced by chicken fed on flaxseed contain long chain omega-3 fatty acids

in the form of docosahexaenoic acid (DHA) (Burns et al., 2014). Cholesterol is almost entirely found in the yolk. An egg contains about 200 mg of cholesterol (Spence et al., 2010). Based on the recent evidence on the reduction of dietary cholesterol, USDA (2015) and Recommended Nutrient Intake (RNI) for Malaysia (National Coordinating Committee on Food and Nutrition, 2017) have removed the restriction of dietary cholesterol intake for a healthy population. However, at present understanding on dietary cholesterol intake and its effect on plasma lipid concentrations, with primary interest in cholesterol concentrations, it is suggested that restrictions in dietary cholesterol intake may be the best care for those with high cholesterol; however inter-individual variability remains unclear (Griffin & Lichtenstein, 2013; Spence & Jenkins, 2018). Hypercholesterolaemic individuals should limit their intake of cholesterol to a maximum of 200 mg/day (Krauss et al., 2001; Kuang et al., 2018).

Legumes, nuts and eggs contribute mainly proteins towards a vegetarian diet. In addition, eggs also contribute to folate and B vitamins in a diet that has very low bio-available folate and B vitamins. Vegetarians, particularly vegans might have inadequate B vitamins from their food intake. Supplementation helps to improve folate and vitamin B12 nutrition status. However, proper dietary advice from nutritionists, dietitians or medical doctors is required before consuming any supplements.

4.3 Scientific Basis

Vegetarians substitute “meat, poultry or fish” with various types of eggs, legumes and nuts. The key nutritional issue for vegetarians is whether substitutions with foods that are appropriate and acceptable to vegetarians, can provide the required nutrients (Philips, 2005). Some nutrients, particularly protein may be difficult to be directly obtained from a vegan diet (Clarys et al., 2014) or vegetarian diets with food allergies. While, there is no published national food allergy prevalence data among Malaysian population, including vegetarians, previous local studies reported that the most common food allergens among atopic children were seafood such as crab, prawn and squid (Zahedi et al., 2011), eggs and cow’s milk (Yadav & Naidu, 2015). It should be noted that childhood food allergies to eggs, milk, wheat and soya typically resolve during childhood, whereas allergies to nuts are persistent (Sicherer & Sampson, 2014). Since vegetarians with food allergy problem have limited choices of food,

consuming allergy-free and nutritious foods can be challenging. Hence, a well-planned vegetarian diet is necessary to ensure the nutritional needs for all life stages are met while optimizing the health benefits (Craig et al., 2021).

One of the major concerns about vegan diet is that plant-based protein sources contain essential amino acids in inadequate amounts for the human body. Plant proteins contain low amount of sulphur compounds, namely methionine and cysteine. For example, soya and other legumes have inadequate methionine for human requirements. Wheat contains inadequate lysine. Intake of a variety of legumes, lentils, nuts and seeds every day is important to provide vegetarians with all the essential amino acids. The human body is able to maintain the pool of amino acids to complement dietary proteins; there is no need to combine different plant proteins at every meal, as long as a

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variety of foods are consumed and daily energy intake is met (Institute of Medicine, 2005). Consumption of a variety of protein rich plant sources also contributes non-haem iron, zinc and calcium to the body. Individuals who do not include milk and eggs in their diet, such as vegans, should consume vitamin B12 fortified foods and use good manufacturing practice (GMP) certified vitamin B12 supplements (BDA, nd; Langan & Goodbred, 2017). A vegetarian diet can easily meet the human dietary protein requirement as long as the energy intake is met (Craig & Mangels, 2009). This is because nearly all food items consumed by vegetarians including vegans contain some, and often much, protein (vegetables, beans, grains, nuts and seeds).

Studies have shown significant associations between consumption of plant proteins and lower risks of overweight, obesity and chronic diseases (Preis et al., 2010; Halkjaer et al., 2011; Kuil et al., 2012; Turner-McGrievy et al., 2017). Most plant protein sources are lower in saturated fat, do not contain cholesterol and high in fibre. They are also good sources of antioxidants as compared to animal protein sources. However, protein deficiency is an area of nutrition concern. Studies have found that protein intake was lower in vegetarians compared with non-vegetarians (Clarys et al., 2013; Wong et al., 2013; Clarys et al., 2014; Gan et al., 2018). About two in five of Malaysian vegetarians did not meet the protein requirement (Ching et al., 2023). Inadequate intake of plant-based proteins among vegetarians was associated with poor bone health and muscle wasting and consequently weight loss. Risk fracture was high among vegetarians who have protein deficiency (Falchetti et al., 2022). Vegetarians who excluded meats were found to be at the highest risk of wrist fracture (Thorpe et al., 2008). Vegans had higher risks of either total or some site-specific fractures, particularly hip fractures. Tong et al. (2020) analysed data collected from 1993 to 2001 and 2010 from the European Prospective Investigation into Cancer and Nutrition Oxford cohort and found that compared with meat eaters after adjustment for socio-economic status, lifestyle confounders and body mass index, the risks of hip fracture were higher in fish eaters (hazard ratio 1.26; 95% CI 1.02 - 1.54), vegetarians (1.25; 1.04 - 1.50), and vegans (2.31; 1.66 - 3.22), equivalent to rate differences of 2.9 (0.6 - 5.7), 2.9 (0.9-5.2), and 14.9 (7.9 - 24.5) more cases for every 1,000 people over 10 years, respectively.

Meat can be replaced by other protein sources in a well-planned vegetarian diet, to provide comparable protein quality to consumers. However, elimination of meat in the diet may affect the bioavailability of

trace minerals (Panebianco, 2007). While meat enhances non-haem iron absorption, egg and soya proteins (independent of the phytic acid in soya) inhibit iron absorption (Hallberg & Hulthen, 2000). The type of iron in plant foods (non-haem iron) is not as easily absorbed as that in animal foods (haem iron). Inclusion of vitamin C rich foods and decrease of phytates and tannins enhance the absorption of non-haem iron (Hallberg et al., 1989). In a crossover 8-week intervention study (n=21 women aged 20 - 42 years old whose serum ferritin concentrations were 6 - 149 µg/L), Hunt & Roughead (1999) reported that participants absorbed 70% lower non haem iron from a lacto-ovo-vegetarian diet than from a non-vegetarian diet. They also reported an associated decrease in faecal ferritin excretion during the lacto-ovo-vegetarian diet which suggested partial physiologic adaptation to increase the efficiency of iron absorption. Their study provided participants with 12.6 mg total iron (0 mg haem iron), 1656 mg phytic acid, 206 mg ascorbic acid, 970 mg calcium during the lacto-ovo-vegetarian diet, and 13.6 mg total iron (1.2 mg haem iron), 542 mg phytic acid, 170 mg ascorbic acid, 952 mg calcium during the non-vegetarian diet period.

Phytic acid that is usually found in legumes and nuts have been shown in vivo and in vitro studies to inhibit the bioavailability of iron, zinc and calcium (Zhang et al., 2020). These conditions may lead vegetarians to be more prone to iron deficiency. Iron deficiency over a prolonged period affects the formation and structure of red blood cells, and limits the ability of the blood to carry oxygen throughout the body. Zhang et al.'s (2020) review showed that various studies found the addition of salts such as NaCl can lead to improvements in bioavailability of iron and zinc. There are several preparation methods that can reduce the phytate content of legumes. Soluble phytates can be removed by soaking dried beans and discarding the soaking water before cooking (Gibson, Perlas, Hotz, 2006). Another method is to partially substitute legumes with sprouted legumes where possible. Germination of legumes increases activity of endogenous phytase enzymes, thus reducing the phytate content of the legumes. (Gibson, Heath, Szymlek-Gay, 2014). However, there are very few in vivo isotope studies that measure the increase in absorbable iron and zinc that might be achieved through these strategies (Gibson, Raboy & King, 2018). In vivo stable isotope studies comparing food treated with newly developed dephytinization technology and native non-dephytinized products showed improved zinc absorption in dephytinized wheat and soy-based complementary foods (Egli, Davidsson, Zeder et al., 2004). Enzymatic

dephytinization of maize porridge using commercially available wheat phytase increased zinc absorption (Thacher, Aliu, Griffin et al., 2009).

Zinc is second only to iron among trace elements in the body. Zinc deficiency is related to poor bioavailability of zinc from vegetarian diets (Hunt, 2003). Dietary zinc intakes and serum zinc concentrations were lower in habitual vegetarian adults compared with non-vegetarian counterparts (Foster, Chu, Petocz, & Samman, 2013; Gorczyca et al., 2013). However, in older adults, children, and women during pregnancy and lactation, there is insufficient evidence to determine whether zinc intake and status are lower in vegetarians compared to omnivores (Foster & Samman, 2015). A meta-analysis reported that dietary zinc intake was lower in pregnant vegetarians when compared to non-vegetarians and both groups consumed lower than recommended amounts (Foster et al., 2015).

Plant foods such as legumes and nuts are important dietary sources of zinc, yet, are also high in phytic acid, an inhibitor of zinc bioavailability (Saunders, Craig & Baines, 2013). Zinc absorption is reduced when consuming plant foods that are high in phytates, while zinc absorption is enhanced when consuming animal protein. Legumes and nuts, although not as good as animal protein, can enhance the zinc absorption among vegetarians. The European Food Safety Authority assumes a bioavailability of 20% for zinc if the diet is predominantly vegetarian (EFSA NDA Panel, 2014). Zinc deficiency can cause poor growth and delayed sex maturation in children, hair loss, poor wound healing, loss of appetite, skin rashes and impaired immune function.

Vitamin B12 and folate which are abundant in animal food sources, are important for red blood cell formation. Vitamin B12 and folate intakes were lower among vegetarians, particularly among strict vegetarians, as compared to non-vegetarians. In addition, Pawlak et al., (2014) documented high prevalence of Vitamin B12 deficiency among vegetarians based on findings from 40 studies reviewed. Deficiency in vitamin B12 can be found among vegans, particularly vegans without

sufficient intake of vitamin B12 fortified plant foods. This deficiency is most severe and dangerous in breastfed infants of vegan mothers (Langan & Goodbred, 2017). Iron and B12 deficiencies may lead to different forms of anaemia. Chronic deficiency of vitamin B12 among strict vegetarians may predispose them to anaemia, especially during pregnancy (Guez et al., 2012). Iron and B12 deficiency anaemia in pregnant women can result in premature delivery, inborn errors of absorption and metabolism, low birth weight in infants and higher infant mortality (Guez et al., 2012; Saunders, Craig, Baines, & Posen, 2013). Anaemia in infancy can result in poorer cognition, school achievement and more behavioural problems in their middle childhood (Grantham-Mcgregor & Ani, 2001). Pregnant (including those planning to become pregnant) and breastfeeding women who are on a vegan diet should give special attention on dietary intake and get advice from a nutritionist, dietitian, or medical doctor.

A cross-sectional analysis involving 689 men (226 omnivores, 231 vegetarians and 232 vegans) from the European Prospective Investigation into Cancer and Nutrition Oxford cohort found that mean serum vitamin B12 was highest among omnivores (281, 95% CI: 270 pmol/l - 292 pmol/l), intermediate among vegetarians (182, 95% CI: 175 pmol/l - 189 pmol/l) and lowest among vegans (122, 95% iCI: 117 pmol/l - 127 pmol/l). They found that there was no significant association between age or duration of practising a vegetarian or vegan diet and serum vitamin B12 (Gilsing, Crowe, Lloyd-Wright et al., 2010). Gilsing et al., also reported that mean concentration of serum folate in vegans was 34% higher than in vegetarians and 88% higher than in omnivores. They also reported that vegetarians who reported taking a folate supplement had significantly higher mean serum folate concentration compared to non-supplement users ($p=0.015$). However, mean serum folate concentrations between supplement and non-supplement users did not differ for omnivores ($p=0.738$) and vegans ($p=0.072$). Very few participants (<1% of omnivores) were found to be folate deficient (6.3 nmol/l) (Gilsing et al., 2010).

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Consume adequate amounts of legumes and nuts and moderate amounts of eggs

4.4 Current Status

Protein foods should contribute between 10 to 20 percent to the total daily energy intake for the Malaysian adult population. For adult males, the protein recommendations are 62 g/day (18 years old - 29 years old), 61 g/day (30 years old - 59 years old) and 58 g/day (≥ 60 y). For adult females, the protein recommendations are 53 g/day (18 years old - 29 years old), 52 g/day (30 years old - 59 years old) and 50 g/day (≥ 60 years old) (NCCFN, 2017). The protein recommendation for vegetarians and non-vegetarians are similar. There is no separate recommendation for the intake of protein for vegetarians in Malaysia. The RNI 2017 concurred with IOM (2005) that available evidence does not support a separate protein recommendation for vegetarians who consume complementary mixtures of plant proteins. To date, Japan is the only country that has a recommendation for protein intake for lacto-ovo-vegetarians, which is 67 g/day compared to 50 g for males aged 50 years old - 69 years old with activity level I and 60 g for females aged 18 years old - 69 years old with activity level II (Nakamoto, Arashi, Noparatanawong et al., 2009). There is evidence from one study conducted in 38 omnivores and 22 vegetarian athletes in the USA that suggested vegetarian athletes would need an additional 10 g protein/day to reach the low end of recommended intake for protein (1.2 g/kg/d) for competitive athletes. An additional 22 g protein/day would be needed to achieve an intake of 1.4 g/kg/d, which is the upper end of the recommended intake range. This study used the DIAAS (Digestible Indispensable Amino Acid Score) method which reflects ileal digestibility of indispensable amino acids to compute the available protein in the diet (Ciuris, Lynch, Wharton et al., 2019). However, this increase in protein recommendation did not take into account protein contribution from fruits and vegetables, which are consumed in amounts that do provide substantial and clinically relevant quantities of digestible protein (Genoni, Craddock, Strutt, 2020).

Small local studies indicated that the contribution of protein to total daily energy intake was within the Malaysian RNI 2017 recommendation. Wong, Shalini, Liana et al., (2013) compared the nutrient intakes of lacto-ovo-vegetarians, vegans and non-vegetarians ($n = 105$, 35 in each group) in Kuala Lumpur. They reported that the percentage of contribution of protein intake to total daily energy was in the range of 13%-16%; $13.65 \pm 3.10\%$ for lacto-ovo vegetarians and $15.38 \pm 4.09\%$ for vegans; these were significantly lower than non-vegetarians who had $18.04 \pm 3.93\%$ ($p < 0.001$). In comparison, these ranges were higher than the findings from a

Taiwanese study (Women: 12%; Men: 11%) (Chiu et al., 2014), but was consistent with an Australian study (12%-16%) where vegans were reported to have a lower protein intake (12%) as compared to lacto-ovo-vegetarians (16%) (Wilson & Bali, 1999). Wong, Shalini, Liana et al., (2013) reported that protein intake was 62.74 ± 23.71 g for lacto-ovo-vegetarians, 76.67 ± 49.02 g for vegans, and 82.82 ± 29.18 g for non-vegetarians ($p < 0.001$). Mean protein intake of male lacto-ovo-vegetarians was lower than the RNI 2017 level for adult males. Cheah, Chang, Tang et al., (2018) compared the nutrient intakes of Hindu and Buddhist vegetarians in Kuching, Sarawak ($n = 114$). Their respondents were mostly lacto-ovo-vegetarians, with the majority being Buddhists (81.6%). Their findings were reported by faith groups rather than by the type of vegetarianism practised. Protein intake in Buddhist participants was 72.5 ± 31.58 g compared to Hindu participants, 102.9 ± 34.76 g ($p < 0.01$). Mean protein intakes in both groups were more than the RNI 2017 recommendation.

There are limited local studies on the consumption of types of protein foods among vegetarians. Khor et al. (2000) indicated that more than 60% of respondents consumed eggs, peanuts and soy products at least once a week ($n=136$, Chinese Malaysians in Kuala Lumpur). Most Buddhists in Malaysia are from the Chinese ethnic group and most Hindus are from the Indian ethnic group. Ethnic related food choices were evident in the reported frequency of intake of lentils and meat analogue. Consumption of lentils are more common among Indians, whilst consumption of meat analogue is more common among Chinese in Malaysia. Cheah, Chang, Tang et al., (2018) reported that only one Buddhist vegetarian respondent (1.1%) reported daily lentil intake and eight (8.6%) respondents reported 1 - 5 times / week, compared to six (28.6%) of Hindu respondents reported daily intake and 12 (57.1%) reported 1 - 5 times / week. For meat analogue, 20 (21.5%) Buddhists reported daily intake and 47 (50.5%) reported 1 - 5 times / week compared to 2 (9.5%) Hindus reported daily intake and 7 (33.3%) reported 1 - 5 times / week.

Wong, Shalini, Liana et al., (2013) reported that lacto-ovo-vegetarian, vegan and non-vegetarian respondents did not have significantly different iron and zinc intakes. Mean daily iron intake was 21.20 ± 9.61 mg for lacto-ovo-vegetarians, 21.43 ± 15.36 mg for vegans, and 19.97 ± 8.25 mg for non-vegetarians. Mean zinc intake was 7.09 ± 5.70 mg for lacto-ovo-vegetarians, 9.90 ± 13.85 mg for

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Consume adequate amounts of legumes and nuts and moderate amounts of eggs

vegans, and 6.51 ± 3.32 for non-vegetarians. Vegans had significantly higher intake of calcium compared to non-vegetarians ($p = 0.043$). Mean calcium intake was 806.77 ± 766.61 mg for lacto-ovo-vegetarians, 842.15 ± 868.70 mg for vegans, and 628.40 ± 505.51 mg for non-vegetarians. Iron and zinc intakes were within RNI 2017 recommendations for all three groups, except for zinc intake of lacto-ovo-vegetarians. All groups did not achieve the RNI 2017 recommendations for calcium. Wong et al, did not perform analysis on the bioavailability of minerals. Mean folate intake was significantly higher for both lacto-ovo-vegetarians (267.92 ± 117.80 µg) and vegans (271.79 ± 13.38 µg) compared to non-vegetarians (161.58 ± 10.39 µg) ($p < 0.001$). Mean vitamin B12 intake was significantly lower for both lacto-ovo-vegetarians (0.82 ± 0.65 µg) and vegans (0.52 ± 1.64 µg) compared to non-vegetarians (2.37 ± 1.42 µg) ($p < 0.001$). Mean vitamin C intake was significantly higher in both lacto-ovo-vegetarians (142.94 ± 78.60 mg) and vegans (111.52 ± 62.11 mg) compared to non-vegetarians (68.96 ± 40.81 mg) ($p < 0.001$). However, the variations in nutrient intake among these three groups were high. Twelve (34.3%) of lacto-ovo-vegetarian respondents reported consumption of supplements, compared to 3 (8.6%) for vegans and none for non-vegetarians. Mean folate and vitamin B12 intake levels for all groups did not meet the RNI 2107 recommendations. Mean vitamin C intake levels for all groups met the RNI 2017 recommendations.

Cheah, Chang, Tang et al, (2018) reported that mean daily iron intake was 23.73 ± 17.24 mg among Buddhist and 44 ± 21.39 mg among Hindu vegetarian respondents in Kuching, Sarawak. Mean calcium intake was 966.7 ± 927.69 mg for Buddhist respondents and 1697 ± 780.87 mg for Hindu respondents. Mean vitamin C intake was 209.1 ± 144.21 mg for Buddhists and 413.59 ± 501.85 mg for Hindus. All intakes were significantly different ($p < 0.01$), and the variations were high. Mean intakes for these nutrients met the RNI 2017 recommendations except for calcium intake in Buddhists.

Although total iron intake in a national sample of British children aged 4 years old to 18 years old were not significantly different between vegetarians and non-vegetarians, vegetarian adolescent girls were more likely to have poorer iron status than their meat-eating counterparts (Thane, Bates, & Prentice, 2003). Nonetheless, the UK National Diet and Nutrition Survey of Young People 1997, showed that a well-balanced vegetarian diet was not associated with lower than average total iron intake (Gregory et al, 2000). In the EPIC-Oxford cohort study, the vegan group had the highest intake of iron when compared to meat eaters and lacto-ovo vegetarians (Sobiecki et al, 2016). However, in Indonesian adolescent girls ($n=176$, mean age 15.2 years old) who were non-vegetarians but consumed a largely plant-based diet in an environment where access to fortified foods are uncommon, prevalence of iron deficiency anaemia was 22.2% and iron intake was 6.59 mg/day, which was lower than the recommended amount (Rahfiludin et al, 2021). Vegetarian female athletes in India ($n = 50$, 18 to 25 years) had significantly lower haemoglobin concentration than their non-vegetarian counterparts despite no significant difference in their iron intake (Kaur & Kaur, 2020). As for Malaysian adults, no significant difference was found in the mean iron intake between non-vegetarians, ovo-lacto-vegetarians, and strict vegetarians in Kuala Lumpur, Malaysia (Wong et al, 2013). These findings from a Western country (UK) with substantial fortification of food products, Southeast Asian countries (Malaysia, Indonesia) where fortified foods are available but might not be commonly consumed by vegetarians or people who consumed a predominantly plant-based diet or a country (India) where vegetarianism is a widely accepted practice showed that iron nutrition status among vegetarians are dependent on food habits, types of foods available and fortification of food items commonly consumed by vegetarians.

4.5 Key Recommendations

Key Recommendation 1

Consume adequate amounts of legumes daily.

How to achieve:

1. Consume a variety of legumes and legume products, for example chickpeas, dhal (lentils), soya beans, peanuts, tempeh, *tauhu* and *taugeh*.
2. Add legumes to rice, porridge, soups, salad, desserts, dishes and snacks.
3. Eat foods rich in Vitamin C during meals to increase absorption of non-haem iron from legumes, particularly for vegans.

Key Recommendation 2

Consume a variety of nuts and seeds daily.

How to achieve:

1. Add nuts and seeds into main dishes, e.g. cashew nuts, almonds and sesame seeds into rice and dishes like stir-fried vegetables.
2. Choose unsweetened and unsalted nuts and seeds as snacks.
3. Use nuts and seeds in your cooking and baking, e.g. almonds, pumpkin seeds and sunflower seeds.

Key Recommendation 3

Lacto-ovo-vegetarians and ovo-vegetarians are advised to consume eggs moderately.

How to achieve:

1. Consume eggs in moderate amount, up to an average of one a day, either whole or in dishes.
2. Poach or boil eggs – these are healthier methods to prepare eggs. Limit deep frying methods.

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Consume adequate amounts of legumes and nuts and moderate amounts of eggs

4.6 References

- Alcorta, A., Porta, A., Tárrega, A., Alvarez, M. D., & Vaquero, M. P. (2021). Foods for plant-based diets: challenges and innovations. *Foods*, 10(2), 293. <https://doi.org/10.3390/foods10020293>.
- Asgar, M. A., Fazilah, A., Hudu, N., & Karim, A. A. (2010) Nonmeat protein alternatives as meat extenders and meat analogs. *Comprehensive Reviews in Food Science and Food Safety*, 9, 513-529. <http://dx.doi.org/10.1111/j.1541-4337.2010.00124.x>.
- Bakaloudi, D. R., Halloran, A., Rippin, H. L., Oikonomidou, A. C., Dardavesis, T. I., Williams, J., Wickramasinghe, K., Breda, J., & Chourdakis, M. (2021). Intake and adequacy of the Vegan diet. A systematic review of the evidence. *Clinical Nutrition*, 40(5), 3503-3521. <https://doi.org/10.1016/j.clnu.2020.11.035>.
- Bda. (n.d.). *Vegetarian, vegan and plant-based diet*. British Dietetic Association (BDA). <https://www.bda.uk.com/resource/vegetarian-vegan-plant-based-diet.html>
- Burns-Whitmore, B., Haddad, E., Sabaté, J., & Rajaram, S. (2014). Effects of supplementing n-3 fatty acid enriched eggs and walnuts on cardiovascular disease risk markers in healthy free-living lacto-ovo-vegetarians: a randomized, crossover, free-living intervention study. *Nutrition Journal*, 13(1), 29. <https://doi.org/10.1186/1475-2891-13-29>.
- Cheah, W. L., Chang, K. H., Tang, M. F., Yuvaraj, L., Aldrin, R. D., & Llah, N. I. A. (2018). Vegetarian practices, body composition and dietary intake among Hindus and Buddhists in Kuching, Sarawak, Malaysia. *South East Asia Journal of Public Health*, 8(1), 3-9. <https://doi.org/10.3329/seajph.v8i1.42266>.
- Chiu, T. H. T., Huang, H., Chiu, Y., Pan, W., Kao, H., Chiu, J. P., Lin, M. N., & Lin, C. L. (2014). Taiwanese vegetarians and omnivores: Dietary composition, prevalence of diabetes and IFG. *PloS ONE*, 9(2), e88547. <https://doi.org/10.1371/journal.pone.0088547>.
- Ching, Y. K., Chin, Y. S., Appukutty, M., Chan, Y. M., Lim, P. Y., & Nasir, K. H. (2023). Interactions of genetic and macronutrient intake with abdominal obesity among middle-aged vegetarians in Malaysia. *Asia Pacific Journal of Clinical Nutrition*, 32(1), 168–182. [https://doi.org/10.6133/apjcn.202303_32\(1\).0020](https://doi.org/10.6133/apjcn.202303_32(1).0020).
- Clarys, P., Deliens, T., Huybrechts, I., Deriemaeker, P., Vanaelst, B., De Keyser, W., Hebbelinck, M., & Mullie, P. (2014). Comparison of nutritional quality of the vegan, vegetarian, semi-vegetarian, pesco-vegetarian and omnivorous diet. *Nutrients*, 6(3), 1318-1332. <https://doi.org/10.3390/nu6031318>.
- Clarys, P., Deriemaeker, P., Huybrechts, I., Hebbelinck, M., & Mullie, P. (2013). Dietary pattern analysis: A comparison between matched vegetarian and omnivorous subjects. *Nutrition Journal*, 12(1), 82-87. <https://doi.org/10.1186/1475-2891-12-82>.
- Craig, W. J., Mangels, A. R., Fresán, U., Marsh, K., Miles, F. L., Saunders, A. V., Haddad, E. H., Heskey, C. E., Johnston, P., Larson-Meyer, E., & Orlich, M. (2021). The safe and effective use of plant-based diets with guidelines for health professionals. *Nutrients*, 13(11), 4144. <https://doi.org/10.3390/nu13114144>.
- Craig, W. J., & Mangels, A. R. (2009). Position of the American Dietetic Association: Vegetarian diets. *Journal of The American Dietetic Association*, 109(7), 1266-1282. <https://doi.org/10.1016/j.jada.2009.05.027>.
- Diez-Espino, J., Basterra-Gortari, F. J., Salas-Salvadó, J., Buil-Cosiales, P., Corella, D., Schröder, H., Estruch, R., Ros, E., Gómez-Gracia, E., Arós, F., Fiol, M., Lapetra, J., Serra-Majem, L., Pintó, X., Babio, N., Quiles, L., Fitó, M., Martí, A., & Toledo E for the PREDIMED Investigators (2017). Egg consumption and cardiovascular disease according to diabetic status: The PREDIMED study. *Clinical Nutrition*, 36(4), 1015-1021. <https://doi.org/10.1016/j.clnu.2016.06.009>.
- Falchetti, A., Cavati, G., Valenti, R., Mingiano, C., Cosso, R., Gennari, L., Chiodini, I., & Merlotti, D. (2022). The effects of vegetarian diets on bone health: A literature review. *Frontiers in Endocrinology*, 13, 899375. <https://doi.org/10.3389/fendo.2022.899375>.

KM4

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- EFSA NDA Panel (EFSA Panel on Dietetic Products, Nutrition and Allergies) (2014). Scientific opinion on dietary reference values for zinc. *EFSA Journal* 12(10), 3844 <https://doi.org/10.2903/j.efsa.2014.3844>.
- Egli, I., Davidsson, L., Zeder, C., Walczyk, T., & Hurrell, R. (2004). Dephytinization of a complementary food based on wheat and soy increases zinc, but not copper, apparent absorption in adults. *Journal of Nutrition*, 134(5), 1077-1080. <https://doi.org/10.1093/jn/134.5.1077>.
- FAO (2011). *Dietary protein quality evaluation in human nutrition*. Report of an FAO Expert Consultation. Food and Nutrition Paper 92. FAO, Italy.
- Food Regulations 1985. P.U. (A) 437/85 incorporating latest amendment – P.U. (A) 200/2017.
- Foster, M., Chu, A., Petocz, P., & Samman, S. (2013). Effect of vegetarian diets on zinc status: A systematic review and meta-analysis of studies in humans. *Journal of the Science of Food and Agriculture*, 93(10), 2362-2371. <https://doi.org/10.1002/jsfa.6179>.
- Foster, M., & Samman, S. (2015). Vegetarian diets across the lifecycle: impact on zinc intake and status. *Advances in Food and Nutrition Research*, 74, 93-131. <https://doi.org/10.1015/bs.afnr.2014.11.003>.
- Foster, M., Herulah, U. N., Prasad, A., Petocz, P., & Samman, S. (2015). Zinc status of vegetarians during pregnancy: A systematic review of observational studies and meta-analysis of zinc intake. *Nutrients*, 7(6), 4512-4525. <https://doi.org/10.3390/nu7064512>.
- Gan, W. Y., Boo, S., Seik, M. Y., & Khoo, H. (2018). Comparing the nutritional status of vegetarians and non-vegetarians from a Buddhist organisation in Kuala Lumpur, Malaysia. *Malaysian Journal of Nutrition*, 24(1), 89-101.
- Genoni, A., Craddock, J. C., & Strutt, E. F. (2020). Limitations of the Digestibility Indispensable Amino Acid Score (DIAAS) and choice of statistical reporting. Comment on “A comparison of dietary protein digestibility, based on DIAAS scoring, in vegetarian and non-vegetarian athletes. *Nutrients*, 2019, 11, 3106”. *Nutrients*, 12(4), 1183. <https://doi.org/10.3390/nu12041183>.
- Gibson, R. S., Heath, A. L. M., & Szymlek-Gay, E. A. (2014). Is iron and zinc nutrition a concern for vegetarian infants and young children in industrialized countries? *American Journal of Clinical Nutrition*, 100(Suppl 1), 459S-468S. <https://doi.org/10.3945/ajcn.113.071241>.
- Gibson, R. S., Perlas, L. A., & Hotz, C. (2000). Improving the bioavailability of nutrients in plant foods at the household level. *Proceedings of the Nutrition Society*, 65(2), 160-168. <https://doi.org/10.1079/pns2006489>.
- Gibson, R. S., Raboy, V., & King, J. C. (2018). Implications of phytate in plant-based foods for iron and zinc bioavailability, setting dietary requirements, and formulating programs and policies. *Nutrition Reviews*, 76(11), 793-804. <https://doi.org/10.1093/nutrit/nuy028>.
- Gilting, A. M. J., Crowe, F. L., Lloyd-Wright, Z., Sanders, T. A. B., Appleby, P. N., Allen, N. E., & Key, T. J. (2010). Serum concentrations of vitamin B12 and folate in British male omnivores, vegetarians and vegans: results from a cross-sectional analysis of the EPIC-Oxford cohort study. *European Journal of Clinical Nutrition*, 64(9), 933-939. <https://doi.org/10.1038/ejcn.2010.142>.
- Gorczyca, D., Prescha, A., & Szeremeta, K. (2013). Impact of vegetarian diet on serum immunoglobulin levels in children. *Clinical Pediatrics*, 52(3), 241-246. <https://doi.org/10.1177/0009922812472250>.
- Grantham-Mcgregor, S., & Ani, C. (2001). A review of studies on the effect of iron deficiency on cognitive development in children. *Journal of Nutrition*, 131(2), 649S-668S. <https://doi.org/10.1093/jn/131.2.649S>.

- Gregory, J., Lowe, S., Bates, C.J., Prentice, A., Jackson, L. V., Smithers, G., Wenlock, R., & Farron, M. (2000). *National Diet and Nutrition Survey: Young people aged 4 to 18 years. Vol 1 Report of the Diet and Nutrition Survey*. London.
- Griffin, J. D., & Lichtenstein, A. H. (2013). Dietary cholesterol and plasma lipoprotein profiles: Randomized-controlled trials. *Current Nutrition Reports*, 2(4), 274–282. <https://doi.org/10.1007%2Fs13668-013-0064-0>.
- Guez, S., Chiarelli, G., Menni, F., Salera, S., Principi, N., & Esposito, S. (2012). Severe vitamin B12 deficiency in an exclusively breastfed 5-month-old Italian infant born to a mother receiving multivitamin supplementation during pregnancy. *BMC Pediatrics*, 12, 85. <https://doi.org/10.1186/1471-2431-12-85>.
- Halkjær, J., Olsen, A., Overvad, K., Jakobsen, M. U., Boeing, H., Buijsse, B., Palli, D., Tognon, G., Du, H., Forouhi, N. G., Wareham, N. J., Feskens, E. J., Sørensen, T. I., & Tjønneland, A. (2011). Intake of total, animal and plant protein and subsequent changes in weight or waist circumference in European men and women: The Diogenes project. *International Journal of Obesity*, 35, 1104–1113. <https://doi.org/10.1038/ijo.2010.254>.
- Hallberg, K., Brune, N., & Rossander, L. (1989). The role of vitamin C in iron absorption. *International Journal for Vitamin and Nutrition Research*, 30, 103–108.
- KM4** Hallberg, L., & Hulthen, L. (2000). Prediction of dietary iron absorption: an algorithm for calculating absorption and bioavailability of dietary iron. *American Journal of Clinical Nutrition*, 71(5), 1147–1160. <https://doi.org/10.1093/ajcn/71.5.1147>.
- Hunt, J. R. (2003). Bioavailability of iron, zinc and other trace minerals from vegetarian diets. *American Journal of Clinical Nutrition*, 78(3 Suppl), 633S–639S. <https://doi.org/10.1093/ajcn/78.3.633s>.
- Hunt, J.R., & Roughead, Z. K. (1999). Nonheme-iron absorption, fecal ferritin excretion, and blood indexes of iron status in women consuming controlled lactoovovegetarian diets for 8 wk. *American Journal of Clinical Nutrition*, 69(5), 944–952. <https://doi.org/10.1093/ajcn/69.5.944>.
- Institute of Medicine (2005). Dietary reference intakes for energy, carbohydrate, fiber, fat, fatty acids, cholesterol, protein and amino acids. From: http://www.nap.edu/download.php?record_id=10490.
- Kaur, A., & Kaur, H. (2020). Study of iron intake and haemoglobin concentration in vegetarian and non-vegetarian female athletes. *European Journal of Molecular & Clinical Medicine*, 7(7), 5960–5965.
- Khor, G. L., Voon, P. C., Tee, E. S., Hsu-Hage, B. H., & Wahlqvist, M. L. (2000). Cardiovascular risk factors among Malaysian urban vegetarians. *Journal of Community Nutrition and Health*, 2(2), 110–118.
- Krajcovicova-Kudlackova, M., Simoncic, R., Bederova, A., Babinska, K., & Beder, I. (2000). Correlation of carnitine levels to methionine and lysine intake. *Physiological Research*, 49, 399–402.
- Krauss, R. M., Eckel, R. H., Howard, B., Appel, L. J., Daniels, S. R., Deckelbaum, R. J., Erdman, J. W. Jr., Kris-Etherton, P., Goldberg, I. J., Kotchen, T. A., Lichtenstein, A. H., Mitch, W. E., Mullis, R., Robinson, K., Wylie-Rosett, J., St Jeor, S., Suttie, J., Tribble, D. L., & Bazzare, T. L. (2001). AHA dietary guidelines. Revision 2000: A statement for healthcare professionals from the Nutrition Committee of the American Heart Association, *Circulation*, 102(18), 2284–2299. <https://doi.org/10.1161/01.CIR.102.18.2284>.
- Kuang, H., Yang, F., Zhang, Y., Wang, T., & Chen G (2018). The impact of egg nutrient composition and its consumption on cholesterol homeostasis. *Cholesterol*, 6303810. <https://doi.org/10.1155/2018/6303810>.
- Kuil, W. a. D., Engberink, M. F., Vedder, M. M., Boer, J. M., Verschuren, W. M. M., & Geleijnse, J. M. (2012). Sources of dietary protein in relation to blood pressure in a general Dutch population. *PLoS ONE*, 7(2), e30582. <https://doi.org/10.1371/journal.pone.0030582>.

- Langan, R. C., & Goodbred, A. J. (2017). Vitamin B12 Deficiency: Recognition and Management. *American Family Physician*, 96(6), 384–389.
- Layman, K. L., & Rodriguez, N. R. (2009). Egg protein as a source of power, strength, and energy. *Nutrition Today*, 44(1), 43–48. <http://dx.doi.org/10.1097/NT.0b013e3181959cb2>.
- Lao, T. D. L., & Le, T. A. H. (2020). Exploring the multiples PCR for detection of animal-derived ingredients in vegetarian foods. *Pharmacophore*, 10(3), 69–74.
- Lea, E., & Worslet, A. (2003). The factors associated with the belief that vegetarian diets provide health benefits. *Asia Pacific Journal of Clinical Nutrition*, 12 (3), 296–303.
- Lee, H. Y., Woo, J., Chen, Z. Y., Leung, S. F., & Peng, X. H. (2000). Serum fatty acid, lipid profile and dietary intake of Hong Kong Chinese omnivores and vegetarians. *European Journal of Clinical Nutrition*, 54(10), 768–773. <https://doi.org/10.1038/sj.ejcn.1601089>.
- Lee, K.W. (2011). *Tabiat makanan di kalangan vegetarian etnik Cina di Kota Kinabalu dan Ipoh. (In Malay). Food habits among ethnic Chinese vegetarians in Kota Kinabalu and Ipoh.* ([BSc Thesis, Universiti Malaysia Sabah].
- Li, Y., Zhou, C., Zhou, X., & Li, L. (2013). Egg consumption and risk of cardiovascular diseases and diabetes: a meta-analysis. *Atherosclerosis*, 229(2), 524–530. <https://doi.org/10.1016/j.atherosclerosis.2013.04.003>.
- Lousuebsakul-Matthews, V., Thorpe, D. L., Knutsen, R., Beeson, W.L., Fraser, G. E., & Knutsen, S. F. (2013). Legumes and meat analogues consumption are associated with hip fracture risk independently of meat intake among Caucasian men and women: The Adventist Health Study-2. *Public Health Nutrition*, 17(10), 2333–2343. <https://doi.org/10.1017/s1368980013002693>.
- Mi, X., Yang, J., Cao, L., Wei, X., Zhu, Y., Li, Q., Liu, X., He, X., Liao, Q., & Yan, Z. (2015). Potential DNA markers as a rapid tracing tool for animal adulterants in vegetarian food. *Food Research International*, 76(4), 926–931. <https://doi.org/10.1016/j.foodres.2015.04.007>.
- Mangels, R., Messina, V., & Messina, M. (2011). *The Dietitian's Guide to Vegetarian Diets: Issues and Applications*. Sudbury, MA: Jones & Bartlett Learning.
- Mariotti, F., & Gardener, C. D. (2019). Dietary protein and amino acids in vegetarian diet - a review. *Nutrients*, 11(11), 2661. <https://doi.org/10.3390/nu11112661>.
- Ministry of Health (2014). Vegetarian practice in Malaysia quick survey. (Unpublished data).
- Miranda, J. M., Anton, X., Redondo-Valbuena, C., Roca-Saavedra, P., Rodriguez, J. A., Lamas, A., Franco, C. M., & Cepeda, A. (2015). Egg and egg-derived foods: effects on human health and use as functional foods. *Nutrients*, 7(1), 706–29. <https://doi.org/10.3390/nu7010706>.
- Nakamoto, K., Arashi, M., Noparatanawong, S., Kamohara, S., Radak, T., Tuchida, M., Miyazaki, K., Watanabe, S., Kudo, H., Tanaka, A. (2009). A new Japanese vegetarian food guide. *Asia Pacific Journal of Public Health*, 21(2), 160–169. <https://doi.org/10.1177/1010539509331595>.
- National Coordinating Committee on Food and Nutrition (NCCFN) (2017). *Recommended Nutrient Intake for Malaysians*. Ministry of Health, Putrajaya, Malaysia.
- Panebianco, S. M. (2007). The merits and pitfalls of vegetarianism. *Explore*, 3(1), 55–58. <https://doi.org/10.1016/j.explore.2006.10.008>.
- Pawlak, R., Lester, S. E., & Babatunde, T. (2014). The prevalence of cobalamin deficiency among vegetarians assessed by serum vitamin B12: A review of literature. *European Journal of Clinical Nutrition*, 68(5), 541–548. <https://doi.org/10.1038/ejcn.2014.46>.

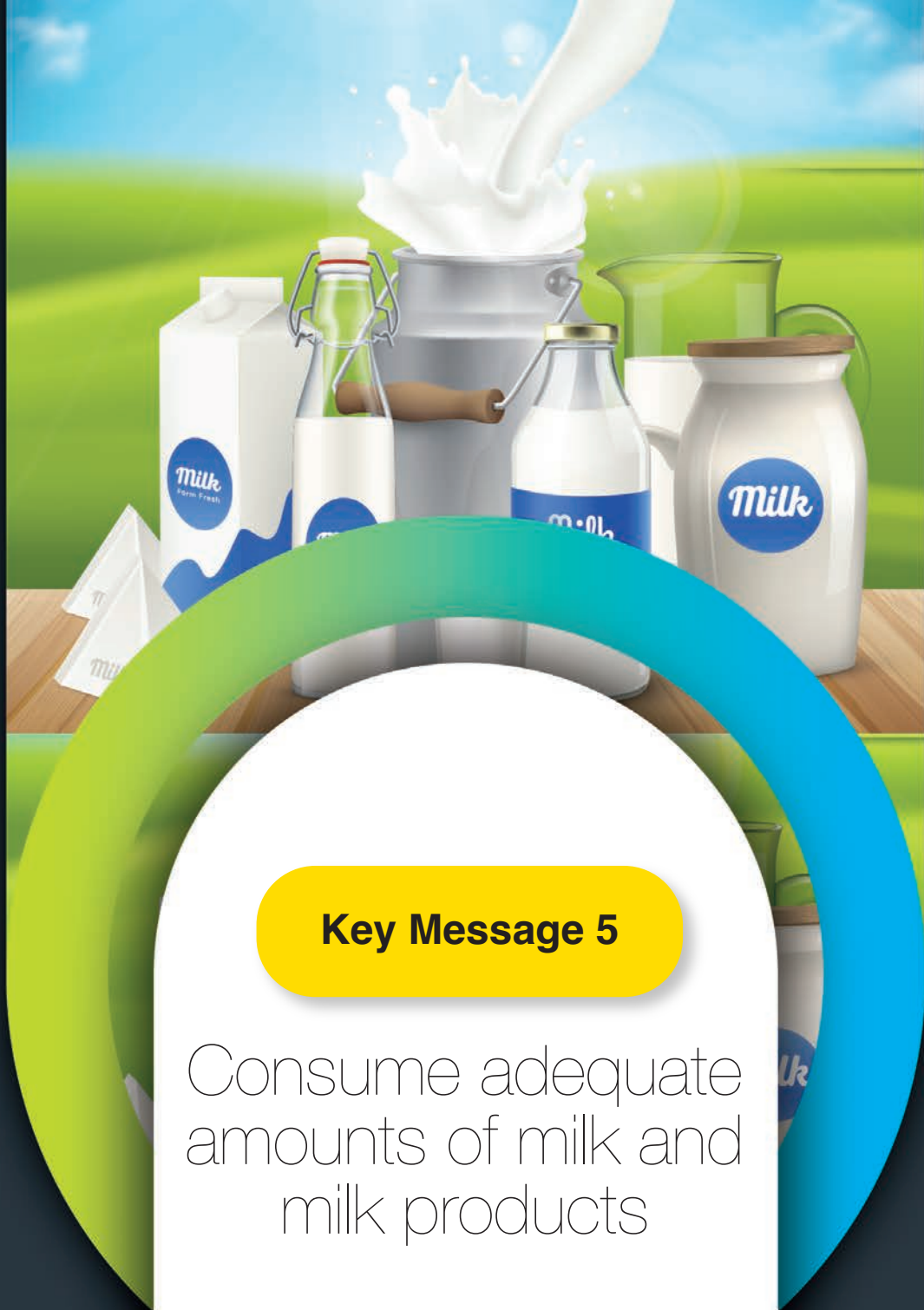
- Phillips, F. (2005). Vegetarian nutrition. *Nutrition Bulletin*, 30(2), 132-167. <https://doi.org/10.1111/j.1467-3010.2005.00467.x>.
- Preis, S. R., Stampfer, M. J., Spiegelman, D., Willett, W. C., & Rimm, E. B. (2010). Dietary protein and risk of ischemic heart disease in middle-aged men. *American Journal of Clinical Nutrition*, 92(5), 1265-1272. <https://doi.org/10.3945/ajcn.2010.29626>.
- Rahfludin, M. R., Aros, S. P., Joko, T., Asna, A. F., Murwani, R., & Hidayanti, L. (2021). Plant-based diet and iron deficiency anemia in Sundanese adolescent girls at Islamic boarding schools in Indonesia. *Journal of Nutrition and Metabolism*, 6469883. <https://doi.org/10.1155/2021/6469883>.
- Réhault-Godbert, S., Guyot, N., & Nys, Y. (2019). The Golden egg: nutritional value, bioactivities, and emerging benefits for human health. *Nutrients*, 11(3), 684. <https://doi.org/10.3390/nu11030684>.
- Richard, C., Cristall, L., Fleming, E., Lewis, E. D., Ricupero, M., Jacobs, R. L., & Field, C. J. (2017). Impact of egg consumption on cardiovascular risk factors in individuals with type 2 diabetes and at risk for developing diabetes: A systematic review of randomized nutritional intervention studies. *Canadian Journal of Diabetes*, 41 (4), 453–463. <https://doi.org/10.1016/j.jcjd.2016.12.002>.
- Ruxton, C. (2010). New evidence and recommendations for the use of eggs in the diet. *Nursing Standard*, 24(37), 47-55. <http://dx.doi.org/10.7748/ns2010.05.24.37.47.c7780>.
- KM4** Satija, A., & Hu, F. B. (2018). Plant-based diets and cardiovascular health. *Trends in Cardiovascular Medicine*, 28(7), 437-441. <https://doi.org/10.1016/j.tcm.2018.02.004>.
- Saunders, A. V., Craig, W. J., & Baines, S. K. (2013). Zinc and vegetarian diets. *The Medical Journal of Australia*, 199 (S4), S17-S21. <https://doi.org/10.5694/mja11.11493>.
- Saunders, A. V., Craig, W. J., Baines, S. K., & Posen, J. S. (2013). Iron and vegetarian diets. *The Medical Journal of Australia*, 199 (S4), S11-S16. <https://doi.org/10.5694/mja11.11494>.
- Sharima-Abdullah, N., Hassan, C. Z., Arifin, N., & Huda-Faujan, N. (2018). Physicochemical properties and consumer preference of imitation chicken nuggets produced from chickpea flour and textured vegetable protein. *International Food Research Journal*, 25(3), 1016-1025.
- Sicherer, S. H., & Sampson, H. A. (2014). Food allergy: Epidemiology, pathogenesis, diagnosis and treatment. *Journal of Allergy and Clinical Immunology*, 133(2), 291-307. <https://doi.org/10.1016/j.jaci.2013.11.020>.
- Sobiecki, J. G., Appleby, P. N., Bradbury, K. E., & Key, T. J. (2016). High compliance with dietary recommendations in a cohort of meat eaters, fish eaters, vegetarians, and vegans: results from the European Prospective Investigation into Cancer and Nutrition-Oxford study. *Nutrition Research*, 36(5), 464-477. <https://doi.org/10.1016/j.nutres.2015.12.016>.
- Spence, J. D., & Jenkins, D. (2018). Cardiovascular benefit of egg consumption is most likely. *Heart*, 104(21), 1805-1806. <http://dx.doi.org/10.1136/heartjnl-2018-313687>.
- Spence, J. D., Jenkins, D. J. A., & Davignon, J. (2010). Dietary cholesterol and egg yolks: not for patients at risk of vascular disease. *Canadian Journal of Cardiology*, 26(9), e336-e339. [https://doi.org/10.1016/s0828-282x\(10\)70456-6](https://doi.org/10.1016/s0828-282x(10)70456-6).
- Thacher, T. D., Aliu, O., Griffin, I. J., Pam, S. D., O'Brien, K. O., Imade, G. E., & Abrams, S. A. (2009). Meals and dephytinization affect calcium and zinc absorption in Nigerian children with rickets. *Journal of Nutrition*, 139(5), 926-932. <https://doi.org/10.3945/jn.108.101030>.
- Thane, C. W., Bates, C. J., & Prentice, A. M. (2003). Risk factors for low iron intake and poor iron status in a national sample of British young people aged 4–18 years. *Public Health Nutrition*, 6(5), 485–496. <https://doi.org/10.1079/phn2002455>.

- Thorpe, D. L., Knutsen, S. F., Beeson, W. L., Rajaram, S., & Fraser, G. E. (2008). Effects of meat consumption and vegetarian diet on risk of wrist fracture over 25 years in a cohort of peri- and postmenopausal women. *Public Health Nutrition*, 11(6), 564–572. <https://doi.org/10.1017/s1368980007000808>.
- Tong, T. Y. N., Appleby, P. N., Armstrong, M. E.G., Fensom, G. K., Knuppel, A., Papier, K., Perez-Cornago, A., Travis, R. C., & Key, T.J. (2020). Vegetarian and vegan diets and risks of total and site-specific fractures: results from the prospective EPIC-Oxford study. *BMC Medicine*, 18(1), 353. <https://doi.org/10.1186/s12916-020-01815-3>.
- Turner-McGrievy, G., Mandes, T., & Crimarco, A. (2017). A plant-based diet for overweight and obesity prevention and treatment. *Journal of Geriatric Cardiology*, 14(5), 369–374. <https://doi.org/10.11909/j.issn.1671-5411.2017.05.002>.
- USDA, US Department of Agriculture. (2005). *Report of the dietary guidelines advisory committee on the dietary guidelines for Americans*. Washington, DC: US Department of Agriculture and Department of Health and Human Services.
- Wilson, A. K., & Bali, M. J. (1999). Nutrient intake and iron status of Australian male vegetarians. *European Journal of Clinical Nutrition*, 53(3), 189–194. <https://doi.org/10.1038/sj.ejcn.1600696>.
- Wong, Y. S., Shalini, S., Farah Liana, M. R., Wan Nurul Amara, W. M. S., Mohd Naquiddin, M. Y., Norhazlina, A. W., & Zaleha, M. I. (2013). Nutrient intake pattern of vegetarians and non-vegetarians in Kuala Lumpur, Malaysia. *Pakistan Journal of Nutrition*, 12(4), 371–376. <https://doi.org/10.3923/pjn.2013.371.376>.
- Yadav, A., & Naidu, R. (2015). Clinical manifestation and sensitization of allergic children from Malaysia. *Asia Pacific Allergy*, 5(2), 78–83. <https://doi.org/10.5415%2Fapallergy.2015.5.2.78>.
- Young, V. R., & Pellett, P. L. (1994). Plant proteins in relation to human protein and amino acid nutrition. *American Journal of Clinical Nutrition*, 59(5 Suppl), 1203S–1212S. <https://doi.org/10.1093/ajcn/59.5.1203s>.
- Young, V. R. (1994). Adult amino acid requirements: The case for a major revision in current recommendations. *Journal of Nutrition*, 124(8 Suppl), 1517S–1523S. https://doi.org/10.1093/jn/124.suppl_8.1517s.
- Zahedi, F. D., Gendeh, B. S., & Husain, S. (2011). Sensitisation to common allergens in children with allergic rhinitis. *Brunei International Medical Journal*, 7(4), 200–206.
- Zhang, Y. Z., Stockmann, R., Ng, K., & Ajlouni, S. (2020). Revisiting phytate-element interactions: implications for iron, zinc and calcium bioavailability, with emphasis on legumes. *Critical Reviews in Food Science and Nutrition*. 62(6), 1696–1712. <https://doi.org/10.1080/10408398.2020.1846014>.

KM4

Consume adequate amounts of legumes and nuts and moderate amounts of eggs





Key Message 5

Consume adequate
amounts of milk and
milk products

Key Message 5

Consume adequate amounts of milk and milk products

*Ms. L. Mageswary Lapchmanan and
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KM5

Consume adequate amounts of milk and milk products

5.1 Terminology

Buttermilk

Commercially available cultured buttermilk is pasteurised and homogenised with 1% to 2% fat and inoculated with bacterial culture. Buttermilk is derived from the liquid that remains after churning butter from milk cream.

Condensed milk or sweetened condensed milk

Condensed milk or sweetened condensed milk is produced by adding sugar to evaporated milk. It must contain a minimum of 8% milk fat and 28% milk solids (Food Regulations, 1985). Due to its high sugar content, condensed milk has a longer shelf life without the need for refrigeration. It is frequently incorporated into beverages such as coffee, tea or

chocolate drinks. Additionally, sweetened condensed milk is utilised in the preparation of desserts like cakes, ice creams, and biscuits. It's important to note that vegetable oil-based sweetened creamer is a non-dairy product and does not contain milk solids.

Evaporated milk

Evaporated milk or unsweetened condensed milk is produced by heating the milk and evaporating it under reduced pressure. It usually contains not less than 8% of milk fat and 28% of total milk solids (Food Regulations, 1985). It is commonly used in producing desserts and in cooking.

Full cream milk powder or dried full cream milk

Full cream milk powder or dried full cream milk is milk or recombined milk from which the water has been removed. This type of milk powder typically contains a milk fat content exceeding 26%.

Filled milk

Filled milk is a product in which the milk fat has been replaced wholly or partially by an edible vegetable oil or fat, or a blend of both, such as palm oil.

Fermented milk or cultured milk

Fermented milk, also known as cultured milk, is a product made by culturing pasteurised milk, sterilised milk, recombined milk, pasteurised cream or reduced cream with lactic acid bacteria. Yoghurt, cultured cream and *lassi* are some examples of fermented or cultured milk products.

Flavoured milk

Flavored milk refers to a beverage derived from milk that incorporates approved flavoring substances like chocolate, vanilla and strawberry. Typically, it contains sugar, a minimum of 2% milk fat and 8% nonfat milk solids (Food Regulations, 1985). Flavored milk undergoes a heat treatment process, either through pasteurization or ultra-high temperature treatment.

Low fat milk

Low fat milk is milk which contains not more than 1.5 g of fat per 100 ml of milk.

Milk

Milk refers to a white liquid in raw or fresh form produced by the mammary glands of animals such as cows, buffalo, goats and sheep. Milk is a complex mixture of lipids, carbohydrates, proteins and many other organic, as well as inorganic compounds. Milk should consist of at least 3.25% milk fat and 8.50% of non-fat milk solids. It can be found in the form of fresh, pasteurised, sterilised and ultra-high temperature (UHT). Powdered milk on the other hand can be found in the form of full-cream, skimmed, malted and filled milk powder (Food Regulations, 1985). Powdered milk or dried milk is manufactured by spray drying and vacuum drying.

Milk product

Milk product is the term used to describe milk-based or milk-related products such as cheese, butter, ghee (clarified butter oil), yogurt, *kefir* and ice cream (Food Regulations, 1985). It is usually milk or a product thereof, whether alone or combined with another agricultural product, that contains no oil or fat other than that of milk (Food Regulations, 1985).

Pasteurised milk

Pasteurised milk is produced by heating for a short time to kill harmful microorganisms and cooling it based on the method of pasteurisation. The pasteurised milk should be stored at 8 degrees Celsius and below, to ensure it is safe for consumption (Food Regulations, 1985).

Plant-based or non-dairy milk alternatives

Plant-based milk or non-dairy milk alternatives are liquids derived from plant sources. They can be categorized into different types based on the plants they are derived from, such as cereals (e.g. oat, rice, corn), legumes (e.g. soy, peanut, cowpea), nuts (e.g. almond, coconut, hazelnut, pistachio, walnut), seeds (e.g. sesame, flax, sunflower, chia seed) and pseudo-cereals (e.g. quinoa) (Sethi et al., 2016). The nutritional composition of plant-based milk alternatives differs from that of dairy milk. However, plant-based milks fortified with protein, essential vitamins and minerals can serve as a replacement for traditional milk. As per the Food Regulation (1985), the term "milk," excluding animal-based products, can only be used for beverages derived from soybeans and coconuts.

Raw milk

Raw milk is milk that has not been pasteurised or homogenised.

Sterilised milk

Sterilised milk is milk which has been filtered, homogenised and heated to a temperature of not less than 100°C and maintained at the temperature for length to a time sufficient.

Skimmed/non-fat milk

Skimmed/ non-fat milk is milk from which milk fat has been removed but with most of the other essential nutrients intact. It is useful for those who want to limit their intake of energy, fat and cholesterol.

Ultra-high temperature (UHT) milk

Ultra-high temperature (UHT) milk is milk which has been subjected to heat treatment by being retained at a temperature of not less than 135°C for at least two seconds to render it commercially sterile and immediately aseptically packed in sterile containers.

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Consume adequate amounts of milk and milk products

5.2 Introduction

Milk is a highly nutritious food that is suitable for individuals of all ages, providing a wide array of essential nutrients necessary for growth, development and overall maintenance of the human body. It contains protein, fat, lactose, macronutrients, vitamins, minerals and energy, making it a valuable component of the human diet. A global review of food-based dietary guidelines (Comerford et al., 2021) showed that the most commonly recommended intake for milk and dairy products for Asia and Pacific regions is 1-2 servings/day, consistent with the latest recommendation by the Malaysian Dietary Guidelines 2020 (NCCFN, 2020). However according to the US Department of Health and Human Services (HHS) and the Department of Agriculture (USDA, 2005), consuming three cups of milk or dairy

products daily contributes more than 10% of the recommended nutrient intake for riboflavin, vitamin A, thiamin, phosphorus, magnesium, zinc, potassium, protein and carbohydrates.

Among vegetarians, only lacto-ovo-vegetarians and lacto-vegetarians include milk and dairy products in their meals. Consequently, vegans, who exclude milk and dairy products from their diet, require guidance to ensure a healthy and balanced diet. Studies have indicated that vegans may experience reduced bone density, increased risk of fractures and other health concerns, such as Vitamin D deficiency, compared to non-vegetarians or lacto-vegetarians (Wang et al., 2008; Weaver, 2009; Iguacel et al., 2019; Menzel et al., 2021).

5.3 Scientific Basis

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Consume adequate amounts of milk and milk products

Milk is a good source of protein and it is one of the main nutrients for growth, repair and synthesis of enzymes and hormones (Joint WHO/FAO/UNU Expert Consultation, 2007). Comparative studies between vegetarian and non-vegetarian subjects reveal significant lower protein intake in the vegetarian diet (Clarys et al., 2013; Clarys et al., 2014). Protein intake in vegetarians is reported to be significantly lower with a range of 64 – 88 g/day (Clarys et al., 2014; Mariotti & Gardner, 2019). However, despite lower protein intake, most vegetarians and vegans still manage to meet the recommended protein intake of around 13%-14% of energy (Clarys et al., 2013; Clarys et al., 2014).

Another area of concern is the low biological value of protein in plant-based proteins and the lack of essential amino acids. Some plant proteins have lower levels of leucine, methionine, lysine and tryptophan in comparison with animal proteins (Craig & Mangels, 2009). Therefore, it is important for vegetarians to consume a wide range of plant-based proteins to obtain all the essential amino acids. Additionally, incorporating milk and milk products into their daily diet is recommended to enhance protein quality. Other vegetarian dietary guidelines recommend adequate energy intake with a combination of different types of plant-based proteins to meet individual needs (Mariotti & Gardner, 2019).

Calcium is a major mineral in milk and dairy products and plays a significant role on bone homeostasis and development. Hence, it is

important that the intake of milk and dairy products be continued from infant to adolescence to support optimal bone mass accumulation. Moreover, adults are encouraged to incorporate milk and dairy products into their diet to reduce the risk of bone-related conditions such as osteoporosis (Ratajczak et al., 2021).

Adequate calcium intake can be achieved in both vegetarians and non-vegetarians through careful diet planning (Yen et al., 2008; Mangels, 2014). For instance, diets rich in fruits and vegetables have positively been associated with positive calcium balance, as assessed by numerous bone metabolism markers in men and women (Calvez et al., 2012; Brondani et al., 2019). This can be attributed to the high potassium and magnesium content in fruits and vegetables, which produce an alkaline ash that helps inhibit bone resorption and reduce calcium loss through urine (Higgs et al., 2017). Additionally, engaging in regular exercise is beneficial as it helps reduce calcium loss from bones, further supporting overall bone health.

Vegans who choose to exclude milk and dairy products from their diet may face difficulties in meeting the recommended calcium intake. This becomes particularly challenging for individuals experiencing rapid growth, such as children and adolescents. Research indicates that avoiding dairy products during young age could result in low peak bone mass attainment and consequently increases the risk of osteoporotic fractures later in life (Feskanich et al., 2014; Rozenberg et al., 2016).

To ensure adequate calcium intake, it is recommended for vegans to regularly incorporate calcium-fortified foods and non-dairy beverages made from sources such as soy, rice, oat, almond, and nuts into their diet. Additionally, calcium can be obtained from plant-based sources like beans and various green vegetables, including kale, *bok choy*, okra, turnip greens, fortified fruit juices, parsley, amaranth, *agathi keerai*, drumstick leaves, and broccoli (Weaver & Plawecki, 1994; Dietitians of Canada, 2014; Sethi et al., 2016), in *ragi* (millets), legumes (cow pea, black gram lentils, dhal, soya bean) and fruits such as dates, figs and raisins. In addition, condiments and spices such as asafetida (*inggu*), cardamom, coriander, cucumin (*kunyit*) and fenugreek seeds (*halba*) are also high in calcium contents. Supplementation may be necessary for certain populations such as children and adolescents due to high demand of calcium requirement if there is a need for additional calcium intake.

Based on the Recommended Nutrient Intake (RNI) for Malaysia (National Coordinating Committee on Food and Nutrition, 2017), a child aged between 1 and 3 years old requires 700 mg of calcium daily and 4 to 9 years old need 1,000 mg/day. During adolescence, the recommended intake increases to 1,300 mg/day. In general, both men and women, including pregnant and lactating women, require 1,000 mg of calcium daily. However, for women above 51 years of age, the recommended intake rises to 1,200 mg/day. Teenagers who are pregnant or lactating are advised to have a higher calcium intake of 1,300 mg/day. By consuming appropriate servings of milk and milk products suitable for their age group, individuals can ensure they meet the recommended calcium intake throughout their lifespan.

Vitamin D is an additional important nutrient for bone health because of its effects to increase the intestinal absorption of calcium. Vegan sources of vitamin D are exposure to sunshine, fortified fat spreads, breakfast cereals, soya milk and vitamin D supplements, which are highly recommended for elderly vegans (Weikert et al., 2020).

There is a significant amount of Vitamin B2 (riboflavin) present in milk. It plays an important role in maintaining normal vision, red blood cells, functioning of the immune system and healthy skin. A glass of fresh cows' milk (250 ml) can provide 0.54 mg of riboflavin that can fulfill 41.5% to 54% of the daily requirements of riboflavin for adolescents and adults.

Vitamin B12 is another important vitamin present in milk and milk products. The main role of vitamin B12 is to maintain healthy nerves and red blood cells, as well as normal energy production and cell division. Vitamin B12 deficiency is common in developing countries, particularly among vegans. Regular intake of milk and milk products improves vitamin B12 deficiency in vegetarians (Naik et al., 2013). Pregnant women predominantly practising vegetarian diet for a long term have an increased risk for vitamin B12 deficiency. Dietary intake of vitamin B12 was lowest in the lacto-ovo-vegetarians and a considerable proportion of women were far below the estimated average requirement for vitamin B12 intake during pregnancy (Corinna et al., 2004). Pawlak et al. (2013) have reported that higher rates of deficiency B12 were noted among vegans compared with vegetarians and among individuals who had adhered to a vegetarian diet since birth, compared with those who had adopted such a diet later in life.

Nevertheless, vegans can obtain vitamin B12 from fortified foods and beverages such as breakfast cereals, non-dairy milk alternatives (e.g.: rice, almond, soya) and fermented foods (Dietitians of Canada, 2014). Since the main sources of vitamin B12 in a lacto-ovo-vegetarian diet are dairy products and eggs, the dietary guidelines for women of childbearing age could focus on increased consumption of milk and dairy products to ensure an adequate supply of vitamin B12. Vitamin B12 supplement must also be recommended for pregnant women and lactating vegetarian mothers, especially vegans.

There were some health concerns brought against milk and dairy products such as the presence of lactose, saturated fat and increased risk of cancer, but the evidence was a lack of strength and mechanistic literature support (Lampe, 2011).

For infants and children with severe lactose intolerant, it is advisable to consume lactose-free milk or plant-based formulas. If milk intolerance persists, they may need to continue to avoid milk products and choose hypoallergenic formulas or soya-based formulas with advice from medical personnel.

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Consume adequate amounts of milk and milk products

5.4 Current Status

Studies and reports on nutritional quality in restrictive diets like vegetarianism are very limited, especially in Malaysia.

Research shows generally there is low milk consumption among Malaysians, in both adults and children. According to the findings from the Malaysian Adult Nutrition Survey (MANS) in 2014, Malaysian adults were only consumed about 1.4 servings of milk and/or dairy products in a day, either fresh or commercially-based milk products, which was at the lower range than the recommended intake of 1-3 servings per day (Institute for Public Health, 2014; Lee & Wan Muda, 2019). Malaysians tend to neglect the importance of milk as they grow older and opt for sugar-sweetened beverages such as tea, coffee and cordials. Almost, 59% of Malaysians drink sugar and sweetened condensed milk daily. Women were shown to be frequent drinkers of full cream milk compared to men who preferred sweetened condensed milk. Urban populations were drinking more full cream milk and the rural population preferred sweetened condensed milk (Norimah et al., 2008).

In a study among lacto-ovo-vegetarians, only 19.9% consumed milk daily, 18.3% drank milk 2 to 3 times a week, 16.2% once a week and 12.5% had milk once a month. On average 33.1% seldom include milk in their diet (Khor et al., 2000). Among vegetarians, although they consume milk and milk products frequently, only 22.9% of the lacto-ovo-vegetarians and 25.7% of strict vegetarians achieved RNI for calcium. Female vegetarians were noted to have inadequate calcium intake, which increases their risk for osteoporosis (Wong, 2013).

Malaysian school children preferred to drink flavoured milk as compared to plain milk, and other preferred beverages were mineral water, chocolate malt drink, fruit juices, yogurt, fruit flavoured cordial, tea and soft drinks (Babolian Hendijani & Ab Karim,

2010). Similarly, intake of milk and milk products among adolescents remains low (Abdul Majid et al., 2016). In both urban and rural children, low milk consumption was due to the existence of competing drinks, effect of packing, sensory properties, belief in health benefits, effect of family members and friends, availability at home and advertisements (Babolian Hendijani & Ab Karim, 2010).

The average daily intake of calcium in adult Malaysians is only 357 mg as compared to the recommended intake of 1,000-1,200 mg/day (NCCFN, 2017). As compared to other food groups, low consumption of milk and milk products contributed to the low calcium intakes (Lee & Wan Muda, 2019). Lacto-vegetarians compared with omnivores; do not seem to be at a greater risk of osteoporosis. Highest calcium consumption was found in semi-vegetarians and pesco-vegetarians, and lowest in vegans with intakes of 1,470, 1,470 and 738 mg/day calcium, respectively. However, omnivores had significantly lower calcium intake (1,199 mg/day) compared to the vegetarians and semi-vegetarians in Belgium (Clarys et al., 2013).

However, vegans are prone to calcium deficiency, especially during the rapid growth phase in children (Weaver & Plawecki, 1994) and overall bone health (Tucker, 2014). Vegans have reduced bone mineral density, increased incidence of fracture and other health risks compared with omnivores or lacto-vegetarians (Weaver, 2009; Tucker, 2014). Even though intake of minerals, calcium, iron and zinc is related to diet planning, the dietary intake of calcium was noted to be significantly higher among vegetarians as compared with non-vegetarians in a study in Kuala Lumpur (Wong, 2013). Careful meal planning, calcium-fortified foods or drinks are essential together with evaluation on the need for supplementation among vegans (Weaver & Plawecki, 1994).

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Consume adequate amounts of milk and milk products

5.5 Key Recommendations

Key Recommendation 1

Consume milk and milk products daily.

How to achieve:

1. Consume 2 to 3 servings of milk and/or milk products every day according to individual needs.
2. Consume milk and/or milk products at breakfast, in between main meals or before bedtime.
3. Add milk and/or milk products in the preparation of meals (e.g. breakfast cereals with milk, milkshakes, crackers with cheese, fruit *lassi*, add milk/yogurt in cooking, add *paneer* (cottage cheese) in fried rice, add cheese on salads and sandwiches).

Key Recommendation 2

Choose milk and milk products wisely.

How to achieve:

1. Drink plain milk instead of flavoured milk.
2. Replace sweetened creamers with plain milk and/or milk products.
3. Add plain milk and/or milk products in beverages and desserts (e.g. tea, coffee and pudding).
4. Replace full cream/whole milk and milk products with low fat milk and milk products if there is a need to reduce calorie intake (not suitable for infants, children and adolescents).

5.5.1 Additional recommendations for vegans

Meeting the recommended intake of calcium and vitamin B12 can be challenging for vegans who exclude milk and milk products from their diet.

However, there are alternative sources of calcium. Vegans can obtain calcium from various plant-based sources, including beans, green vegetables like kale, *bok choy*, okra, turnip greens, parsley, *agathi keerai*, drumstick leaves, and broccoli. Calcium-fortified foods like soy, rice, oat, almond milk, and nuts can also contribute to calcium intake. Other calcium-rich options include *ragi* (millets), legumes such as cowpea, black gram lentils, dhal, soybeans, and

fruits like dates, figs, and raisins. Condiments and seasonings like asafoetida, cardamom, coriander, cumin, and fenugreek seeds can also provide calcium.

As for vitamin B12, vegans can obtain it from fermented foods like tempeh, *miso*, *kimchi*, and *sauerkraut*. It is also recommended for vegans to consume vitamin B12 fortified foods and beverages such as breakfast cereals, almond milk and soya milk. In the case of vegan pregnant women and lactating mothers, a vitamin B12 supplement is encouraged to ensure adequate intake.

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5.6 References

- Abdul Majid, H., Ramli, L., Ying, S.P., Su, T.T., Jalaludin, M.Y., & Abdul Mohsein, N.A.S. (2016). Dietary Intake among Adolescents in a Middle-Income Country: An Outcome from the Malaysian Health and Adolescents Longitudinal Research Team Study (the MyHeARTs Study). *PLoS One*, 11(5), e0155447. <https://doi.org/10.1371/journal.pone.0155447>.
- Babolian, H.R., & AB Karim, M.S. (2010). Factors affecting milk consumption among school children in urban and rural areas of Selangor, Malaysia. *International Food Research Journal* 17: 651-660.
- Brondani, J.E., Comim, F.V., Flores, L.M., Martini, L.A., & Premaor, M.O. (2019). Fruit and vegetable intake and bones: A systematic review and meta-analysis. *PLoS One*, 31 14 (5), e0217223. doi: 10.1371/journal.pone.0217223.
- Calvez, J., Poupin, N., Chesneau, C., Lassale, C., & Tome'D. (2012). Protein intake, calcium balance and health consequences. *Eur J Clin Nutr* (Full text) 66, 281–295. <https://doi.org/10.1038/ejcn.2011.196>
- Clarys, P., Deliens, T., Huybrechts, I., Deriemaeker, P., Vanaelst, B., De Keyzer, W., Hebbelinck, M., & Mullie, P. (2014). Comparison of nutritional quality of the vegan, vegetarian, semi-vegetarian, pesco-vegetarian and omnivorous diet. *Nutrients*, 6, 1318-1332. doi: 10.3390/nu6031318.
- Clarys, P., Deriemaeker, P., Huybrechts, I., Hebbelinck, M., & Mullie, P. (2013). Dietary pattern analysis: A comparison between matched vegetarian and omnivorous subjects. *Nutrition Journal* 12: 18. doi: 10.1186/1475-2891-12-82.
- Comerford, K.B., Miller, G.D., Boileau, A.C., Masiello Schuette, S.N., Giddens, J.C., Brown, K.A. (2021). Global Review of Dairy Recommendations in Food-Based Dietary Guidelines. *Front Nutr.* 25;8:671999. doi: 10.3389/fnut.2021.671999.
- Corinna, K., Ingrid, H., Pieter, C.D., Ulrike, A.H., Sunitha, N.W., Indrika, D.R., Sindy, G., Jan, L., & Claus, L. (2004). Long-term ovo-lacto vegetarian diet impairs vitamin B-12 status in pregnant women. *Journal of Nutrition* 134: 3319 –3326. doi: 10.1093/jn/134.12.3319.
- Craig, W.J., & Mangels, A.R. (2009). Position of the American Dietetic Association: Vegetarian diets. *Journal of American Dietetic Association*, 109, 1266-1282. doi: 10.1016/j.jada.2009.05.027.
- Dietitians of Canada. (2014). *Healthy Eating Guidelines for Vegan*. <<https://www.dietitians.ca/Downloads/Factsheets/Guidelines-for-Vegans.aspx>>.
- Feskanich, D., Bischoff-Ferrari, H.A., Frazier, A.L., & Willett, W.C. (2014). Milk Consumption During Teenage Years and Risk of Hip Fractures in Older Adults. *JAMA Pediatr* (Full text)168(1), 54-60. doi: 10.1001/jamapediatrics.2013.3821.
- Food Regulations 1985. P.U. (A) 437.85. Incorporating latest amendment – P.U. (A) 200/2017.
- Higgs, J., Derbyshire, E., & Styles, K. (2017). Nutrition and osteoporosis prevention for the orthopaedic surgeon: A wholefoods approach. *EFORT Open Reviews* 2: 300-308. doi: 10.1302/2058-5241.2.160079. eCollection 2017 Ju.
- Iguacel, I., Miguel-Berges, M.L., Gómez-Bruton, A., A Moreno, L., & Julián, C. (2019). Veganism, vegetarianism, bone mineral density, and fracture risk: A systematic review and meta-analysis. *Nutr. Rev* 77:1–18. doi: 10.1093/nutrit/nuy045.
- Institut Kesihatan Umum (2014). *National Health and Morbidity Survey 2014 - Malaysian Adult Nutrition Survey (MANS)*. Ministry of Health Malaysia, Putrajaya.

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- Joint WHO/FAO/UNU Expert Consultation (2007). Protein and amino acid requirements in human nutrition. *World Health Organ Tech Rep Ser* 935: 1-265.
- Khor, G.L., Voon, P.C., Tee, E.S., Hsu-Hage, B.H., & Wahlqvist, M.L. (2000). Cardiovascular risk factors among Malaysian urban vegetarians. *Journal of Community Nutrition and Health* 2 (2): 110-118.
- Lampe, J.W. (2011). Dairy products and cancer. *J Am Coll Nutr* Oct;30(5 Suppl 1):464S-70S. doi: 10.1080/07315724.2011.10719991.
- Lee, Y.Y., & Wan Muda, W.A.M. (2019). Dietary intakes and obesity of Malaysian adults. *Nutrition Research and Practice*, 13(2), 159. <https://doi.org/10.4162/nrp.2019.13.2.159>.
- Mangels, A.R. (2014). Bone nutrients for vegetarians. *Am J Clin Nutr* Jul;100 Suppl 1:469S-75S. doi: 10.3945/ajcn.113.071423.
- Mariotti, F., & Gardner, C.D. (2019). Dietary Protein and Amino Acids in Vegetarian Diets-A Review. *Nutrients* 11(11):2661. doi:10.3390/nu11112661.
- Menzel, J., Abraham, K., Stangl, G.I., Ueland, P.M., Obeid, R., Schulze, M.B., Herter-Aeberli, I., Schwerdtle, T., & Weikert, C. (2021). Vegan Diet and Bone Health—Results from the Cross-Sectional RBVD Study. *Nutrients* 13, 685. <https://doi.org/10.3390/nu13020685>.
- Molloy, A.M., Kirke, P.N., Brody, L.C., Scott, J.M., & Mills, J.L. (2008). Effects of folate and vitamin B12 deficiencies during pregnancy on fetal, infant, and child development. *Food and Nutrition Bulletin* 29 (2) Supplement: S101-106. ; discussion S112-5. doi: 10.1177/15648265080292S114. PMID: 18709885.
- Naik, S., Bhide, V., Babhulkar, A., Mahalle, N., Parab, S., Thakre, R., & Kulkarni, M. (2013). Daily milk intake improves vitamin B-12 status in young vegetarian Indians: An intervention trial. *Nutrition Journal* 12: 136. <https://doi.org/10.1186/1475-2891-12-136>.
- National Coordinating Committee on Food and Nutrition (2017). *Recommended Nutrient Intake for Malaysians*. Ministry of Health, Putrajaya, Malaysia.
- National Coordinating Committee on Food and Nutrition (2020). *Malaysian Dietary Guidelines 2020*. Ministry of Health, Putrajaya, Malaysia.
- National Health & Medical Research Council (NHMRC) (2003). *Dietary Guidelines for Children and Adolescents in Australia*. Commonwealth of Australia, Australia.
- Norimah, A.K., Safiah, M., Jamal, K., Siti, Haslinda., Zuhaida, H., Rohida, S., Fatimah, S., Siti Norazlin., Poh, B.K., Kandiah, M., Zalilah, M.S., Wan Manan, W.M., Fatimah, S., & Azmi, M.Y. (2008). Food Consumption Patterns: Findings from the Malaysian Adult Nutrition Survey (MANS). *Malaysian Journal of Nutrition* 14 (1): 25 – 39. Epub 2008 Mar 15. PMID: 22691762.
- Pawlak, R., Parrott, S.J., Raj, S., Cullum-Dugan, D., & Lucas, D. (2013). How prevalent is vitamin B12 deficiency among vegetarians? *Nutrition Reviews* 71 (2): 110-117. doi: 10.1111/nure.12001. Epub 2013 Jan.
- Piccoli, G.B., Clari, R., Vigotti, F.N., Leone, F., Attini, R., Cabiddu, G., Mauro, G., Castelluccia, N., Colombi, N., Capizzi, I., Pani, A., Todros, T., & Avagnina, P. (2015). Vegan–vegetarian diets in pregnancy: Danger or panacea? A systematic narrative review. *BJOG: An International Journal of Obstetrics & Gynaecology* 122: 623–633. doi: 10.1111/1471-0528.13280. Epub 2015 Jan 20.
- Ratajczak, A.E., Zawada, A., Rychter, A.M., Dobrowolska, A., & Krela-Ka 'zmierczak, I. (2021). Milk and Dairy Products: Good or Bad for Human Bone? Practical Dietary Recommendations for the Prevention and Management of Osteoporosis. *Nutrients* 2021, 13, 1329. <https://doi.org/10.3390/nu13041329>.

- Rozenberg, S., Body, J.J., Bruyère, O., Bergmann, P., Brandi, M.L., Cooper, C., Devogelaer, J.P., Gielen, E., Goemaere, S., Kaufman, J.M., Rizzoli, R., & Reginster, J.Y. (2016). Effects of dairy products consumption on health: benefits and beliefs-a commentary from the Belgian Bone Club and the European Society for clinical and economic aspects of osteoporosis, osteoarthritis and musculoskeletal diseases. *Calcified Tissue International* 98: 1-17. doi: 10.1007/s00223-015-0062-x.
- Sethi, S., Tyagi, S.K., & Anurag, R.K. (2016). Plant-based milk alternatives an emerging segment of functional beverages: A review. *Journal of Food Science Technology* 53 (9): 3408-3423. doi: 10.1007/s13197-016-2328-3. Epub 2016 Sep 2.
- Tucker, K.L. (2014). Vegetarian diets and bone status. *The American Journal of Clinical Nutrition* 100 (Suppl): 329S–335S. doi: 10.3945/ajcn.113.071621. Epub 2014 Jun 4.
- US Department of Health and Human Services and US Department of Agriculture (2005). *Dietary Guidelines for Americans* (2005). 6th Ed. Washington, DC: US Government Printing Office.
- Wang, Y.F., Chiu, J.S., Chuang, M.H., Chiu, J.E., & Lin, C.L. (2008). Bone mineral density of vegetarian and non-vegetarian adults in Taiwan. *Asia Pacific Journal of Clinical Nutrition* 17 (1): 101-106.
- Weaver, C.M., & Plawecki, K.L. (1994). Dietary calcium: Adequacy of a vegetarian diet. *The American Journal of Clinical Nutrition* 59: 1238S-1241S. doi: 10.1093/ajcn/59.5.1238S.
- KM5** Weaver, C.M. (2009). Should dairy be recommended as part of a healthy vegetarian diet? Point. *The American Journal of Clinical Nutrition* 89 (Suppl): 1634S-1637S. doi: 10.3945/ajcn.2009.26736O.
- Weikert, C., Trefflich, I., Menzel, J., Obeid, R., Longree, A., Dierkes, J., Meyer, K., Herter-Aeberli, I., Mai, K., Stangl, G.I., Müller, S.M., Schwerdtle, T., Lampen, A., & Abraham, K. (2020). Vitamin and Mineral Status in a Vegan Diet. *Dtsch Arztebl Int* Aug 31;117(35-36):575-582. doi: 10.3238/arztebl.2020.0575.
- Wong, Y.S., Shalini, S., Farah Liana, M.R., Wan Nurul Amera, W.M.S., Mohd Naquiddin, M.Y., Norhazlina, A.W., & Zaleha, M.I. (2013). Nutrient intake pattern of vegetarians and non-vegetarians in Kuala Lumpur, Malaysia. *Pakistan Journal of Nutrition* 12 (4): 371-376. doi: 10.3923/pjn.2013.371.376.
- Yen, C.E., Yen, C.H., Huang, M.C., Cheng, C.H., & Huang, Y.C. (2008). Dietary intake and nutritional status of vegetarian and omnivorous preschool children and their parents in Taiwan. *Nutrition Research* 28 (7): 430-436. <https://doi.org/10.1016/j.nutres.2008.03.012>.



Key Message 6

Include appropriate amounts and types of fat in the diet

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6.1 Terminology

Cholesterol

Cholesterol is a waxy, fat-like substance found in all animals including people. Cholesterol is an essential part of cells in the body and is used to make certain hormones and digest fats. The primary source of cholesterol in the body is the liver (blood or serum cholesterol), while dietary cholesterol which comes from foods of animal origin provides a much smaller amount by comparison (Whitney & Sharon, 2005).

Essential fatty acids

Essential fatty acids (EFAs) refer to the polyunsaturated omega-6 fatty acid, linoleic acid (C18:2) and the polyunsaturated omega-3 fatty acid, alpha-linolenic acid (C18:3). These fatty acids

cannot be synthesised by the human body and must therefore be provided by the diet. "Omega" in fatty acid nomenclature refers to the position of the first carbon-carbon double bond counting from the methyl (CH₃) end of the molecule (Vaclavik & Christian, 2014).

Hydrogenation

Hydrogenation process is used to treat or combine an unsaturated (fatty acid) compound with hydrogen, chemically. The chemical structure of the unsaturated fatty acids after the partial hydrogenation process turns them into trans-fatty acids. Liquid vegetable oils are often hydrogenated to turn them into solids (Gupta, 2017).

Saturated fatty acids

Saturated fatty acids (SFAs) from dietary fats and oils do not carry any carbon-carbon double bonds in the fatty acid chain. SFAs common to the human diet have 12 to 18 carbons, namely lauric acid (C12:0), myristic acid (C14:0), palmitic acid (C16:0) and stearic acid (C18:0). SFAs of 12 to 16 carbons are known to raise blood cholesterol levels, whilst stearic acid and SFAs less than 12 carbons are considered neutral. Saturated fat is found in significant amounts in animal foods, such as the visible fat on meat and chicken, butter, cream and full fat dairy products, and in plant foods like coconut oil (DeBruyne et al., 2012; Vaclavik & Christian, 2014).

Trans fatty acids

Trans fatty acids (TFAs) are produced industrially through partial hydrogenation of liquid plant oils to become solid fats. Elaidic acid (C18:1 trans-9), a common industrial TFA is produced during formulation of margarines, vanaspati (vegetable ghee), shortenings and bakery products. Naturally occurring TFAs also occur as vaccenic acid (C18:1 trans-11) in the stomach of dairy cattle.

Unsaturated fatty acids

There are two types of unsaturated fatty acids, namely monounsaturated fatty acids (MUFA) and polyunsaturated fatty acids (PUFA). The predominant MUFA in the diet is oleic acid (C18:1) whilst the PUFA consists of two groups, the omega-3 fatty acid [e.g., alpha-linolenic acid (ALA, C18:3), eicosapentaenoic acid (EPA, C20:5) and docosahexaenoic acid (DHA, C22:6)] and the omega-6 fatty acid [e.g., linoleic acid (LA, C18:2) and arachidonic acid (AA, C20:4)]. LA and ALA are known as essential fatty acids (EFA) as they cannot be synthesised by humans and therefore must be included in the diet.

Vanaspati

Vanaspati is a vegetable substitute for ghee and is produced by hydrogenation of vegetable oils such as soya bean, sunflower, cottonseed, canola and palm oil. It is mainly produced in India, Iran, Pakistan and some other Arabic and African countries. Other names of *vanaspati* include vegetable ghee, culinary hydrogenated vegetable oil and grainy fat (Farmani et al., 2007; Downs et al., 2013). *Vanaspati* is generally used as an all-purpose culinary fat in frying, baking and cooking applications (Labbe et al., 2009). *Vanaspati* contains high amounts of TFAs, and are therefore associated with increased risk of coronary heart disease in adults (Farmani & Golitabar, 2015).

6.2 Introduction

Fats supply energy and essential fatty acids and serve as a carrier for the absorption of the fat-soluble vitamins A, D, E and K and carotenoids. Fats serve as building blocks of membranes and play a key regulatory role in numerous biological functions such as maintaining a regular heartbeat, providing an anti-inflammatory function, regulating cholesterol and contributing to brain and eye development in developing babies. The optimum amount of fat in our diet depends on our stage of life. The recommended total fat intake is between 25% and 30% of calories for both vegetarian and non-vegetarian adults (NCCFN, 2017).

Fats (solid at 25°C) and oils (liquid at room temperature) are also important for vegetarians. They perform the vital functions as outlined above. For vegetarian diets, sources of fat mainly are from plants such as fruits, seeds, nuts and vegetables. These include pumpkin seeds, sunflower seeds, flax seeds, walnuts, almonds, chickpeas, kidney beans, tofu, tempeh, soya milk, potatoes, avocados, mangoes, chikoo, virgin olive oil, canola oil, flaxseed oil, rapeseed oil, soya oil, sesame oil and peanut oil. Table 6.1 shows the energy and types of fat content for 1 tablespoon of selected oils in Malaysia.

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Table 6.1: Energy and types of fat content for 1 tablespoon of selected edible oils

Type of oils	Energy (kcal)	Fat (g)	Sat (g)	PUFA (g)	MUFA (g)
Corns (n=3)	124	13.9	2.10	7.6	4.0
Canola (n=4)	125	14.0	1.60	3.9	7.2
Soybean (n=2)	128	14.0	2.10	8.2	3.2
Sunflower (n=5)	124	13.9	1.60	7.6	4.3
Canola + Corns (n=1)	123	13.6	1.30	5.0	7.3
Canola + Soybean (n=1)	130	14.0	1.50	6.0	5.0
Canola + Soya + Olein (n=1)	122	14.0	4.60	4.6	4.6
Canola + red palm fruit	122	14.0	2.10	3.8	8.1
Canola + Sunflower (n=6)	126	14.0	1.40	5.1	7.2
Sunflower + palm olein (n=1)	126	14.0	1.20	12.7	1.2
Peanuts (n=1)	122	13.50	5.90	1.5	6.1
Sesame (Chinese) (n=1)	114	12.0	1.80	4.8	4.8
Palm olein (n=6)	125	14.0	6.30	1.8	6.3
Palm olein + peanut + sesame (n=2)	127	14.0	6.10	1.9	6.0
Mustard seed oil (n=1)	130	14.0	0.74	3.2	10.5
Sesame seeds & Jaggery (n=1)	120	14.0	2.00	5.4	4.4
Olive oil (original, extra virgin/ organic) (n=4)	124	13.8	2.00	1.0	10.9
Rice bran (n=1)	125	13.7	3.30	4.7	5.8

Source: Nutrition Division, Ministry of Health, unpublished data, 2015

6.3 Scientific Basis

Fats and oils are an essential part of a diet, however the type and total amount of fat consumed makes a difference to health. High intake of saturated fats and cholesterol increases the risk of unhealthy blood lipid levels, which, in turn, may increase the risk of coronary heart disease (CHD). Whereas, a low intake of fats and oils increases the risk of inadequate intakes of vitamin E and essential fatty acids and

may contribute to unfavourable changes in high-density lipoprotein (HDL) blood cholesterol and triglycerides. Compared to non-vegetarian diets, vegetarian diets tend to be lower in total fat, saturated fat, monounsaturated fat, cholesterol and α -linolenic acid (ALA) (an omega-3 fatty acid) (Rizzo et al, 2013; Wong et al, 2013; Elorinne et al, 2016; Gan et al, 2018; Ching, Y.K, Chin et al, 2019).

6.3.1 Total fat

A recent finding from the Malaysian Adults Nutrition Survey (MANS) 2015 concluded that total fat intake of Malaysian adults was 29% with meat consumption constituted in the top ten most consumed foods (MANS, 2015). So, it may be reasonable to expect that omitting meat from the diet could significantly reduce the total fat and saturated fatty acid in the diet. The Recommended Nutrient Intake (RNI) for fat intake for Malaysians is between 25% to 30% of total energy intake (NCCFN, 2017).

While a dietary recommendation on reducing total dietary fat intake, specifically to 30% of total energy was intended to address mortality from cardiovascular diseases, this evidence was not fully supported by a number of meta-analysis that have reviewed randomised controlled trials (RCTs) and epidemiological studies (Hooper et al., 2012; Harcombe et al., 2016; Zhu et al., 2019). However, these meta-analyses were not conducted among vegetarians per se and moreover, the role of specific fatty acids was suggested as more important in the development of cardiovascular diseases as compared to total dietary fat intakes.

6.3.2 Saturated fatty acids

SFAs are mainly found in animal products such as butter, cream and hard cheese and also in palm oil and coconut products. In Malaysia, palm oil is extensively used as vegetable oil in home food preparation and in the production of local biscuits, pastries and confectioneries.

The percentage of energy from saturated fat in both European male and female meat-eaters were found to be higher than fish-eaters and vegetarians, while the lowest SFA intake was reported in vegans (Bradbury et al., 2014). Another study reported that the SFA intake of lacto-ovo vegetarians (7.7 g) was lower than omnivorous (12.3 g) (Vinagre et al., 2014).

Similar finding was also reported by Al-Ma'aitah & Tayyem (2021), where the lacto-ovo vegetarians Jordanian adults consumed less amount of SFA compared to non-vegetarians. Furthermore, a cross-sectional study reported that the mean SFA intake of vegetarians adults was 7 g less than the non-vegetarian (Bowman, 2020). In Adventist Health Study-2 cohort, vegans and lacto-ovo vegetarians had a lower proportion of saturated fatty acids when compared to non-vegetarians (Miles et al., 2019).

SFAs have harmful effects on health such as weight gain, which is associated with diabetes, and raising blood cholesterol levels. A pooled analysis of 13 epidemiologic studies showed a significant relationship between dietary SFA intake and increased risk of coronary events and coronary mortality in middle-aged and older men and women (Micha & Mozaffarian, 2010). It is thus advisable to limit SFA consumption to not greater than 10% of total calories to prevent the risk of raising blood cholesterol level. A meta-analyses of prospective cohort studies reported pooled relative risk estimates comparing extremes of intake of SFA of 1.07 (95% confidence interval 0.96 to 1.19; P=0.22) for CHD, 0.81 (0.62 to 1.05; P=0.11) for stroke and 1.00 (0.89 to 1.11; P=0.95) for cardiovascular diseases (CVD) (Siri-Tarino et al., 2010). Modest cardiovascular benefits of reducing intake of SFA while increasing intake of PUFA were found in several intervention trials, but most trials lasted only up to two years and examined surrogate outcomes (Mozaffarian et al., 2010). A meta-analysis of randomized trials suggested a 17% reduction in risk of CVD in studies that reduced SFA intake from about 17% to about 9% of energy (0.83, 0.72 to 0.96) (Hooper et al., 2015). While the latest meta-analysis of observational studies indicated lack of correlation between SFA and all-cause mortality, the evidence is rather heterogeneous with a few methodological limitations. Hence, guidelines for saturated fat intake must carefully consider the health effects of alternative macronutrients suggested as a replacement (Souza et al., 2015).

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Table 6.2: Maximum daily amounts of saturated fat to keep saturated fat below 10% of total calorie intake

Total calorie intake (kcal)	Limit on saturated fat intake (g)
1,500	17 g or less
2,000	22 g or less
2,500	28 g or less
2,800	31 g or less

Table 6.2 shows the maximum amounts of SFA (grams) to be consumed to keep its intake below 10% of total calorie intake, for selected calorie levels. This table may be useful when combined with label-reading guidance.

6.3.3 Cholesterol

Similar to saturated fats, dietary cholesterol also has an ability to alter unfavourable blood lipid levels particularly serum cholesterol and low-density lipoprotein cholesterol (LDL-C). A meta-analysis study reported that a reduction of over 20% in total serum cholesterol concentration can result in a 25% fall in mortality from CHD (Hooper et al., 2012). A longitudinal study of 34670 Swedish women found that greater dietary cholesterol intakes were positively associated with higher risk of cerebral infarction (RR= 1.29; 95% CI: 1.05, 1.58) (Larsson et al., 2012). The findings gathered from this review suggested that in dietary modification, this degree of cholesterol lowering is probably beneficial in reducing cardiovascular risks.

Nevertheless, a recent meta-analysis by Berger et al. in 2015 suggested that a higher intake of dietary cholesterol was not associated with cardiovascular diseases. However, this meta-analysis was limited by lack of evidence from observational studies, specifically those of cohort findings. Evidence from intervention trials, on the other hand, indicated an effect of greater dietary cholesterol intake (≤ 900 mg) on total cholesterol and LDL-C (Berger et al., 2015).

Nonetheless, vegetarian diets are typically lower in cholesterol compared with omnivorous diets. For example, three cohort studies that comprised large numbers of vegetarian participants in the US and Europe reported a lower cholesterol intake in vegetarians than in non-vegetarians (Appleby & Key, 2016). Similarly, a lower level of cholesterol intake was observed in vegetarians compared to non-vegetarians in two studies conducted in the US and Malaysia, respectively (Wong et al., 2013; Jaceldo-Siegl et al., 2017). Furthermore, vegans consumed 27 mg of cholesterol compared to 297mg of cholesterol by non-vegetarians (Wong et al., 2013).

6.3.4 Trans fatty acids

To date, there are only limited number of evidence have suggested that vegetarians or vegans had a lower intake of trans fatty acids (TFA) compared to those of non-vegetarians (Lee et al., 2000; Rizzo et al., 2013; Chamorro et al., 2020). For instance, a study found that vegetarians had a markedly lower serum TFA level than non-vegetarians (0.03% compared with 0.50% of total TFA, respectively) (Lee et al., 2000). A greater level of TFA in non-vegetarians was mainly due to the frequent consumption of TFA-containing foods, particularly processed food including commercial baked goods, shortening, microwave popcorn, frozen pizza and stick margarine (Lee et al., 2000).

Greater intakes of TFA were linked to CHD risk in many epidemiological studies including the Nurses' Health Study (Curtis et al., 2016), The Zutphen study (Oomen et al., 2001), the Health Professionals Follow-up study (Mozaffarian et al., 2006), the Alpha-Tocopherol Beta-Carotene Trial (Simila et al., 2013) as well as in a review of prospective observational studies (Teegala et al., 2009). A 23% overall increased risk of CHD for each additional 2% energy intake from a total of TFA was reported when all other available evidence were combined (Mozaffarian et al., 2006).

Due to the adverse health effects, WHO recommends less than 1% of TFA consumption of total energy intakes (WHO, 2018). However, the levels of TFAs in Malaysian foods are somewhat low. This is because most Malaysian food items are produced using palm oil rather than hydrogenated PUFA which are commonly used abroad (Filippou et al., 2014). Currently, TFA labelling is not mandatory in Malaysia.

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6.3.5 Essential fatty acids

The sources of omega-3 fatty acids include ALA and marine-derived omega-3 fatty acids eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA). ALA, the starter EFAs, generates an even longer chain EPA and DHA (omega-3) while linoleic acid generates arachidonic acid (AA) (omega-6). EPA is the precursor of the series 3 eicosanoids which oppose the negative effects (vasoconstriction, fever, pain and inflammation) of the series 2 eicosanoids derived from AA (Calder, 2012; Harris, 2014). These EFAs are the omega-6 linoleic acid (LA) and the omega-3 alpha linolenic acid (ALA), which are required for cell membrane structure and function and for the synthesis of “local hormones” called eicosanoids (Tony Ng et al, 2019). Thus, EPA-derived eicosanoids are ‘good’ while those derived from AA are ‘bad’.

The biological importance of omega-6 fatty acids relates to growth (Semba et al, 2017), fertility and strengthening of skin and red blood cell structure, whereas omega-3 fatty acids are important for cell structure and function of the retina and nervous system. A deficiency causes poor wound healing, impaired vision and hearing, liver and kidney abnormalities and early death (Calder, 2012; Silva et al, 2018). Skin problems are the consequences of adult weight loss diets that are low in fat or in young infants and children who are not consuming enough fat. All these connect to the human body’s specific requirement for linoleic acid or ALA, and hence regarded as EFAs (Harris, 2014).

Diets that do not include fish and eggs generally lack omega-3 fatty acids, EPA and DHA, which are important for cardiovascular health and brain function (Matsumoto et al, 2019). Low dietary intakes of omega-3 fatty acids (EPA and DHA) have been observed in vegetarians, and typically none in vegans (Saunders et al, 2013). Compared with non-vegetarians, vegetarians, particularly vegans tend to have lower plasma levels of EPA and DHA (Saunders et al, 2013). Vegetarians can obtain omega-3 fatty acids from plant sources like seeds (flax, chia, camelina, canola and hemp), walnuts and their oils, as well as microalgae-based DHA supplements (Saunders et al, 2013). The omega-3 fatty acid recommendation to achieve nutritional adequacy, is 0.6%-1.2% of energy for ALA; and up to 10% of this can be provided by EPA or DHA. The evidence supports a dietary recommendation of ~500 mg/day of EPA and DHA for CVD risk reduction.

A large body of evidence from epidemiological and controlled clinical studies including the well-known Spain Multicenter Randomized Controlled Trial (Estruch et al, 2013), Nurses’ Health Study (Curtis et al, 2016), Japan Public Health Center-Based Study Cohort I (Hamasaki et al 2018), Diet and Reinfarction Trial (DART) (Burr L, 2007) and Japan EPA Lipid Intervention Study (Sasaki et al, 2012), demonstrated the protective effect of ALA, EPA and DHA and fish intake [unsaturated fats (MUFA and PUFA)] reduced the risk of CVD in healthy subjects and hypercholesterolemic patients with heart disease.

In a meta-analysis of randomized controlled trials (RCTs) and prospective cohort studies on EPA and DHA intake and CHD risk, a beneficial relationship between EPA and DHA, and CHD risk was observed, with a 6% reduced risk of any CHD event among RCTs (although not statistically significant) and a significant 18% reduced risk was found among the prospective cohort studies (Alexander et al, 2017). Innes & Calder (2018) conducted a systematic review involving 18 papers on the differential effects of EPA and DHA on cardiometabolic risk factors revealed that both EPA and DHA are effective at lowering triglycerides level and inhibiting platelet function, but had no effect on total cholesterol and fibrinolytic function; DHA increases the cardioprotective HDL2 subfraction and lowering heart rate and blood pressure in normotensive individuals.

Evidence from prospective cohort studies and secondary prevention trials suggest that omega-3 and omega-6 should be consumed in appropriate proportions for one’s cardiovascular health. Any imbalances between these fatty acids can cancel out the benefits of omega-3 fatty acids (Shahidi & Ambigaipalan, 2018). Excessive amounts of omega-6 PUFAs and a very high omega-6/omega-3 PUFAs ratio, found in typical Western diets, promote the development of CVD, cancer, inflammatory and autoimmune diseases; whereas increased levels of omega-3 PUFAs (lower omega-6/omega-3 ratio), exert beneficial effects. A ratio of 2-5/1 (omega-6/omega-3 PUFAs) is recommended. A lower ratio of omega-6/omega-3 PUFAs is more desirable in reducing the risk of many chronic diseases (Shahidi & Ambigaipalan, 2018). A typical diet which provides proportionately high amounts of unsaturated fats with balanced omega-6/omega-3 PUFAs ratio is the Mediterranean diet, commonly consumed by people in Italy, Spain, Portugal, Greece, Croatia and Cyprus. It contains olives, legumes, unrefined cereals and moderate amounts of fish (Deckelbaum & Torrejon, 2012; Estruch et al, 2013).

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6.4 Current Status

Vegetarian diets are in general lower in total fat, SFA, and cholesterol than are non-vegetarian diets. With respect to EFAs intake and balance, vegetarian diets appear to offer no advantages over omnivorous dietary patterns. Vegetarian, especially vegan diets are relatively low in ALA compared with linoleic acid (LA) and provide little, if any, EPA and DHA (Burns-Whitmore, 2019). In view of these findings, guidance needs to be given to vegetarians in obtaining the appropriate amounts and types of fat in their diet.

A cross-sectional comparison between three types of diet practices was carried out in Kuala Lumpur in 2013, with 35 respondents in each group (non-vegetarian, lacto-ovo-vegetarian and strict vegetarian). Despite the finding that the total energy intakes of the three dietary groups were similar, the non-vegetarians derived a significantly higher proportion of their energy from total fat (31%) compared to vegetarians (24%-25%) (Wong et al, 2013). In this study, vegetarians showed a healthy contribution of fat percentage in their daily consumption that was recommended by RNI for Malaysia (25%-30% for fat) compared to non-vegetarians. This was similarly reported by Gan et al. (2018) whereby a greater number of vegetarians (76%) consumed less or equal than 30% compared to vegetarians (62%) (Gan et al, 2018). In line with that, another small-scale cross-sectional study in 2016 among vegetarian adults in Kuala Lumpur reported a total fat intake of 22% (Ho, 2016). Moreover, similar findings were also reported by studies by Lin (2008) and Deriemaeker et al. (2010).

Similarly, findings from two cross-sectional studies among American, Canadian and Belgium adults concurred to those studies conducted in Kuala Lumpur. For example, the Adventist-Health-Study-2 conducted among American and Canadian participants reported that mean energy intake of total animal derived fats was highest in non-vegetarians (35%) compared to semi-vegetarian (33%), pesco-vegetarian (33%), lacto-ovo-vegetarian (33%) and vegans (30%) (Rizzo et al, 2013).

Studies comparing the diet quality of vegetarians and non-vegetarians in Belgium found that vegans (2,383 kcal, 25% fat) had the lower energy intake and better fat intake profile when compared to the omnivorous (2,985 kcal, 36% fat), while vegetarians, semi-vegetarians and pesco-vegetarians contain similar amounts of fat (Clarys et al., 2013; Clarys et al., 2014). Similarly, the nutrient intake profile of the dietary patterns in EPIC-Oxford cohort revealed that the energy from total fat in meat-eaters women (31%) was relatively higher than the fish-eaters (30%), vegetarians and vegan women (30%). However, similar energy from total fat (30%-31%) was observed in men of all diet groups (Sobiecki et al., 2016). Based on the results from AHS-2, it was indicated that non-vegetarians (34%) had higher total fat intake than vegetarians (31%-32%), and vegans has the lowest total fat intake of 28% (Segovia-Siapco & Sebaté, 2019). Additionally, the VeChi Diet Study conducted on German children and adolescents found that all diet groups' fat intake of between 30%-40% met the German reference for fat intake, despite the higher fat intake in omnivorous than vegetarians and vegans (Weder et al., 2019; Alexy et al., 2021). The overall total energy intake from fats among vegans reported in several studies is in accordance with the recommended fat intake in WHO guidelines (<30%) (Bakaloudi et al., 2021).

Cholesterol intake was significantly different between all the three dietary groups, with the non-vegetarians consuming the most (297 mg), followed by ovo-lacto-vegetarian (128 mg) and strict-vegetarians the least (27 mg) (Wong et al, 2013). This data indicates that Malaysian vegetarians on average consumed between 24% to less than 30% of their total energy intake from fat. However, since evidence on nationwide macronutrients intakes among vegetarians is sparse, no conclusion currently can be made on total fat and different types of fatty acids among Malaysian vegetarians.

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6.5 Key Recommendations

Key Recommendation 1

Keep total daily fat intake between 25% to 30% of total energy daily.

How to achieve:

1. Limit deep frying, shallow frying and batter (coated with flour) frying when cooking.
2. Change *nasi minyak**, *nasi beriani***, *sambal tumis****, fried rice to tomato rice and plain rice with less calories.
3. Choose a low fat vegetarian dish when eating out i.e. fresh fruits and stir fried vegetables.

**Nasi Minyak* is a dish of rice cooked with little ghee, evaporated milk, spices and pandan leaves.

***Nasi beriani* is a dish made with Indian spices, vegetables, rice, and usually some type of meat, or in some cases without any meat.

****Sambal Tumis* or stir fried chilli paste, is basically the cooked chilli paste that served as a sauce with dishes or as the actual spicy gravy of the dishes.

Key Recommendation 2

Limit the intake of foods high in saturated fats, cholesterol and trans-fatty acids (TFAs).

How to achieve:

1. Limit the use of coconut oil, coconut milk or cream, palm kernel oil, milk fat (ghee or butter) and *vanaspati* (vegetable ghee) as they are high in saturated fats.
2. Limit intake of margarines and shortenings made from hydrogenated or hardened fats.
3. Read the ingredients list on food labels so that foods made with partially hydrogenated fats and oils or hardened fats or oils e.g. bakery products can be avoided.
4. Avoid reusing cooking oil more than twice as this will increase TFAs.

Key Recommendation 3

Increase the intake of unsaturated fats [monounsaturated fatty acids (MUFA) and polyunsaturated fatty acids (PUFA)] within the recommended levels.

How to achieve:

1. Use blended oil by mixing one part palm oil with one part of omega-6 PUFA oil (1-to-1 of e.g. corn oil/ soya bean oil/ sunflower oil) for stir-frying, baking and grilling.
2. Choose food rich in omega-3 (PUFA), e.g. flaxseed and canola oil, chia, pumpkin seeds, sesame and walnut.
3. Limit the use of processed salad dressing e.g. mayonnaise or thousand island sauces.

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6.6 References

- Alexander, D. D., Miller, P. E., Van Elswyk, M. E., Kuratko, C. N., & Bylsma, L. C. (2017). A meta-analysis of randomized controlled trials and prospective cohort studies of eicosapentaenoic and docosahexaenoic long-chain omega-3 fatty acids and coronary heart disease risk. *Mayo Clinic Proceedings. Mayo Clinic*, 92(1), 15–29. <https://doi.org/10.1016/j.mayocp.2016.10.018>.
- Alexy, U., Fischer, M., Weder, S., Längler, A., Michalsen, A., Sputtek, A., & Keller, M. (2021). Nutrient intake and status of German children and adolescents consuming vegetarian, vegan or omnivore diets: Results of the VeChi Youth Study. *Nutrients*, 13(5), 1707. <https://doi.org/10.3390/nu13051707>.
- Al-Maaitah, A., Tayyem, R., Bakaloudi, D. F., Halloran, A., Rippin, H. L., Oikonomidou, A. C., Dardavesis, T. I., Williams, J., Wickramasinghe, K., Breda, J., & Chourdakis, M. (2021). Comparison of nutritional status between lacto-ovo vegetarian and non-vegetarian Jordanian adults. *Nutrition and Food Science*, 51, 3503–3521.
- Appleby, P. N., & Key, T. J. (2016). The long-term health of vegetarians and vegans. *Proceedings of the Nutrition Society*, 75, 287–293.
- Bakaloudi, D. R., Halloran, A., Rippin, H. L., Oikonomidou, A. C., Dardavesis, T. I., Williams, J., Wickramasinghe, K., Breda, J., & Chourdakis, M. (2021). Intake and adequacy of the vegan diet. A systematic review of the evidence. *Clinical Nutrition (Edinburgh, Scotland)*, 40(5), 3503–3521. <https://doi.org/10.1016/j.clnu.2020.11.035>.
- Berger, S., Raman, G., Vishwanathan, R., Jacques, P. F., & Johnson, E. J. (2015). Dietary cholesterol and cardiovascular disease: a systematic review and meta-analysis. *The American Journal of Clinical Nutrition*, 102(2), 276–294. <https://doi.org/10.3945/ajcn.114.100305>.
- Bowman, S. A. (2020). A vegetarian-style dietary pattern is associated with lower energy, saturated fat, and sodium intakes; And higher whole grains, legumes, nuts, and soy intakes by adults: National Health and nutrition examination surveys 2013-2016. *Nutrients*, 12(9), 2668. <https://doi.org/10.3390/nu12092668>.
- Bradbury, K. E., Crowe, F. L., Appleby, P. N., Schmidt, J. A., Travis, R. C., & Key, T. J. (2014). Serum concentrations of cholesterol, apolipoprotein A-I and apolipoprotein B in a total of 1694 meat-eaters, fish-eaters, vegetarians and vegans. *European Journal of Clinical Nutrition*, 68(2), 178–183. <https://doi.org/10.1038/ejcn.2013.248>.
- Burns-Whitmore, B., Froyen, E., Heskey, C., Parker, T., & San Pablo, G. (2019). Alpha-linolenic and linoleic fatty acids in the vegan diet: Do they require Dietary Reference Intake/Adequate Intake Special Consideration? *Nutrients*, 11(10), 2365. <https://doi.org/10.3390/nu11102365>.
- Burr, M. L. (2007). Secondary prevention of CHD in UK men: the Diet and Reinfarction Trial and its sequel. *The Proceedings of the Nutrition Society*, 66(1), 9–15. <https://doi.org/10.1017/S0029665107005241>.
- Calder, P. C. (2012). Mechanisms of actions of (n-3) fatty acids. *Journal of Nutrition*, 142, 592S–599S.
- Chamorro, R., Gonzalez, M. F., Aliaga, R., Gengler, V., Balladares, C., Barrera, C., Bascuñan, K. A., Bazinet, R. P., & Valenzuela, R. (2020). Diet, plasma, erythrocytes, and spermatozoa fatty acid composition changes in young vegan men. *Lipids*, 55(6), 639–648. <https://doi.org/10.1002/lipd.12265>.
- Chan, H. M. (2016). *Associations between behavioral and psychosocial factors with body weight status among Indian adult vegetarians in Kuala Lumpur*.
- Ching, Y. K., Chin, Y. S., Appukutty, M., Ramanchadran, V., Yu, C. Y., Ang, G. Y., Gan, W. Y., et al. (2019). Interaction of Dietary Linoleic Acid and α - Linolenic Acids with rs174547 in FADS1 Gene on Metabolic Syndrome Components among Vegetarians. *Nutrients*, 11(7), 1686. MDPI AG. Retrieved from <http://dx.doi.org/10.3390/nu11071686>.

KM6

Include appropriate amounts and types of fat in the diet

- Clarys, P., Deliens, T., Huybrechts, I., Deriemaeker, P., Vanaelst, B., De Keyzer, W., Hebbelinck, M., & Mullie, P. (2014). Comparison of nutritional quality of the vegan, vegetarian, semi-vegetarian, pesco-vegetarian and omnivorous diet. *Nutrients*, 6, 1318–1332.
- Clarys, P., Deriemaeker, P., Huybrechts, I., Hebbelinck, M., & Mullie, P. (2013). Dietary pattern analysis: A comparison between matched vegetarian and omnivorous subjects. *Nutrition Journal*, 12.
- Curtis, C. J., Clapp, J., Goldstein, G., & Angell, S. Y. (2016). How the Nurses' Health Study helped American take the trans-fat out. *Am J Public Health*, 106(9), 1537–1539.
- Debruyne, L. K., Pinna, K., & Whitney, E. N. (2012). Lipids. In: (8th ed) Nutrition and Diet Therapy.
- Deckelbaum, R. J., & Torrejon, C. (2012). The omega-3 fatty acid nutritional landscape: health benefits and sources. *The Journal of Nutrition*, 142(3), 587S-591S. <https://doi.org/10.3945/jn.111.148080>.
- Deriemaeker, P., Alewaeters, K., Hebbelinck, M., Lefevre, J., Philippaerts, R., & Clarys, P. (2010). Nutritional status of Flemish vegetarians compared with non-vegetarians: a matched samples study. *Nutrients*, 2(7), 770–780. <https://doi.org/10.3390/nu2070770>.
- Downs, S. M., Gupta, V., Ghosh-Jerath, S., Lock, K., Thow, A. M., & Singh, A. (2013). Reformulating partially hydrogenated vegetable oils to maximise health gains in India: is it feasible and will it meet consumer demand? *BMC Public Health*, 13(1). <https://doi.org/10.1186/1471-2458-13-1139>.
- Elorinne, A.-L., Alfthan, G., Erlund, I., Kivimäki, H., Paju, A., Salminen, I., Turpeinen, U., Voutilainen, S., & Laakso, J. (2016). Food and nutrient intake and nutritional status of Finnish vegans and non-vegetarians. *PloS One*, 11(2), e0148235. <https://doi.org/10.1371/journal.pone.0148235>.
- Estruch, R., Ros, E., Salas-Salvadó, J., Covas, M.-I., Corella, D., Arós, F., Gómez-Gracia, E., Ruiz-Gutiérrez, V., Fiol, M., Lapetra, J., Lamuela-Raventós, R. M., Serra-Majem, L., Pintó, X., Basora, J., Muñoz, M. A., Sorlí, J. V., Martínez, J. A., & Martínez-González, M. A. (2013). Primary prevention of cardiovascular disease with a Mediterranean diet. *The New England Journal of Medicine*, 368(14), 1279–1290. <https://doi.org/10.1056/nejmoa1200303>.
- Farmani, J., Hamed, M., Safari, M., & Madadlou, A. (2007). Trans-free Iranian vanaspati through enzymatic and chemical transesterification of triple blends of fully hydrogenated soybean, rapeseed and sunflower oils. *Food Chemistry*, 102(3), 827–833. <https://doi.org/10.1016/j.foodchem.2006.06.015>.
- Farmani, Jamshid, & Gholitabar, A. (2015). Characterization of vanaspati fat produced in Iran. *Journal of the American Oil Chemists' Society*, 92(5), 709–716. <https://doi.org/10.1007/s11746-015-2641-4>.
- Filippou, A., Teng, K.-T., Berry, S. E., & Sanders, T. A. B. (2014). Palmitic acid in the sn-2 position of dietary triacylglycerols does not affect insulin secretion or glucose homeostasis in healthy men and women. *European Journal of Clinical Nutrition*, 68(9), 1036–1041. <https://doi.org/10.1038/ejcn.2014.141>.
- Gan, W. Y., Boo, S., Seik, M. Y., & Khoo, H. E. (2018). Comparing the nutritional status of vegetarians and non-vegetarians from a Buddhist organisation in Kuala Lumpur, Malaysia. *Malaysian Journal of Nutrition*, 24(1), 89–101.
- Gupta, M. K. (2017). Chapter 7 - hydrogenation. In M. K. Gupta (Ed.), *Practical guide to vegetable oil processing* (pp. 171–215). AOCS Press.
- Hamazaki, K., Iso, H., Eshak, E. S., Ikehara, S., Ikeda, A., Iwasaki, M., Hamazaki, T., Tsugane, S., Tsugane, S., Sawada, N., Iwasaki, M., Sasazuki, S., Yamaji, T., Shimazu, T., Goto, A., Hidaka, A., Hanaoka, T., Ogata, J., Baba, S., ... Sato, Y. (2018). Plasma levels of n-3 fatty acids and risk of coronary heart disease among Japanese: The Japan Public Health Center-based (JPHC) study. *Atherosclerosis*, 272, 226–232. <https://doi.org/10.1016/j.atherosclerosis.2017.12.004>.

- Harcombe, Z., Baker, J. S., DiNicolantonio, J. J., Grace, F., & Davies, B. (2016). Evidence from randomised controlled trials does not support current dietary fat guidelines: a systematic review and meta-analysis. *Open Heart*, 3(2), e000409. <https://doi.org/10.1136/openhrt-2016-000409>.
- Harris, W. S. (2014). Achieving optimal n-3 fatty acid status: the vegetarian's challenge... or not. *The American Journal of Clinical Nutrition*, 100 Suppl 1(suppl_1), 449S-52S. <https://doi.org/10.3945/ajcn.113.071324>.
- Ho, M. T. (2016). *Factors associated with body weight status: Comparison between Chinese vegetarians and non-vegetarians (Unpublished BSc Thesis)*.
- Hooper, L., Martin, N., Abdelhamid, A., & Davey Smith, G. (2015). Reduction in saturated fat intake for cardiovascular disease. *Cochrane Database of Systematic Reviews*, 6, CD011737. <https://doi.org/10.1002/14651858.CD011737>.
- Hooper, L., Summerbell, C. D., Thompson, R., Sills, D., Roberts, F. G., Moore, H. J., & Davey Smith, G. (2012). Reduced or modified dietary fat for preventing cardiovascular disease. *Cochrane Database of Systematic Reviews*, 5, CD002137. <https://doi.org/10.1002/14651858.CD002137.pub3>.
- Innes, J. K., & Calder, P. C. (2018). The differential effects of eicosapentaenoic acid and docosahexaenoic acid on cardiometabolic risk factors: A systematic review. *International Journal of Molecular Sciences*, 19(2). <https://doi.org/10.3390/ijms19020532>.
- Jaceldo-Siegl, K., Lütjohann, D., Sirirat, R., Mashchak, A., Fraser, G. E., & Haddad, E. (2017). Variations in dietary intake and plasma concentrations of plant sterols across plant-based diets among North American adults. *Molecular Nutrition & Food Research*, 61(8). <https://doi.org/10.1002/mnfr.201600828>.
- Labbe, M. R., Stender, S., Skeaff, M., & Tavella, M. (2009). Approaches to removing trans fats from the food supply in industrialized and developing countries. *Eur J Clin Nutr*, 63, S50–S67.
- Larsson, S. C., Virtamo, J., & Wolk, A. (2012). Dietary fats and dietary cholesterol and risk of stroke in women. *Atherosclerosis*, 221(1), 282–286. <https://doi.org/10.1016/j.atherosclerosis.2011.12.043>.
- Malaysian Adult Nutrition Survey (MANS) (2015). *Dietary Intake of Adults Aged 18 to 59 years*. Ministry of Health Malaysia, Kuala Lumpur.
- Matsumoto, S., Beeson, W. L., Shavlik, D. J., Siapco, G., Jaceldo-Siegl, K., Fraser, G., & Knutsen, S. F. (2019). Association between vegetarian diets and cardiovascular risk factors in non-Hispanic white participants of the Adventist Health Study-2. *J Nutr Sci*, 8(e6). <https://doi.org/10.1017/jns.2019.1>.
- Mensink, R. P., & Katan, M. B. (1990). Effect of dietary trans fatty acids on high-density and low-density lipoprotein cholesterol levels in healthy subjects. *The New England Journal of Medicine*, 323(7), 439–445. <https://doi.org/10.1056/NEJM199008163230703>.
- Micha R Mozaffarian, D. (2010). Saturated fat and cardio-metabolic risk factors, coronary heart disease, stroke, and diabetes: A fresh look at the evidence. *Lipids*, 45(10), 893–905.
- Miles, F. L., Lloren, J. I. C., Haddad, E., Jaceldo-Siegl, K., Knutsen, S., Sabate, J., & Fraser, G. E. (2019). Plasma, urine, and adipose tissue biomarkers of dietary intake differ between vegetarian and non-vegetarian diet groups in the Adventist health study-2. *The Journal of Nutrition*, 149(4), 667–675. <https://doi.org/10.1093/jn/nxy292>.
- Mozaffarian, D., Katan, M. B., Ascherio, A., Stampfer, M. J., & Willett, W. C. (2006). Trans fatty acids and cardiovascular disease. *Obstetrical & Gynecological Survey*, 61(8), 525–526. <https://doi.org/10.1097/01.ogx.0000228706.09374.e7>.

- Mozaffarian, D., Micha, R., & Wallace, S. (2010). Effects on coronary heart disease of increasing polyunsaturated fat in place of saturated fat: a systematic review and meta-analysis of randomized controlled trials. *PLoS Medicine*, 7(3), e1000252. <https://doi.org/10.1371/journal.pmed.1000252>.
- National Coordinating Committee on Food and Nutrition (NCCFN) (1999). Malaysian Dietary Guidelines, Ministry of Health Malaysia, Kuala Lumpur.
- National Coordinating Committee on Food and Nutrition (NCCFN) (2010). *Key Message 8: Limit Intake of Foods High in Fats and Minimize Fats and Oils in Food Preparation*. In *Malaysian Dietary Guidelines*. (pp. 111-124). Ministry of Health Malaysia, Putrajaya.
- National Coordinating Committee on Food and Nutrition (NCCFN) (2017). Recommended Nutrient Intakes for Malaysia (RNI). A Report of the Technical Working Group on Nutrition Guidelines. Ministry of Health Malaysia, Putrajaya.
- Rizzo, N., Jaceldo-Siegl, K., Sabate, J., & Fraser, G. (2013). Nutrient profiles of vegetarian and non-vegetarian dietary patterns. *Journal of the Academy of Nutrition and Dietetics*, 113, 1610–1619.
- Sasaki, J., Yokoyama, M., Matsuzaki, M., Saito, Y., Origasa, H., Ishikawa, Y., Oikawa, S., Itakura, H., Hishida, H., Kita, T., Kitabatake, A., Nakaya, N., Sakata, T., Shimada, K., & Matsuzawa, Y. (2012). Relationship between coronary artery disease and non-HDL-C, and effect of highly purified EPA on the risk of coronary artery disease in hypercholesterolemic patients treated with statins: sub-analysis of the JELIS. *J Atheroscler Thromb*, 19(2), 194–204.
- Saunders, A. V., Davis, B. C., & Garg, M. L. (2013). Omega-3 polyunsaturated fatty acids and vegetarian diets. *The Medical Journal of Australia*, 199(S4), S22-6. <https://doi.org/10.5694/mja11.11507>.
- Segovia-Siapco, G., & Sabaté, J. (2019). Health and sustainability outcomes from vegetarian dietary patterns: A revisit of the EPIC-Oxford and the Adventist Health Study-2 cohorts. *European Journal of Clinical Nutrition*, 72, 60–70.
- Semba, R. D., Trehan, I., Li, X., Salem, N., Moaddel, R., Ordiz, M. I., Maleta, K. M., Kraemer, K., & Manary, M. J. (2017). Low serum omega-3 and omega-6 polyunsaturated fatty acids and other metabolites are associated with poor linear growth in young children from rural Malawi. *Am J Clin Nutr*, 106(6), 1490–1499.
- Shahid, F., & Ambigaipalan, P. (2018). Omega-3 polyunsaturated fatty acids and their health benefits. *Annu Rev Food Sci Technol*, 25(9), 345–381.
- Silva, J. R., Burger, B., Kühl, C. M. C., Candreva, T., dos Anjos, M. B. P., & Rodrigues, H. G. (2018). Wound healing and omega-6 fatty acids: From inflammation to repair. *Mediators of Inflammation*, 2018, 1–17. <https://doi.org/10.1155/2018/2503950>.
- Similä, M. E., Kontto, J. P., Männistö, S., Valsta, L. M., & Virtamo, J. (2013). Glycaemic index, carbohydrate substitution for fat and risk of CHD in men. *The British Journal of Nutrition*, 110(9), 1704–1711. <https://doi.org/10.1017/S0007114513000858>.
- Siri-Tarino, P. W., Sun, Q., Hu, F. B., & Krauss, R. M. (2010). Saturated fatty acids and risk of coronary heart disease: modulation by replacement nutrients. *Current Atherosclerosis Reports*, 12(6), 384–390. <https://doi.org/10.1007/s11883-010-0131-6>.
- Sobiecki, J. G., Appleby, P. N., Bradbury, K. E., & Key, T. J. (2016). High compliance with dietary recommendations in a cohort of meat eaters, fish eaters, vegetarians, and vegans: results from the European Prospective Investigation into Cancer and Nutrition-Oxford study. *Nutrition Research (New York, N.Y.)*, 36(5), 464–477. <https://doi.org/10.1016/j.nutres.2015.12.016>.

- Souza, D., Mente, R. J., Maroleanu, A., Cozma, A., Ha, A. I., Kishibe, V., & Anand, T. (2015). Intake of saturated and trans unsaturated fatty acids and risk of all-cause mortality, cardiovascular disease, and type 2 diabetes: systematic review and meta-analysis of observational studies. *BMJ*, 351.
- Teegala, S. M., Willett, W. C., & Mozaffarian, D. (2009). Consumption and health effects of trans fatty acids: a review. *Journal of AOAC International*, 92(5), 1250–1257. <https://doi.org/10.1093/jaoac/92.5.1250>.
- Tony Ng, K. W., Appukutty, M., Shyam, S., Voon, P. T., & Selvaduray, K. R. (2019, January 1). *Chapter 65 - Cooking Oils in Health and Sports* (D. Bagchi, S. Nair, & C. K. Sen, Eds.). ScienceDirect; Academic Press. <https://www.sciencedirect.com/science/article/abs/pii/B9780128139226000655>.
- Vaclavik, V., & Christian, E. W. (2014). Part IV - Fats in Food. In *Essentials of Food Science* (pp. 238–239).
- Vinagre, J. C., Vinagre, C. C. G., Pozzi, F. S., Zácari, C. Z., & Maranhão, R. C. (2014). Plasma kinetics of chylomicron-like emulsion and lipid transfers to high-density lipoprotein (HDL) in lacto-ovo vegetarian and in omnivorous subjects. *European Journal of Nutrition*, 53(3), 981–987. <https://doi.org/10.1007/s00394-013-0602-y>.
- Weder, S., Hoffmann, M., Becker, K., Alexy, U., & Keller, M. (2019). Energy, macronutrient intake, and anthropometrics of vegetarian, vegan, and omnivorous children (1–3 years) in Germany (VeChi Diet Study). *Nutrients*, 11(4), 832. <https://doi.org/10.3390/nu11040832>.
- Whitney, E., & Rolfes, S. (2016). *Understanding Nutrition*. Cengage Learning.
- Wong, Y. S., Shalini, S., & Liyana Mr Wan Nurul Amara, F. (2013). Nutrient intake patterns of vegetarians and non-vegetarians in Kuala Lumpur, Malaysia. *Malaysia. Pakistan Journal of Nutrition*, 12(4), 371–376.
- World Health Organization (2018). *Nutrition: Trans fat*. Who.int. Retrieved May 19, 2023, from <https://www.who.int/news-room/questions-and-answers/item/nutrition-trans-fat>.
- Zhu, Y., Bo, Y., & Liu, Y. (2019). Dietary total fat, fatty acids intake, and risk of cardiovascular disease: a dose-response meta-analysis of cohort studies. *Lipids in Health and Disease*, 18(1), 91. <https://doi.org/10.1186/s12944-019-1035-2>.

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Include appropriate amounts and types of fat in the diet



Key Message 7

Make effective use
of information on
food labels

Key Message 7

Make effective use of information on food labels

Ms. Nurul Hidayati Mohd Nasir, Assoc. Prof. Dr. Norazmir Md. Nor, Assoc. Prof. Dr. Gan Wan Ying and Mr. Ng Chee Kai

7.1 Terminology

Allergen

Allergen is an otherwise harmless substance capable of triggering a response that starts in the immune system and results in an allergic reaction in certain individuals. In the case of foods, it is a protein found in food capable of triggering a response in individuals sensitised to it (Codex Alimentarius, 2020).

Food additive

Any safe substance that is intentionally introduced into or on food in small quantities to affect the food's keeping quality, texture, consistency, appearance, odour, taste, alkalinity, or acidity, or to serve any other technological function in the manufacture, processing, preparation, treatment, packing, packaging, transport, or storage of the food, and that

results or may be reasonably expected to result directly or indirectly in the substance or any of its by-products becoming a component of, or otherwise affecting the characteristics of, the food, and includes any preservative, colouring substance, flavouring substance, flavour enhancer, antioxidant and food conditioner, but shall not include added nutrient, incidental constituent or salt (MOH, 1985).

Front-of-pack nutrition labelling (FOP-NL)

Front-of-pack nutrition labelling (FOP-NL) is any system that presents simplified nutrition information on the front-of-pack of pre-packaged foods. It can include symbols/ graphics, text or a combination thereof, that provide information on the overall nutritional value of the food and/ or on the nutrients (FAO/WHO, 2019).

Hypersensitivity

Hypersensitivity refers to any discomfort (example: irritation, swelling, vomiting, asthma, chest pain, fainting), which may at times lead to death due to reaction of the body's immunity system (MOH, 1985).

Ingredient

Any substance, including a food additive and added nutrient, is used to manufacture or prepare food and present in the final product, although possibly in a modified form (MOH, 1985).

Nutrition information

Nutrition information on food labels includes nutrition information panel (NIP), nutrition claims and front-of-pack nutrition labelling (FOP-NL).

Pre-packaged

Food which is packaged or made up in advance in a container, ready for the offer to the consumer, or for catering purposes (Codex Alimentarius, 2018).

Processed foods

Edible parts of plants after separation from nature or modified/ preserved by minimal processes or modified with the addition of salt, sugar, oils or fats to preserve and enhance their sensory qualities. These include canned or bottled vegetables or legumes (pulses) preserved in brine; whole fruit preserved in syrup; most freshly baked breads; and simple cheeses to which salt is added (Monteiro et al, 2019a).

7.2 Introduction

7.2.1 Main components of a food label

In a rapidly developing country like Malaysia, the need for pre-packaged and processed foods is likely to increase. The food industries in the country are indeed growing at a rapid pace. Vegetarians too can rely on pre-packed foods for some of their nutrient needs when selected appropriately. Therefore, this chapter can be useful for providing information for practising vegetarians to make appropriate choices of pre-packaged foods.

To protect the public against health hazards and frauds in the preparation, sale and use of foods, food manufacturers and distributors must abide by the Malaysian Food Act 1983 and Food Regulations 1985. Food labelling is an important component of the Food Act and Food Regulations. The general rules for the labelling of pre-packaged foods are laid out in Part IV (Regulation 9-18) of the Food Regulations 1985 (MOH, 1985).

The primary objective of a food label on pre-packaged food is to inform consumers about the product's characteristics at the point of purchase. The label provides information that describes what is contained in a package to guide consumers in making appropriate choices. The main information on a food label includes the appropriate name of the product. These ingredients are used to prepare the food, food additives used, net weight or volume, expiry date and name, and address of the manufacturer. Where appropriate, food labels also provide information on food hypersensitivity in a product. Some food labels also state which country the product originated from.

In addition, food labels are required to provide information on the product's nutrient content through the nutrient information panel. Labels may also carry other nutrition information such as nutrient content, nutrient function and other functions. Such information helps consumers choose products that can form part of an overall healthy eating plan.

In addition, food labels may provide nutrition information, some of which are mandatory. Food labels are required to provide information on the product's nutrient content through the nutrient information panel. Labels may also carry other nutrition information such as nutrient content, nutrient function and other function claims. In addition, manufacturers may also include front-of-pack nutrition labelling. Such nutrition information helps consumers make informed choices of packaged products that can form part of an overall healthy eating plan.

There are no specific labelling requirements for vegetarian foods in Malaysia. Nevertheless, as mentioned above, most of the information required to be on the food label is relevant and useful in helping vegetarians choose appropriate products. This is particularly relevant when vegetarians avoid certain food groups or ingredients.

This key message explains which of the different components on a food label are particularly useful/ relevant for vegetarians and how such information can be utilised effectively in assisting the decision-making on food choices.

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7.2.2 Ingredient listing

One of the regulatory requirements of a food label is ingredient listing. Pre-packaged food products must have a list of ingredients on the packaging except for single-ingredient products. Ingredients are listed in descending order of proportions by weight. This means that the ingredients listed earlier in the list are found in greater amounts than those lower in the list.

Ingredient listing is useful to vegetarians. They can scan through the list and avoid products containing animal origin ingredients. Consumers can easily recognise these ingredients, as the law requires that the animal's common name must be given if the ingredients are of animal origin. In addition, it is also a requirement that if the product contains edible fat or oil, the common name of the animal or vegetable source of the fat or oil must be stated.

7.2.3 Listing of food additives

Malaysian consumers are becoming more aware of their dietary intakes in terms of ingredients and the food additives contained therein. Hence, besides the declaration of ingredients in pre-packaged food, the law requires that food additives used in a product

be declared on the label. Any substance added to food as an additive must have been previously evaluated and approved for safety in use and controlled under the Malaysian Food Regulations 1985. Only additives in the list approved in the Food Regulations may be used.

Starting from 2024, Malaysian Food Regulations 1985 require the International Numbering System (INS) to declare food additives on the label. This is a harmonized naming system for food additives as an alternative to the use of the specific name, which may be lengthy. INS numbers are assigned to identify each food additive. INS numbers generally correspond to E numbers for the most compound, e.g. INS102, Tartrazine, is also E102.

Consumers should be aware that some of the food additives are animal-based. The law requires that if a food additive is derived from animal origin, the common name of such animal should be stated on the label. Vegetarians can look out for such a statement and avoid purchasing products with animal-based food additives.

The following are examples of some statements that indicate the presence of animal-based food additives.

Example 1: Biscuit

Ingredients: Wheat flour, sugar, margarine, eggs, milk protein, anticaking agent (INS170(i)) (egg) and omega 3 (fish).

Example 2: Processed Cheese

Ingredients: Cheese [Milk, Salt, Starter Culture, Enzyme (Rennet)], Colors (INS160a, 101), Butter (Milk), Water, Emulsifier (470(iii)(beef), Salt, Thickener (INS1405).

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Table 7.1: List of several food additives that may not be suitable for vegetarians

Additives	Function/ Source
1. Cochineal/ Carmine	This red colouring is isolated from crushed insects; <i>Coccus cacti</i> .
2. Edible Bone Phosphate	From animal bones.
3. Bees wax	Made by bees but does not contain insects.
4. Shellac	Natural polymer derived from lac beetles.
5. Lanolin	A wax from sheep excreted by the skin of sheep and extracted from the wool.
6. Lactitol	Made from milk sugar.
7. Lysozyme	From chicken eggs.
8. Riboflavin	Yellow food colour which can be isolated from milk but can be commercially produced from micro-organisms, such as yeast.
9. Canthaxanthin	Commercially prepared from mushrooms/ flamingo feathers, mainly produced synthetically from carotene.
10. Lactic Acid & Lactates	All fermented products (dairy and non-dairy) contain this as a result of bacterial fermentation. Commercially produced by bacterial fermentation on starch and molasses.
11. Ascorbyl Palmitate	Combination of the fatty acid palmitate with ascorbic acid occurs naturally in most fruits and vegetables. Palmitic acid is produced from fat, the main source is vegetable fat but it can be obtained from animal fat.
12. Lecithin	Commercially isolated (mainly) from soya beans or egg yolk (may be made from caged hens).
13. Glycerol/ Glycerin	A natural carbohydrate alcohol. Commercially produced either synthetically from propene or by bacterial fermentation of sugars.
14. Fatty Acid Salts	Derived mainly from plant origin, but animal fats may be used.
15. Sorbitans	Produced from sorbitol and varying acids, these are normal fatty acids derived from vegetable or animal origin.
16. Sodium inosinate, Sodium salt of inosinic acid	A natural acid mainly present in animals. It is commercially prepared from meat/ fish but may also be produced by bacterial fermentation of sugars.
17. Glycines	Originally isolated from gelatine but can be prepared synthetically.

(Source: Vegetarian Society of the United Kingdom, 2016)

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7.2.4 Hypersensitivity listing

Consumers with food allergies need to have clear information about the ingredients used in the foods they buy to successfully avoid the foods that they know they are sensitive to. While numerous foods have been identified as sources of allergic reactions, 90% of the allergic reactions associated with foods are caused by one of these eight foods: milk, eggs, fish, crustacean shellfish, peanuts, tree nuts, wheat and soy (Anderson et al., 2010).

The Malaysian Food Regulations 1985 requires food industries to disclose any of the top five foods or food groups that may cause hypersensitivity as below:

1. Cereal containing gluten including wheat, rye, barley and oat
2. Nut and nut products, including peanut and soya bean
3. Fish and fish products
4. Milk and milk products (including lactose)
5. Egg and egg products

Hypersensitivity labelling is not specifically directed towards vegetarians. However, some of the hypersensitivity as listed above are of animal origin. Hence, hypersensitivity labelling indirectly assists the vegetarian consumer to scrutinise the declaration of foods that may cause hypersensitivity when purchasing pre-packaged foods. The following are examples of some hypersensitivity labelling:

Example 1: Canned Mushroom Soup

Ingredients: Water, Mushrooms, Vegetable Oil (Corn, Canola and Soybean), Thickener (Modified Corn Starch), Wheat Flour, Salt, Cream (Milk), Whey, Soy Protein Concentrate, Flavor Enhancer (INS631) (Fish), Flavor Enhancer (Yeast Extract), Dried Garlic, Natural Flavoring.

Example 2: Gummy Candy

Ingredients: Glucose Syrup (From Wheat), Sugar, Gelatin (Beef), Dextrose (Wheat), Acidity Regulator (INS330), Artificial and Natural Flavor, Palm Oil, Palm Kernel Oil, Carnauba Wax (INS903), Glazing agent (INS901), Color (INS102, INS129 & INS133).

Example 3: Premix Coffee

Ingredients: Sugar, Creamer [Sodium Caseinate (Milk Protein)], Instant Coffee, Emulsifiers (INS322(i)) (Soya).

Example 4: Ice cream

Ingredients: Cream (Milk), Nonfat Milk, Sugar, Corn Syrup, Chocolate Chips, Whey Powder, Stabiliser/Emulsifier Blend (INS471, 468 Gum, 412, 407), Natural Flavour, Colour (INS133, INS 160a(ii)).

Example 5: Tuna Sandwich Spread

Ingredients: Tuna Flakes, Crunchy Carrots, Fresh Celery, Pickle Relish, Mayonnaise (Eggs), Soya Bean Oil

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7.2.5 Nutrition information on food labels

Although not specifically for vegetarians, nutrition information on food labels is also useful in making appropriate food choices. These include nutrients declared in the nutrition information panel (NIP) and nutrition claims. It also includes front-of-pack nutrition labelling (FOP-NL) which comprises energy icon and healthier choice logo (Refer to Key Message 14 of the Malaysian Dietary Guidelines 2020). Vegetarians should effectively use such information on food labels to understand the energy and content of nutrients.

Healthy eating habits are the key to promoting nutritional wellbeing, including preventing non-communicable diseases such as diabetes, hypertension and stroke. Making healthier food choices by reading food labels enable healthful dietary practices (Christoph & An, 2018; Fernandes et al, 2016; Songet al, 2021). For example, a meta-analysis done by Shangguan et al. (2019) showed that food labeling decreased consumer intakes of energy by 6.6% (95% CI=-8.8%, -4.4%), total fat by 10.6% (95% CI=-17.7%, -3.5%), and other unhealthy dietary options by 13.0% (95% CI=-25.7%, -0.2%), while increasing vegetable consumption by 13.5% (95% CI=2.4%, 24.6%). A cross-sectional population-based study of 1817 young adults participating in the Project Eating and Activity in Teens and Young Adults-IV reported that nutrition label users consumed significantly more fruits, vegetables and whole grains and fewer sugar-sweetened beverages compared with non-users (Christoph et al, 2018). However, there are very few studies to demonstrate that reading food labels among vegetarians leads to better dietary intakes.

7.2.6 Food labelling issue faced by vegetarians

The main purpose of food labelling is to assist consumers in making appropriate food choices; however, the current system does not allow vegetarians to have full access to the information they need to make these decisions. Vegetarians found it difficult to identify foods that suited their lifestyles. Vegetarians have to read food labels to avoid animal products. When they have queries about specific food products, they must directly contact the manufacturer. Misleading labels or lack of listing all ingredients makes vegetarians more susceptible to unconsciously breaking the rules of not consuming foods of animal origin. Some issues that vegetarians usually face include using “natural ingredients”, whereby the food might be plant or animal-based sources such as lecithin (Fields, 2011).

Many of the natural flavours are derived from animal sources. It is impossible to differentiate which food comes from plants or animals without confirmation from the manufacturer. Some companies also might just put natural or artificial spices and flavourings as “spices”, “flavours”, “natural flavour” or “artificial flavours”.

The current nutrition labels and claims regulation in Malaysia was gazetted in 2003. However, there is no definition in the law of the terms “vegetarian” or “vegan”. Several vegetarian and vegan organisations in North America and Europe have introduced vegetarian and vegan icons on packaged food products and displayed them in restaurants. Many private companies have also developed these icons for their food packages. For example, an icon states “suitable for vegans” in words or as a logo on food packages. The V-Label is a recognised, registered symbol for labelling vegan and vegetarian products and services in Argentina, Austria, Denmark, Finland, France, Germany, Netherlands, Switzerland, Turkey and South Africa (European Vegetarian Union, 2021). However, it is difficult to find these vegetarian and vegan icons on most of the food packages in Malaysia as there is no officially recognised icon for food manufacturers to include on their food packages uniformly. Therefore, vegetarians should read the food labels on food products carefully before buying them to avoid animal products.

Elsewhere in the world, there has been increased interest in developing regulations directed towards improved labelling of vegetarian foods. Packaged food products sold in India must be labelled with a mandatory mark to distinguish between vegetarian and non-vegetarian foods. The Food Safety and Standards Authority of India (FSSAI), via its notification in September 2021, has notified the Food Safety and Standards (Vegan Foods) Regulations in September 2021. As per the regulation, food products or ingredients to be called VEGAN shall not have involved animal testing for evaluating the final product/ ingredient; and shall not contain any animal-derived GMOs or products prepared using an animal-derived gene for manufacturing the ingredients or products (FSSAI, 2021).

Taiwan probably has the best vegetarian labelling laws in the world. In 2009, in response to the number of vegetarians in Taiwan and the disturbing survey which found real meat (by-products) in many “vegetarian” fake meats, the government implemented a new labelling system for vegetarian foods. Vegetarian food is divided into five categories: food containing eggs, milk, or both milk and eggs, as well

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as green onions, garlic, green leeks, red leeks and white onions and those containing none of the ingredients above (Joshi, 2009). According to the labelling rules for packaged vegetarian food, all packaged foods claiming to be vegetarian should indicate the type of vegetarian ingredients. For example, “vegan” (food products without egg or dairy), “vegetarian and egg” or “vegetarian and dairy”. There are also serious fines for companies breaching this policy or being caught selling non-vegetarian food as a vegetarian.

European countries have no legal definition for ‘vegetarian’ or ‘vegan’ (Alcorta et al., 2021). The “Mandatory food labelling Non-Vegetarian/ Vegetarian/ Vegan” initiative was adopted by the European Commission in 2018. It mandated the use of pictorial labels on all food products to help vegetarians and vegans identify suitable food products and reduce ambiguity (European Union, 2018). The Vegetarian Society (British) aims to influence food producers to remove non-vegetarian ingredients such as gelatin or cheese produced using animal rennet from their products (Vegetarian Society of the United Kingdom, 2016). The Vegetarian Society sought manufacturers to become accredited and mark their food products with the Society’s trademark symbol.

7.3 Current Status

Studies on the usage of food labels were mainly conducted among consumers without any dietary restrictions. A nationwide cohort of 42,750 Thai adults found that 89.0% read food labels, of which 64.4% were frequent users (Rimpeekool et al., 2017). A cross-sectional study on Filipino adults conducted by Sy and Bullecce (2020) found that 87.7% read nutrition labels when they purchased a food product. In Israel, 59.3% of adults aged 21 and above reported that the information on food labels

often or always affects their decision to purchase a food product (Shahrabani, 2021). In Malaysia, the Malaysian Adult Nutrition Survey reported that 45% of adults read food labels (Ambak et al., 2018). Another local study conducted by Norazlan Shah et al. (2013) on 165 university students in Kuantan reported that 57.6% read the food labels before purchasing a food product. However, there are no studies on food labels among vegetarians.

7.4 Key Recommendations

Key Recommendation 1

Use information on food labels to assist in selection of vegetarian food.

How to Achieve:

1. Check the ingredient list and look out for ingredients derived from animal sources.
2. Be aware of common terms that indicate foods of animal sources.
3. Scrutinise carefully additives used in food products. Look out for additives derived from animal sources. For example, it “contains an enzyme (rennet from a cow) as permitted food conditioner” or “enzyme (rennet from a cow)”.

Key Recommendation 2

Use nutrition information on food labels for healthier choices.

How to Achieve:

1. Consider how the energy and nutrients contained in food contribute to the total intake of the body by referring to the “per serving” column in NIP and FOP-NL (if the latter is available on the label).
2. Compare the energy and nutrients on the label of the different brands of similar food by referring to the “per 100 g or 100 ml” column in NIP.
3. Opt for products with lower content of sodium (salt) and sugar by referring to the NIP and, if available on the label, nutrition claims and FOP-NL.

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7.5 References

- Alcorta A, Porta A, Tárrega A, Alvarez MD & Vaquero MP (2021). Foods for plant-based diets: challenges and innovations. *Foods* 10(2): 293.
- Ambak R, Tupang L, Hasim MH, Che Salleh N, Zulkafly N, Salleh R, Ahmad MH & Naidu BM (2018). Who do not read and understand food label in Malaysia? Findings from a population study. *Health Science Journal* 12(1): 548.
- Anderson J, Young L & Perryman S (2010). Understanding the Food Label. From: <http://www.ext.colostate.edu/pubs/foodnut/09365.pdf>
- Christoph MJ & An R (2018). Effect of nutrition labels on dietary quality among college students: a systematic review and meta-analysis. *Nutrition Reviews* 76(3):187-203.
- Christoph MJ, Larson N, Laska MN & Neumark-Sztainer D (2018). Nutrition facts panels: who uses them, what do they use, and how does use relate to dietary intake? *Journal of Academy of Nutrition and Dietetics* 118(2):217-228.
- Codex Alimentarius (2018). General Standard for the Labelling of Pre-packaged Foods. CXS 1-1985 (Rev. 2018). Food and Agriculture Organization and World Health Organization, Rome.
- Codex Alimentarius (2020). Code of Practice on Food Allergen Management for Food Business Operators. CXC 80-2020. Food and Agriculture Organization and World Health Organization, Rome.
- FAO/WHO (2019). Proposed Draft Guidelines on Front-of-Pack Nutrition Labelling – CX/FL 19/45/6. Codex Committee on Food Labelling. Joint FAO/WHO Food Standards Programme, Rome.
- Fernandes AC, Oliveira RC, Proença RPC, Curioni CC, Rodrigues VM & Fiates GMR (2016). Influence of menu labeling on food choices in real-life settings: a systematic review. *Nutrition Reviews* 74(8):534-548.
- Fields JL (2011). Vegan 101: Reading labels to avoid animal products. From: <http://jlgoesvegan.com/vegan-101-reading-labels-to-avoid-animal-products/>.
- FSSAI (2021). Draft notification of the Food Safety and Standards (Vegan Foods) Amendment Regulations, 2021. Food Safety and Standards Authority of India From: https://fssai.gov.in/upload/uploadfiles/files/Draft_Notification_FSS_Vegan_Food_08_09_2021.pdf.
- Joshi M (2009). Taiwan to enact world's strictest law on veggie food labeling. From: <http://www.topnews.in/taiwan-enact-worlds-strictest-law-veggie-food-labeling-2175647>.
- MOH (1985). Part IV Labelling. Food Regulations 1985. Ministry of Health Malaysia.
- Norazlan Shah H, Muhammad I, Hasmira MD, Mashita M, Norfazilah MR, & Fazlyla Nadya MF (2013). The use of nutrition label on food purchasing decision among university students in Kuantan, Malaysia. *Health and the Environment Journal* 4 (1): 1-10.
- European Union (2018). Commission Decision (EU) 2018/1701 on the Proposed Citizens' Initiative Entitled 'Mandatory Food Labelling Non-Vegetarian/Vegetarian/Vegan'. *Official Journal of the European Union*. From: <https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:32018D1701&from=EN>.
- European Vegetarian Union (2021). V-label. From <https://www.euroveg.eu/v-label/>.

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- Rimpeekool W, Kirk M, Yiengprugsawan V, Banwell C, Seubsman S-A & Sleigh A (2017). Nutrition label experience and consumption of transitional foods among a nationwide cohort of 42,750 Thai adults. *British Food Journal* 119(2):425-439.
- Shahrabani S (2021). Determinants of Israeli consumers' decision to use food label information more frequently: a national survey study. *Israel Journal of Health Policy Research* 10:25.
- Shangguan S, Afshin A, Shulkin M, Ma W, Marsden D, Smith J, Saheb-Kashaf M, Shi P, Micha R, Imamura F, Mozaffarian D & Food PRICE (Policy Review and Intervention Cost-Effectiveness) Project (2019). A meta-analysis of food labeling effects on consumer diet behaviors and industry practices. *American Journal of Preventive Medicine* 56(2):300-314.
- Song J, Brown MK, Tan M, MacGregor GA, Webster J, Campbell NRC, Trieu K, Mhurchu CN, Cobb LK & He FJ (2021). Impact of color-coded and warning nutrition labelling schemes: A systematic review and network meta-analysis. *PLoS Medicine* 18(10):e1003765.
- Sy DC & Bullecce ER (2020). Prevalence and factors associated with nutrition label use among selected Filipino adults. *Acta Medica Philippina* 54(5): 612-619.
- Vegetarian Society of the United Kingdom (2016). Food labelling – Vegetarian & vegan. From: <https://www.vegsoc.org/foodlabelling>.

Malaysian Dietary Guidelines For Vegetarians
Focus Group Discussion on the
Key Messages, Key Recommendations &
How To Achieve
30th June 2023

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A circular arrangement of various fruits and vegetables surrounding the central text. The items include: a whole red tomato, a sliced tomato, two strawberries (one whole, one sliced), a whole carrot, a whole chili pepper, a sliced chili pepper, a whole eggplant, a whole onion, a whole banana, a sliced banana, a whole pear, a sliced pear, a whole kiwi, a sliced kiwi, a whole cucumber, a whole lettuce head, a whole apple, a sliced apple, a whole orange, a sliced orange, a whole egg, a sliced egg, a whole lime, a sliced lime, a whole lemon, a sliced lemon, a whole grape, a sliced grape, a whole strawberry, a sliced strawberry, a whole carrot, a sliced carrot, a whole chili pepper, a sliced chili pepper, a whole eggplant, a sliced eggplant, a whole onion, a sliced onion, a whole banana, a sliced banana, a whole pear, a sliced pear, a whole kiwi, a sliced kiwi, a whole cucumber, a sliced cucumber, a whole lettuce head, a sliced lettuce head, a whole apple, a sliced apple, a whole orange, a sliced orange, a whole egg, a sliced egg, a whole lime, a sliced lime, a whole lemon, a sliced lemon, a whole grape, a sliced grape.

Malaysian Dietary Guidelines For Vegetarians
Consensus Meeting
8th February 2023

Consensus Meeting of The Malaysian Dietary Guidelines For Vegetarians (8th February 2023)

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